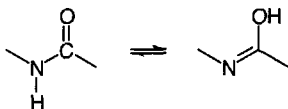


β -lactamase

four-membered β -lactam ring (see **lactam**). They include the **penicillins** and many of the **cephalosporins**.

β -lactamase any member of a group of constitutive or inducible enzymes (EC 3.5.2.6); *systematic name*: β -lactam-hydrolase. They are produced by certain strains of various bacterial species and can catalyse the hydrolysis of the **lactam** linkage in one or more β -lactam antibiotics, leaving a substituted β -amino acid; they thereby inactivate the antibiotic. The enzymes are of varying specificity, some acting more rapidly on penicillins (**penicillinases**), some more rapidly on cephalosporins (**cephalosporinases**); the latter were formerly classified separately as EC 3.5.2.8. The β -lactamases can be inhibited by various substances, such as **clavulanic acid**. There are several classes: class A (or type I) enzymes have seven motifs; class B (or type II) enzymes have a zinc cofactor. Class A example, Men-1 broad spectrum (hydrolyses penicillins and many cephalosporins) from *Escherichia coli*; database code BLM1_ECOLI, 263 amino acids (28.18 kDa). Class B example (hydrolyses carbapenems and penicillins) from *Bacteroides fragilis*; database code BLAB_BACFR, 249 amino acids (27.23 kDa).

lactam-lactim tautomerism a form of **tautomerism** occurring in lactam-lactim interconversion as a result of migration of a hydron between the nitrogen atom and the oxygen atom. It represents a special case of amide-imidol tautomerism.



lactase a trivial name for any β -D-galactosidase. Some hydrolyse certain other glycosides, including α -L-arabinosides, β -D-fucosides, and β -D-glucosides.

lactase deficiency a condition of humans in which there is a congenital or acquired deficiency of lactase (i.e. β -galactosidase) in the mucosa of the small intestine leading to failure to digest lactose and to **lactose intolerance**. The condition is rare in infants and young children but is more common in healthy adolescents and adults; its frequency varies considerably in different population groups: it is more prevalent in native populations of Australia and Oceania, east and south-east Asia, tropical Africa, and the Americas (45–95%) than in those of central and northern European ancestry (<25%).

lactate 1 α -hydroxypropionate; 2-hydroxypropanoate; the anion, $\text{CH}_3\text{-CH(OH)-COO}^-$, derived from **lactic acid**. It occurs naturally as D(-)-lactate (i.e. (R)-2-hydroxypropanoate) and L(+)-lactate (i.e. (S)-2-hydroxypropanoate). L(+)-lactate is formed by anaerobic glycolysis in animal tissues, and DL-lactate is found in sour milk, molasses, and certain fruit juices. Bacterial **lactic-acid fermentations** produce either the D(-) or L(+) enantiomer or the racemate (DL) depending on the individual organism. 2 any salt or carboxylic ester of lactic acid. 3 to secrete milk; see **lactation**.

lactate dehydrogenase any of several enzymes that interconvert **lactate** and pyruvate. L-Lactate dehydrogenase (EC 1.1.1.27; *abbr.*: LD or LDH) acts on L-lactate and NAD^+ . Example from human (heart muscle form): database code LDHH_HUMAN, 333 amino acids (36.47 kDa); this is a homotetramer, but some are heterotetramers of the heart muscle (H) and skeletal muscle (M) forms (the five isoenzymes are: LD₁, H₄; LD₂, H₃M; LD₃, H₂M₂; LD₄, HM₃; and LD₅, M₄). In clinical chemistry the assay of LD₁ (known also as α -hydroxybutyrate dehydrogenase or HBD) can be of use in diagnosing myocardial infarction, in which it reaches a peak at 2–3 days after the infarct, and in hemolytic crisis of sickle-cell anemia, since this isoenzyme is the predominant form in cardiac muscle and red cells. D-Lactate dehydrogenase (EC 1.1.1.28)

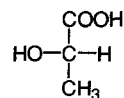
acts on D-lactate and NAD^+ . L-Lactate dehydrogenase (cytochrome) (EC 1.1.2.3) acts on L-lactate and ferricytochrome *c* and is a flavohemoprotein (FMN is coenzyme). D-Lactate dehydrogenase (cytochrome) (EC 1.1.2.4) acts on D-lactate and ferricytochrome *c* and is a flavoprotein (FAD is coenzyme). D-Lactate dehydrogenase (cytochrome *c*-553) (EC 1.1.2.5) acts on D-lactate and ferricytochrome *c*-553 and is present in *Desulfovibrio vulgaris*. These enzymes are very widely distributed, catalysing the interconversion of pyruvate and lactate.

lactation 1 the secretion of milk by the mammary gland. 2 a complete period of milk secretion from about the time of parturition to that of weaning. See also **lactate** (def. 3).

lacteal 1 of, relating to, or resembling milk; milky. 2 (of lymph vessels) containing or conveying **chyle**. 3 any blind-ended lymph vessel that carries chyle from a villus of the small intestine to the thoracic duct.

lactic pertaining to, or derived from, milk.

lactic acid α -hydroxypropionic acid; 2-hydroxypropanoic acid; a reasonably strong acid ($\text{pK}_a = 3.86$ at 25 °C). In biological systems it normally exists largely as its anion, **lactate** (def. 1). It is also useful as a preservative of foodstuffs.



lactic-acid bacterium any bacterium in which lactate is the main metabolic product. Lactic-acid bacteria belong chiefly to the genera *Lactobacillus* and *Streptococcus*, and some are of commercial importance, e.g. in the manufacture of fermented foods.

lactic-acid cycle an alternative name for **Cori cycle**.

lactic-acid fermentation a general name for **heterolactic fermentation** and/or **homolactic fermentation**.

lactic acidosis a clinical condition characterized by a persistent raised blood lactate concentration, usually above 5 mmol L⁻¹, with a lowered blood pH. The condition may be caused by severe exercise, cardiovascular insufficiency, hypoxia, shock, drug intoxication, and some inborn errors of metabolism. Compare **hyperlactatemia**.

lactide any intermolecular cyclic ester formed by self-esterification from two (or more) molecules of a hydroxy carboxylic acid. Compare **lactone**.

lactifer a special cell, found in the bark of the rubber tree (*Hevea*), in which rubber latex accumulates.

lactim any intramolecular cyclic carboximidic acid produced by the formal removal of a molecule of water between the amino and the carboxyl groups of an amino carboxylic acid other than an α -amino acid. A prefix (as in β -lactim) may be used to designate the position of the amino group in the parent compound. A lactim commonly exists in tautomeric equilibrium with its corresponding **lactam**.

lactitol 4-O- β -D-galactopyranosyl-D-glucitol; a disaccharide used as a sweetener.

lacto+ or (before a vowel) **lact+** *comb. form* indicating milk.

lactobacillic acid (1R-*cis*)-2-hexylcyclopropanedecanoic acid; 11,12-methyleneoctadecanoic acid; a lipid constituent of various microorganisms, including *Lactobacillus arabinosus*.

Lactobacillus a genus of Gram-positive, anaerobic or facultatively aerobic, straight or curved bacilli or coccobacilli. Their metabolism is fermentative and all species form lactate as the major end product of glucose fermentation. Various species of lactobacilli are important in the preparation of certain fermented foods, e.g. cheese, yoghurt, sauerkraut.

Lactobacillus bulgaricus factor an alternative name for **pan-tetheine**.

Lactobacillus casei factor an alternative name for **folic acid**.

lactobionic acid the **aldonic acid** obtained by the oxidation of **lactose**.

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lactochrome

lactochrome a former name for **riboflavin**.

lactoferrin or **lactotransferrin** an iron(III)-binding protein present in milk. Its level in human plasma may be elevated in pancreatitis.

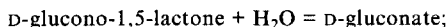
lactoflavin or **lactoflavine** a former name for **riboflavin**.

lactogen 1 any agent that stimulates the secretion of milk. 2 **placental lactogen** an alternative name for **choriomammatotropin**. —lactogenic adj.

lactogenic hormone an alternative name for **prolactin**.

β -lactoglobulin see **beta lactoglobulin**.

lactonase 1 EC 3.1.1.17; recommended name: gluconolactonase; systematic name: D-glucono-1,5-lactone lactonohydrolase; other name: aldonolactonase. An enzyme that catalyses the reaction:

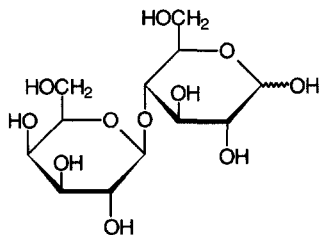


it also acts on a wide range of hexono-1,4-lactones. Example from *Zymomonas mobilis*: database code GNL_ZYMMO, 320 amino acids (34.73 kDa); homodimer. 2 **1,4-lactonase** or **γ -lactonase** EC 3.1.1.25; an enzyme that specifically hydrolyses 1,4-lactones having 4–8 carbon atoms to the corresponding 4-hydroxyacids.

lactone any intramolecular cyclic ester produced by the formal removal of a molecule of water between the hydroxyl and the carboxyl groups of a molecule of a hydroxy carboxylic acid other than an α -hydroxy acid. A prefix (as in β -lactone) may be used to designate the position of the hydroxyl group in the parent compound. Compare **lactide**.

lactoperoxidase a special name for **peroxidase** (def. 2) from milk. It also catalyses the oxidation of iodide to iodine, hence it is useful in radioiodine labelling of proteins and other biological materials under mild conditions.

lactose or (formerly) **milk sugar** the trivial name for the disaccharide 4-O- β -D-galactopyranosyl-D-glucose. It constitutes roughly 5% of the milk of almost all mammals (human milk 6.7%; cow's milk 4.5%). It occurs rarely in plants, e.g. in the anthers of *Forsythia* spp. See also **allolactose**.



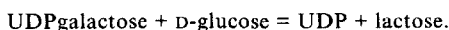
lactose fermenter jargon term for any genus, species, or strain of microorganism that ferments lactose.

lactose intolerance a syndrome occurring in humans in response to the ingestion of lactose (in milk) due to a congenital or acquired **lactase deficiency** in the mucosa of the small intestine. The failure to hydrolyse lactose leads to accumulation of the disaccharide within the intestine and consequent osmotically induced net fluid secretion into the gut, rapid intestinal transit, and acid stools due to bacterial fermentation of lactose. The condition is particularly prevalent in the Orient.

lactose operon see **lac**.

lactose permease an H^+ -lactose symporter in *Escherichia coli*. It has 12 transmembrane helices but is not very closely related to most other members of the superfamily. Database code LACY_ECOLI, 417 amino acids (46.45 kDa).

lactose synthase EC 2.4.1.22; systematic name UDPgalactose:D-glucose 4- β -D-galactotransferase; other name: UDPgalactose-glucose galactosyltransferase. An enzyme that catalyses the reaction:



In mammary gland the protein comprises *N*-acetylglucosamine

synthase, the A protein, and α -lactalbumin (see **lactalbumin**), the B protein, both of which are required for lactose synthesis.

lactose-tolerance test a procedure used in the investigation of suspected **lactase deficiency**. Blood-glucose concentrations are estimated before an oral dose of lactose (1 g per kg body mass) and thereafter at 30-min intervals for 2 h. In normal individuals an increase of at least 30 mg per 100 ml (0.167 mM) in the blood-glucose concentration above the fasting value is observed.

lactosuria the presence of lactose in the urine. It is not uncommon during late pregnancy and lactation, or if the flow of milk is prevented.

lactosyl either of the glycosyl groups formed from lactose by detaching the anomeric (α or β) hydroxyl group on C-1.

lactotransferrin an alternative name for **lactoferrin**.

lactotroph any of a class of irregular, granulated, acidophilic cells of the anterior pituitary gland that secrete prolactin. Their number increases greatly during pregnancy.

lactotrophic describing any agent that induces normal growth and development of the mammary gland and sustains it.

lactotropic describing any agent that acts on the mammary gland to induce changes in its metabolism.

lactotropic hormone an alternative name for **prolactin**.

lactotropin an alternative name for **prolactin**.

lactoyl the acyl group, $\text{CH}_3\text{-CH(OH)-CO-}$, derived from lactic acid.

lactulose the trivial name for 4-O- β -D-galactopyranosyl-D-fructofuranose; a semisynthetic disaccharide, prepared from lactose, that is sweeter than lactose but less sweet than sucrose. Concentrated solutions of lactulose are useful clinically as an osmotic laxative by virtue of the inability of enzymes of the small intestine to hydrolyse lactulose to absorbable monosaccharides. On ingestion it therefore passes unchanged into the colon, where it acts partly directly to increase the water content of the bowel by osmosis and partly indirectly through breakdown by commensal saccharolytic bacteria to simple organic acids, with resultant lowering of colonic pH and further osmotic uptake of water.

lactyl a former term for **lactoyl**.

lacY a gene in *Escherichia coli* for **lactose permease**.

L1 adhesion molecule a type I membrane glycoprotein involved in neuron-neuron adhesion, neurite fasciculation, outgrowth of neurites, etc. It is a member of the immunoglobulin superfamily, and has fibronectin domains. Example (precursor) from human: database code CAML_HUMAN, 1257 amino acids (139.77 kDa).

Laetrile see **amygdalin**.

laevo- a variant spelling (esp. Brit.) of **levo-**.

laevo+ a variant spelling (esp. Brit.) of **levo+**.

laevodopa a variant spelling (esp. Brit.) of **levodopa**.

laevorotation a variant spelling (esp. Brit.) of **levorotation**.

laevulose a variant spelling (esp. Brit.) of **levulose**.

LAF abbr. for lymphocyte-activating factor (i.e. **interleukin 1**).

lagging strand the DNA strand that is synthesized by the joining together of short **Okazaki fragments**, giving the effect of continuous 3' to 5' synthesis in the direction of fork movement during the discontinuous replication of duplex DNA. Compare **leading strand**.

lag phase 1 the initial phase of the growth of a bacterial culture during which the rate of increase of cell numbers remains static, before rising to a value determined by environmental conditions. The extent of the lag phase is determined by the cultural history of the cells of the inoculum and the chemical and physical conditions of the medium into which they are placed. 2 a period of comparatively low activity after the start of an enzymic reaction that involves enzymes that act at a surface, e.g. phospholipases. It is due to the slow formation of the complex between the enzyme and the substrate interface. Once this has been formed the reaction accelerates.

lake to lyse erythrocytes, especially by suspending them in a hypotonic medium.



Laki-Lorand factor

Laki-Lorand factor *abbr.*: LLF or L-L factor; an alternative name for factor XIII; see **blood coagulation**.

lambda *symbol*: λ (lower case) or Λ (upper case); the eleventh letter of the Greek alphabet. For uses see **Appendix A**.

lambda chain or **λ chain** *symbol*: λ ; one of the two types of **light chain** of human immunoglobulins, the other type being a kappa chain. Example, V region (4A) from human: database code LV0A_HUMAN, 117 amino acids (12.37 kDa).

lambda phage or **λ phage** a **temperate phage** that infects *Escherichia coli* and contains linear duplex DNA, which circularizes during infection; the genome is 48 514 bp. Important gene clusters include those for the head and tail genes, recombination (*att*, *int*, *xis*, α , β , γ), lysis, and regulatory genes. Of the regulatory genes (*cIII*, *N*, *cI*, *cro*, *cII*, *O*, *P*, *Q*), two are immediate-early genes, *cro* and *N*; *cro* prevents synthesis of the repressor (see **prophage**) and turns off expression of immediate-early genes, including itself, after its product, *cro*, binds to *O*. Lambda phage is an example of the antitermination mechanism of phage control (see **antiterminator**); *N* codes for an antitermination factor that allows transcription to proceed through the terminator to the delayed-early genes, *cII* and *cIII*, and *Q*, codes for an antitermination factor that allows host RNA polymerase to proceed to the late (lytic) genes. The lytic cycle is repressed by expression of *cI*, the product of which acts at the operator *O* to prevent use of the promoter *P*. Lambda phage has been widely used as a vector in cDNA technology; foreign DNA can be inserted into the chromosome, and some foreign genes can be expressed using the lambda promoters. The linear DNA has cohesive ends that, when the DNA circularizes, associate to form the **cos site** (cohesive end site), which is important in packaging. The *cos* site has been exploited in the formation of **cosmids**.

lambda pipette *colloquial term* for a pipette used for the transfer of volumes in the microlitre range.

lambda repressor or **λ repressor** the product of gene *cI*; it is responsible for maintenance of phage λ **prophage**. Its product binds to the operator associated with the RNA polymerase promoter, thus preventing RNA polymerase from initiating transcription. The phage genome then cannot enter the lytic cycle. Mutants in this gene cannot maintain **lysogeny**. Example from bacteriophage λ : database code RPC1_LAMBD, 236 amino acids (26.05 kDa); and from related phage 434: database code S32822, 210 amino acids (22.96 kDa). 3-D structures, including complexes with **operators**, are known.

Lambert-Beer law another name for **Beer-Lambert law**.

Lambert's law (of absorption) a physical law in which **Bouguer's law** was re-expressed; put into modern terms, it states that the fraction of monochromatic light (or other electromagnetic radiation) absorbed by a system is independent of the incident **radiant power**, Φ_0 . The law holds only if Φ_0 is small and other factors such as reflection, scattering, and photochemical reactions are negligible. See also **Beer-Lambert law**. [After Johann Heinrich Lambert (1728–77), German mathematician and physicist who stated it in 1760.]

lamella (*pl.* lamellae) any thin layer, membrane, or platelike structure, e.g. one of the series of double membranes that carry the photosynthetic pigments in chloroplasts, one of the thin layers of calcified bone matrix, or one of the spore-bearing gills of the fruiting bodies of mushrooms or related fungi. —**lamellar** or **lamellate** *adj.*

lamellar phase one type of **liquid-crystalline phase**.

lamin any of a group of intermediate-filament proteins that form the fibrous matrix (**nuclear lamina**) on the inner surface of the nuclear envelope. They are classified as lamins A, B, and C. Examples, all from *Mus musculus*: lamin A: database code LAMA_MOUSE, 665 amino acids (74.21 kDa); lamin B₁: database code LAM1_MOUSE, 584 amino acids (66.75 kDa); lamin C: database code LAMC_MOUSE, 574 amino acids (65.45 kDa). See also **gliadin**.

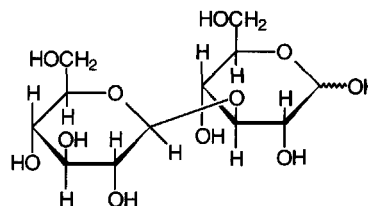
lamina (*pl.* laminae) 1 any thin plate or layer, e.g. of bone or mineral. 2 (*in botany*) a the blade of a leaf or petal. b the flat-

tened part of the thallus of certain algae. See also **basal lamina**, **nuclear lamina**. —**laminar** *adj.*

laminar flow a form of nonturbulent flow of a fluid in which parallel layers (laminae) of the fluid are moving in the same direction but with different velocities.

laminar-flow cabinet another name for **safety cabinet**.

laminaribiose the disaccharide, 3-*O*- β -D-glucopyranosyl-D-glucose; a unit of the seaweed polysaccharide **laminarin**.



pyranose form

laminarin a β -1,3-glucan that functions as a reserve polysaccharide in some brown algae (*Laminaria* spp.). It contains **laminaribiose**.

laminin a glycoprotein that is the major constituent of basement membranes. It consists of three chains, bound to each other by disulfide bonds into a cross-shaped molecule having the appearance of one long and three short arms with globules at each end (the three chains are wound together in a helical conformation for two thirds of their length, but diverge at the N-terminal end into the three arms that form the apex and side arms of a cross). The molecule interacts with procollagen type IV. Laminin mediates the attachment, migration, and organization of cells into tissues during embryonic development by interacting with other extracellular-matrix components. The precursor sequences of the three chains (all human) are: A, database code LMA_HUMAN, 3075 amino acids (336.78 kDa); B1, database code LMB1_HUMAN, 1786 amino acids (197.84 kDa); B2, database code LMB2_HUMAN, 1609 amino acids (177.41 kDa). Laminins of varying composition and sequence have been categorized as laminins 1 to 5. Laminin 5 is also known as **kalinin**.

laminin receptor any of a number of proteins, especially **integrins** (see also **VLA**) that bind laminin, interacting with it to promote cell adhesion and migration.

Lamm equation a general hydrodynamic equation describing the mass transport of a two-component system during ultracentrifugation. It is:

$$dc_2/dt = (1/x) d\{x[D(dc_2/dx) - c_2\omega^2x]\}/dx,$$

where c_2 is the concentration of solute molecules at distance x from the axis of rotation, D is the translational diffusion constant, ω is the angular velocity, and s is the sedimentation coefficient.

lampbrush chromosome any of the large diplotene chromosomes found in the oocytes of many animals. They are up to 1 mm long and are covered with loops projecting in pairs from most chromomeres; each loop has a core of DNA and is the site of active gene expression.

lamprin a self-aggregating extracellular cartilage protein of lampreys, aggregates of which form fibres rich in beta structure. It is a major structural protein isolated from the annular cartilage of the lamprey, similar in amino-acid composition to **elastin**, but it does not contain desmosine or isodesmosine. Examples (precursors) from *Petromyzon marinus*: lamprin 1.9-12/10: database code LAMP_PETMA, 139 amino acids (13.24 kDa); lamprin 1.8-10: database code LAMA_PETMA, 119 amino acids (11.32 kDa); these two are seemingly formed by alternative splicing from the same gene.



Lan

Lan symbol for lanthionine.

lan+ comb. form indicating wool. [From Latin *lana*, wool.]

Landsteiner, Karl (1868–1943), Austrian-born US pathologist and immunologist distinguished for his discoveries of the human blood groups. He designated A, B, and O (in 1901), factors M and N (in 1927), and factor Rh (in 1940); Nobel Laureate in Physiology or Medicine (1930) 'for his discovery of human blood groups'.

Landsteiner-Weiner antigen *abbr.*: LW antigen; *see* **Rhesus factor**.

lane any of a set of parallel tracks resulting from the simultaneous separation of similar samples by one-dimensional zone electrophoresis or chromatography in thin-layer media.

Langendorff perfused heart a nonworking rodent heart preparation maintained *in vitro* by perfusion into the aorta with an oxygenated fluid so that the fluid circulates primarily into the coronary arteries. It is useful for studying the metabolism of heart muscle. A number of improvements of the original preparation have been made since its introduction. [After O. Langendorff (1853–1909), German physiologist.]

Langerhans, Paul (1847–88), German physician and anatomist who first described the **islets of Langerhans**.

Lang-Levy pipette a type of constriction micropipette (5–500 μ L) that is self-adjusting and very reproducible.

Langmuir, Irving (1881–1957), US physical chemist and inventor; Nobel Laureate in Chemistry (1932), for his discoveries and investigations in surface chemistry.

Langmuir adsorption isotherm an equation describing the fraction of the surface of an adsorbent covered with a monomolecular layer of an adsorbed gas, arrived at from a consideration of the number of gas molecules striking and leaving the surface in a given time. It is:

$$V/V_m = k_1 p / (1 + k_2 p),$$

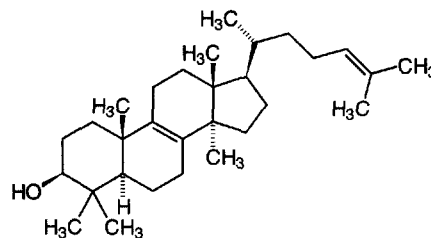
where V is the volume of gas adsorbed, V_m is the volume of gas required to cover the whole surface, p is the gas pressure, and k_1 and k_2 are constants for the system. *Compare* **Hill–Langmuir equation**.

Langmuir–Blodgett multilayer a thin film of oriented amphiphilic molecules, consisting of superposed monolayers built up by successive dipping and withdrawing of a solid substrate, e.g. a glass slide, through an insoluble monolayer of an appropriate amphiphilic substance (or mixture of such substances) spread on the surface of an aqueous solution and kept at constant pressure. [After I. Langmuir and Katharine B. Blodgett (1898–1979).]

Langmuir trough an apparatus for measuring the dependence of the surface force exerted by a film of a substance (of low vapour pressure), floating on a liquid with which it is immiscible, on the area of the film. The apparatus consists of a rectangular tank containing the subnatant liquid, e.g. water, on which a known amount of the immiscible substance, e.g. an oil or a higher fatty acid, is spread between two parallel barriers to form a film. One of the barriers is fixed and the other may be moved so as to compress the film to a predetermined area. The force exerted on the film to achieve any given area is measured by a system of levers and either a torsion wire or balancing masses.

lanolin or **wool fat** the fatlike secretion of the sebaceous glands of sheep that is deposited onto the wool fibres. It is a wax consisting of a complex mixture of esters and polyesters of over 30 high-molecular-mass aliphatic, steroid, and triterpenoid alcohols and a similar number of fatty acids. Lanolin normally contains 25–30% water and is widely used as an ointment base. *See also* **lan+**.

lanosterol *kryptosterol*; lanosta-8,24-dien-3 β -ol; a steroid alcohol (triterpene) originally obtained from **lanolin**. It occurs in the biosynthetic pathway from squalene 2,3-epoxide to cholesterol in animals and fungi, but not in photosynthetic tissues, where it is replaced by **cycloartenol**. *See also* **lanosterol synthase**.



lanosterol

lanosterol synthase EC 5.4.99.7; *other names*: 2,3-epoxysqualene-lanosterol cyclase; oxidosqualene-lanosterol cyclase (*abbr.*: OSC); an enzyme of the pathway for the synthesis of sterols. It catalyses the cyclization of (*S*)-2,3-epoxysqualene to lanosterol. Example from *Saccharomyces cerevisiae*: database code ERG7_YEAST, 730 amino acids (83.33 kDa).

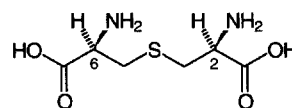
lantern *see* **firefly lantern**.

lanthanoid or **lanthanide** or (formerly) **lanthanon** any member of the series of 15 metallic elements with atomic numbers 57 (lanthanum) to 71 (lutetium) inclusive that occur together in group 3 and period 6 of the IUPAC **periodic table**; sometimes the term is restricted to the 14 elements following lanthanum. Lanthanoid is now the preferred name. Together with the elements scandium and yttrium, the lanthanoids are known also as **rare-earth elements** or (more loosely) **rare earths** (the latter term referring strictly to their oxides). Despite such names, however, they do mostly occur widely in nature, albeit in small amounts. Apart from the element of atomic number 61 (promethium), all lanthanoids have at least one stable nuclide and occur naturally, the principal source being the mixed phosphate ore monazite. All nuclides of promethium, however, are radioactive, and (except for traces possibly occurring as a fission product of natural uranium) all are artificial. The lanthanoids are strongly electropositive, and their chemical properties are similar, due usually to the filling of an inner electron subshell (4f) progressively across the series. Like the **actinoids**, they thus represent a series of inner transition elements within the series of **transition elements** embraced by groups 3 to 11 inclusive. Some paramagnetic lanthanoid ions induce strong shifts in the positions of many NMR spectral peaks and have been used as relaxation and shift probes in structural studies employing NMR spectroscopy.

lanthionine (2*S*,6*R*)-lanthionine, *meso*-lanthionine, *S*-(alanin-3-yl)-L-cysteine, 2,2'-diamino-3,3'-thiobis(propionic acid); *symbol* (recommended):



or **Lan**; a di(α -amino acid) found in **lantibiotic** polypeptides. It is formed as a post-translational modification, the first step involving dehydration of a Ser to the α,β -unsaturated amino-acid residue, 2,3-didehydroalanine. A bond is then formed between the =CH₂ group thus formed and the sulfur atom of a cysteine, inversion of configuration of the α carbon of the Ser-derived moiety occurring in the process. Chick embryo protein contains the (2*R*,6*R*) diastereoisomer. Lanthionine is also formed artefactually by the mild alkaline treatment of wool, hence its name (*see* **lan+**).



(2*S*,6*R*)-lanthionine



lantibiotic

lantibiotic any of a number of polypeptides, of 1.89–4.63 kDa, that contain **lanthionine** or methylanthionine (*threo*- β -methylanthionine, (2S,3S,6R)-3-methylanthionine) and have antibiotic action. They are produced by and primarily act on Gram-positive bacteria. In addition to lanthionines, they contain the amino acids 2,3-didehydroalanine (*abbr.*: Dha) and 2,3-didehydrobutyrine (2-aminocrotonic acid, (*Z*)-2-aminobut-2-enoic acid, *abbr.*: Dhb) resulting from dehydration of serine and threonine residues respectively. These amino acids are formed post-translationally from normally coded amino acids. Lantibiotics are broadly divided into type A, elongated amphiphilic peptides, and type B, compact globular peptides. Type-A peptides kill bacterial cells by forming pores in the cytoplasmic membrane, type-B peptides possibly acting by enzyme inhibition.

LAP (*in clinical chemistry*) *abbr.* for leucine aminopeptidase (LAS preferred).

LA-PF4 *abbr.* for low-affinity platelet factor 4; a peptide derived from **platelet basic protein precursor**. It stimulates DNA synthesis, mitosis, glycolysis, intracellular cyclic AMP accumulation, prostaglandin E₂ secretion, and synthesis of hyaluronic acid and sulfated glycosaminoglycan. *Other name*: connective-tissue activating peptide III (*abbr.*: CTAP-III).

Laplace transform method a method that enables solutions of linear differential equations with constant coefficients to be found by reference to tables of Laplace transform pairs, with a minimum of mathematical work. The Laplace transform of a function $y = F(t)$ is defined by:

$$F(s) = \int_0^{\infty} e^{-st} F(t) dt$$

in which s is usually a complex quantity and must have a positive real term, so that e^{-st} will be zero when $t = \infty$. [After Pierre Simon, Marquis de Laplace (1749–1827), French astronomer and mathematician.]

Largactil *see* **chlorpromazine**.

large calorie an alternative name for **kilocalorie** or (*in nutrition*) **Calorie**.

large-patch repair (of DNA) a name sometimes given to the process of excision repair of duplex DNA in some mutant organisms, e.g. *polA⁻ Escherichia coli*, in which the newly synthesized polynucleotide 'patch' is considerably larger than in the wild type.

large T-antigen *see* **T antigen** (def. 1).

lariat a lasso-shaped structure formed from the fragment of RNA to be excised during the process of splicing out the introns in the primary transcript of eukaryotic genes.

lariat debranching enzyme an enzyme that cleaves the 2',5'-phosphodiester bond formed during pre-mRNA splicing between the 5'-terminal guanylyl residue of the intron and an adenyl residue near the 3' splice site, at the branch point of excised intron lariats. It is involved in the turnover of excised introns. Example from yeast: database code DBR1_YEAST, 405 amino acids (47.69 kDa).

Larmor frequency the angular frequency of **Larmor precession** of an atomic electron.

Larmor precession the **precession** of a charged particle moving under the influence of a central force, e.g. of an orbiting electron in an atom, when acted on by an external magnetic field. [After Joseph Larmor (1847–1942), Irish-born physicist.]

LAS (*in clinical chemistry*) *abbr.* for leucine aminopeptidase (preferred alternative to LAP).

lasalocid or **X-537A** any of five closely related lipophilic antibiotics, lasalocid A–E, isolated from cultures of *Streptomyces lasaliensis*. Lasalocid A ($M_r = 591$) is a carboxylic ionophore for virtually every known cation, but transports monovalent cations about ten times more effectively than divalent cations.

lased light an imprecise term for laser light; *see* **laser** (def. 2).

laser 1 any device for producing an intense, coherent, monochromatic, parallel beam of light (or other electromagnetic radiation) by stimulated emission. A laser consists of an optically transparent cylinder containing an active medium.

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bounded by a full mirror at one end and by a partial mirror at the other end. Some chemical species of the medium are excited above their ground state by application of electromagnetic radiation from an external source; the excited species revert to their ground state with the emission of a photon, which in turn can collide with other excited species to produce stimulated emission of other photons. The radiation is reflected backwards and forwards up and down the cylinder and the beam emerges through the partial mirror. The frequency of the radiation is determined by the nature of the active medium; many different types of laser are available with a wide range of power outputs. 2 having the characteristic quality of, being, or using, light emitted by a **laser** (def. 1), as in laser densitometry, laser ionization (micro) mass spectrometry, laser light, laser microprobe mass analysis, laser nephelometry, laser photolysis, laser (light) scattering, laser spectrophotometry, and laser velocimetry. [Acronym from light amplification by stimulated emission of radiation.]

Latapie mincer a device for mincing tissues for metabolic studies in which both the rate of rotation of the cutting disks and the rate at which the tissue is forced against these disks may be controlled by varying the rates at which two cranks are turned, thus enabling minces of different degrees of fineness to be obtained. Sterile conditions may be maintained during the mincing if desired.

late enzyme *see* **late gene**.

late gene any gene of a bacteriophage or other viral genome that is expressed significantly, as a **late enzyme** or other **late protein**, only after replication of the genome has started. *Compare* **early gene**.

latency 1 or **crypticity** the state or quality of being **latent**. 2 an alternative term for **latent period**.

latent 1 existing in a potential, dormant, or suppressed form but usually capable of being expressed or evoked. 2 (*in pathology*) (of an infection or disease) not yet manifest or apparent.

latent enzyme any enzyme whose activity only becomes manifest when the conditions are changed. The term is used especially of a particulate enzyme that becomes active when the particles are disrupted.

latent heat *symbol*: L ; the change in the heat content of a (given amount of) substance when it changes phase without change in its temperature.

latent image (*in photography*) the invisible image produced by the action of light, etc. on the photosensitive emulsion that is made visible by development.

latent period or **latency** 1 the time between the arrival of a stimulus at an irritable tissue and the beginning of the appropriate response. 2 the time between the adsorption of virulent bacteriophage onto a bacterium and lysis with liberation of newly formed virions.

late protein *see* **late gene**.

lateral phase separation the phenomenon of coexistence of solid and liquid regions in a lipid bilayer at temperatures between the **pretransition temperature** and the **transition temperature**.

lathyrism any of several pathological states that result from ingesting any of several toxins, characteristically found in plants of the genus *Lathyrus*. In humans one kind of lathyrism is characterized by a spastic paraplegia of the lower extremities associated with degenerative changes in the lateral pyramidal tracts of the spinal cord. It is caused by ingestion of neurotoxins, e.g. β -cyano-L-alanine and L - α , γ -diaminobutyric acid. These are present in several species of plants of the genus *Lathyrus* and in the common vetch (*Vicia sativa*). In experimental animals a defect of cross-linking of collagen is caused by feeding sweet pea (*Lathyrus odoratus*) seeds; the toxic substance was first identified as β -(γ -L-glutamyl)aminopropionitrile but it was later shown that β -aminopropionitrile alone is fully active. Turkeys fed *L. odoratus* died of hemorrhage from rupture of the aorta.

lathyrigen any member of a group of substances that on ingestion induce the disease **lathyrism**.

Latin square

Latin square (*in statistics*) a square array of n columns and n rows containing n different elements and in which no element occurs more than once in any row or column. It is used as a basis of experimental procedures when it is desired to allow for two sources of variability while investigating a third.

LATS *abbr. for* long-acting thyroid stimulator.

lattice **1** any regular array of objects or points in two or three dimensions. **2** (*in crystallography*) any regular, three-dimensional array of fixed points about which atoms, ions, or molecules vibrate in a crystal. **3** (*in immunology*) the three-dimensional aggregate suggested to be built up in an antigen-antibody reaction (*see* **lattice hypothesis**).

lattice energy (of a crystal) the energy that would be required to separate the ions of a crystal to an infinite distance from each other.

lattice hypothesis a hypothesis describing antigen-antibody reactions in terms of the building up of large antigen-antibody aggregates or three-dimensional lattices. The hypothesis requires both antigen and antibody to have valencies greater than one.

Lau *symbol for* the dodecanoyl (i.e. **lauroyl**) group.

Laue, Max Theodor Felix von (1879–1960), German physicist, Nobel Laureate in Physics (1914) 'for his discovery of the diffraction of X-rays by crystals'.

Laue equation any of a set of three equations describing the occurrence, positions, and intensities of X-ray beams after diffraction by a crystal. For three crystal axes, x , y , and z , with separations of chemical entities along them of a , b , and c , respectively, the equations are:

$$a(\cos\alpha - \cos\alpha_0) = h\lambda,$$

$$b(\cos\beta - \cos\beta_0) = k\lambda,$$

and

$$c(\cos\gamma - \cos\gamma_0) = l\lambda,$$

where α_0 , β_0 , and γ_0 are the angles that the incident beam makes with the x , y , and z , axes, respectively, and α , β , and γ are the corresponding angles for the diffracted beam; λ is the wavelength of the radiation and h , k , and l are integers corresponding to the **Miller indices** of the planes considered to be diffracting the radiation. *See also* **Bragg's law**. [After M. F. T. von Laue.]

Laue photograph or **Laue pattern** the spot diagram produced when a heterogeneous beam of X-rays is diffracted by passage through a crystal.

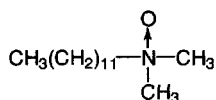
laurate **1** the trivial name for dodecanoate, $\text{CH}_3[\text{CH}_2]_{10}\text{COO}^-$, the anion derived from lauric acid (i.e. dodecanoic acid), a saturated straight-chain higher fatty acid. **2** any mixture of free lauric acid and its anion. **3** any salt or ester of lauric acid. Dodecyl laurate is one of the main components of laurel oil, the oil obtained from berries of the laurel plant, *Laurus nobilis*. Laurate is present in appreciable amounts in hydrolysates of phosphoglycerides of plants and animals.

Laurell technique an alternative name for **rocket immunoelectrophoresis**. [After C.-B. Laurell, Swedish biochemist.]

lauroyl or (formerly) **lauryl** *symbol*: Lau; the trivial name for dodecanoyl, $\text{CH}_3[\text{CH}_2]_{10}\text{CO}-$, the acyl group derived from lauric acid (i.e. dodecanoic acid). It occurs in natural lipids, mainly as acylglycerols.

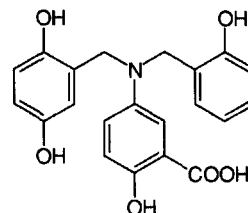
lauryl **1** *symbol*: Dod; a common name for dodecyl, $\text{CH}_3[\text{CH}_2]_{10}\text{CH}_2-$, the alkyl group derived from dodecane; use deprecated. **2** a former trivial name for dodecanoyl (lauroyl now used).

lauryldimethylamine oxide *abbr.*: LDAO; *N,N*-dimethyl-*N*-dodecylamine oxide; a water-soluble, zwitterionic (at pH 7 or above) detergent, aggregation number 76, CMC 1–2 mM.



lauryl sulfate, sodium salt an alternative name for **sodium dodecyl sulfate**.

lavendustin any of several substituted amino salicylic acids that are used as protein tyrosine kinase inhibitors. Lavendustin A is {5-amino-[(*N*-2,5-dihydroxybenzyl)-*N'*-2-hydroxybenzyl]}aminosalicylic acid and lavendustin C is 5-(*N*-2,5-dihydroxybenzyl)aminosalicylic acid. Lavendustin A inhibits protein tyrosine kinase activity with little effect on cyclic AMP-dependent kinase or protein kinase C. Lavendustin C is similar but also inhibits Ca^{2+} /calmodulin kinase II.



lavendustin A

lawn a layer of microorganisms uniformly distributed in or over the surface of a solid culture medium.

law of conservation of energy *see* **thermodynamics, laws of**.

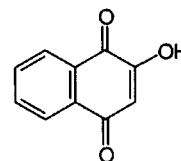
law of independent assortment *see* **Mendel's laws of heredity**.

law of mass action *see* **mass action**.

Lawrence, Ernest Orlando (1901–58), US physicist; Nobel Laureate in Physics (1939) 'for the invention and development of the cyclotron and for results obtained with it, especially with regard to artificial radioactive elements'.

laws of thermodynamics *see* **thermodynamics**.

lawsone 2-hydroxy-1,4-naphthoquinone; the active dyeing compound of **henna**.



layer line one of the parallel lines of spots obtained on the photographic plate in the rotating crystal method of X-ray diffraction analysis.

lb *symbol for* pound.

LBF *abbr. for* *Lactobacillus bulgaricus* factor (*see* **pantetheine**).

LC *abbr. for* liquid chromatography.

LCAD *abbr. for* long-chain acyl-CoA dehydrogenase; *see* **beta-oxidation system**.

LCAT *abbr. for* lecithin-cholesterol acyl transferase.

LCD *abbr. for* liquid-crystal display.

L cell an alternative name for **EG cell**.

L chain *see* **light chain** (def. 1).

LCK a gene family encoding nonreceptor tyrosine kinases (e.g. p56^{lck}) of the Src family (*see* **src**), originally detected in the LSTRA murine lymphoma cell line. The product is lymphocyte-specific and associates with the cytoplasmic domains of CD4 and CD8. The mitogen-activated protein (MAP) kinase is a substrate for p56^{lck}; it is a myristoylated protein with SH2 and SH3 domains. Example from human: database code KLSK_HUMAN, 508 amino acids (57.80 kDa).

LCR *abbr. for* locus control region.

LD *abbr. for* 1 lethal dose. **2** (*in clinical chemistry*) L-lactate dehydrogenase (alternative to **LDH**); *see also* **HBD**. **3** low density.

LD₅₀ *abbr.* for median lethal dose.

LDH *abbr.* for L-lactate dehydrogenase (alternative to LD).

LDL *abbr.* for low-density lipoprotein.

LDL receptor a cell-surface receptor that binds apolipoprotein B and so internalizes **low-density lipoprotein** (LDL) from the plasma, leading to processing that regulates cholesterol and LDL synthesis, and thence plasma cholesterol concentration. It is a type I membrane protein. The number of LDL receptors on liver cells is regulated by cholesterol synthesis: when cholesterol levels rise, the number of receptors falls. Defects in this receptor are associated with certain forms of type II **hyperlipidemia**. The amino end of the extracellular domain contains seven or eight 40-residue repeats; each repeat has about six Cys residues, all of which are involved in disulfide bonds; following these repeats is a region of about 350 residues that is homologous with part of the epidermal growth factor precursor. Example (precursor) from human: database code LDLR_HUMAN, 860 amino acids (95.27 kDa).

Le *abbr.* for Lewis antigen.

leaching (microbial) a process for the solubilization of metals, mostly from poor ores, by lithotrophic bacteria. *See also* lithotroph.

lead *symbol*: Pb; a very soft, bluish-white metallic element of group 14 of the IUPAC **periodic table**. It has a low melting point and high relative density; proton number 82; relative atomic mass 207.2. It is commonly used for shielding from ionizing radiation.

leader peptidase EC 3.4.99.36; *other name*: signal peptidase I, phage-procoat-leader peptidase; an enzyme that catalyses the cleavage of N-terminal leader sequences from secreted proteins.

leader peptide *an alternative name for signal peptide*.

leader sequence or **lead sequence** *an alternative name for signal sequence (see signal peptide)*.

leading strand the DNA strand that is synthesized continuously, in the 5' to 3' direction, towards the replication fork during the discontinuous replication of duplex DNA. *Compare* lagging strand.

lead screw a threaded rod the rotation of which is used to drive the tool carriage of a lathe, the plunger of a syringe, etc.

leaf (*pl.* leaves) **1** the principal organ of photosynthesis and transpiration in higher plants. It typically consists of a flattened blade-like lamina, often relatively broad, attached to the plant stem directly or by a stalk. **2** any thin, flattened object that resembles a leaf. *See also* leaflet.

leaflet **1** a any small or young **leaf** (def. 1). **b** an individual segment of a compound leaf. **2** any small, thin, flattened object or structure that resembles the leaf of a plant, especially a lipid bilayer or a portion of a biological membrane.

leaky (*in genetics*) describing any mutation that fails to shut off completely the activity of a gene so that some residual expression of the gene remains.

Le antigen *symbol for* Lewis antigen.

least squares a statistical method for determining the 'best' value of an unknown quantity relating one or more sets of measurements or observations. It is based on minimizing the sum of the squares of the deviations of the experimentally determined values from the calculated values, and is used especially to find a curve that best fits a given set of data.

leaving group any atom or group, charged or uncharged, that departs during a substitution or displacement reaction from what is regarded as the residual or main part of the substrate of the reaction.

Lebedev-Saft or **Lebedev juice** an aqueous extract of air-dried yeast. It will ferment glycerone (dihydroxyacetone) and, in the presence of phosphate, produces a hexose phosphate. [After Alexander Nikolaevich Lebedev (1881–1938), Russian botanist who described the preparation in 1911.]

Le Chatelier principle or **Le Chatelier–Braun principle** a principle stating that when a system in equilibrium is subject to stress, i.e. to a change in its conditions, the system will adjust

itself to annul, as far as possible, the effect of the stress. [First formulated (in 1885) by Henry (Louis) Le Chatelier (1850–1936), French chemist, and independently in 1886 by Karl Ferdinand Braun (1850–1918), German physicist.]

lecithin *an old trivial name, still in frequent use, for any 3-sn-phosphatidylcholine*.

lecithinase any of a number of distinct enzymes acting on lecithins that are now usually considered among the **phospholipases**.

lecithin-cholesterol acyl transferase *abbr.*: LCAT; EC 2.3.1.43; *recommended name*: phosphatidylcholine-sterol O-acyltransferase; *other name*: phospholipid-cholesterol acyltransferase. An enzyme that catalyses a reaction between phosphatidylcholine and cholesterol to form a cholesteryl ester and 1-acylglycerophosphocholine. This enzyme is central to the extracellular metabolism of plasma lipoproteins; it acts on free cholesterol in the high-density lipoprotein (HDL) precursor, transferring a fatty acyl group to cholesterol, thus contributing to the altered class composition of the mature HDL; some of the resulting cholesterol esters are partly taken up by low-density lipoprotein. The enzyme is a glycoprotein, and is activated by apolipoprotein A1. It is deficient in Norum and fish eye diseases. Example (precursor) from human: database code LCAT_HUMAN, 440 amino acids (49.52 kDa).

lectin any of a group of specific agglutinins and other antibody-like (glyco)proteins of nonimmune origin, defined (by IUB) as 'sugar-binding protein or glycoprotein of non-immune origin which agglutinates cells and/or precipitates glycoconjugates'. Lectins are widely distributed in nature, being found mainly in seeds but also in other parts of certain plants, and in many other organisms from bacteria to mammals. Lectins bear at least two sugar-binding sites; they bind specific sugars and thereby precipitate certain polysaccharides, glycoproteins, and glycolipids, and/or agglutinate animal and plant cells. Some can distinguish between normal and malignant cells, and many, e.g. phytohemagglutinin, are mitogenic. They are widely used experimentally, especially **concanavalin A**, as tools in carbohydrate biochemistry, for studying cell surfaces and for inducing **transformation** (def. 3) of lymphocytes. Plant lectins are also known as **phytoagglutinins**.

LED *abbr.* for light-emitting diode.

Lederberg, Joshua (1925–), US geneticist notable for his work on sex factors in bacteria and for discovering bacterial transduction; Nobel Laureate in Physiology or Medicine (1958) 'for his discoveries concerning genetic recombination and the organization of the genetic material of bacteria' [prize shared with G. W. Beadle and E. L. Tatum].

leghemoglobin or (*esp. Brit.*) **leghaemoglobin** a hemoglobin-like red pigment found in the root nodules of leguminous plants. An autoxidizable hemoprotein, it sequesters dioxygen to prevent inhibition of **nitrogenase** in the bacteroids. It is thus essential for symbiotic nitrogen fixation. Example, leghemoglobin I (monomer) from pea: database code LGB1_PEA, 147 amino acids (15.82 kDa).

legumain EC 3.4.22.34; *other names*: bean endopeptidase; vicilin peptidohydrolase; phaseolin; an endopeptidase from legumes that hydrolyses Asn–Xaa bonds in small molecule substrates such as Boc-Asn–l-OPhNO₂. Example (precursor) from *Phaseolus vulgaris*: database code CYSP_PHAVU, 362 amino acids (40.17 kDa).

legumin a major storage protein of the seeds of both leguminous and nonleguminous plants. It consists of two chains, an acidic α chain and a basic β chain which are linked by a single disulfide bond. These are derived from a common precursor; example (precursor) from *Pisum sativum*: database code LEG2_PEA, 520 amino acids (59.27 kDa). Residues 23–335 form the α chain, and 336–520 the β chain.

Leigh theory a theory that enables the use of electron spin resonance data to estimate distances between two interacting spins embedded in a rigid lattice. [After J. S. Leigh.]

Leloir, Luis Federico (1906–87), Argentinian biochemist noted



for his isolation of uridine diphosphate glucose and his discovery of the role of sugar nucleotides in interconversion of sugars and in polysaccharide formation; Nobel Laureate in Chemistry (1970) 'for his discovery of sugar nucleotides and their role in the biosynthesis of carbohydrates'.

LEM *abbr.* for light-effect mediator.

length *symbol:* *l* or *L*; a scalar physical quantity indicating linear extent; one of the seven SI base physical quantities. The SI base unit of length is the **metre**.

lens **1** a piece of glass or other transparent and refracting material, or two or more such pieces joined together, for converging or diverging a beam of light in order to form optical images. **2** any electrostatic or electromagnetic device for focusing or otherwise altering the direction of movement of a beam of electrons or other elementary charged particles; any similar device acting on sound waves. **3** (*in zoology*) any of various transparent structures responsible for focusing light onto photoreceptors. The **crystalline lens** of vertebrates is a near-spherical structure in the eye, lying in the aqueous humour behind the pupil. It focuses or assists in focusing light rays onto the retina. Enclosed within a collagenous capsule, it consists largely of cells containing proteins called **crystallins**.

lenticular **1** having the shape of a biconvex **lens** (def. 1). **2** of or pertaining to a **lens**.

lentinan a polysaccharide isolated from the edible mushroom *Lentinus edodes*. It is a β -1,3-glucan, having for every five β -1,3 linear linkages, two glucopyranoside units in β -1,6 linkage.

lentinarin *see callose*.

lentropin a protein, present in the vitreous humour of the eye, that stimulates lens-fibre differentiation. It is related to **insulin-like growth factors**.

Lepidoptera a large order of insects that includes butterflies, skippers, and moths. The adults are characterized by a prominent coiled proboscis and two pairs of large, scaly, membranous wings. The larvae are caterpillars. —**lepidopteran** *adj.*

leporine of, pertaining to, or resembling a rabbit or hare.

leprachaunism an autosomal recessive inherited disease characterized by small stature and elfin-like appearance, with motor and mental retardation and severe failure to thrive. Most patients die in early childhood, though a few, who are compound heterozygotes for two different alleles, survive childhood. The condition can be caused by any of a number of mutations in the gene encoding the insulin receptor. These involve changes in single amino acids, those so far found being between residues 55 and 487.

leptin a protein encoded by a gene, *ob*, first identified as the *obese* mutation (*see ob/ob*) in mice. Leptin may act as a signal in the regulation of adipose mass, possibly by regulating appetite and also possibly by regulating energy expenditure. It appears to be expressed exclusively in white adipose tissue. The predicted sequence is largely hydrophilic and has structural features indicative of a secreted protein. Example from human (precursor): database code OB_HUMAN, 167 amino acids (18.64 kDa). *See also* **leptin receptor**.

leptin receptor a receptor family that binds **leptin** and is presumed to mediate its intracellular effect. It is probably encoded by the gene *db* (for *diabetes*), and has at least six alternatively spliced forms. It is expressed at high levels in the hypothalamus relative to other tissues. A mutant form of the receptor which lacks the cytoplasmic region is likely to be defective in signal transduction and may be responsible for obesity in *db* mice. Example from human (precursor): database code OBR_HUMAN, 1165 amino acids (132.45 kDa). *See also* **diabetes mutation, ob/ob**.

leptocene the first phase of prophase I in **meiosis**.

leptotene *see meiosis*.

Lesch-Nyhan syndrome an X-chromosome-linked, inherited disorder of male humans in which there is an almost complete deficiency of **hypoxanthine phosphoribosyltransferase** (EC 2.4.2.8), one of the two key enzymes of the **salvage pathway** of purine metabolism. Clinically the condition is characterized

by hyperuricemia, excessive uric-acid biosynthesis, and certain neurological features including self-mutilation, spasticity, and mental retardation. Death usually occurs in the second or third decade of life. [After M. Lesch (1939–) and William Leo Nyhan (1926–), US physician and biochemist.]

lesion **1** (*in pathology*) a zone of tissue with impaired function resulting from damage by disease or wounding. **2** (*in biochemistry*) any deleterious disturbance of a metabolic pathway resulting from chemical interference or genetic abnormality.

+less *suffix forming adjectives and adverbs* without, lacking; unable to do or to be done. In biochemistry it is used especially of auxotrophic mutants to indicate the substance they cannot synthesize, e.g. thymineless mutant.

let23 a gene in *Caenorhabditis elegans* for a receptor **tyrosine kinase** of the epidermal growth factor family. It is probably activated by the *lin3* product. The product of *let23* is a type I membrane glycoprotein. Precursor: database code LT23_CAEEL, 1323 amino acids (150.34 kDa).

lethal dose *abbr.*: LD; the amount of a drug or other agent that when administered to an organism causes the organism's death. *See also* **median lethal dose**.

lethal factor any **lethal mutation**.

lethal gene any gene carrying a **lethal mutation**. Such mutations may only be fatal in the homozygous state.

lethal mutant any organism carrying a **lethal mutation**.

lethal mutation any **mutation** that may bring about the death of the organism before its maturity. The consequences of such a mutation may depend on whether the organism is homozygous or heterozygous for the gene in question, and possibly also on the conditions (*see conditional lethal mutation*).

lethal synthesis any process by which a highly toxic compound is synthesized in an organism from a nontoxic precursor. The classic example is the formation of **fluorocitrate** when e.g. kidney homogenates are incubated with **fluoroacetate**. The fluorocitrate thus formed inhibits **aconitase**.

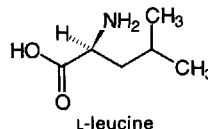
LETS protein *abbr.* for large external transformation sensitive (glyco)protein (i.e. cellular **fibronectin**).

Leu *symbol* for a residue of the α -amino acid L-leucine (alternative to L).

leuc+ a variant spelling of **leuk+**. *See* **leuko+**.

leucinate **1** leucine anion; the anion, $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{COO}^-$, derived from **leucine**. **2** any salt containing leucine anion. **3** any ester of leucine.

leucine the trivial name for α -aminoisocaproic acid; α -amino- γ -methylvaleric acid; 2-amino-4-methylpentanoic acid; $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{COOH}$; a chiral α -amino acid. L-leucine (*symbol:* L or Leu), (S)-2-amino-4-methylpentanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: CUA, CUC, CUG, or CUU; UUA or UUG. In mammals it is an essential amino acid and is ketogenic. Residues of D-leucine (*symbol:* D-Leu or DLeu), (R)-2-amino-4-methylpentanoic acid, occur in a number of peptide antibiotics, e.g. gramicidins A, B, C, and D and polymyxins D₁, D₂, E₁, and E₂.



leucine aminopeptidase *abbr.* (*in clinical chemistry*): LAP (preferred) or LAS; the common name for aminopeptidase (cytosol) (EC 3.4.11.1), an enzyme that, despite its name, in fact acts on most amino-acid residues. It is widely distributed in animal tissues, its level in the serum being useful clinically in monitoring hepatobiliary disease processes and placental well-being. The recommended name is leucyl aminopeptidase.

leucine and valine biosynthesis the biosynthesis of leucine



[5-leucine]enkephalin

and valine initially follows a common pathway that commences with the formation of acetolactate from two molecules of pyruvate, a reaction catalysed by acetolactate synthase. Acetolactate is converted via α,β -dihydroxyisovalerate (2,3-dihydroxy-3-methylbutanoate) to α -ketoisovalerate (2-oxo-3-methylbutanoate). At this point the pathway branches. α -Ketoisovalerate can be converted by transamination with glutamate to valine. Alternatively, it can be converted successively to α -isopropylmalate (2-isopropylmalate), β -isopropylmalate (3-isopropylmalate), and α -ketoisocaproate (4-methyl-2-oxopentanoate), and thence by transamination with glutamate to leucine. *See also* **3-isopropylmalate dehydratase**, **3-isopropylmalate dehydrogenase**.

[5-leucine]enkephalin *abbr.*: [Leu⁵]enkephalin or [Leu]enkephalin; *the recommended trivial name for the pentapeptide Tyr-Gly-Gly-Phe-Leu. See* **enkephalin**.

leucine-rich repeat *abbr.*: LRR; short sequence motifs in proteins, of average length 24 residues, containing a high proportion of leucines, typically five to seven at positions 2, 5, 7, 12, 16, 21, and 24 within the repeat. Such arrays form an amphipathic sequence, with leucine as the predominant hydrophobic residue. It is thought they participate in protein-protein or protein-lipid interaction. The first was discovered in α_2 -glycoprotein, but they have since been found in other proteins. The number of such repeats can vary from one, in platelet glycoprotein Ib β to 30, in e.g. **chaoptin**. Structure analysis has shown that in ribonuclease inhibitor protein these repeats comprise 90% of the protein, and that they correspond to β - α structural units, arranged to form a parallel β sheet with a parallel α helix, such that the β sheet has one surface exposed to solvent so that the protein acquires a non-globular shape.

leucine zipper a set of leucines (normally four or five) repeating every seventh amino-acid residue in a protein structure. Such an arrangement was originally found in DNA-binding proteins, but it is now known to occur also in other proteins, especially adjacent to proposed transmembrane regions, mediating dimerization. The abbreviation ZIP is used e.g. in designating the **bZIP** family of proteins. *See also* **potassium channels**.

leucinium leucine cation; the cation, $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}(\text{NH}_3^+)-\text{COOH}$, derived from **leucine**.

leucino the alkylamino group, $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}(\text{COOH})-\text{NH}-$, derived from **leucine**.

leuco+ a variant spelling of **leuko+**.

leucocyte a variant spelling of **leukocyte**.

leucyl the acyl group, $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{CO}-$, derived from **leucine**.

leucyl aminopeptidase EC 3.4.11.1; *other names*: leucine aminopeptidase; peptidase S; cytosol aminopeptidase; an enzyme that catalyses the hydrolysis of an aminoacyl-peptide to an amino acid and the residual peptide. It thus releases the N-terminal amino acid of a protein, provided the amino group is free; it has been so used in protein sequence determination. Its action is most rapid when the N-terminal amino acid is leucine; it liberates other residues at varying rates, and acts poorly at an Xaa-Pro bond. It requires zinc. Example from *Rickettsia prowazekii*: AMPL_RICPR, 500 amino acids (53.94 kDa).

[Leu⁵]enkephalin or **[Leu]enkephalin** *abbr. for* [5-leucine]enkephalin.

leuko+ a variant form of **leuko+** (before a vowel).

leukemia or (*esp. Brit.*) **leukaemia** or (*sometimes*) **leukosis** any of a group of neoplastic diseases of the leukocytes, usually involving increase in the number of leukocytes in the blood. Leukemias are classified according to the type of leukocyte affected and whether the disease is acute or chronic.

leukemia inhibitory factor *abbr.*: LIF; *also called* **differentiating-stimulating factor** or **differentiation-inducing factor**; a **cytokine** produced by fibroblasts, T cells, and macrophages. It is a monomer of 179 amino acids that maintains the pluripotent phenotype of embryonic stem cells, and potentiates interleukin-3-dependent proliferation of hemopoietic progenitors.

leuko or **leuco** describing the colourless derivative formed

versibly from a coloured substance, often by reduction; e.g. leuko compound, leuko dye.

leuko+ or **leuco+** or (*before a vowel*) **leuk+** or **leuc+** 1 white or colourless. 2 denoting of, pertaining to, or acting upon a leukocyte or leukocytes.

leukocidal or **leucocidal** describing an agent that kills leukocytes.

leukocidin or **leucocidin** any extracellular bacterial product that can kill leukocytes of certain animal species and that may also be toxic or lytic towards other cell types. Leukocidins are produced by pathogenic strains of *Staphylococcus* and *Streptococcus*.

leukocyte any white blood cell, i.e. one not containing an oxygen-transporting pigment such as hemoglobin. Vertebrate leukocytes include the **polymorphonuclear leukocytes**, **lymphocytes**, and **monocytes**; they are distinguished by size, shape, and staining characteristics. —**leukocytic** or **leucocytic** *adj.*

leukocytosis or **leucocytosis** a gross increase in the number of leukocytes in the blood. It often occurs as a response to an infection. *See also* **leukemia**.

leukokinin or **leucokinin** a specific cytophilic immunoglobulin G (IgG) obtained by fractionating human plasma IgG and constituting about 2% of the total IgG. It binds specifically and reversibly to autologous polymorphonuclear leukocytes, causing 2- to 2.5-fold stimulation of their phagocytic activity. A proteinase, **leukokininase**, cleaves from leukokinin a tetrapeptide, **tuftsin**, which is thought to bear all the activity.

leukopenia or **leucopenia** an abnormal decrease or deficiency of leukocytes circulating in the blood. *Also (esp. Brit.)* **leukopaenia** or **leucopaenia**.

leukoplast or **leucoplast** or **leukoplastid** or **leucoplastid** any of the colourless **plastids** found in plant cells. They include **aleuroplasts**, **amyloplasts**, and **elaioplasts**. Leukoplasts may develop into **chloroplasts**.

leukopoiesis or **leucopoiesis** the processes by which leukocytes are formed in the body. —**leukopoietic** or **leucopoietic** *adj.*

leukopterin or **leucopterin** 2-amino-4,6,7-trihydroxypteridine; a white pigment found in the wings of butterflies, especially white-winged butterflies.

leukosis or **leucosis** 1 a condition in which there is an abnormal proliferation of one or other of the leukocyte-forming tissues. 2 an alternative name for **leukemia**.

leukosulfakinin a sulfated myotrope 11-residue peptide isolated from head extracts of the cockroach. It exhibits sequence homology with human gastrin II and cholecystokinin.

leukotaxin or **leucotaxin** a nitrogen-containing material, found in injured tissue, that when injected into an animal causes increased capillary permeability and migration of leukocytes to the site of injection.

leukotoxin or **leucotoxin** 1 any compound toxic to leukocytes. 2 cytochrome P450-derived linoleic acid peroxide; any of a group of substances observed in high concentrations in burned skin samples that may have a bacteriocidal function. **Leukotoxin A** is (12Z)-9,10-epoxyoctadec-12-enoic acid, and **leukotoxin B** is (9Z)-12,13-epoxyoctadec-9-enoic acid (*abbr.*: EpODE) or, its methyl ester (*abbr.*: EpOME).

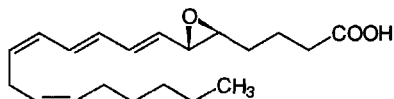
leukotriene *abbr.*: LT; any member of a family of pharmacologically active substances derived from polyunsaturated fatty acids, notably arachidonic acid, some of which contain a peptide moiety based on cysteine. They may be regarded as local hormones that are not stored, but synthesized in response to specific stimuli. They are formally derived from eicosanoic acid and contain a set of three conjugated double bonds (hence 'triene'); one or two additional double bonds, not conjugated to the others, may also be present – a subscript numeral after the abbreviation indicates the total number of double bonds, depending on the polyunsaturated fatty acid from which each is derived, e.g. LTA₄ from arachidonic acid, LTB₅ from eicosapentaenoic acid. They are products of the 5-lipoxygenase pathway. LTC₄ consists of glutathione linked to



leukotriene A

an eicosanoid; loss of glutamic acid from this leads to LTD₄, and loss of glycine then leads to LTE₄. These three, sometimes collectively known as **peptidoleukotrienes**, are potent spasmogenic agents that make up the active components of **slow-reacting substance of anaphylaxis** (SRS-A). See also individual entries below.

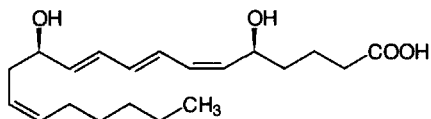
leukotriene A *abbr.*: LTA; any of a group of **leukotrienes** characterized as unstable 5,6-epoxides. Leukotriene A₄ (*abbr.*: LTA₄) is (7E,9E,11Z,14Z)-(5S,6S)-5,6-epoxyeicosa-7,9,11,14-tetraen-1-oate. Leukotriene A₃ (*abbr.*: LTA₃) is similar, but lacks the double bond at the C-14, while leukotriene A₅ (*abbr.*: LTA₅) has an additional, 17Z, double bond. LTA₄ is derived biosynthetically from arachidonate via 5-hydroperoxyeicosatetraenoate, and is the precursor either of leukotriene B₄ (see **leukotriene B**) by the action of highly specific **leukotriene-A₄ hydrolase** (EC 3.3.2.6), or of **leukotriene C₄** by the action of leukotriene-C₄ synthase (EC 2.5.1.37).



leukotriene A₄

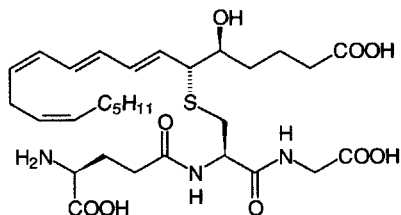
leukotriene-A₄ hydrolase *abbr.*: LTA₄ hydrolase; EC 3.3.2.6; an enzyme that converts (7E,9E,11Z,14Z)-(5S,6S)-5,6-epoxyeicosa-7,9,11,14-tetraenoate (LTA₄) to (6Z,8E,10E,14Z)-(5S,12R)-5,12-dihydroxyeicosa-6,8,10,14-tetraenoate (LTB₄). Example from human: database code LKHA_HUMAN, 610 amino acids (69.07 kDa).

leukotriene B *abbr.*: LTB; any of a group of **leukotrienes** characterized as (5S,12R)-5,12-dihydroxy polyunsaturated acids. Leukotriene B₄ (LTB₄) is (6Z,8E,10E,14Z)-(5S,12R)-5,12-dihydroxyeicosa-6,8,10,14-tetraen-1-oate. Leukotriene B₅ (LTB₅) has an additional, 17Z, double bond. LTB₄ is a product of leukotriene-A₄ hydrolase (see **leukotriene A**); it is a major product of the 5-lipoxygenase pathway, and is a potent aggregatory, chemokinetic, and chemotactic agent for polymorphonuclear leukocytes (PMNLs) *in vitro* and induces increased vascular permeability and PMNL infiltration *in vivo*.



leukotriene B₄

leukotriene C₄ *abbr.*: LTC₄; a **leukotriene** comprising a glutathione moiety linked through its thio group to C-6 of an eicosanoid. LTC₄ is formed from LTA₄ and glutathione by leukotriene-C₄ synthase (EC 2.5.1.37). Among other actions, it is a potent mediator of smooth muscle contractility, and of vascular tone and permeability, being a potent vasoconstrictor



in a variety of vascular beds including the coronary and cerebral circulations. It is the predominant leukotriene in the central nervous system.

leukotriene D₄ *abbr.*: LTD₄; a **leukotriene** comprising a cysteinylglycine moiety linked through its thio group to an eicosanoid moiety. It is a rapidly formed metabolite of **leukotriene C₄** (LTC₄), resulting from the action of membrane-bound γ -glutamyl transferase, which removes glutamic acid. Its actions are similar to those of LTC₄.

leukotriene E₄ *abbr.*: LTE₄; a **leukotriene** comprising a cysteine moiety linked to an eicosanoid moiety. It is derived metabolically from **leukotriene D₄** (LTD₄) by the action of specific membrane-bound dipeptidases, which remove glycine. It has a somewhat longer metabolic half-life than LTD₄, but is further metabolized to a number of more polar products, notably those in which the terminal methyl group of the eicosanoid moiety is oxidized to a carboxyl group. It activates receptors in human airways and has been implicated in airway hyper-responsiveness in humans.

leukovirus or **leucovirus** any tumorigenic **retrovirus**.

leukovorin or **leucovorin** see **folinic acid**.

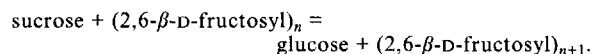
leumorphin an endogenous 29-amino-acid opioid peptide that corresponds to residues 228–256 of preproenkephalin B; the N-terminal 13 residues correspond to rimorphin (dynorphin B) and the N-terminal five to Leu-enkephalin (see **enkephalin**).

leupeptin any of several modified tripeptide protease inhibitors produced by various actinomycetes. Commonly *N*-acetyl-Leu-Leu-argininal is supplied for use in the laboratory; other variants have propionyl in place of acetyl, and valine or isoleucine in place of one or both leucines.

Leu-prolide an analogue of **gonadotropin-releasing hormone** in which residues 5 and 10 (glycine and glycineamide) are replaced by D-Leu and ethylamine respectively.

levan a common name for any (2→6)- β -D-fructofuranan; such compounds are produced mostly by bacteria. As in **inulin** each chain terminates in a (1→2)- α -D-glucopyranosyl residue, but it may also contain (2→1)- β branch points.

levansucrase the recommended name for sucrose 1-fructosyl transferase, EC 2.4.1.10; an enzyme that catalyses the reaction:



See also **sucrase** (def. 3).

levarterenol see **norepinephrine**.

levigate (*in chemistry*) 1 to grind (something) into a smooth powder or paste. 2 to separate fine from coarse particles by grinding and suspending them in a fluid. —**levigation** *n*.

Levi-Montalcini, Rita (1909–), Italian cell biologist; Nobel Laureate in Physiology or Medicine (1986) jointly with S. Cohen 'for their discoveries of growth factors'.

levo- or (*esp. Brit.*) **laevo-** *symbol*: *l-*; a prefix, no longer used, to denote a compound that is **levorotatory**; the levorotatory enantiomer of compound A is now designated as (–)-A.

levo+ or (*before a vowel*) **lev+** *comb. form* denoting on or towards the left. Also (*esp. Brit.*) **laevo+**, **laev+**. Compare **dextro+**.

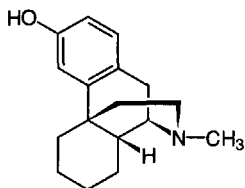
levodopa or (*Brit.*) **laevodopa** (–)-3-hydroxy-L-tyrosine; a naturally occurring amino acid that has been used in the treatment of **Parkinson's disease** easing symptoms such as rigidity, postural problems, and slowness of movement. See **3,4-dihydroxyphenylalanine**.

levorotatory or (*esp. Brit.*) **laevorotatory** 1 *symbol*: (–) or (*formerly*) *l-*; describing a chemical compound that rotates the plane of polarization of a transmitted beam of plane-polarized light or other electromagnetic radiation to the left, i.e. in an anticlockwise direction as viewed by the observer looking towards the light source. 2 describing any leftwards or anticlockwise rotation. Compare **dextrorotatory**. —**levorotation** or **laevorotation** *n*.

levorphanol 17-methylmorphinan-3-ol; a synthetic analogue of morphine. It is an orally active analgesic that binds stereo-

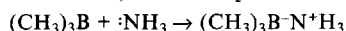


specifically with high affinity to opioid receptors (K_d 10^{-9} M). Five to six times more active than morphine, it is a controlled substance under the US Code of Federal Regulations. **Dextrophan**, the dextrorotatory form, does not bind to opioid receptors, is not analgesic, but is a very effective antitussive.

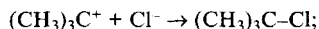


levulose or (esp. Brit.) **laevulose** an old name, still sometimes used in pharmacy, for D-(–)-fructose, the levorotatory component of invert sugar. Compare **dextrose**.

Lewis acid any chemical species that is able to accept an electron pair and is thus able to react with a **Lewis base** to form an adduct (i.e. Lewis adduct). For example:



or



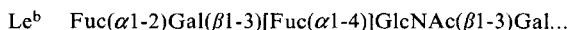
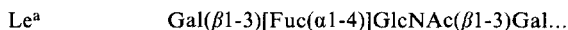
i.e.

Lewis acid + Lewis base $\rightarrow$ Lewis adduct.

[After Gilbert Newton Lewis (1875–1946), US chemist.]

Lewis adduct the adduct formed between a **Lewis acid** and a **Lewis base**.

Lewis antigen *abbr.*: Le antigen; *symbol*: Le; one of the serologically distinguishable human **blood-group substances**. Lewis specificities are carried on glycosphingolipids and glycoproteins. Primarily a system of soluble antigens in secretions (e.g. saliva) and plasma, the Lewis antigens on red cells are adsorbed passively from plasma, which must be constantly present for them to be maintained. Three Lewis phenotypes are common amongst Caucasians, namely $\text{Le}(\text{a}^+\text{b}^-)$, $\text{Le}(\text{a}^-\text{b}^+)$, and $\text{Le}(\text{a}^-\text{b}^-)$; the very rare $\text{Le}(\text{a}^+\text{b}^+)$ is detected in Polynesians and Orientals. The carbohydrate structures providing Le^{a} and Le^{b} specificity are shown below; these partial structures are linked through other carbohydrate moieties to a **ceramide**, thus forming a glycosphingolipid, or to a protein, thus forming a glycoprotein.



The structures of the Le^{c} , Le^{d} , Le^{x} , and Le^{y} determinants are also known. The Lewis antigens are named after a Mrs H. D. G. Lewis, in whom they were first identified. See also **galactoside 3(4)-L-fucosyltransferase**.

Lewis base any chemical species that is able to supply an electron pair and is thus able to react with a **Lewis acid** to form an adduct (i.e. Lewis adduct).

Lewis formula or **Lewis structure** a depiction of molecular structure in which the valency electrons are shown as dots so placed that each pair of dots represents two electrons. When placed between two atoms each pair of dots also represents a single covalent bond, whereas dots placed adjacent to only one atom represent the non-bonded outer-shell electrons considered to be associated with that atom alone. (e.g. $\text{H} \cdot \ddot{\text{C}} \cdot \text{H}$). A double bond is represented by two pairs of dots (e.g. $\ddot{\text{O}} :: \ddot{\text{O}}$), and so on. Formal charges are attached to atoms as appropriate.

Lewis transferase see **galactoside 3(4)-L-fucosyltransferase**.

lexA a gene in *Escherichia coli* encoding a repressor that regulates the genes induced by **SOS repair**. In the latter, following DNA damage, the *lexA* product, LexA, is hydrolysed by RecA protein (see **recA**), thereby allowing the SOS genes to be activated.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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ated. *lexA* is itself regulated autogenously. LexA binds to operators as a dimer. Database code LEXA_ECOLI, 202 amino acids (22.33 kDa).

lexicographer one who compiles a dictionary, defined by Samuel Johnson (the first English lexicographer) as 'a harmless drudge'. An alternative definition is from *The Enlarged Devil's Dictionary* by Ambrose Bierce, compiled and edited by Ernest J. Hopkins: 'A pestilent fellow who, under the pretense of recording some particular stage in the development of a language, does what he can to arrest its growth, stiffen its flexibility and mechanize its methods.' See also the Preface to this present work.

Leydig cell a type of steroid-secreting, interstitial cell that occurs in large numbers between the seminiferous tubules of the testis. The cells secrete androgens, especially testosterone, and this activity is stimulated by luteinizing hormone, for which the cells have specific receptors. [After Franz von Leydig (1821–1908), German anatomist and zoologist.]

LFA 1 a leukocyte cell-surface glycoprotein involved in the adhesion of leukocytes to target cells and thus in a variety of immune phenomena including leukocyte–endothelial cell interaction, cytotoxic T-cell-mediated killing, and antibody-dependent killing by granulocytes and monocytes. It binds to **ICAM**. LFA 1 is a type I membrane glycoprotein belonging to the LEUCAM family of **integrins**; it is a dimer of the α_L and β_2 subunits. Example from human (alpha precursor); database code ITAL_HUMAN, 1170 amino acids (128.70 kDa).

L form (of a bacterium) a defective bacterial cell, of indefinite or spherical shape, in which the cell wall is defective or absent. L forms may develop spontaneously or in response to a variety of stimuli, e.g. thermal or osmotic shock, or the presence of antibiotics that inhibit cell-wall synthesis. They may revert to their original condition on removal of the stimulus or they may reproduce as L forms. On solid media, L forms have a characteristic colonial appearance resembling that of **mycoplasmas** (with which they were originally confused).

lg *symbol* for decadic, or common or Briggsian, **logarithms** (alternative to $\log_{10}$).

LH *abbr.* for luteinizing hormone.

LHC *abbr.* for light-harvesting complex.

LHCP *abbr.* for light-harvesting chlorophyll protein.

LH/FSH-RF *abbr.* for luteinizing hormone/foolicle-stimulating hormone releasing factor (i.e. **gonadotropin releasing hormone**).

LHRF or **LH-RF** *abbr.* for luteinizing hormone releasing factor (see **gonadotropin-releasing hormone**).

LHRH or **LH-RH** *abbr.* for luteinizing hormone releasing hormone (see **gonadotropin releasing hormone**).

Li *symbol* for lithium.

Libby, Willard Frank (1908–80), US chemist; Nobel Laureate in Chemistry (1960) for 'his method to use carbon 14 for age determination in archaeology, geology, geophysics, and other branches of science'.

+liberin *word ending* denoting a hypothalamic factor (or hormone) that promotes the release (and perhaps also the biosynthesis) of the particular pituitary hormone indicated; e.g. lulliberin, thyroliberin. Compare **+statin**.

library of genes see **gene library**.

librate to oscillate about an equilibrium position. —**libration** *n*.

Librium see **chlordiazepoxide**.

licorice see **glycyrrhiza**.

Liebermann–Burchard reaction the reaction of acetic anhydride in chloroform with unsaturated steroids (e.g. cholesterol) or triterpenes in the presence of concentrated sulfuric acid to produce a blue-green colour. It is used especially in the estimation of cholesterol.

LIF *abbr.* for leukemia inhibitory factor.

life cycle the totality of functional and morphological stages through which any organism passes between any one developmental stage and the identical stage in the next generation. Less commonly it refers to the totality of the various stages through which an individual organism passes between birth and death.

Lig

light

Lig symbol for the lignoceroyl group, $\text{CH}_3\text{--}[\text{CH}_2]_{22}\text{--CO--}$.

ligancy a less common alternative name for **coordination number**.

ligand **1** (in a simple inorganic compound) any individual atom, group, or molecule that is attached covalently to the characteristic or central atom or moiety. **2** (in an organic compound) any individual atom or chemical group that is attached covalently to a specified carbon atom; *see also* **point ligand**. **3** (in a coordination entity) **a** any individual atom, ion, or molecule that is attached coordinately to one central metal atom (**unidentate ligand**). **b** an ion, molecule, or molecular grouping that has more than one coordinating group (**bi-**, **tri-**, or **multi-dentate ligand**). In both cases the ligand is an electron donor in the formation of one or more coordinate bonds. **4** an ion of either sign that can associate reversibly with one or more (charged or uncharged) atoms or characteristic molecular groupings in another (usually larger) molecule (as in the hydration of a conjugate base, the coordination of a metal cation by a protein, etc.). **5** a molecule (or a class of molecules) that has become linked covalently to another molecule to form a conjugate, whether in a single-step reaction or by stepwise addition of its components (e.g. an oligosaccharide moiety in a glycoprotein). **6** a molecule (or part of one) that is bound or is able to bind selectively and stoichiometrically, whether covalently or not, to one or more specific sites on another molecule (as in the combination of antigen with antibody, of hormone with receptor, of substrate with enzyme, etc.). [German: bound element or group, from Latin *ligandus*, gerundive of *ligare* to bind.]

liganded bound by or as a ligand or ligands; ligated; *see* **ligate** (def. 2).

ligand exchange (chromatography) a column-chromatographic technique in which a cation-exchange resin is loaded with a complexable metal ion and the resulting loaded resin, which retains the capability of the metal ion to coordinate to an appropriate complexing agent or ligand, is used for the sorption of that ligand from a solution or a gas phase. Elution is then effected by displacement with, i.e. exchange for, another ligand. The resin is commonly a chelating resin (e.g. a styrene-divinylbenzene copolymer to which iminodiacetate groups are attached) and the ion is usually of a transition metal (e.g. copper or nickel), such ions tending to be bound strongly to these resins. Separation of components of mixtures occurs by virtue of the differences in the stabilities of the various metal-ligand complexes formed.

ligand-field theory (in inorganic chemistry) a theory, closely related to the valence-bond theory, that deals with the effects of ligands on the energy levels of the central atom or ion in complexes and crystals. It is especially useful in interpretation of the spectra of inorganic complexes.

ligand-gated describing an ion channel in a cell membrane that is caused to open by the action of an agonist binding to the receptor that governs the channel. Channel opening can be direct, when the receptor acts as an ionophore, or indirect, involving a second messenger regulated by a G-protein.

ligandin a name formerly given to a basic dimeric protein of ≈ 25 kDa per dimeric subunit (*see* **glutathione transferase**). It is abundant in rat and human liver, kidney, and small intestine. It binds with high affinity to heme, bilirubin, and other organic anions, polycyclic aromatic carcinogens, and various metabolites. In addition it catalyses the conjugation of glutathione with a variety of electrophilic substrates, shows selenium-independent glutathione-peroxidase activity and steroid isomerase activity, and covalently binds activated metabolites of several carcinogens. Ligandin corresponds to the α class of glutathione transferase. Ligandin is present in the pars recta of the proximal renal tubules but does not normally pass into the urine: its presence in the urine, **ligandinuria**, has been used to follow the evolution of various forms of kidney injury, while its presence in serum provides a sensitive indicator in liver injury. Also variously called Y protein, organic anion binding protein, basic azo-dye binding protein, cortisol

metabolite binding protein (binder protein I), glutathione-transferase B.

liganding binding by or as a ligand or ligands; ligating; *see* **ligate** (def. 2).

ligandinuria *see* **ligandin**.

ligand-mediated chromatography a variant of **affinity chromatography** that employs an additional component, the affinity ligand, which is a ligand both for the reactive component attached to the insoluble matrix and for the substance to be purified. Because the ligand may readily be renewed after completion of each separation, the method is particularly useful when the substance used as the ligand easily becomes degraded or inactivated during the course of a purification.

ligand-receptor assay a form of **saturation analysis** for a hormone or other agonist wherein the specific binding agent used is an isolated preparation of the naturally occurring cellular receptor substance for the agonist. An advantage of the assay is that the values obtained are considered to be a direct reflection of the biological properties of the substance assayed.

ligase **1** or (formerly) **synthetase** the systematic class name for enzymes of class EC 6, all of which catalyse the ligation of molecules of two substances with concomitant breaking of a diphosphate linkage in a nucleoside triphosphate. The recommended names of particular ligases are formed in some instances, especially those of enzymes of subclass EC 6.4, by using the generic name carboxylase, e.g. pyruvate carboxylase, EC 6.4.1.1. In the remaining instances the generic names **ligase** (def. 2) or **synthase** are used, according to whether the name of an individual enzyme is to be based on the names of the reactants or on the name of the product, respectively. **2** a generic name for many enzymes of class EC 6 (*see* **ligase** (def. 1)); when added to the names of the two reactants whose ligation is catalysed by a particular enzyme it forms the recommended name of that enzyme; e.g. tyrosine-tRNA ligase (EC 6.1.1.1), acetate-CoA ligase (EC 6.2.1.1). *See also* **synthetase** (def. 2). **3** *see* **DNA ligase**.

ligate **1** to join together, especially with a ligature. **2** (in chemistry) to join (molecules or molecular fragments) together with a bond; to coordinate. **3** to bind as or to a **ligand** or ligands. **4** (in surgery) to tie off (a blood vessel or duct) with a ligature so as to occlude. —**ligation** *n.*; **ligatable** *adj.*; **ligative** *adj.*

ligatin a filamentous plasma membrane protein for the attachment of peripheral glycoproteins to the external cell surface. It acts as a trafficking receptor for phosphoglycoproteins, localizing them, after internalization, within endosomes. Example (fragment) from human: database code LIGA_HUMAN, 75 amino acids (8.47 kDa).

light **1** electromagnetic radiation capable of producing a visual sensation in the eye. It comprises wavelengths in the range 380–780 nm (frequencies 385–789 THz), although the term is sometimes loosely extended to include some ultraviolet and infrared radiation in adjacent parts of the spectrum. **2** (in chemistry) describing any (especially metallic) element whose density is relatively small (usually $< 5000 \text{ kg m}^{-3}$). **3** (in physics) of less than the usual mass. *Compare* **heavy**.

light band a region of a muscle **sarcomere** that stains less densely in light microscopy and low-magnification electron microscopy relative to the **dark bands**; together, these differentially staining bands give the striated appearance of skeletal muscle. The light bands are relatively isotropic in polarized light; they represent the regions occupied by **thin filaments**. *See also* **thick filament**, **Z line**.

light chain **1** or **L chain** or (formerly) **B chain** symbol: L; the shorter of the two main types of polypeptide chain of an **immunoglobulin** of any class. Of ≈ 22 kDa (human), each light chain is linked at its C-terminal cysteine residue by a disulfide bond to the constant region of a **heavy chain**. In any given animal species, light chains may be distinguished serologically into two types, κ and λ , which occur in immunoglobulin molecules of every class in proportions that vary only with the

light-effect mediator

species (about 65% κ in the human), and which have different but homologous sequences. The C-terminal half of each type of light chain consists of a **constant region** and the N-terminal half is a **variable region** possibly related to the antibody-combining site. *See also* **Bence-Jones protein**. **2** any of the four 15–27 kDa polypeptide chains that, with two heavy chains of ≈ 200 kDa, constitute a molecule of **myosin**.

light-effect mediator *abbr.*: LEM; any of a number of substances that, following their light-induced reduction, activate some of the enzymes of the Calvin cycle (*see* **reductive pentose phosphate cycle**) in photosynthesis. Not all of these mediators have been identified, but it is thought that they are cyst(e)ine-containing proteins that become oxidized in the dark and reduced again in the light. An important member of the group is **thioredoxin**.

light-emitting diode *abbr.*: LED; a device consisting of a semiconductor diode (made, e.g., of gallium arsenide) in which light is emitted at a p–n junction when electrons and holes recombine in a manner proportional to the bias current. It is used to give a visual display of numbers, etc. in calculators, counters, and other pieces of apparatus.

light guide a device consisting of one or more cylinders or strips of transparent material along which light can be made to travel, by means of total internal reflection, in a desired direction and with little loss.

light-harvesting chlorophyll protein or **light-harvesting chlorophyll-binding protein** *abbr.*: LHCP; a chlorophyll-binding protein that is abundant in green plants, particularly in those grown at low light intensities. It consists of a 25 kDa polypeptide, encoded by a nuclear gene and synthesized in the cytoplasm, to which is bound most of the chlorophyll *b* and xanthophylls together with some of the chlorophyll *a*. It appears to be capable of transferring the energy of absorbed light photons to both photosystem I and photosystem II. Example, *Lycopersicon esculentum* (tomato) chlorophyll *a-b* binding protein of light-harvesting complex I (type I precursor): database code CB11_LYCES, 246 amino acids (26.57 kDa); residues 1–44 are a transit peptide, 45–246 LHCP. *See also* **light-harvesting complex**.

light-harvesting complex *abbr.*: LHC; a **thylakoid** membrane complex of chlorophylls *a* and *b* together with chlorophyll *a-b* binding proteins. In addition, LHCs contain a number of other proteins, the function of which is speculative, together with accessory pigments. The LHCs capture and transfer energy to photosystems I and II. Example, chlorophyll *a-b* binding protein of pea: database code CB21_PEA, 228 amino acids (24.30 kDa). It is regulated by phosphorylation on threonine residues. *See also* **light-harvesting chlorophyll protein**.

light meromyosin *abbr.*: L meromyosin or LMM; the smaller, 120 kDa, fragment, also called F3 fragment, produced when **myosin** is subjected to a brief treatment with trypsin or certain other peptidases. It contains much of the helical, rod-shaped tail (>90% α helical) and forms large ordered aggregates at low ionic strength. Light meromyosin has no ATPase activity and cannot bind actin.

light path **1** the path followed by a beam of light when passing through an optical device or a particular part of one. **2** or **path length** the length of such a path.

light pen an input/output device for use with a computer that, when pointed at a spot on a visual display unit, can sense whether or not the spot is illuminated; the output current from the light pen can then be used to control the operations of the computer.

light reactions of photosynthesis or **light phase of photosynthesis** those reactions of **photosynthesis** in higher plants and algae in which light energy is used to generate NADPH and ATP.

light repair an alternative name for **photoreactivation**.

light-scattering a technique used for the determination of the relative molecular mass, M_r , in solution of a macromolecule whose size is large relative to the wavelength of the light used.

limit dextrin

The value of M_r of the macromolecule can be calculated from the wavelength of the incident light, the angle and intensity of the scattered light, the refractive indices of the solvent and solute, and the concentration of the solute. *See also* **Rayleigh scattering**.

light strand or **L strand** **1** a polynucleotide strand not labelled with heavy isotope. **2** the strand in a naturally occurring polynucleotide duplex that is of a lower density than its complementary strand. *Compare* **heavy strand**.

light trapping **1** the process of capturing the energy of incident light in a photosynthetic pigment system. **2** the act or process of using focused light in a liquid medium so arranged as to trap phototactic organisms in a restricted part of that medium.

lignan any member of a class of phenylpropanoid dimers in which the phenylpropane units are linked tail-to-tail and thus having a 2,3-dibenzylbutane skeleton (*compare* **neolignan**). Until recently such compounds had been found only in higher plants, but certain types have now been identified in humans and other mammals as products of the action of gut flora on dietary lignans of plant origin. Guaiaretic acid is structurally one of the simplest; **podophyllotoxin** and more particularly its semisynthetic derivatives **Etoposide** and **Teniposide** have defined actions in clinical medicine, with antitumour and possibly antiviral actions. *See also* **lignin**.

lignify to make or become woody; to accumulate or deposit lignin in the cell walls (of plant tissues). —**lignification** *n*.

lignin any random phenylpropanoid polymer formed in higher plants by the dehydrogenative radical polymerization of various 4-hydroxycinnamyl alcohols, in which the residues are joined in differing proportions and by several different linkages, some of which are nonhydrolysable. Lignin is of high molecular mass, is insoluble in water, and is found in the cell walls of vascular plants (especially of the supporting and conducting tissues), on which it confers strength, rigidity, and resistance to degradation. It is one of the most abundant biopolymers. *See also* **lignan**.

ligninase an enzyme that catalyses breakdown of lignin; strictly most lignin-degrading systems rely on peroxide formed from **oxygenases** to cleave carbon–carbon bonds. An example is diarylpropane peroxidase, EC 1.11.1.14; *systematic name*: diarylpropane:oxygen,hydrogen peroxide oxidoreductase (C–C bond-cleaving); *other name*: ligninase I. It is a hemoprotein enzyme that catalyses a reaction between 1,2-bis(3,4-dimethoxyphenyl)propane-1,3-diol and H_2O_2 to form 3,4-dimethoxybenzaldehyde, 1-(3,4-dimethylphenyl)ethane-1,2-diol, and $4H_2O$. Example from *Phanerochaete chrysosporium*: database code LIG2_PHACH, 371 amino acids (39.33 kDa).

lignocellulose a covalent adduct of **lignin** and **cellulose**, found in the walls of xylem cells in woody tissues of plants. —**lignocellulosic** *adj.*, *n*.

lignocerate **1** the trivial name for tetracosanoate, $CH_3-[CH_2]_{22}-COO^-$, the anion derived from lignoceric acid (i.e. tetracosanoic acid), a saturated straight-chain higher fatty acid. **2** any mixture of lignoceric acid and its anion. **3** any salt or ester of lignoceric acid.

lignoceroyl *symbol*: Lig; the trivial name for tetracosanoyl, $CH_3-[CH_2]_{22}-CO-$, the acyl group derived from lignoceric acid (i.e. tetracosanoic acid). It occurs naturally as acylglycerols in peanut and rapeseed oils, and in certain sphingomyelins.

lignoceroyl-CoA synthase *see* **long-chain-fatty-acid-CoA ligase**.

limit (*in mathematics*) a value to which a function or the sum of a series approaches as an independent variable approaches a specified value or infinity.

limit dextrin the highly branched core that remains after exhaustive treatment of amylopectin or glycogen with α - and/or β -amylases. It is formed because these enzymes cannot hydrolyse the (1 $\rightarrow$ 6) glycosidic linkages present.

limit dextrinase *see* α -dextrin endo-1,6- α -glucosidase, oligo-1,6-glucosidase.

limit dextrinosis an alternative name for type III **glycogen disease**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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limit digest

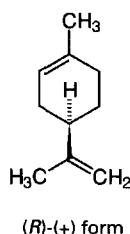
limit digest a digest in which no further degradation can occur under the particular conditions adopted.

limiting current (in polarography) the maximal value of the current reached in circumstances when the steady-state current/potential characteristic has a positive slope.

limiting rate symbol: V ; the recommended name for the extrapolated rate of an enzyme-catalysed reaction when the enzyme is saturated with substrate(s). It is equal to the product of the stoichiometric concentration of active centres, $[E]_0$, and the catalytic constant, k_0 . It can be determined by a number of methods, including graphical methods such as the **Lineweaver-Burk plot**. Other names: maximum rate; maximal rate; maximum velocity; maximal velocity.

limiting viscosity number the recommended name for intrinsic viscosity.

limonene *p*-mentha-1,8-diene; a compound that occurs in many essential oils (e.g. oils of lemon, orange, and dill); both enantiomers and the racemic mixture (i.e. dipentene) occur naturally.



LIMS abbr. for laboratory information management systems; normally, systems involving automatic electronic data capture and computerized manipulation of data.

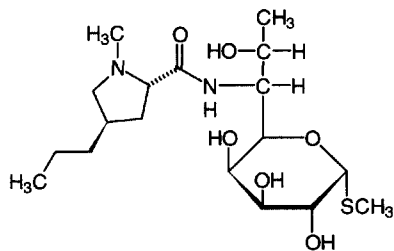
Lin symbol for the linoleoyl group, $\text{CH}_3\text{--}[\text{CH}_2]_3\text{--}[\text{CH}_2\text{CH}=\text{CH}]_2\text{--}[\text{CH}_2]_7\text{--CO--}$ (all-*Z* isomer).

lin3 a gene in *Caenorhabditis elegans* that encodes an **epidermal growth factor**-like type I membrane protein. The protein has two versions (produced by alternative splicing), and is essential for vulval development. Database code LIN3_CAEEL, 438 amino acids (48.87 kDa). See also **let23**.

lin14 a gene involved in embryonic development in *Caenorhabditis elegans* that codes for a nuclear protein. Regulatory sequences in the 3' untranslated region generate a temporal switch during development. There are three forms of the product: A, database code B40581, 537 amino acids (59.09 kDa); B1, database code A40581, 539 amino acids (59.28 kDa); B2, database code C40581, 515 amino acids (56.65 kDa).

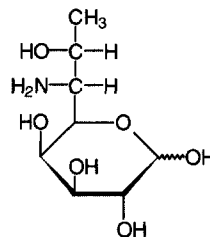
linalool 3,7-dimethyl-1,6-octadien-3-ol; a compound that occurs in linaloe oil and other essential oils; both enantiomers occur naturally. It is used in perfumery.

lincomycin an antibiotic produced by *Streptomyces lincolnensis*. It consists of an amide formed between a substituted pyrrolidine-2-carboxylic acid and methyl 1-thio- α -lincosamine. Lincomycin is active mainly against Gram-positive bacteria; it inhibits protein synthesis by interacting with the 50S ribosomal subunit, inhibiting the early steps of peptide bond formation prior to the formation of polysomes.



lincosamides generic name for a group of antibiotics structurally related to **lincomycin**.

lincosamine 6-amino-6,8-dideoxy-D-erythro-D-galacto-octose; the carbohydrate component of **lincomycin**.



line 1 (in spectroscopy) a very narrow band of frequencies of electromagnetic radiation that is noticeably more intense or less intense than adjacent parts of the spectrum; the image of such a line in a spectrogram. **2** (in biology) several generations of a family. See also **cell line**, **lineage**. —**lineal** or **linear** adj.

LINE abbr. for long interspersed nucleotide element (or sequence or repeat); part of the moderately repetitive DNA of mammalian genomes. Any highly repeated DNA sequence interspersed in a mammalian genome and characteristically several kilobases in length. LINES are **retrotransposons** resulting from the action of RNA polymerase II (see **RNA polymerase**). Mammalian genomes may contain up to 50 000 copies of particular LINES. They are regarded as processed **pseudogenes**. The function of LINES is unknown. Compare **SINE**. See also **Alu sequence**.

lineage lineal descent, especially from a common ancestor.

linear accelerator a device for producing high-energy particles, in which charged particles are accelerated in a straight line along a long evacuated tube by potential differences applied to a series of circular electrodes along the tube.

linear chain a chain of atoms in a molecule, or a chain of residues in a macromolecule, that is neither branched nor covalently closed.

linear correlation any relationship between two variables such that a graphical plot of one variable against the other is a straight line.

linear density gradient a gradient of linearly and continuously increasing density.

linear dichroism the **dichroism** (def. 1) occurring when linearly polarized light is absorbed by a sample containing partially or completely oriented anisotropic molecules.

linear equation any equation containing any number of variables that is of the first degree, i.e. one in which the variables are all raised to the power of one.

linear growth the growth of a culture such that the number of cells (or the cell mass) increases in direct proportion to the duration of growth.

linearity the condition of being in a straight line.

linearize or **linearise** **1** to make linear, i.e. into a straight line. **2** to convert a branched or cyclic form of a (macro)molecule to a linear form, i.e. one with only two ends. —**linearization** or **linearisation** *n*.

linearly polarized or **plane-polarized** describing light (or other electromagnetic radiation) in which the electric vectors of the vibrations all lie in a given plane containing the light beam. See **polarized light**.

linear velocity the distance moved in a straight line in unit time.

line formula a two-dimensional representation of a molecular entity in which atomic symbols are joined by lines representing single bonds, without any indication or implication of stereochemistry.

line immunoelectrophoresis a one-dimensional single **electroimmunodiffusion** technique, useful for the qualitative and



line spectrum

quantitative comparison of the antibody spectra of different polyvalent antisera. Samples of antigens to be compared are moulded into adjoining rectangular sample gels along one edge of a gel containing a mixture of the corresponding antibodies, into which the antigens are forced at right angles by electrophoresis. Precipitation lines form parallel to the origin at a distance of antigen migration proportional to the antigen/antibody ratio. Precipitation lines from the samples against the same antibody are continuous and can be directly compared.

line spectrum any spectrum consisting of discrete spectral lines.

Lineweaver-Burk plot a method of plotting enzyme kinetic data. The Michaelis equation is rearranged to give:

$$1/v = K_m/[VS] + 1/V,$$

where K_m is the Michaelis constant, $[S]$ the concentration of the substrate S , v the (initial) rate of reaction at a given value of $[S]$, and V the limiting rate of reaction at infinite $[S]$. If the Michaelis equation holds for the system, a plot of $1/v$ against $1/[S]$ gives a straight line of slope K_m/V , with intercepts on the $1/v$ axis at $1/V$ and on the $1/[S]$ axis at $-1/K_m$. See **Michaelis kinetics**. [After Hans Lineweaver and Dean Burk.]

linewidth the width of a spectral line, usually taken as the distance, in terms of wavelength, frequency, or magnetic flux difference, between the two points of half-peak intensity (half-maximum height).

link (in chemistry) an informal term for **bond** (def. 1). See also **linkage** (def. 2).

linkage 1 (in genetics) an association of two or more non-allelic genes so that they do not show independent **assortment**; i.e. they tend to be transmitted together usually due to physical association on the same chromosome. **2** (in chemistry) an informal term for **bond** (def. 1). See also **link**.

linkage disequilibrium the preferential association of linked genes/DNA markers in a population; i.e. the tendency for some alleles at a locus to be associated with certain alleles at another locus on the same chromosome with frequencies greater than would be expected by chance alone.

linkage group any group of genes whose members show a marked tendency to segregate as a unit. See also **genophore**.

linkage map a map of a chromosome showing in linear order the relative positions of the known genes on that chromosome.

linked genes genes that exhibit **linkage** (def. 1).

linker 1 any small synthetic single-chain oligodeoxynucleotide that is six or more residues long and has a self-complementary base sequence. When self-annealed it forms a stable duplex with blunt ends and containing a **restriction site**. Such linkers are useful for gene splicing. **2** see **cross-link**.

linker DNA see **chromatin, nucleosome**

linking number *symbol:* L ; a topological property of closed circular duplex DNA, defined as the number of turns one strand makes about the other. It must be an integer, and is positive if the turns are right-handed and negative if they are left-handed. It can change only when one or both strands are nicked. It comprises the sum of the **writhing number** (W) and the **twist number** (T), so that a change (ΔL) in the linking number (L) is equal to $\Delta W + \Delta T$.

linoleate 1 the trivial name for (*all-Z*)-octadeca-9,12-dienoate, $\text{CH}_3\text{--}[\text{CH}_2]_3\text{--}[\text{CH}_2\text{--CH=CH}]_2\text{--}[\text{CH}_2]_7\text{--COO}^-$, (*all-Z* isomer) the anion derived from linoleic acid, (*all-Z*)-octadeca-9,12-dienoic acid, a diunsaturated straight-chain higher fatty acid essential in the diet of mammals. It occurs naturally as acylglycerols in many vegetable oils and in the lipids of many animal tissues. **2** any mixture of free linoleic acid and its anion. **3** any salt or ester of linoleic acid. See also **linoleic family**.

linoleate synthase see **phosphatidylcholine desaturase**.

linoleic family (of polyunsaturated fatty acids) a series of polyenoic acids occurring in mammals, in which the hydrocarbon chain terminates with the alkenyl grouping, $\text{CH}_3\text{--}[\text{CH}_2]_4\text{--CH=CH--}$ (as in linoleic acid; see **linoleate**) and

that are synthesized from linoleic acid by chain elongation and/or further desaturation (but which in mammals cannot be synthesized from the corresponding saturated acids or from (9,12,15)-linolenic, oleic, or palmitoleic acids – the latter represent three other series of polyenoic acids). The linoleic family is also known as the $\omega 6$, or *n-6*, family because of the position of the double bond nearest the terminal methyl group. The series includes (6,9,12)-linolenic acid, eicosa-11,14-dienoic acid, eicosa-5,11,14-trienoic acid, eicosa-8,11,14-trienoic acid, and arachidonic acid.

linolenate 1 or (9,12,15)-linolenate or (formerly) **α -linolenate** the alternative trivial names for (*all-Z*)-octadeca-9,12,15-trienoate; $\text{CH}_3\text{--}[\text{CH}_2\text{--CH=CH}]_3\text{--}[\text{CH}_2]_7\text{--COO}^-$ (*all-Z* isomer); the anion derived from (9,12,15)-linolenic acid, (*all-Z*)-octadeca-9,12,15-trienoic acid, a triunsaturated straight-chain higher fatty acid synthesized by plants (notably found in linseed oil) but essential in the diet of mammals (see **linolenic family**). The name equally refers to any mixture of (*all-Z*)-octadeca-9,12,15-trienoic acid and its anion, or any salt or ester of this acid. **2** (6,9,12)-linolenate or (formerly) **γ -linolenate** the trivial name for (*all-Z*)-octadeca-6,9,12-trienoate; $\text{CH}_3\text{--}[\text{CH}_2]_3\text{--}[\text{CH}_2\text{--CH=CH}]_3\text{--}[\text{CH}_2]_4\text{--COO}^-$ (*all-Z* isomer), the anion derived from (6,9,12)-linolenic acid (*all-Z*)-octadeca-6,9,12-trienoic acid, a triunsaturated straight-chain higher fatty acid belonging to the **linoleic family**. It occurs naturally as acylglycerols in certain plant-seed oils, especially that of the evening primrose (*Oenothera* spp.), and also in small amounts in the lipids of many animal tissues, especially human breast milk. It is the biosynthetic precursor of arachidonate and the prostaglandins. The name refers equally to any mixture of (*all-Z*)-octadeca-6,9,12-trienoic acid and its anion or any salt or ester of the acid.

linolenic family (of polyunsaturated fatty acids) a series of polyenoic acids occurring in mammals, in which the hydrocarbon chain terminates with the alkenyl grouping $\text{CH}_3\text{--CH}_2\text{--CH=CH--}$ (as in (9,12,15)-linolenic acid; see **linolenate**). The linolenic family is also known as the $\omega 3$ or *n-3* family because of the position of the terminal double bond three carbon atoms from the methyl end. The longer-chain members of the family can be synthesized in mammals only from dietary (9,12,15)-linolenic acid by chain elongation and/or further desaturation (but not from the corresponding saturated acids or from linoleic, oleic, or palmitoleic acids – the latter represent the other three series of polyenoic acids). If animals are deprived of a source of these fatty acids in the diet, certain tissues, such as the eye, retain existing stores of these fatty acids tenaciously, even in offspring of such animals when the restricted diet is continued for more than one generation. Under these conditions, long-chain acids of the **linoleic family**, e.g. (4,7,10,13,16,-)docosapentaenoic (C22:5 *n-6*) acid, are synthesized in appreciable amounts and presumably partially support the function supplied by long-chain $\omega 3$ acids, though the nature of this is unknown. The series includes eicosapentaenoic (C20:5 $\omega 3$) and docosahexaenoic (C22:6 $\omega 3$) acids, which are found in high concentration in fish oils; these $\omega 3$ acids may have value in protecting against heart disease, a role that, if real, would be unrelated to their function as essential fatty acids, since the amount considered necessary for this purpose greatly exceed those needed to fulfil their role as essential fatty acids.

linolenoyl 1 or (9,12,15)-linolenoyl or (formerly) **α -linolenoyl** *symbol:* αLnn or (sometimes) Lnn ; the alternative trivial names for (*all-Z*)-octadeca-9,12,15-trienoyl; $\text{CH}_3\text{--}[\text{CH}_2\text{--CH=CH}]_3\text{--}[\text{CH}_2]_7\text{--CO--}$ (*all-Z* isomer); the acyl group derived from (*all-Z*)-octadeca-9,12,15-trienoic acid. **2** (6,9,12)-linolenoyl or (formerly) **γ -linolenoyl** *symbol:* γLnn ; the trivial names for (*all-Z*)-octadeca-6,9,12-trienoyl; $\text{CH}_3\text{--}[\text{CH}_2]_3\text{--}[\text{CH}_2\text{--CH=CH}]_3\text{--}[\text{CH}_2]_4\text{--CO--}$ (*all-Z* isomer); the acyl group derived from (*all-Z*)-octadeca-6,9,12-trienoic acid. See **linolenate**.

linoleoyl *symbol:* Lin ; the trivial name for (*all-Z*)-octadeca-9,12-dienoyl; $\text{CH}_3\text{--}[\text{CH}_2]_3\text{--}[\text{CH}_2\text{--CH=CH}]_2\text{--}[\text{CH}_2]_7\text{--CO--}$ (*all-Z* isomer); the acyl group derived from linoleic acid. See **linoleate**.



liothyronine

liothyronine a pharmaceutical name for L-3,5,3'-triiodothyronine.

lip+ a variant form of **lipo+** (before a vowel).

lipaemia a variant spelling (*esp. Brit.*) of **lipemia**.

lipase 1 (*in enzyme nomenclature*) a general term for any acylglycerol carboxylic ester hydrolase, of EC sub-subclass 3.1.1. The term may be incorporated in the enzyme name as the suffix '+lipase'. 2 *abbr. (in clinical chemistry):* LPS; an alternative name for **triacylglycerol lipase**. See also **hormone-sensitive lipase**, **lipoprotein lipase**, **pancreatic lipase**.

lipemia or **lipidemia** the condition when there is a greater than normal content of lipid in the blood, e.g. after a fatty meal or in certain pathological states. Also (*esp. Brit.*) **lipaemia**, **lipidaemia**.

lipid or (formerly) **lipide** or **lipoid** or **lipin(e)** 1 any member of a large and diverse group of oils, fats, and fatlike substances that occur in living organisms and that characteristically are soluble in **lipid solvents** but only sparingly soluble in aqueous solvents. Lipids constitute one of the four major classes of compounds found in living tissues (the others being carbohydrates, proteins, and nucleic acids) and they include: (1) fatty acids; (2) neutral fats (i.e. triacylglycerols), other fatty-acid esters, and soaps; (3) long-chain (or fatty) alcohols and waxes; (4) sphingoids and other long-chain bases; (5) glycolipids, phospholipids, and sphingolipids; and (6) carotenes, polyphenols, sterols (and related compounds), terpenes, and other isoprenoids. See individual entries for details. 2 any oily, fatty, or fatlike material that occurs in living organisms. 3 of, concerned with, for, or relating to **lipid** (def. 2); consisting of or containing **lipid** (def. 1, 2); **lipoid** (def. 1) or **lipoidal**. —**lipidic** *adj.*

lipid A the glycolipid moiety of bacterial **lipopolysaccharide**.

lipidaemia a variant spelling (*esp. Brit.*) of **lipidemia**; see **lipemia**.

lipid bilayer any layer, two molecules thick, of amphipathic lipid molecules. If it is surrounded by a polar environment, the polar parts of the lipid molecules are oriented outwards towards the environment and the nonpolar parts inwards into the interior of the bilayer. The two layers of lipid are referred to as **leaflets**, e.g. the inner and outer leaflets of the bilayer. See also **black lipid membrane**.

lipide a variant spelling of **lipid**.

lipidemia an alternative term for **lipemia**.

lipidosis or **lipoidosis** (*pl. lipidoses or lipoidoses*) any clinical disorder, also called **lipid storage disease**, that results in the excessive deposition within specific tissues of particular types of lipids. Many lipidoses are inherited disorders of galactosphingolipid metabolism.

lipid-soluble a vague term applied to various classes of nonpolar substances, e.g. sterols, certain vitamins, carotenoids, and polyphenols, that are commonly present in small amounts in association with the major lipid.

lipid solvent any relatively nonpolar solvent that can be used to extract lipids from tissues or other materials.

lipid storage disease an alternative term for **lipidosis**.

lipin or **lipine** a former term for **lipid** (def. 1), especially one containing nitrogen and/or phosphorus.

Lipmann, Fritz Albert (1899–1986), German-born US biochemist; Nobel Laureate in Physiology or Medicine (1953) 'for his discovery of co-enzyme A and its importance for intermediary metabolism' [prize shared with H. A. Krebs].

lipo+ or (before a vowel) **lip+** *comb. form* denoting lipid, lipid-containing, or lipid-like.

lipoamide or (sometimes) **lipoyllysine** a **lipoyl** group when in amide linkage to the ε-amino group of a specific lysine residue in dihydrolipoamide acetyltransferase (EC 2.3.1.12) and dihydrolipoamide succinyl transferase (EC 2.3.1.61), which form, respectively, parts of the **pyruvate dehydrogenase complex** and the **2-oxoglutarate dehydrogenase complex**.

lipoamide reductase (NADH) EC 1.8.1.4; *recommended name:* dihydrolipoamide dehydrogenase; *other names:* E3 component

ponent of α-ketoacid dehydrogenase complexes; lipoyl dehydrogenase; dihydrolipoamide dehydrogenase; diaphorase. An enzyme that catalyses the oxidation by NAD⁺ of dihydrolipoamide to form lipoamide and NADH. FAD is a coenzyme. It is typically a homodimer, occurring in the mitochondrial matrix (in eukaryotes), with a redox-active S–S bond similar to glutathione reductase. Example (precursor) from yeast (the first 21 amino acids form the mitochondrial transit peptide); database code DLDH_YEAST, 499 amino acids (53.95 kDa).

lipoate 1 1,2-dithiolane-3-pentanoate, the anion derived from **lipoic acid** (i.e. thioctic acid). 2 any salt or ester of lipoic acid.

lipocaic a substance extractable from the pancreas that was said to be lipotropic in pancreatectomized animals.

lipocalin any of a large group of ligand-binding proteins, particularly one that binds hydrophobic ligands, e.g. biliverdin pheromone. The family includes retinol-binding protein, α-microglobulin, insect bilin-binding proteins, and **beta lactoglobulin**. The name derives from the fact that the ligand is enclosed by the protein as a flower is enclosed by the calyx (i.e. the sepals). All members of the family have a similar β-barrel structure of eight barrels crossing in the centre; different ligand-binding properties are determined by loops on the periphery. See also **Appendix E**.

lipochrome a former name for any of various lipid-soluble biological pigments.

lipocortin see **annexin**.

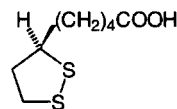
lipofection a lipid-mediated DNA-transfection technique that makes use of unilamellar liposomes that contain a cationic lipid and spontaneously react with DNA.

lipofuscin any of various brownish lipid-soluble pigments that are deposited in animal cells, especially in senility.

lipogenesis the formation of fat. The term is applied especially to the processes by which higher fatty acids are synthesized from nonlipid sources. —**lipogenic** *adj.*

lipogenic amino acid any amino acid that can provide carbon atoms for the biosynthesis of higher fatty acids.

lipoic acid or α-lipoic acid *symbol:* Lip(S₂); 1,2-dithiolane-3-pentanoic acid; thioctic acid; the naturally occurring form is the (R)-(+)-enantiomer. Lipoic acid provides the reversibly reducible moiety, **lipoyl**, of **lipoamide** and is a growth factor for certain microorganisms. The reduced form is dihydrolipoic acid, HS–CH₂–CH₂–CH(SH)–[CH₂]₄–COOH (*symbol:* Lip(SH)₂); the redox potential of the system is –0.29 V at pH7. Lipoic acid is an essential component of the α-ketoacid dehydrogenase complexes. See **pyruvate dehydrogenase complex** and **2-oxoglutarate dehydrogenase complex**.



lipoid 1 or **lipoidal** being or resembling an oil or a fat; oily, fatty. 2 a former term for **lipid** (def. 1, 2).

lipolysis the breakdown of lipids, especially the (enzymic) hydrolysis of acylglycerols. —**lipolytic** *adj.*

lipolytic hormone an alternative name for **lipotropin**.

lipomodulin or **annexin I** a 40 kDa regulatory protein isolated from rabbit neutrophils that, in its nonphosphorylated form, is an inhibitor of phospholipase A₂ and that on phosphorylation by a cyclic-AMP-dependent protein kinase loses this inhibitory property. It has been suggested that lipomodulin thereby regulates the release of arachidonate from leukocytes, or other cells, that may occur in response to chemoattractants and other external stimuli.

lipophilic tending to dissolve in, having a strong affinity for, or readily mixing with lipids or substances of low polarity. See also **hydrophilic**, **lyophilic**. —**lipophilicity** *n.*

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lipophilin

lipophilin another name for myelinproteolipid. See **myelin protein**.

lipophorin any member of the major class of lipid-transporting proteins found in the hemolymph of insects. In locusts the apoprotein is synthesized in the fat body as an 85 kDa polypeptide. See also **apolipophorin**.

lipophosphoglycans see **GPI anchor**.

lipopolysaccharide *abbr.*: LPS; any of a group of related, 10 kDa, structurally complex components of the outer leaflet of the outer membrane of Gram-negative bacteria. Lipopolysaccharide molecules consist of three covalently linked regions: lipid A, core oligosaccharide, and an O side chain. The innermost layer, lipid A, which is responsible for the toxicity of the lipopolysaccharide, consists of six fatty acyl chains (sometimes hydroxylated) linked in various ways to two glucosamine residues. The branched core oligosaccharide contains ten saccharide residues, several of them unusual, and has a structure that appears to be similar in closely related bacterial strains. The outermost O side chain, which is highly variable and determines the antigenic specificity of the organism, is made up of many (≈ 50) repeating units of a branched tetrasaccharide containing further unusual sugar residues.

lipopolysaccharide N-acetylglucosaminyltransferase EC 2.4.1.56; an enzyme of the lipopolysaccharide core biosynthetic pathway in enteric bacteria; it catalyses a reaction between UDP-N-acetyl-D-glucosamine and lipopolysaccharide to form UDP and N-acetyl-D-glucosaminyl lipopolysaccharide. Example from *Salmonella typhimurium*: database code RFAC_SALTY, 381 amino acids (43.10 kDa). There is little similarity between the given example and the corresponding enzyme in *Escherichia coli*. See also **glycolipid**.

lipoprotein any conjugated, water-soluble protein in which the nonprotein moiety consists of a lipid or lipids. The lipid may be triacylglycerol, cholesterol, or phospholipid, or a combination of these. Compare **proteolipid**.

lipoprotein(a) one or two molecules of apolipoprotein(a) linked by a disulfide bridge to apolipoprotein B-100; it may be a physiological inhibitor of plasminogen activation. Levels in human plasma (which range from undetectable to 100 mg dL⁻³) show correlation with the incidence of atherosclerosis.

lipoprotein lipase EC 3.1.1.34; *abbr.*: LPL; *other names*: clearing factor lipase; diglyceride lipase; diacylglycerol lipase. An enzyme that catalyses the hydrolysis of triacylglycerol to diacylglycerol and fatty-acid anion. Its function, after uptake of dietary lipid, is to hydrolyse triacylglycerols in chylomicrons and very-low-density lipoproteins, to diacylglycerols and fatty-acid anions; the fatty acids are then taken into peripheral tissues for further metabolism; it will also hydrolyse diacylglycerols. It acts in the presence of apolipoprotein C-II on the luminal surface of vascular endothelium. It is a **GPI-anchor** dimer, and is released by heparin, etc. Example (precursor) human: database code LIPL_HUMAN, 475 amino acids (53.10 kDa).

lipoprotein signal peptidase see **premurein-leader peptidase**.

liposome 1 a natural lipid globule suspended in the cytoplasm of some cells. 2 any small, roughly spherical, artificial vesicle consisting of a continuous bilayer or multibilayer of complex lipids enclosing some of the suspending medium. Liposomes are formed by allowing complex lipids to 'swell' in aqueous solution sometimes with the aid of sonication. They are used experimentally as models of biological membranes and therapeutically for entrapment of drugs, enzymes, or other agents with a view to their more effective delivery to target cells. Originally, such a structure was termed a cellule (when bilayered), smectic mesophase, spherule, or spherulite. See also **liquid crystal**.

lipotauroine the 7,13-dihydroxy-2-trans-octadecenoyl amide of taurine. It is found in the lipid fraction of *Tetrahymena thermophila*.

lipoteichoic acid any of a group of **teichoic acids** that contain lipid, are present in the membranes of all species of Gram-positive bacteria so far examined, and, unlike wall teichoic acids, are extractable with hot water or phenol.

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lipotropic or **lipotropic** tending to prevent the accumulation of, or to remove, abnormal amounts of lipid (in a tissue or organ, e.g. adipose tissue, liver).

lipotropic agent or **lipotropic substance** any substance capable of preventing or correcting fatty infiltration of the liver caused by choline deficiency.

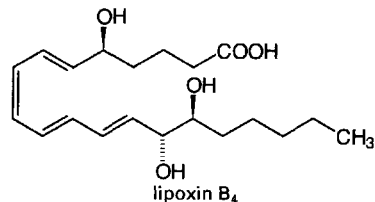
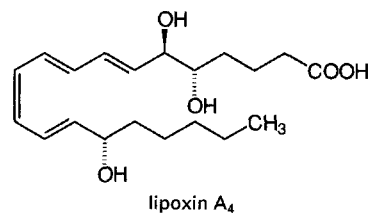
lipotropic hormone or **lipotropic hormone** a former name for **lipotropin**.

lipotropin or **lipotropic hormone** (*abbr.*: LPH) either of two polypeptides, β - and γ -lipotropin, that can be isolated from the anterior and intermediary lobes of mammalian pituitary glands and are characterized by their ability to elicit lipolysis in adipose tissue. β -Lipotropin corresponds in sequence to the 91 C-terminal amino-acid residues of **proopiomelanocortin**, and γ -lipotropin corresponds to the 58 N-terminal residues of β -lipotropin. In fact, they are of no greater lipotropic potency than corticotropin and α -melanotropin, and whether they have any intrinsic physiological function is still uncertain. Both β - and γ -lipotropins are found in normal human blood. *Other names*: lipolytic hormone; adipokinetic hormone; *former names*: lipotrophin; lipotropic hormone.

lipotropism the state or quality of being lipotropic.

lipovitellin a ≈ 135 kDa phospholipoprotein that, together with phosvitin, constitutes the bulk of the yolk proteins of the eggs of oviparous vertebrates. These two proteins are synthesized and secreted by the liver as a single large molecule, **vitellogenin**, which is taken up by the ovary and split into the two components.

lipoxin *abbr.*: LX; any of a group of eicosanoids containing a conjugated tetraene structure and having three hydroxyl groups. They are generated by the action of **lipoxygenase** enzymes on polyunsaturated fatty acids (arachidonate and eicosapentaenoate). The major lipoxins, A₄ and B₄, are derived from arachidonate; lipoxin A₄ (LXA₄) is (7E,9E,11Z,13E)-(5S,6R,15S)-5,6,15-trihydroxy-eicosa-7,9,11,13-tetraen-1-oate; lipoxin B₄ (LXB₄) is (6E,8Z,10E,12E)-(5S,14R,15S)-5,14,15-trihydroxy-eicosa-6,8,10,12-tetraen-1-oate. LXA₄ has been found to cause contraction of guinea-pig lung strips, but not ileum; LXA₄ and LXB₄ cause dilatation of arterioles.



lipoxygenase 1 EC 1.13.11.12; *systematic name*: linoleate:oxygen 13-oxidoreductase; *other names*: lipoxidase; carotene oxidase. An enzyme that catalyses the oxidation by dioxygen of linoleate to (9Z,11E)-(13S)-13-hydroperoxyoctadeca-9,11-dienoate (*abbr.*: 13(S)-HpODE); iron is a cofactor. 2 any member of a group of dioxygenase enzymes that catalyse the oxidation of polyunsaturated fatty acids to a particular corresponding hydroperoxide. Such enzymes are



widely distributed in plants and in animals from invertebrates to humans. They are found variously in leukocytes, mast cells, platelets, and lung, their positional specificities varying from tissue to tissue; notably a 12-lipoxygenase predominates in platelets, and a 5-lipoxygenase in neutrophils. They are important in the conversion of arachidonate to 5- and 12-hydroperoxyeicosatetraenoates (the 5-hydroperoxy compound being the biosynthetic precursor of the leukotrienes). Lipoxygenase activity has also been observed upon a variety of polyunsaturated fatty acids, indicating that such enzymes exist in a wide variety of forms. The configuration at the oxidized carbon is most often *S*, but *R* products are known, e.g. of sea urchin egg lipoxygenase. Of the three main lipoxygenase activities, the function of 12-lipoxygenase (EC 1.13.11.31) remains to be clarified; 5-lipoxygenase (EC 1.13.11.34) is important in yielding precursors for **leukotriene** biosynthesis; 15-lipoxygenase (EC 1.13.11.33) yields precursors for the **lipoxins**. All three, however, are involved in the synthesis of hydroperoxy- and dihydroxy-eicosatetraenoates. Compounds that inhibit both lipoxygenase and prostaglandin-endoperoxide synthase include **curcumin**, **eicosatetraenoic acid**, and phenidone (1-phenyl-3-pyrazolidinone). Compounds that selectively inhibit lipoxygenases without inhibiting prostaglandin-endoperoxide synthase include **eicosatrienoic acid** and **esculetin**. See also **dihydroxyeicosatetraenoate**, **hydroperoxyeicosatetraenoate**, **hydroperoxyoctadecadienoate**, **hydroxyeicosatetraenoate**.

lipoyl the acyl group, 1,2-dithiolane-3-pentanoyl, derived from **lipoic acid**. It provides the reversibly reducible moiety of **lipoamide**.

lipoyllysine an alternative name for **lipoamide**.

Lip(S₂) symbol for a residue of (the oxidized form of) **lipoic acid**.

Lip(SH)₂ symbol for a residue of dihydrolipoic acid, the reduced form of **lipoic acid**.

lipstatin an irreversible inhibitor of several lipases, e.g. pancreatic, gastric, carboxyl ester, or bile-stimulated lipase of milk, isolated from *Streptomyces toxytricini*. It blocks fat absorption selectively.

lipuria the presence of lipid or oily droplets in the urine.

liquefy or **liquify** to become, or cause to become, liquid. —**liquefaction** or **liquifaction** *n*.

liquid 1 a state of matter, intermediate between a solid and a gas, in which the molecules are relatively free to move with respect to each other but are restricted by intermolecular forces strong enough to prevent spontaneous expansion or significant compression. 2 any substance that is liquid at normal room temperature and pressure. 3 of, pertaining to, being, or behaving as a liquid. Compare **fluid**.

liquid air a pale-blue liquid containing principally liquid nitrogen and liquid oxygen, b.p. -194.35 °C (78.80 K), obtained by compressing and cooling gaseous air. It is used as a refrigerant and in the preparation of nitrogen and oxygen.

liquid chromatography *abbr.*: LC; any form of **chromatography** in which a liquid is used as the mobile phase.

liquid crystal or **mesomorphic state** or **mesophase** any new phase, resembling both liquid and solid phases, that appears when certain pure liquids are cooled, or pure solids are heated. These new phases have a translucent or cloudy appearance. In liquids of asymmetric molecules the molecular axes are arranged at random. In liquid crystals, however, there is some alignment, which occurs in three main distinguishable modes (certain such mesomorphic substances displaying more than one). In **nematic liquid crystals** the long axes of the molecules are aligned parallel to each other but the molecules are not arranged in layers. These scatter light strongly and appear translucent. In **cholesteric liquid crystals** the molecular axes are aligned and the molecules are arranged in layers but the orientation of the axes changes in a regular way on going from one layer to the next. Because the distance between any two layers with the same molecular orientation is of the order of the wavelength of visible light such crystals display vivid iridescence. **Smectic liquid crystals** are formed by certain mol-

ecules that have chemically dissimilar parts, e.g. polar and nonpolar; the chemically similar parts attract one another and the molecules form layers and align themselves in one direction. See also **liquid-crystalline phase**.

liquid-crystal display *abbr.*: LCD; any display of characters or numbers in a calculator, clock, or other device, based on **liquid crystal** cells whose reflectivity is changed by application of an electric field

liquid-crystalline phase any of several types of phase adopted by polar lipids at moderately elevated temperatures. Two major types exist, the lamellar and hexagonal phases. In the lamellar phase, also known as the *L_α* phase, the lipids form sheets of bilayers with the polar groups oriented outwards and the fatty-acyl moieties oriented into the hydrophobic core between the two bilayers. Such a phase is likely to include many such bilayers stacked on top of each other, with their polar groups apposed. In the hexagonal phase, the lipids form a spherical aggregate, in which the polar headgroups may be oriented either into the centre of the sphere, when this is known as the *H₁* phase, or towards the surface of the sphere, the *H₂* phase. In liquid-crystalline phase, the lipid adopts a looser packing, with greater mobility of the fatty-acyl chains than in the crystalline state. At low temperatures, in the absence of water, lipids exist in a crystalline state. If the lipid is heated to a point that attains the **transition temperature**, a temperature characteristic of each pure compound, the lipid undergoes a transition to one of the liquid-crystalline phases. On further heating, it melts completely. Addition of water at low temperature to the crystalline state causes formation of a gel phase, with water layers between the apposed polar headgroups. This gel phase will, on heating to the transition temperature, undergo transition to a liquid-crystalline phase similar to those described above.

liquid-gel chromatography a type of liquid chromatography in which the stationary phase is a gel. It includes **gel-permeation chromatography** and **ion-exchange chromatography**.

liquid glucose an alternative name for **corn syrup**.

liquid-junction potential the electric potential, not experimentally measurable, that arises across the boundary of two solutions of different compositions, e.g. between two half-cells. It is a complex function of the activities and transference numbers of the several ions in the transition layer.

liquid-liquid chromatography *abbr.*: LLC; a type of liquid chromatography in which the stationary phase is a liquid supported on a solid.

liquid medium any liquid **culture medium**.

liquid nitrogen nitrogen in the liquid state. It has a b.p. of -195.79 °C (77.36 K), and is useful as a cooling and freezing agent.

liquid oxygen oxygen in the liquid state. It has a b.p. of -182.96 °C (90.19 K).

liquid paraffin a colourless, odourless fraction of higher liquid hydrocarbons obtained from petroleum. Light liquid paraffin has a density of 0.83–0.86 g ML⁻¹ and heavy liquid paraffin has a density of 0.86–0.89 g ML⁻¹. It is used for (hot) oil baths, and the purified material is used as a suspending medium in infrared spectroscopy.

liquid-scintillation counting *abbr.*: LSC; a technique for counting particulate radiation. The (radioactive) sample is dissolved or suspended in, or placed in close approximation to, a solvent containing one or more substances that are fluorescent and emit a pulse of light (scintillation) when excited by an incident ionizing particle or photon. The light is detected by photomultipliers, in a liquid-scintillation counter.

liquid-scintillation spectrometry **liquid-scintillation counting** of particulate radiation of one or more selected ranges of energies.

liquid-solid chromatography *abbr.*: LSC; a type of liquid chromatography in which the mobile phase is a liquid and the stationary phase a solid.

liquor 1 (in pharmacy) a (usually aqueous) solution of a pure

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liquorice

substance. **2** a solution in a chemical manufacturing process. **3** the water used in brewing.

liquorice or (esp. US) **licorice** or **glycyrrhiza** the dried roots and rhizome of *Glycyrrhiza glabra*. This preparation contains glycyrrhizin, the glucoside of glycyrrhetic acid; it resembles **aldosterone** and has effects similar to the **mineralocorticoids**.

LISP a high-level computer language used for artificial intelligence. [Acronym from 'list processing'.]

liter the US spelling of **litre**.

lithiasis (in pathology) the presence of calculi (stones), e.g. in the biliary or urinary tracts.

lithium symbol: Li; an element of group 1 of the IUPAC periodic table; atomic number 3; relative atomic mass 6.939. It is a soft, silvery white, alkali metal of such high reactivity that it must be stored in an inert atmosphere or under oil. It is not found free in nature, occurring in the minerals spodumene and lepidolite. Lithium carbonate, Li_2CO_3 (commonly referred to as 'lithium') is used to control the symptoms of manic-depressive illness. Li^+ inhibits inositol phosphate phosphatase, thereby inhibiting the recycling of inositol, and reducing its availability for phosphatidylinositol synthesis.

lithogenesis (in pathology) the processes leading to formation of calculi (stones). —**lithogenic** adj.

lithotroph any member of either of two groups of organisms, chemoautotrophs (see **chemotroph**) and **photolithotrophs**, that use CO_2 as a carbon source.

litre or (US) **liter** symbol: L or l; a non-SI unit of volume, now identical to the cubic decimetre; i.e. $1 \text{ L} = 1 \text{ dm}^3 = 10^{-3} \text{ m}^3$.

little gastrin see **gastrin**.

liver one of the largest organs of the vertebrate body, situated in the abdominal cavity and formed as a diverticulum of the gut. It is highly vascular; venous blood carrying the products of digestion enters via the hepatic portal vein, while arterial blood enters via the hepatic artery. Blood leaves via the hepatic vein. The liver functions as a gland by secreting **bile** and is an important site of metabolism of carbohydrates, proteins, and fats; it stores carbohydrate as **glycogen**, is active in **gluconeogenesis**, and is important in the regulation of the blood glucose level. It is also an important site for the synthesis of **urea** and **ketone bodies**. The liver synthesizes **fibrinogen**, **prothrombin**, **serum albumin**, and other blood proteins and plays an important part in **detoxication**. The major cells in the liver are the **hepatocytes**. The liver also contains **Kupffer cells** of the reticulo-endothelial system, cells derived from the bile duct cannaliculae, and fibroblasts, which are particularly important in the pathogenesis of liver cirrhosis. See also **regeneration** (def. 2).

LLC abbr. for liquid-liquid chromatography.

LLD factor abbr. for *Lactobacillus lactis* Dorner factor; a growth factor for this organism, identical to **vitamin B₁₂**.

LLF or **L-L factor** abbr. for Laki-Lorand factor, i.e. factor XIII (see **blood coagulation**).

lm symbol for lumen (def. 1).

L meromyosin abbr. for light meromyosin.

LMM abbr. for light meromyosin.

ln symbol for natural, or Napierian, **logarithm** (alternative to **log_e**).

αLnn or (sometimes) **Lnn** symbol for the linolenoyl or (9,12,15)-linolenoyl or (formerly) α-linolenoyl group; see **linolenoyl** (def. 1).

γLnn symbol for the (6,9,12)-linolenoyl or (formerly) γ-linolenoyl group; see **linolenoyl** (def. 2).

lnRNA abbr. for low-molecular-weight nuclear RNA (see **small nuclear RNAs**).

load (in metabolic studies) the amount (of a specified substance) given in a metabolic test.

loading test or **load test** a metabolic test in which the subject is given a dose of a specific substance and the concentration in blood and/or urine of the substance, or of some related metabolite, is determined as a function of time. Such a test is useful for determining the capacity of a particular metabolic pathway in the subject.

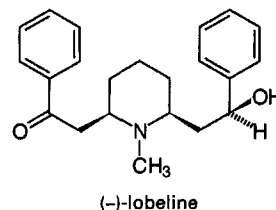
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lobelia **1** any plant of the genus *Lobelia*. **2** or **Indian tobacco** the dried leaves, shoot tips, and seeds of *Lobelia inflata*, which contain the alkaloid **lobeline**.

lobeline 2-[6-(β-hydroxyphenethyl)-1-methyl-2-piperidyl]acetophenone; an alkaloid derived from **lobelia** (def. 2) with actions on the central and peripheral nervous systems that resemble those of **nicotine**. It is responsible for some of the properties of herbal cigarettes.



Lobry de Bruyn-van Ekenstein transformation a non-enzymic transformation occurring in carbohydrates, usually in alkaline conditions, that includes epimerization of both aldoses and ketoses and aldose-ketose isomerization. [After Cornelius Adrian Lobry de Bruyn and W. Alberda van Ekenstein (1857–1937), who first described it in 1895.]

localized melting the (reversible) destabilization of tertiary structure in a limited region of an ordered biopolymer through interaction with a specific agent, e.g. an enzyme.

locant a prefix or infix used in organic chemical nomenclature to identify the position of a substituent in a parent structure or of an isotopic modification to a compound. Locants are commonly Arabic numerals, but Greek letters and italicized symbols of heteroatoms are also used. In isotopically modified compounds the italicized name of the function containing the modification may be used if it has no number, e.g. in [*methyl*-¹⁴C]methionine.

loci the plural of **locus**.

lock-and-key model a model for the mechanism of an enzyme-substrate combination, of a hormone-receptor interaction, or of an antibody-antigen reaction. In this model, which is based on the analogy of the complementarity of a lock and its key, one reactant is envisaged binding to the other on a site to which it is structurally complementary.

Locke's solution or **Locke solution** or **Ringer-Locke solution** any one of several balanced salt solutions modified from **Ringer's solution**, initially for the perfusion of mammalian heart and other tissues in physiological experiments. Such solutions have usually contained (in grams per litre, approx.) NaCl (9.0–9.5), KCl (0.20–0.42), CaCl₂ (anhydrous, 0.20–0.24), NaHCO₃ (0.1–0.3), and glucose (1.0–2.5). [After Frank Spiller Locke (1871–1949), British physiologist, who devised them in 1894 and 1900.]

lock-washer structure a putative intermediate structure formed during the assembly of subunits of the coat protein of tobacco mosaic virus into the helically organized structure of the fully formed virion. It is believed to arise after the primary subassembly, which is a two-layered disk with 17 subunits per layer, has self-aggregated into a cylinder; a dislocation in each disk then occurs so that it becomes a two-turn flattened proto-helix, of similar shape to a lock washer of the sprung type.

loco disease see **locoweed**.

locoweed any of various plants belonging to the genera *Astragalus* or *Oxytropis*, found mainly in western USA. They contain L-2-amino-3-(methylseleno)propionic acid (methylselenocysteine), a nonprotein amino acid, and may cause poisoning (**loco disease**) when ingested by livestock. [From Spanish *loco*, mad.]

locus (pl. **loci**) **1** (in genetics) the specific position, in all homologous chromosomes, of a particular gene or one of its alleles. **2** (in mathematics) the curve or line traced by a point moving so as to satisfy a particular set of conditions.



locus control region

locus control region *abbr.*: LCR; a group of five regions upstream of the structural genes for the β chain of hemoglobin in human and mouse. These regions are essential for the expression of all the genes of the β -like gene cluster.

lod score *abbr.* for log of the odds score; a statistical test to determine whether a set of linkage data indicates linkage or not, the $\log_{10}$ of the odds favouring linkage. For genetic disorders that are not X-linked, a lod score of +3 (1000:1 odds of linkage) indicates linkage, whilst a score of -2 is odds of 100:1 against linkage.

Loewi, Otto (1873–1961), German-born US physiologist; Nobel Laureate in Physiology or Medicine (1936) jointly with H. H. Dale 'for their discoveries relating to chemical transmission of nerve impulses'.

log 1 *symbol* for logarithm. 2 *abbr.* for logarithmic.

logarithm *abbr. and symbol*: log; the power to which a fixed number (the base) must be raised in order to produce a specified number; i.e. in the expression $a = b^n$, a is the specified number, b is the base, and n is the logarithm. In **decadic** (common) or **Briggsian**, **logarithms** the base is 10; the common logarithm of a is written as $\log_{10}a$, or $\lg a$. In **natural**, or **Napierian**, **logarithms** the base is e ($\approx 2.718\ 28\dots$); the natural logarithm of a is written as $\log_e a$ or $\ln a$. In **binary logarithms** the base is 2, and the binary logarithm of a is written as $\log_2 a$ or $\lg a$. In converting between them, $\log_e a = 2.303 \times \log_{10} a$, and $\log_2 a = 3.322 \times \log_{10} a$.

logarithmic or **logarithmical** *abbr.*: log; 1 of, pertaining to, using, or containing **logarithms** of numbers or variables. 2 describing a relationship in which one variable is proportional to the logarithm of another.

logarithmic growth a type of growth pattern shown by bacteria or other cells in which the number of cells, or the cell mass, increases logarithmically (exponentially) with time. The rate of increase at any time is proportional to the number of cells, or the cell mass, present.

logarithmic phase *abbr.*: log phase; the phase of growth of bacterial or other cells during which they undergo **logarithmic growth**.

logarithmic plot *abbr.*: log plot; a graph in which one axis has a **logarithmic** (def. 1) scale and the other a linear scale.

logical operation an operation on **binary** numbers or statements used in computing and also in applications such as databases and library searches. There are three common binary operators (AND, OR, XOR) and one **unary operator** (NOT).

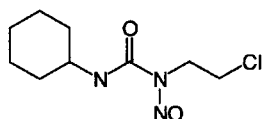
log-log plot double logarithmic plot; a graph on which both variables are plotted logarithmically.

Lohmann reaction a name sometimes given to the reversible reaction, catalysed by **creatine kinase**, in which ATP and creatine are formed from ADP and phosphocreatine. [After Karl Lohmann (1898–1978), German biochemist.]

Lohmann's enzyme an alternative name for **creatine kinase**.

lombricine the 2-guanidoethanol phosphoester of phosphoserine. In earthworms, the guanidine group can undergo phosphorylation to phospholombricine, which acts as a **phosphagen**.

lomustine 1-(2-chloroethyl)-3-cyclohexyl-1-nitrosourea; a chloroethylnitrosourea derivative with actions resembling **carmustine**.



London forces *see* **dispersion forces**.

lone pair or **nonbonding electrons** a pair of electrons of opposite spins occupying the same orbital in the valence shell of an atom.

long-acting thyroid stimulator *abbr.*: LATS; an IgG immunoglobulin found in the serum of a large proportion of thyrotoxic patients. It binds competitively with thyrotropin to thyrotropin receptors in thyroid plasma membrane, stimulates membrane adenylate cyclase, and causes longer-lasting stimulation of thyroid-hormone secretion than thyrotropin itself.

long-chain describing an aliphatic compound of any type with a chain length greater than ten carbon atoms (i.e. C_{10}).

long-chain alcohol or **fatty alcohol** any aliphatic compound with a chain length greater than C_{10} that possesses a terminal hydroxymethyl group. They are components of **ether lipids**, and esterified with long-chain fatty acids they form wax esters, often major components of plant cuticular waxes.

long-chain-alcohol O-fatty-acyltransferase EC 2.3.1.75; *other name*: wax synthase; an enzyme that catalyses the formation of a long-chain fatty acyl ester, i.e. a wax, from acyl-CoA and a long-chain alcohol with release of CoA.

long-chain base any aliphatic compound with a chain length greater than C_{10} that possesses a terminal basic group.

long-chain fatty acid any aliphatic compound with a chain length greater than C_{10} that possesses a terminal carboxyl group.

long-chain-fatty-acid-CoA ligase EC 6.2.1.3; *other names*: acyl-activating enzyme; acyl-CoA synthetase; fatty acid thio-kinase (long-chain); lignoceroyl-CoA synthase; an enzyme that catalyses the reaction between ATP and a long-chain carboxylic acid and CoA to form an acyl-CoA with release of AMP and pyrophosphate. It activates free fatty acids formed, e.g., by phospholipase action for further metabolism. Examples from yeast: long-chain-fatty-acid-CoA ligase 1 (long-chain acyl-CoA synthetase 1; fatty acid activator 1); database code LCF1_YEAST, 700 amino acids (77.78 kDa); long-chain-fatty-acid-CoA ligase 2 (long-chain acyl-CoA synthetase 2; fatty acid activator 2); database code LCF2_YEAST, 744 amino acids (83.34 kDa); long-chain-fatty-acid-CoA ligase 3 (long-chain acyl-CoA synthetase 3; fatty acid activator 3); database code LCF3_YEAST, 694 amino acids (77.86 kDa).

long-chain-3-hydroxyacyl-CoA dehydrogenase EC 1.1.1.211; *other name*: β -hydroxyacyl-CoA dehydrogenase; an enzyme of the **beta-oxidation system** that catalyses the oxidation by NAD^+ of (S)-3-hydroxyacyl-CoA to 3-oxoacyl-CoA with the formation of NADH. *See* **3-hydroxyacyl-CoA dehydrogenase**.

longitudinal relaxation time an alternative name for **spin-lattice relaxation time**.

long-patch repair *see* **short-patch repair**.

long-range interaction describing an interaction between residues that are situated relatively far apart in a polypeptide or polynucleotide sequence.

long terminal repeat *abbr.*: LTR; a double-stranded DNA sequence, generally several hundred base pairs long, that is repeated at the two ends of the DNA of retroviruses and retrotransposons.

loop 1 the round or oval figure produced by a line, wire, thread, etc. that curves round to cross (or nearly cross) itself. 2 (*in electronics*) any closed electric or magnetic circuit through which a signal can circulate. 3 (*in computing*) any set of instructions in a program that is carried out repeatedly until some specified condition is satisfied.

loopback DNA an alternative name for **hairpin DNA**; *see* **hairpin**.

loop injector a device for introducing samples into a high-pressure liquid chromatographic column without interruption of the solvent flow to the column. It consists of a loop of metal tubing of small volume, which is filled with the sample; this can then, by an appropriate valve, be flushed by eluent directly onto the column.

loop of Henle or **Henle's loop** the U-shaped part of a nephron lying in the renal medulla. It comprises a thin descending tubule and an ascending tubule formed of both a thin and a



lophotoxin

thick segment. It plays a role in the selective reabsorption of fluid and solutes. [After Friedrich Gustav Jakob Henle (1809–85), German anatomist and pathologist.]

lophotoxin a cyclic diterpenoid, isolated from Pacific gorgonians (sea fans and whips) of the genus *Lophogorgia*, that irreversibly inactivates the nicotinic acetylcholine receptor on intact BC3H-1 cells.

Lorentz correction factor a correction factor for the effect of solvent polarizability on optical rotation by solutions of proteins or other macromolecules. It is given by: $3/(n^2 + 2)$, where n is the refractive index of the solvent. [After Hendrik Antoon Lorentz (1853–1928).]

Lorentzian lineshape function an equation for the lineshape of a nuclear magnetic resonance absorption signal:

$$g(\omega) = T_2/[1 + (\omega - \omega_0)^2 T_2^2],$$

where $g(\omega)$ is a measure of the intensity of the resonance at a frequency ω , ω_0 is the actual Larmor frequency of the resonating nucleus, and T_2 is the spin-spin relaxation time.

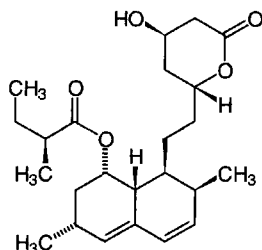
loricrin the major protein of the cornified cell envelope of terminally differentiated epidermal keratinocytes. It is a substrate for transglutaminase, and exists as monomers cross-linked by disulfide and *N*-(γ -glutamyl)lysine isodipeptide bonds. Example from human: database code LORI_HUMAN, 316 amino acids (26.01 kDa).

losartan 2-butyl-4-chloro-1-(*p*-(*O*-1*H*-tetrazol-5-ylphenyl)benzyl)imidazol-5-methanol; an antihypertensive drug that blocks the binding of angiotensin II (see **angiotensin**) to the AT₁ receptor. Affinity for this receptor is much greater than for the AT₂ receptor. It is oxidized by a cytochrome P450 to a carboxylic acid metabolite that is largely responsible for most of its angiotensin II antagonistic effects.

Loschmidt constant *symbol:* N_L ; the number of molecular entities in unit volume of an ideal gas at 1 atm and 273.15 K. It is equal to the **Avogadro constant** divided by the molar volume, and has the value $2.686\,763 \times 10^{25} \text{ m}^{-3}$. [After Josef Loschmidt (1821–95), Austrian physicist.]

Loschmidt number or **Loschmidt's number** the numerical value of the **Loschmidt constant**.

lovastatin or **mevinolin** 6 α -methylcompactin; a metabolite from some fungi (e.g. *Monascus ruber*, *Aspergillus terreus*) that is a potent inhibitor of **hydroxymethylglutaryl-CoA reductase**. It is useful clinically as an antihypercholesterolemic; it reduces plasma levels of total low-density-lipoprotein (LDL), and very-low-density-lipoprotein cholesterol, but it can increase high-density-lipoprotein cholesterol. Its effects result in the induction of up-regulation of LDL receptors on hepatocytes. Two proprietary names are: Mevacor; Mevinacor.



low-complexity region a region in the amino-acid sequence of a protein in which a particular amino acid or a small number of different amino acids are enriched. See, for example, **leucine zipper**.

low-density lipoprotein *abbr.:* LDL; one of the classes of lipoproteins found in blood plasma in many animals (data normally relate to humans). LDLs are also known as β lipoproteins due to their electrophoretic mobility intermediate between high-density or α lipoprotein, and the origin, LDL

particles have diameter 19.6–22.7 nm, solvent density for isolation (g mL^{-1}) 1.019–1.063, hydrated density (g mL^{-1}) 1.034. Their approximate composition (% by weight) is 13% unesterified cholesterol, 39% esterified cholesterol, 17% phospholipid, 11% triacylglycerol, 20% protein. Their apolipoprotein composition (% by weight total apolipoprotein) is 98% B-100 with traces of different C and E apolipoproteins. They are synthesized in plasma from very-low-density lipoproteins and intermediate density lipoproteins through the action of lipoprotein lipase; they transport cholesterol to peripheral tissues. They have the highest content of cholesterol of any plasma lipoprotein, and have been linked to the incidence of coronary heart disease in epidemiological studies. See also **apolipoprotein** (for individual apolipoproteins), and **LDL receptor**.

low-density-lipoprotein receptor see **LDL receptor**.

low-energy compound a name sometimes applied improperly to certain compounds that undergo reaction leading to release or transfer of, e.g., phosphate, where the standard free energy change for the hydrolysis of the bond is $<20 \text{ kJ mol}^{-1}$. It is used especially in making a distinction from a **high-energy compound**. The use of the term does not meet the highest standards of thermodynamic correctness.

low-lipid lipoprotein (*sometimes*) an alternative name for **high-density lipoprotein**.

Lowry method a colorimetric method for the estimation of small amounts (down to 10 μg) of protein. It is a combination of the **biuret reaction** for peptide bonds and the colour formed by reaction of **Folin-Ciocalteu reagent** with tyrosine and tryptophan residues in the protein. [After Oliver Howe Lowry (1910–1996), US pharmacologist and biochemist.]

low-spin complex any of the ligand complexes of transition-metal ions having a maximal number of paired *d*- or *f*-shell electrons in the lower energy levels. This state occurs when the pairing energy is less than the crystal-field splitting energy. Compare **high-spin complex**.

LPG *abbr.* for lipophosphoglycan (see **GPI anchor**).

LPH *abbr.* for lipotropic hormone (i.e. **lipotropin**).

LPL *abbr.* for lipoprotein lipase.

LPS *abbr.* for 1 lipopolysaccharide. 2 lipase (def. 2).

Lr *symbol* for lawrencium.

L region any region in a (carcinogenic) polycyclic aromatic hydrocarbon that consists of two *para*-carbon atoms, e.g. the *meso*-9,10-carbon atoms of anthracene, which display the highest free-valence indices. The lack of such a region has been suggested to correlate with a lack of carcinogenicity of some hydrocarbons. Compare **K region**.

LRF *abbr.* for luteinizing hormone releasing factor (see **gonadotropin-releasing hormone**).

LRH *abbr.* for luteinizing hormone releasing hormone (see **gonadotropin-releasing hormone**).

LRR *abbr.* for leucine-rich repeat.

LSC *abbr.* for 1 liquid-scintillation counting. 2 liquid-solid chromatography.

LSD *abbr.* for lysergic acid diethylamide; see **lysergic acid**.

L7/L12 site a component of EF-Tu translocase (see **EF-T**) activity involving in bacteria the **ribosomal proteins** L7 and L12.

L strand *abbr.* for light strand.

LT *abbr.* for 1 leukotriene; the subtype may be indicated by a third letter, as in LTA (leukotriene A). 2 lymphotoxin.

LTR *abbr.* for long terminal repeat.

L-type pentose pathway see **pentose phosphate pathway**.

Lu *symbol* for lutetium.

Lubrol a proprietary name for a polyoxyethylene nonionic detergent; the structure of Lubrol PX is the same as C_{12}E_9 (see **C₁₂E₉**). It is useful for solubilizing membrane-bound enzymes such as adenylate cyclase.

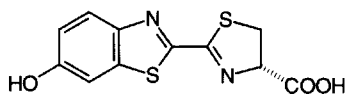
luciferase any member of a group of monooxygenase enzymes that catalyse bioluminescent reactions. **Firefly luciferase** (EC 1.13.12.7; *recommended name:* *Photinus luciferin* 4-monooxygenase (ATP hydrolysing)) occurs in the lantern of the American firefly, *Photinus pyralis*, and may be used with firefly



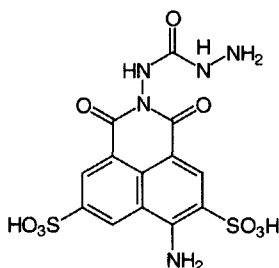
luciferin

luciferin for estimation of low concentrations (10^{-6} – 10^{-10} mol L⁻¹) of ATP. It catalyses a reaction between *Photinus* luciferin, O₂, and ATP to form oxidized *Photinus* luciferin, CO₂, H₂O, AMP, pyrophosphate, and *hν*; magnesium is a cofactor. It covalently binds the adenylyl group (AMP) after reaction with ATP and release of pyrophosphate. It is a peroxisomal enzyme. Example from *Luciola lateralis* (firefly): database code LUCI_LUCLA, 548 amino acids (60.06 kDa). **Bacterial luciferase** (EC 1.14.14.3; *recommended name*: alkanal monooxygenase (FMN-linked)) occurs in *Photobacterium fischeri* and catalyses a reaction between RCHO, reduced FMN, and O₂ to form RCOOH, FMN, H₂O, and *hν*. It may be used together with NAD(P)H dehydrogenase (FMN), EC 1.6.8.1, from luminescent bacteria for estimation of low concentrations (10^{-7} – 10^{-9} mol L⁻¹) of NAD(P)H. For examples *see* **Lux proteins**.

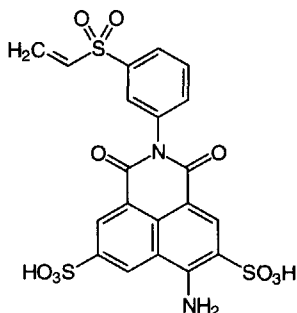
luciferin any bioluminescent substance that is a substrate for a **luciferase** enzyme. Luciferins from different sources often bear no structural resemblance to one another. Firefly luciferin, obtained from the lantern of the American firefly, *Photinus pyralis*, is D-(–)-4,5-dihydro-2-(6-hydroxy-2-benzothiazolyl)-4-thiazolecarboxylic acid.



Lucifer Yellow one of several dyes based on the structure 6-amino-2,3-dihydro-1,3-dioxo-benz[*d,e*]isoquinoline-5,8-disulfonic acid. Lucifer Yellow CH has been used as a marker for movement between cells, and for ultrastructure tracing; Lucifer Yellow VS covalently labels proteins rapidly under mild conditions.



Lucifer Yellow CH

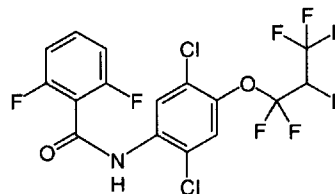


Lucifer Yellow VS

Lucite a proprietary name for poly(methyl methacrylate).

lufenuron *N*-[2,5-dichloro-4-(1,1,2,3,3,3-hexafluoropropoxy)-

phenylaminocarbonyl]-2,6-difluorobenzamide; an inhibitor of insect development, used as a once-per-month pill to control flea populations on domestic dogs, by inhibiting egg development.

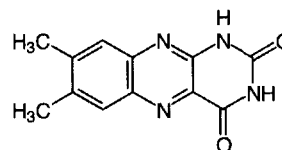


luffin a type I ribosome-inactivating protein.

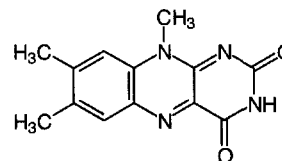
luliberin the *recommended name* for a hypothalamic decapeptide amide now known to be identical with **gonadotropin-releasing hormone**.

lumen 1 (*pl. lumens*) *symbol*: lm; the SI derived unit of **luminous flux**, equal to the quantity of visible electromagnetic radiation emitted per second in a cone of unit solid angle (one steradian, sr) by a uniform point source of unit luminous intensity (one candela, cd); i.e. 1 lm = 1 cd sr. 2 (*pl. lumina*) the cavity or passageway of a tubular organ, such as a blood vessel, duct, or length of gut. 3 the volume enclosed within the membranes of a mitochondrion or of the endoplasmic reticulum, or of microsomes prepared from cells. 4 the space enclosed by the walls of a plant cell that no longer contains a protoplast. 5 the bore of a catheter, hollow needle, or other tubular device. —**luminal** *adj.*

lumichrome 7,8-dimethylalloxazine; a compound showing a blue fluorescence, formed by the photolysis of riboflavin in acid or neutral solution.



lumiflavin 7,8,10-trimethylisoalloxazine; a compound showing a yellow-green fluorescence, formed by the photolysis of riboflavin in alkaline solution.



lumina the plural of **lumen** (def. 2).

luminal proteins or (*sometimes*) **ER-resident proteins** proteins that are retained within the cisternae of the **rough endoplasmic reticulum**, e.g. protein disulfide isomerase. Such proteins possess at their C terminus four characteristic amino acids, Lys-Asp-Glu-Leu (**KDEL**) in animal cells and His-Asp-Glu-Leu (**HDEL**) in yeast. The proteins move to the Golgi and are returned to the endoplasmic reticulum.

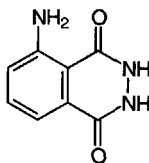
luminescence the production of visible light by any mechanism that does not depend upon the system having a temperature appreciably above the ambient. It comprises **fluorescence** and **phosphorescence**. —**luminescent** *adj.*



luminogenic

luminogenic 1 producing **luminescence**. 2 of or pertaining to a nonluminescent material that can be acted on by an enzyme to produce a luminescent substance.

luminol 5-amino-2,3-dihydro-1,4-phthalazinedione; a substance useful in the detection or determination of trace amounts of oxidizing agents by virtue of the intense bluish luminescence it emits on oxidation in alkaline solution.



luminometer an instrument for measuring **luminescence**. Given appropriate conditions, it may be used also to measure the concentrations of enzymes, substrates, etc. that participate in, or can be coupled to, light-producing reactions.

luminometry the measurement of **luminescence** with a luminometer. — **luminometric** *adj.*

luminophore 1 any luminescent substance, whether a fluor or a phosphor. 2 a molecular grouping responsible, or thought to be responsible, for the **luminescence** of a larger molecule; a luminescent chromophore.

luminous 1 emitting visible light; bright, shining. 2 (*in photometry*) (of a physical quantity relating to light) evaluated by the visual sensation of brightness to the eye of an observer rather than by an absolute measurement of radiant energy.

luminous flux *symbol:* Φ_v or Φ or P_v or P ; a measure of the rate of flow of radiant energy within the visible region of the electromagnetic spectrum. The SI derived physical quantity is the arithmetic product of the **luminous intensity** of a source of visible light and the solid angle, expressed in steradians, into which the light is emitted. The SI derived unit of luminous flux is the **lumen** (def. 1).

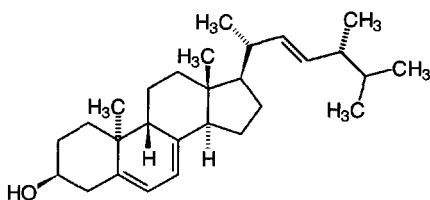
luminous flux density the **luminous flux** per unit cross-sectional area of a beam of light.

luminous intensity *symbol:* I_v or I ; one of the seven SI base physical quantities, defined as the amount of visible light emitted per second per steradian by a point source in a given direction. The SI base unit of luminous intensity is the **candela**.

lumirhodopsin an intermediate in the light-induced modification of **rhodopsin** prior to its breakdown into opsin and all-*trans*-retinal. It has an absorption maximum at 497 nm.

lumisome a membrane-bound particle, ≈ 200 nm in diameter and isolable from homogenates of various species of luminescent marine coelenterates, that contains all the proteins necessary for the production of the typical green bioluminescence observable *in vivo*. It is probably the cytoplasmic organelle responsible for emission of light by such organisms.

lumisterol $9\beta,10\alpha$ -ergosta-5,7,22-trien-3 β -ol; a compound produced by the UV irradiation of **ergosterol**, from which it differs in the spatial arrangement of the methyl group at C-10.



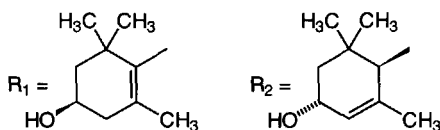
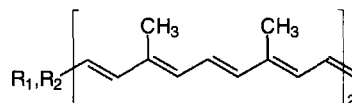
Lundh test a test for pancreatic function in which the pancreas is stimulated by a test meal (corn oil, milk powder, and glucose), after which trypsin activity is measured in duodenal juice.

Luria, Salvador Edward (1912–91), Italian-born US physician and biologist, Nobel Laureate in Physiology or Medicine (1969) jointly with M. Delbrück and A. D. Hershey 'for their discoveries concerning the replication mechanism and the genetic structure of viruses'.

luteal of, relating to, or characterized by the development of a corpus luteum or corpora lutea.

luteal phase *see* **estrous cycle**.

lutein 3,3'-dihydroxy- α -carotene, one of the most widespread naturally occurring xanthophylls (i.e. carotene alcohols; *see* **carotenoid**), found in many higher plants, some algae, egg yolk, and corpora lutea.



luteinize or **luteinise** to cause the formation of corpora lutea in; to undergo transformation into a corpus luteum. — **luteinization** or **luteinisation** *n.*

luteinizing hormone or **luteinising hormone** *abbr.:* LH; one of the two gonadotropic **glycoprotein hormones** secreted by the anterior pituitary. Of molecular mass 28.5 kDa (human), it consists of α and β subunits, the α subunit being almost identical within a given species to that of other glycoprotein hormones and the β subunit conferring hormonal specificity; the carbohydrate content is 15.5% (human). In the female it promotes ovulation and luteinization of the ovarian follicle, where it stimulates the conversion of progesterone to pregnenolone, thereby increasing the formation of further progesterone. In the male it stimulates testicular interstitial (Leydig) cell function by promoting the production of androgens, especially testosterone. Example, β chain (precursor) from *Bos taurus*: database code LSHB_BOVIN, 141 amino acids (15.20 kDa). *Recommended name:* lutropin; *former name:* interstitial cell-stimulating hormone (*abbr.:* ICSH).

luteinizing hormone/follicle-stimulating hormone releasing factor *abbr.:* LH/FSH-RF *see* **gonadotropin-releasing hormone**.

luteinizing hormone releasing factor *abbr.:* LHRF *see* **gonadotropin-releasing hormone**.

luteinizing hormone releasing hormone *abbr.:* LHRH or LH-RH or LRH; *see* **gonadotropin-releasing hormone**.

luteo+ *comb. form* indicating 1 yellow. 2 of the **corpus luteum**.

luteolysin an alternative name for **prostaglandin F_{2a}**.

luteolysis degeneration or regression of the **corpus luteum**. — **luteolytic** *adj.*

luteotropic hormone or **luteotrophic hormone** *abbr.:* LTH; a former name for **prolactin**.

luteotropin or **luteotrophin** a former name for **prolactin**.

lutropin the recommended name for **luteinizing hormone**.

lux *symbol:* lx; the SI derived unit of illuminance, equal to 1 lumen per square metre; i.e. 1 lx = 1 lm m⁻².

Lux proteins the counterpart in marine bacteria of the **luciferase** system. They mediate bioluminescence and are products of genes regulated by the *lux* operon. Examples are from *Vibrio* spp. LuxA (catalytic subunit) and LuxB constitute bacterial luciferase (EC 1.14.14.3). It catalyses a reaction between reduced FMN, RCHO, and O₂ to form FMN, RCOOH, H₂O, and *h ν* . Examples from *V. fischeri*: LUXA, database code LUXA_VIBFI, 354 amino acids (40.30 kDa); LUXB, database



code LUXB_VIBFI, 326 amino acids (37.31 kDa). LuxC is an **acyl-CoA reductase** required to form the 'RCHO' above. Example from *V. harveyi*: database code LUXC_VIBHA, 477 amino acids (54.78 kDa). LuxD is an acyl transferase component of the fatty acid reductase system required for aldehyde biosynthesis (EC 2.3.1.-). Example from *V. harveyi*: database code LUXD_VIBHA, 305 amino acids (34.17 kDa). LuxE is an acyl-protein synthetase (EC 6.2.1.19) that activates tetradecanoic acid. LuxG is the flavin reductase (EC 1.-.-.-), Example from *V. fischeri*: database code LUXG_VIBFI, 236 amino acids (26.47 kDa). LuxI and LuxJ are part of the regulatory system; they are required for the synthesis of OHHL (*N*-(3-oxohexanoyl)-L-homoserine lactone), an autoinducer molecule that binds to LuxR. Example from *V. fischeri*: database code LUXJ_VIBFI, 193 amino acids (21.99 kDa). LuxR is the transcriptional activator. Example from *V. fischeri*: database code LUXR_VIBFI, 250 amino acids (28.52 kDa).

LVP *abbr.* for lysine vasopressin.

LW antigen *abbr.* for Landsteiner-Weiner antigen. *See* **Rhesus factor**.

Lwoff, André Michel (1902–94), French microbiologist; Nobel Laureate in Physiology or Medicine (1965) jointly with F. Jacob and J. L. Monod 'for their discoveries concerning genetic control of enzyme and virus synthesis'.

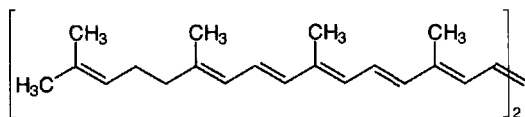
lx *symbol* for lux.

LX *abbr.* for lipoxin.

lyase 1 *the systematic name* for any enzyme of EC class 4 that catalyses the cleavage of C–C, C–O, C–N, and other bonds, leaving double bonds, or conversely adding a group to a double bond. **2** **17,20-lyase** *the common name* for 17 α -hydroxyprogesterone aldolase (EC 4.1.2.30).

lyate ion the anion produced by the autoprotolysis of a protogenic solvent; e.g. the hydroxide ion is the lyate ion of water.

lycopene the characteristic red pigment of tomatoes and some other ripe fruit. It is an acyclic carotenoid, and the biosynthetic precursor of the cyclic carotenes in plants.



lycopodium 1 any clubmoss of the genus *Lycopodium*. **2** the spores of various clubmosses, e.g. *L. clavatum*. They are used as a covering for pills and suppositories, in pyrotechnics, and in foundry work.

lymph 1 the pale-yellow, clear, or cloudy fluid that is contained within the vessels of the **lymphatic system**. It contains a little protein, and derives from the interstitial fluid, which is drained by the lymphatic vessels and ultimately discharged into the bloodstream, e.g. via the thoracic duct. The composition of lymph varies in different parts of the body, that from the liver and intestine containing more protein, and the latter more fat in the form of chylomicrons. Lymph also contains varying numbers of cells, mostly small lymphocytes, and coagulation factors. **2** the exudation from a sore.

lymphatic 1 of, pertaining to, or containing lymph; of or pertaining to the lymphatic system. **2** a lymphatic vessel.

lymphatic system the extensive network of capillary-like lymphatic vessels in vertebrates that convey the lymph from the peripheral tissues to the venous system. **Lymph nodes** occur at various points along the vessels.

lymph node or **lymph gland** any of numerous small swellings distributed throughout the lymphatic system of vertebrates. Each node consists largely of lymphocytes, together with macrophages and dendritic cells held together in a loose reticulum. Lymph nodes receive lymph from the peripheral tissues via afferent lymphatic vessels and pass it on via efferent vessels to more centrally placed lymph nodes, and ultimately to

the thoracic duct. The lymph nodes act as filters through which the lymph must pass and where foreign material comes into intimate contact with macrophages and lymphocytes. They are active centres for phagocytosis and the site of initiation of humoral and cell-mediated immune reactions.

lympho+ or (*before a vowel*) **lymph+** *comb. form* indicating lymph or the lymphatic system.

lymphoblast any **blast cell** of the **lymphoid cell series**. —**lymphoblastic** *adj.*

lymphocyte a type of **leukocyte** found in vertebrates and having various functions in specific immunity. A lymphocyte is a spherical cell, 7–12 μ m in diameter, with a large, round nucleus, often indented when seen by electron microscopy, and scanty cytoplasm containing scattered ribosomes but little endoplasmic reticulum and few other organelles. Lymphocytes occur in lymph nodes, other tissues, lymph, and blood, and they circulate between these tissues. Mature, differentiated lymphocytes comprise two main classes, **B lymphocytes** and **T lymphocytes**. Lymphocytes without the characteristics of either of these classes are termed **null lymphocytes**.

lymphocyte-activating factor *abbr.*: LAF; *an alternative name* for **interleukin 1**.

lymphocyte transformation *see* **transformation** (def. 3).

lymphocytosis an abnormal increase in the number of lymphocytes, especially in blood.

lymphoid of or resembling lymph, or lymphatic tissue.

lymphoid cell series a cell series comprising all cells that morphologically resemble **lymphocytes**, their precursors (e.g. lymphoblasts), and cells that may be derived from them (e.g. plasma cells).

lymphoid tissue any vertebrate tissue that is made up predominantly of **lymphocytes**, e.g. lymph, lymph nodes, spleen, thymus, Peyer's patches, adenoids, pharyngeal tonsils, and, in birds, bursa of Fabricius and cecal tonsils.

lymphokine any member of a varied group of nonimmunoglobulin, soluble factors released by primed lymphocytes on a second interaction with the specific activator (which may be a soluble antigen, a cell-associated antigen, or a mitogen). They play a role in **cell-mediated immunity** by modulating the behaviour of other types of cell within the same organism (e.g. macrophages, polymorphonuclear leukocytes), and are now considered to be a subgroup of the **cytokines**. Specific lymphokines include: chemotactic factor for macrophages, lymphotactin, lymphotoxin (i.e. **tumour necrosis factor** β), macrophage-activating factor (MAF), macrophage migration enhancement factor, **macrophage migration inhibitory factor** (MIF), (antigen-independent) mitogenic factor, skin-reactive factor, and **transfer factor**. Lymphokines that can act between different populations of leukocytes are known also as **interleukins**. *See also* **colony-stimulating factor**.

lymphoma any malignant tumour, or condition allied to a tumour, arising from some or all of the cells of lymphoid tissue. The term includes lymphatic leukemias, Hodgkin's disease, and reticuloses.

lymphoreticular tissue *an old general term* for tissues consisting largely of lymphocytes and macrophages.

lymphotoxin *abbr.*: LT; *an alternative name* for **tumour necrosis factor** β ; *see also* **lymphokine**.

LYN a protooncogene encoding a **tyrosine kinase** belonging to the **src** family: Database code KLYN_HUMAN, 512 amino acids (58.51 kDa).

Lynen, Feodor (1911–79), German biochemist, Nobel Laureate in Physiology or Medicine (1964) jointly with K. E. Bloch 'for their discoveries concerning the mechanism and regulation of the cholesterol and fatty acid metabolism'.

lyo+ *comb. form* indicating looseness, dispersion, or dissolution.

lyochrome *a former name* for **flavin**.

lyocytosis cell lysis by extracellular digestion, e.g. during insect metamorphosis.

lyoenzyme *an old term* for any enzyme existing in the cell in soluble form.

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lyoglycogen

lyoglycogen an old name for the fraction of a sample of glycogen that is extractable by water on a steam bath.

lyolysis an alternative term for **solvolysis**.

Lyon hypothesis (in genetics) the hypothesis that gene dosage compensation in mammals is accomplished by random inactivation of one of the two X chromosomes in all somatic cells of females. [After Mary Francis Lyon (1925–), British geneticist.]

lyonium ion any cation produced by the protonation of a solvent molecule, e.g. H_3O^+ , CH_3OH_2^+ .

lyophile 1 (in chemistry) a **lyophilic** compound. 2 having lyophilic properties.

lyophilic or **lyophile** (of a substance or surface) solvent preferring; easily wetted or solvated by the solvent present; the term comprehends **hydrophilic** when the solvent is water. Compare **lyophobic**.

lyophilic sol any sol in which the disperse phase has an attraction for the continuous phase.

lyophilization or **lyophilisation** strictly, the process of rendering lyophile or lyophilic. The term is now usually specifically applied to freeze-drying in which, by use of a vacuum, water is removed from ice by sublimation. This use bears little relationship to the true meaning of lyophile, except in so far as it maintains proteins subjected to it in a lyophilic state. — **lyophilize** or **lyophilise** *vb*.

lyophilizer or **lyophiliser** an apparatus for effecting lyophilization; a freeze-drier.

lyophobic or **lyophobe** (of a substance or surface) solvent rejecting; resistant to being wetted or solvated by the solvent present; the term comprehends **hydrophobic** when the solvent is water. Compare **lyophilic**.

lyophobic sol any sol in which the disperse phase has no attraction for the continuous phase. Lyophobic sols cannot be formed by spontaneous dispersion in the medium.

lyotropic or **lyotrope** of or associated with the difference between the internal pressure of a solution compared with that of the solvent caused by the solute.

lyotropic mesophase any mesophase (i.e. **liquid crystal**) whose phase transitions are readily effected by changes in concentration.

lyotropic series or **Hofmeister series** a series of ions arranged in order of their **lyotropic** effects, especially their ability to cause precipitation of a lyophilic sol (to salt-out).

lypressin *see* [8-lysine]vasopressin.

Lys symbol for a residue of the α -amino acid L-lysine (alternative to K).

lys+ a variant form of **lyso+** (before a vowel).

lysate any solution or other preparation containing the products of **lysis** of cells. The term is applied especially to lysed bacterial or blood cells.

+lysate *comb. form* denoting a mixture resulting from decomposition, breakdown, or lysis (of or by something indicated); e.g. autolysate, cytolysate, hemolysate, hydrolysate.

Lys-bradykinin *abbr.* for lysylbradykinin (i.e. kallidin); *see* bradykinin.

lyse to effect **lysis** (of a cell, etc.); to undergo lysis.

+lyse *comb. form* denoting decompose, breakdown, or lyse (of or by something indicated); e.g. autolyse, cytolysate, electrolyse, hemolyse, hydrolyse.

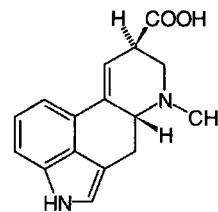
lysergamide lysergic acid amide. All known **ergot alkaloids** are derivatives of D-lysergamide, the simplest being **ergonovine**.

lysergic acid 9,10-didehydro-6-methylargoline-8 β -carboxylic acid an alkaloid obtained from **ergot**; it is a tetracyclic indole derivative. Lysergic acid diethylamide (*abbr.*: LSD) is a powerful hallucinogenic substance. (Illustrated top right.)

lysin any agent, such as an antibody, enzyme, toxin, etc., that is able to lyse cells.

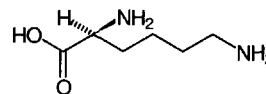
+lysin *comb. form* denoting an (indicated) agent that effects, or can effect, lysis, specifically the lysis of cells (cytolysis).

lysinate 1 lysine anion; the anion, $\text{H}_2\text{N}-[\text{CH}_2]_4-\text{CH}(\text{NH}_2)-\text{COO}^-$, derived from **lysine**. 2 any salt containing lysine anion. 3 any ester of lysine.



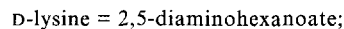
lysergic acid

lysine the trivial name for α,ϵ -diaminocaproic acid; 2,6-diaminohexanoic acid; $\text{H}_2\text{N}-[\text{CH}_2]_4-\text{CH}(\text{NH}_2)-\text{COOH}$; a chiral α -amino acid. L-Lysine (*symbol*: K or Lys), (S)-2,6-diaminohexanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: AAA or AAG. In mammals it is an essential dietary amino acid and is ketogenic. D-Lysine (*symbol*: D-Lys or DLys), (R)-2,6-diaminohexanoic acid, is a substrate for **lysine racemase** and D-lysine 5,6-aminomutase. The conversion of D-lysine to L-pipecolic acid, and of the latter to L-lysine, occurs in *Nicotiana glauca* and in *Neurospora crassa*.



L-lysine

D-lysine 5,6-aminomutase EC 5.4.3.4; *other name*: D- α -lysine mutase; an enzyme that catalyses the reaction:



cobalamin is a cofactor.

lysine(arginine) carboxypeptidase EC 3.4.17.3; *other names*: carboxypeptidase N; arginine carboxypeptidase; kininase I; anaphylotoxin inactivator; an enzyme that hydrolyses peptidyl-L-lysine (or L-arginine) to peptide and L-lysine (or L-arginine). It inactivates bradykinin complement components, C3a and C5a by removal of the C-terminal Arg. Example (precursor) from human: database code CBPN_HUMAN, 458 amino acids (52.23 kDa).

lysine (base) a lysine residue when the ϵ -amino group is unprotonated.

lysine hydroxylase *see* procollagen-lysine 5-dioxygenase.

lysine intolerance a genetically determined metabolic defect of humans in which the ingestion of relatively large amounts of lysine (in protein) causes hyperlysinemia, hyperammonemia, and coma. It is possibly attributable to deficient activity of lysine dehydrogenase (L-lysine:NAD $^+$ oxidoreductase, EC 1.4.1.15).

lysine 6-monooxygenase *see* monooxygenase.

lysine racemase EC 5.1.1.5; an enzyme catalysing the reaction:
L-lysine = D-lysine.

lysine vasopressin *see* [8-lysine]vasopressin.

[8-lysine]vasopressin *abbr.*: [Lys 8]vasopressin; *nonproprietary name*: lypressin; the molecular form of **vasopressin** in which the variable eighth position in its amino-acid sequence is occupied by a lysine residue. This is the form of the hormone present in pigs and a few other related mammals. It is sometimes incorrectly designated as lysine vasopressin.

lysinium 1 the usual term for lysinium(1+); the monocation of **lysine**. In theory, the term denotes any ion or mixture of ions formed from lysine and having a net charge of +1, although the species $\text{H}_3\text{N}^+-[\text{CH}_2]_4-\text{CH}(\text{NH}_3^+)-\text{COO}^-$ generally predominates in practice. 2 the systematic name for lysinium(2+); the dication of lysine.

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lysino

lysino 1 N^2 -lysino the alkylamino group, $H_2N-[CH_2]_4-CH(COOH)-NH-$, derived from lysine by loss of a hydrogen atom from its α -amino group. 2 N^6 -lysino the alkylamino group, $HOOC-CH(NH_2)-[CH_2]_4-NH-$, derived from lysine by loss of a hydrogen atom from its ϵ -amino group.

lysis 1 (in chemistry) the breaking down of a substance by the splitting of its molecules into two parts; the rupture of a covalent bond. 2 (in biology) the disintegration of cells or cell organelles by rupture of their outer membranes. 3 (in medicine) the gradual diminution in the severity of the symptoms of a disease; remission.

+**lysis** *comb. form* denoting breaking down, decomposition, disintegration, or dissolution. The first element of the word may indicate: (1) the agent, e.g. electrolysis, hydrogenolysis, biolysis; (2) the substance or object affected, e.g. glycolysis, lipolysis; or (3) some other characteristic, e.g. autolysis, catalysis, heterolysis, homolysis. —+**lytic** *adj.*

lyso+ or (before a vowel) **lys+** 1 *comb. form* indicating a dissolving or loosening; denoting lysis or the products of lysis. 2 *prefix* (to the name of a phospholipid) denoting removal of one of its two acyl groups; a locant may be used to designate the site of (hydro)lysis, e.g. 2-lyso designates deacylation at position 2. The term originated from the fact that such compounds are hemolytic, but it is now used to indicate a limited deacylation.

lysochrome any substance that will dissolve in lipids and colour them.

lysogen 1 any **lysogenic bacterium**. 2 any antigen that stimulates **lysin** production.

lysogenesis 1 an alternative term for **lysogeny**. 2 the production of a **lysin** or **lysis**.

lysogenic 1 pertaining to, or capable of producing or undergoing **lysis** (def. 1). 2 of, or relating to **lysogeny**. —**lysogenicity** *n.*

lysogenic bacterium any bacterium carrying a temperate bacteriophage as a **prophage** integrated into the bacterial genome. It has the potential to produce and release infective bacteriophages as a stable, heritable trait, which may become activated spontaneously or in response to certain stimuli. Moreover, it is immune to **lytic** infection by the same or closely related phages.

lysogenic conversion *see* **bacteriophage conversion**.

lysogenic virus any virus capable of becoming a **prophage**.

lysogenize or **lysogenise** to convert a nonlysogenic bacterial strain into a lysogenic strain by infection with a temperate bacteriophage, with resultant incorporation of the phage into the bacterial genome. —**lysogenization** or **lysogenisation** *n.*

lysogeny or **lysogenesis** the state created in a bacterial cell when the prophage of a temperate bacteriophage is incorporated into the bacterial genome as a stable, heritable character. Such a cell can be activated to produce and release mature phage, with accompanying cell lysis. *See also* **lysogenic bacterium**.

Lys(5-OH) *symbol* for a residue of the α -amino acid 5-hydroxy-L-lysine (less-preferred alternative to Hyl or 5Hyl).

Lysol the proprietary name for a disinfectant consisting of a mixture of *o*-, *m*-, and *p*-cresols solubilized with an excess of a potassium soap.

lysolecithin a former name for **lysophosphatidylcholine**.

lysolecithin acyltransferase *see* **1-acylglycerophosphocholine O-acyltransferase**.

lysophosphatidic acid any **phosphatidic acid** deacylated at positions 1 or 2.

lysophosphatidyl any **phosphatidyl** group deacylated at positions 1 or 2.

lysophosphatidylcholine or (formerly) **lysolecithin** any **phosphatidylcholine** (i.e. lecithin) deacylated at positions 1 or 2.

lysophospholipid any **phospholipid** deacylated at positions 1 or 2.

lysoplasmic acid any derivative of *sn*-glycero-3-phosphate that has an *O*-(1-alkenyl) residue on carbon-1, the hydroxyl group in position 2 being unsubstituted.

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lysoplasmic acid any **lysoplasmic acid** group in which the hydroxyl group in position 2 is unsubstituted.

lysosomal carboxypeptidase B *see* **cysteine-type carboxypeptidase**.

lysosomal carboxypeptidase C *see* **lysine pro-X carboxypeptidase**.

lysosomal Pro-X carboxypeptidase EC 3.4.16.2; *other names*: proline carboxypeptidase; angiotensinase C; lysosomal carboxypeptidase C; an enzyme that catalyses the cleavage of a Pro-[Xaa bond to release a C-terminal amino acid. Example from human: database code A47352, 496 amino acids (55.74 kDa).

lysosomal storage disease any of a group of hereditary diseases, with widely varying manifestations, in which there is deficiency of a single lysosomal enzyme associated with abnormal deposits (of substrate) within membrane-bound vesicles. The group is subdivided into the following classes: type II **glycogen disease**, **mucopolysaccharidoses**, **sphingolipidoses**, **mucopolipidoses**, and a few others.

lysosome any of a group of related cytoplasmic, membrane-bound organelles that are found in most animal cells and that contain a variety of hydrolases, most of which have their maximal activities in the pH range 5–6. The contained enzymes display latency if properly isolated. About 40 different lysosomal hydrolases are known and lysosomes have a great variety of morphologies and functions. —**lysosomal** *adj.*

lysothaphin EC 3.5.1.-; an enzyme that hydrolyses polyglycine links in peptidoglycans. Zinc is a cofactor. Its physiological role is likely to be in the remodelling of the bacterial sacculus during growth. Example (prepro precursor) from *Staphylococcus simulans*: database code LSTP_STASI, 389 amino acids (42.19 kDa). The C terminus has some similarity to **autolysin**.

lysozyme EC 3.2.1.17; *systematic name*: peptidoglycan *N*-acetylmuramoylhydrolase; *other name*: muramidase. An enzyme that hydrolyses β -1,4-linkages between *N*-acetylmuramic acid and 2-acetamido-2-deoxy-D-glucose residues in peptidoglycan heteropolymers (mucopolysaccharide or mureopeptide) of prokaryote cell walls. It occurs in tears, other exocrine secretions, and in very large amounts in the white of eggs of birds. The antibacterial action of lysozyme depends on the cleavage of the β (1→4) glycosidic linkage between alternating units of *N*-acetylmuramic acid and *N*-acetylglucosamine that form long-chain mucopolysaccharides in the cell walls of bacteria such as *Micrococcus luteus*. There are a number of types of lysozyme, designated A–G. They belong to the **lactalbumin/lysozyme family**. Example of egg-white lysozyme from *Gallus gallus* (domestic fowl): database code NRL_4LYM, 129 amino acids (14.30 kDa); 3-D structure known (first enzyme of which this was true).

[**Lys**⁶]**vasopressin** *abbr. for* [8-lysine]vasopressin.

lysyl the acyl group, $H_2N-[CH_2]_4-CH(NH_2)-CO-$, derived from **lysine**.

lysylbradykinin or **Lys-bradykinin** an alternative name for **kallidin**; *see* **bradykinin**.

lysyl endopeptidase EC 3.4.21.50; *other names*: *Achromobacter* proteinase I; lysyl bond specific proteinase. An extracellular bacterial serine proteinase that preferentially cleaves at Lys-[Xaa, including Lys-[Pro.

lysyl oxidase EC 1.4.3.13; *recommended name*: protein-lysine 6-oxidase; *systematic name*: protein-L-lysine:oxygen 6-oxidoreductase (deaminating). An enzyme that catalyses a reaction between peptidyl-L-lysyl-peptide, H_2O and dioxygen to form peptidyl-allysyl-peptide, NH_3 , and H_2O_2 . Example (precursor) from human: database code LYOX_HUMAN, 417 amino acids (46.89 kDa). *See also* **allysine**.

+**lyte** *comb. form* indicating a substance that can be decomposed, broken down, or lysed.

lytic 1 of, pertaining to, or resulting from **lysis**. 2 of, pertaining to a **lysin**.

+**lytic** *see* +**lysis**.

lyticase a jargon term for a combination of enzymes, with min-

lytic cycle

imal amounts of nucleases, that displays two major activities: β -1,3-glucanase and a highly specific alkaline protease. It is prepared from *Oerskovia xanthineolytica* and is useful in removing cell walls from yeasts and fungi, causing extremely rapid cell lysis.

lytic cycle the sequence of events occurring when a sensitive bacterium is infected by a virulent bacteriophage. The phage diverts the metabolism of the cell to the exclusive production of progeny phage, resulting in the lysis of the host cell with liberation of phage.

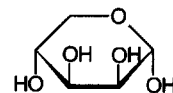
α -lytic endopeptidase EC 3.4.21.12; a bacterial serine proteinase, a homologue of chymotrypsin, that hydrolyses proteins, especially at bonds adjacent to L-Ala and L-Val residues in bacterial cell walls, elastin, and other proteins. Example from *Lysobacter enzymogenes*; database code NRL_2ALP, 198 amino acids (19.84 kDa); 3-D structure known.

lytic virus any virus including bacteriophages whose multiplication in a host cell leads to lysis of that cell.

Lyx symbol for a residue (or sometimes a molecule) of the aldopentose **lyxose**.

lyxo+ prefix (in chemical nomenclature) indicating a particular configuration of a set of three (usually) contiguous >CHOH groups, as in the acyclic forms of D- or L-lyxose. See **monosaccharide**.

lyxose symbol: Lyx; a trivial name for the aldopentose, *lyxopentose*; the C-2 epimer of **xylose**, rare in nature.



α -D-lyxose (pyranose form)

Mm

m symbol for 1 metre. **2 milli+** (SI prefix denoting 10^{-3} times).

m+ prefix (in peptide hormone nomenclature) denoting mouse, murine; derived, or as if derived, from a mouse.

m_r symbol for relative electrophoretic mobility.

m symbol for 1 mass; the various associated terms are distinguished by subscripts: **m_e**, symbol for electron rest mass; **m_n**, symbol for neutron rest mass; **m_p**, symbol for proton rest mass; **m_u**, symbol for atomic mass constant. **2 molality** (alternative to *b*). **3 (bold italic)** magnetic moment.

m- prefix (in chemical nomenclature) denoting *meta-*. Compare *o-*, *p-*.

M symbol for 1 mega+ (SI prefix denoting 10^6 times). **2** a residue of the α -amino acid L-methionine (alternative to Met). **3** a residue of an incompletely specified base in a nucleic-acid sequence that may be either adenine or cytosine. **4** mesomeric effect. **5** a metal (in chemical formulae). **6 molar** (concentration), i.e. moles per litre (alternative to *M* or *M*; mol dm^{-3} is recommended). **6** 6-mercaptopurine ribonucleoside, 6-thioinosine.

M symbol for molar (concentration), i.e. moles per litre (alternative to *M* or *M*; mol dm^{-3} is the recommended form).

α_2 -M abbr. for α_2 -macroglobulin.

M symbol for 1 mass, especially molar mass; related terms are designated by subscripts: **M_m**, symbol for mass-average molar mass (see **average molar mass**); **M_n**, symbol for number-average molar mass (see **average molar mass**); **M_o**, symbol for mean relative residue mass; **M_r**, symbol for relative molecular mass; **M_w**, symbol for weight-average molar mass; **M_z** symbol for Z-average molar mass (see **average molar mass**). **2 molar** (concentration), i.e. moles per litre (alternative to *M* or *M*; no longer recommended; use mol dm^{-3}). **3 (or M_v)** luminous exitance. **4 (or M_e)** radiant exitance. **5 (bold italic)** magnetization. **6 (bold italic)** moment of a force.

MAA abbr. for macroaggregated (human serum) albumin.

MAC abbr. for 1 membrane attack complex. **MAC 1** is a cell-surface glycoprotein of monocytes, macrophages, and granulocytes that is implicated in adhesive interactions of these cells as well as in mediating the uptake of complement-coated particles. It probably recognizes the RGD peptide in complement component C3b. It is a heterodimer of α_M and β_2 chains (see **integrin**). Examples: (human) MAC 1 α_M (precursor): database code ITAM_HUMAN, 1153 amino acids (127.16 kDa); MAC 1 β_2 (precursor): database code ITB2_HUMAN, 769 amino acids (84.69 kDa). **2** mammalian artificial chromosome.

MACIF abbr. for membrane attack complex inhibition factor.

MAC-IP abbr. for MAC-inhibitory protein (i.e. **CD59**).

mackerel any marine fish of the subfamily Scombroidei. These fish are of importance as food fish, and their flesh is rich in oils with a relatively high content of *n-3 fatty acids*; typical composition (major fatty acids, expressed as percentage of total fatty acids): 14:0, 8%; 16:0, 16%; 18:0, 2%; 16:1 (*n-7*) 8%; 18:1 (*n-9*), 8%; 18:1 (*n-7*), 4%; 18:4 (*n-3*), 2%; 20:1 (*n-9*), 11%; 20:5 (*n-3*), 8%; 22:1 (*n-11*), 13%; 22:6 (*n-3*), 8%.

Macleod, John James Richard (1876–1935), British physiologist and biochemist; Nobel Laureate in Physiology or Medicine (1923) jointly with F. G. Banting 'for the discovery of insulin'.

maconha see **cannabis**.

macro+ or (before a vowel) **macr+** comb. form 1 denoting large, long, or great in size or duration; large-scale. **2** denoting larger than normal, overdeveloped. Compare **micro+**.

macroaggregate any macromolecular aggregate large enough to be visible to the unassisted eye.

macrocycle any **macrocyclic** compound.

macrocyclic 1 (in chemistry) describing any compound containing a ring structure consisting of a large number (usually 15 or more) atoms. **2** (in biology) having a long life cycle.

macrocytic anemia anemia in which the circulating erythro-

cytes are larger than normal. It is often due to deficiency of vitamin B₁₂ (cobalamin) or folic acid.

macroglobulin any plasma globulin of >400 kDa. Macroglobulins include **immunoglobulin M**, **α_2 -macroglobulin**, and many lipoproteins.

α_2 -macroglobulin abbr.: α_2 -M; a glycoprotein found in vertebrate plasma that strongly inhibits proteinases of all classes. In humans it constitutes about one third of the α_2 -globulins; in hypoalbuminemia its synthesis is enhanced, when it constitutes a major proportion of the increased α_2 -globulin band on electrophoresis strips. Human α_2 -M is a 726 kDa molecule comprising two pairs of disulfide-linked, identical 185 kDa subunits, which also contain heterodetic peptide rings containing reactive thiol ester linkages, each formed between the γ -carboxyl group of a glutamic residue and the thiol group of a cysteine residue in the same peptide chain. It can inhibit all four classes of **proteinase** by a mechanism involving a **bait region**. This contains specific sites, cleavage of which induces a conformational change that results in trapping of the proteinase: following cleavage in the bait region a thiol-ester bond is hydrolysed and mediates the covalent binding of the protein to the proteinase; subsequently ϵ -amino groups of the proteinase react with the thiol-ester linkages in the inhibitor to form stable amide links. The entrapped proteinase is then able to act on substrates of low molar mass only. Example, subunit precursor from human: database code A2MG_HUMAN, 1474 amino acids (163.09 kDa).

macroglobulinemia or (esp. Brit.) **macroglobulinaemia** a condition in humans in which there is an increase in the concentration of macroglobulins, especially immunoglobulin M, in the blood. It is often associated with lymphocyte tumours.

macro histone 2A abbr.: macro H2Aa; the counterpart of histone H2A that replaces this in certain nucleosomes, with an implied role in nucleosome positioning. Example from rat: database code H2AM_RAT, 367 amino acids (38.91 kDa).

macroion an ion of a macromolecule, especially one that does not pass through a dialysis membrane.

macrolide any of a large group of structurally related antibiotics produced by *Streptomyces* spp. They contain a large lactone ring (see **+olide**) with few or no double bonds and no nitrogen atoms, linked glycosidically to one or more sugar moieties. The macrolides include the carbomycins, the **erythromycins**, oleandomycin, **oligomycins**, and the spiramycins. They are active mainly against Gram-positive bacteria and they inhibit the early stages of protein synthesis.

macroligand (sometimes) any macromolecular ligand.

macrometabolite (sometimes) any metabolite present in a system in reasonably high concentration, especially as opposed to a trace metabolite, i.e. one present in extremely small concentrations.

macromolecule any molecule composed of a very large number of atoms, operationally defined as any molecule of mass greater than about 10 kDa (and ranging up to 10^2 MDa or more) that is incapable of passing through the pores of dialysis tubing as generally used. The term includes nucleic acids and most polysaccharides and proteins. Compare **micromolecule**. —**macromolecular adj.**

macronutrient any element that is required by living organisms in relatively large quantities for normal growth. Compare **micronutrient**.

macrophage any cell of the **mononuclear phagocyte system** that is characterized by its ability to phagocytose foreign particulate and colloidal material. Macrophages occur in connective tissue, liver (Kupffer cells), lung, spleen, lymph nodes, and other tissues. They contain prominent lysosomes, stain with vital dyes, and play an important part in nonspecific immune reactions.

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macrophage-activating factor

macrophage-activating factor *abbr.*: MAF; *see* lymphokine.

macrophage-colony stimulating factor *see* colony stimulating factor.

macrophage inflammatory protein *abbr.*: MIP; any of a number of cytokines produced by B and T cells, monocytes, mast cells, and fibroblasts. MIP 1 α is a chemoattractant for monocytes, T cells, and eosinophils; it also inhibits hematopoietic stem cell production; example from *Mus musculus*: database code M1A_MOUSE, 92 amino acids (10.34 kDa); residues 1–23 are the signal peptide. MIP 1 β is a chemoattractant for monocytes and T cells and is an adhesion molecule, binding to β_1 (VLA family) integrins; example from *M. musculus*: database code M1B_MOUSE, 92 amino acids (10.17 kDa); residues 1–23 are the signal peptide.

macrophage migration enhancement factor *see* lymphokine.

macrophage migration inhibitory factor *abbr.*: MIF; *other name*: delayed early response protein 6; a lymphokine that is released by primed lymphocytes on incubation with the priming antigen. It inhibits the migration of macrophages out of macrophage-rich tissue such as spleen, and is probably a mediator of macrophage participation in inflammation. Example from human: database code MIF_HUMAN, 114 amino acids (12.30 kDa).

macrophage system *see* mononuclear phagocyte system.

macropipette (*sometimes*) a pipette of similar construction to a standard plunger-type micropipette but of capacity in the range 0.25–5 mL.

macropore any pore (in a gel or porous solid) whose width is greater than about 50 nm. *Compare* mesopore, micropore. —**macroporous** *adj.*

macroreticular describing a three-dimensional polymer with a very open structure, the cross links being widely spaced.

macroscopic 1 of a size that is visible to the unaided eye. *Compare* microscopic (def. 1). 2 concerned with large units. *Compare* microscopic (def. 2). 3 relating to the behaviour of atoms or molecules in bulk. *Compare* microscopic (def. 3). —**macroscopically** *adv.*

macroscopic equilibrium constant any constant describing the overall equilibrium between two chemical entities that may be interconverted through two or more alternative intermediates. *Compare* microscopic equilibrium constant.

macrosolute an operational term describing any solute of a size larger than will pass through a membrane of a specified pore size or permeability limit.

macrotetrolide or **macrotetralide** any of a group of antibiotics characterized structurally by the presence of a 32-membered ring built up of four hydroxy-acid residues and containing four ether and four ester bonds; the group includes nonactin and related antibiotics. Some similar compounds have an open-chain structure, e.g. monensin and nigericin.

macula adherens an alternative term for desmosome.

MadCAM *abbr.* for mucosal addressin cell adhesion molecule; MadCAM 1, *M*, 58–66 kDa, is a member of the immunoglobulin superfamily of adhesion molecules. It is constitutively expressed on endothelial cells of post-capillary venules. It mediates lymphocyte homing to mucosal lymphoid organs and lamina propria venules, and binds L-selectin and integrin $\alpha_4\beta_7$. Example from *Mus musculus*: database code 533601, 405 amino acids (43.65 kDa).

mad cow disease an informal name for bovine spongiform encephalopathy.

madder the ground root of *Rubia tinctoria*, used as a plant dyestuff; the principal component is ruberythric acid.

MAF *abbr.* for macrophage-activating factor; *see* lymphokine.

magainin any of several antibiotic peptides from *Xenopus* skin, whose pore-forming activity (*see* PFP) results in the permeabilization of bacterial membranes. They have a wide spectrum of action against bacteria, protozoa, and fungi. Magainin I has the sequence

GIGKFLHSAGKFGKAFVGEIMKS;

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magnetic immuno-



magainin II is (Lys¹⁰,Asn²²)-magainin I. An analogue with greater antibiotic potency is (Ala^{8,13,18})-magainin II amide. *Compare* dermaseptin.

magic spot I *see* magic spot (compound), ppGpp.

magic spot II *see* magic spot (compound), pppGpp.

magic spot (compound) *abbr.*: MS; either of the guanosine polyphosphates produced when stringent strains of *Escherichia coli* are starved of an amino acid required for protein synthesis. MS I is guanosine 5'-diphosphate 2'-(or 3')-diphosphate (*symbol*: ppGpp); MS II is guanosine 5'-triphosphate 2'-(or 3')-diphosphate (*symbol*: pppGpp).

magnesium *symbol*: Mg; a metallic element – an alkaline earth metal – of group 2 of the IUPAC periodic table; atomic number 12; relative atomic mass 24.305. It occurs naturally only in the combined state and is a mixture of stable nuclides of mass 24, 25, and 26. Magnesium is one of the most abundant elements of the Earth's crust and an essential component of all living material. It forms a divalent cation, Mg²⁺, which is a component of chlorophyll, occurs in bone, and is an essential cofactor for many enzymes, including the majority of enzymes utilizing ATP. It is a major biological cation, the fourth most abundant cation in the human, much of it being in the skeleton; the range in normal human plasma is 0.8–1.2 mmol L⁻¹. An excretion mechanism in the kidney is partly regulated by parathyroid hormone. *See also* magnesium-28.

magnesium-28 the artificial radioactive nuclide of magnesium, ²⁸Mg, of half-life 21.2 h. It emits a moderate-energy beta particle (i.e. electron) (0.46 MeV max.) and gamma radiation of various energies (1.35, 0.95, 0.40, and 0031 MeV).

magnesium porphyrin a general term for chlorophyll.

magnet any body that can produce an appreciable magnetic field external to itself. Magnets may be either permanent or temporary. —**magnetic** *adj.*

magnetic circular dichroism spectroscopy circular dichroism spectroscopy in the presence of a magnetic field.

magnetic constant *symbol*: μ_0 ; an alternative name for permeability of vacuum.

magnetic dipole moment *symbol*: *m* or μ (*italic bold*); a vector quantity indicating the strength of a magnet: the product of a magnet's dipole moment and the ambient magnetic flux density gives the torque on the magnet. The SI unit is the A m² (ampere square metre = joule per tesla (J T⁻¹)).

magnetic field 1 a field of force existing in the neighbourhood of either a permanent magnet or a circuit carrying an electric current. Another magnet will experience a couple when placed in a magnetic field. 2 an alternative name for magnetic field strength.

magnetic field strength or **magnetic field** or (*formerly*) **magnetic intensity** *symbol*: *H* (*italic bold*); a vector quantity, the magnetizing force at any point in a medium. It is measured by a couple that would act on a small magnet of unit moment placed at that point with its axis at right angles to the direction of the magnetic field. It is the magnetic flux density, *B*, divided by the permeability, μ ; i.e. $H = B/\mu$. The SI derived unit $H = A m^{-1}$.

magnetic flux *symbol*: Φ ; a measure of the total strength of a magnetic field, defined as the scalar product of the magnetic flux density and the area. The SI derived unit is the weber.

magnetic flux density or **magnetic induction** *symbol*: *B* (*italic bold*); a pseudovector quantity, the magnetic flux passing through unit area of a magnetic field orthogonal to the magnetic force. The SI derived unit is the tesla.

magnetic force the force exerted by a magnetic field on a magnetic pole or a moving electric charge. It may be either attractive or repulsive.

magnetic immunoassay *abbr.*: MAIA; a variant of radioimmunoassay in which the antibody is bound to a particulate, inert, magnetic support material – e.g. cellulose containing embedded grains of iron oxide, Fe₃O₄ – and separation of antigen-antibody complexes from free antigen is effected with the aid of a powerful magnet.



magnetic induction

magnetic induction an alternative term for magnetic flux density.

magnetic intensity a former name for magnetic field strength.

magnetic lens an alternative name for a magnetic electron lens.

magnetic moment 1 symbol: μ ; a property of a charged particle arising from its spin; $\mu = eL/2mc$, where c is the velocity of light, L is the angular momentum of the particle, e is its charge, and m is its mass. The SI unit is joule per tesla. The electron magnetic moment (symbol: μ_e) is equal to $9.284\,7701 \times 10^{-24} \text{ J T}^{-1}$. The proton magnetic moment (symbol: μ_p) is equal to $1.410\,607\,61 \times 10^{-26} \text{ J T}^{-1}$. 2 an alternative name for magnetic dipole moment.

magnetic permeability see permeability, permeability of vacuum.

magnetic pole either of the two regions of a magnet into which the lines of force of the magnetic field converge or from which they diverge. They are designated north (or north-seeking) and south (or south-seeking) according to whether they are attracted to the Earth's north or south pole.

magnetic resonance see electron-spin resonance spectroscopy, nuclear magnetic resonance.

magnetic resonance imaging *abbr.*: MRI; an imaging technique that uses nuclear magnetic resonance (NMR) to investigate the state of tissues in the intact body; it is so-called to avoid the use of the term 'nuclear' in the treatment of patients. It has the advantage that it is non-invasive. A magnet large enough to surround a part of the body is used in conjunction with computer analysis to provide an image resulting from the NMR signals. The technique can distinguish extracellular water from that inside cells; this distinction is used to reveal certain types of tissue damage.

magnetic resonance zeugmatography see zeugmatography.

magnetic shielding 1 protection from the effects of an external magnetic field. 2 (in nuclear magnetic resonance spectroscopy) the reduction of the applied magnetic field at an atomic nucleus (as compared with that applied to the whole sample) brought about by the electron cloud of the molecule.

magnetic stirrer any apparatus for stirring a liquid in which a follower – usually a small bar magnet coated in glass or plastic – may be caused to oscillate or rotate within the liquid under the influence of a more powerful magnet outside the container. Motion of the external magnet is controlled by a variable-speed electric motor. The term is sometimes applied specifically to the follower.

magnetic susceptibility symbol: X or (sometimes) X_m or K ; the ratio of the magnetization, M , produced in a substance to the magnetic field strength, H , applied to it, i.e.

$$X = M/H = \mu_r - 1,$$

where μ_r is the relative permeability of a vacuum. Paramagnetic materials have positive magnetic susceptibilities, diamagnetic materials have negative magnetic susceptibilities. (Note: the symbol μ_m is reserved for molar magnetic susceptibility.)

magnetism 1 the property of producing, or being affected by, magnetic flux. 2 the branch of physics concerned with magnets and magnetic phenomena.

magnetite or magnetic iron oxide or black iron oxide ferrous-ferrous oxide; iron(II)-diiron(III) oxide; Fe_3O_4 ; a strongly magnetic black mineral form of iron oxide.

magnetization or magnetisation 1 the act or process of magnetizing. 2 symbol: M (italic bold); the magnetic moment per unit volume of a magnetized body; i.e. the difference between the ratio of the magnetic flux to the permeability of vacuum and the magnetic field strength. The SI unit is the ampere per metre.

magnetization-transfer nuclear magnetic resonance spectroscopy a procedure in nuclear magnetic resonance spectroscopy in which one of the resonances is saturated by a strong radiofrequency field, and the changes in the other resonances are observed. It is useful in measuring the rates of chemical exchange processes.

magnetize or magnetise to induce a magnetic flux density in (something); to cause (something) to exhibit magnetism (def. 1).

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magnetogyric ratio see gyromagnetic ratio.

magneton a unit of magnetic moment. The Bohr magneton (symbol: μ_B) refers to an electron of mass m_e ; the nuclear magneton (symbol: μ_N) refers to a proton of mass m_p . They are related by the equation: $\mu_N = (m_e/m_p)\mu_B$.

magnetophoresis the movement of magnetizable particles through a fluid under the action of a magnetic field. It is useful for separating of dispersed cells to which ferromagnetic microspheres have been conjugated. —**magnetophoretic** *adj.*

magnification (in microscopy) the apparent linear enlargement of an object when viewed through a lens, system of lenses (as in a microscope, telescope, etc.), or other instrument. It is given by the ratio of the apparent diameter of the object, as seen using the lens, to its real diameter, as seen unaided.

Magnogel a proprietary name for a bead-form support material useful in immunosorption and related procedures. It is similar to Ultrogel but contains embedded particles of iron oxide, Fe_3O_4 , that enable the beads to be drawn to and held on the walls of the container by magnetism.

MAIA *abbr.* for magnetic immunoassay.

Maillard reaction a nonenzymic reaction in which a reducing sugar combines at its anomeric carbon atom with an amino group of an amino acid, peptide, or protein to form first a Schiff base and then, by an Amadori rearrangement, the corresponding ketoamine. The reaction is the initial step of a series that may occur during the processing and/or storage of food and that can result in a loss of nutritive value and the formation of a brown coloration; hence it is also known as the browning reaction. It may also be the reaction leading to the formation of hemoglobin A_{1c} in the erythrocyte. [After Louis Camille Maillard (1878–1936).]

major groove the larger and deeper of the two spiral grooves on the surface of a double-helical B-DNA molecule. Compare minor groove.

major histocompatibility complex *abbr.*: MHC; a complex of genetic loci occurring in higher vertebrates that encodes a family of cellular antigens, known in mice as histocompatibility-2 antigens (*abbr.*: H-2 antigens; see histocompatibility antigen) and in humans as human leukocyte-associated antigens (*abbr.*: HLA antigens; see HLA histocompatibility system). The MHC antigens are cell-surface glycoproteins and may be divided into two classes, designated I and II. In transplantation reactions, cytotoxic T lymphocytes respond mainly against foreign MHC glycoproteins of class I together with antigen, while the response against class II and antigen is mainly by helper T lymphocytes.

Class I MHC antigens (or transplantation antigens) are glycoproteins found on the surface of most nucleated somatic cells that enable the immune system to distinguish self from foreign cells. They are encoded by three separate loci (H-2K, H-2D, and H-2L in mice; HLA-A, HLA-B, and HLA-C in humans). The antigenic activity resides in a single polypeptide chain, the α subunit, of $\approx 45 \text{ kDa}$ (roughly 345 amino acids) inserted into the cell membrane with a short hydrophilic C-terminal segment inside the cell, a short hydrophobic segment traversing the membrane, and a large carbohydrate-containing N-terminal segment. This N-terminal segment is folded into three separate domains (two of which contain a single intrachain disulfide bond) and exposed to the cell exterior, where it is noncovalently linked with the β subunit, an 11.5 kDa molecule also called β_2 microglobulin, which is homologous with a single Ig-like domain (see immunoglobulin fold). In humans, these proteins are also referred to as HLA class I.

Class II MHC antigens are found on the surface of only certain cell types; these include most B lymphocytes, some T lymphocytes, and some macrophages and macrophage-like cells. They are encoded by murine I-region (*abbr.*: Ir) genes and human HLA-D genes, and are two-chain proteins having a $\approx 33 \text{ kDa}$ α chain and a 28 kDa β chain; the C-terminal regions of both subunits (α and β) have transmembrane and intracellular domains. Celiac disease is due to an allelic variant in one



major tranquilizer

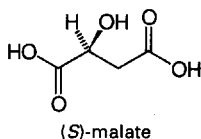
class-II gene. Example, α from human B-lymphoblastoid cell line: database code HUMMHDARN, 229 amino acids (25.83 kDa); β -1 from human B-lymphocyte cell line: database code HUMMHDRBWB, 94 amino acids (11.40 kDa).

major tranquilizer a term, now deprecated, for any drug used to treat major psychosis, e.g. schizophrenia or mania; the term **neuroleptic** is now recommended.

MAK *abbr.* for methylated albumin-kieselguhr; a substance used as a medium for chromatography.

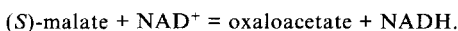
Mal 1 (when linked to one other group) *symbol* for the univalent maleyl group. **2** (when linked to two other groups) *symbol* for the bivalent maleoyl group.

malate 1 a trivial name for hydroxysuccinate; hydroxybutanedioate; $^-OOC-CH_2-CH(OH)-COO^-$; the dianion of malic acid, a chiral hydroxydicarboxylic acid. The (+)-enantiomer, (S)-malate (also designated L-malate), is an important intermediate in metabolism as a component of both the tricarboxylic-acid cycle and the glyoxylate cycle. **2** any mixture of free malic acid and its mono- and dianions. **3** any salt or ester of malic acid.

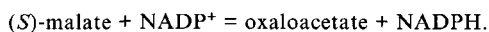


malate-aspartate cycle or **malate cycle** an intracellular metabolic cycle, also known as the malate-aspartate shuttle or malate shuttle, that can transfer reducing equivalents from the cytoplasm to the mitochondria in liver and other tissues. Oxaloacetate formed inside the mitochondria is there converted by a transaminase into aspartate, which passes through the mitochondrial membrane on a specific carrier; a similar transaminase in the cytoplasm converts this aspartate back into oxaloacetate, which then reacts with cytoplasmic NADH, in a reaction catalysed by cytoplasmic **malate dehydrogenase**, to form (S)-malate and NAD^+ . The malate thereupon passes back into the mitochondria, where it is dehydrogenated again to oxaloacetate, with concomitant reduction of mitochondrial NAD^+ to NADH and completion of the cycle.

malate dehydrogenase 1 *abbr.*: MDH or (in clinical chemistry) MD; EC 1.1.1.37; *recommended name*: malate dehydrogenase; *systematic name*: (S)-malate: NAD^+ oxidoreductase; *other name*: malic dehydrogenase. An important and widespread enzyme that interconverts malate and oxaloacetate according to the reaction:

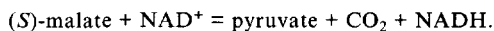


In yeast, e.g., there are three isoenzymes: cytoplasmic, glyoxysomal, and of the mitochondrial matrix. Example from yeast mitochondrial matrix, homodimer (precursor): database code MDHM_YEAST, 334 amino acids (35.53 kDa). **2** EC 1.1.1.82; *recommended name*: malate dehydrogenase ($NADP^+$); *systematic name*: (S)-malate: $NADP^+$ oxidoreductase. A chloroplast enzyme required for both C_3 and C_4 pathways (see **reductive pentose phosphate cycle**, **Hatch-Slack pathway**) to allow plants to circumvent the problem of photorespiration. It catalyses the reaction:



Example (precursor) from maize: database code MDHC_MAIZE, 432 amino acids (46.81 kDa). See also **malate dehydrogenase (decarboxylating)**.

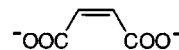
malate dehydrogenase (decarboxylating) any of several enzymes that bring about the oxidation of malate with simultaneous decarboxylation of the product to pyruvate. They include: (1) Malate dehydrogenase (oxaloacetate decarboxylating), EC 1.1.1.38; *systematic name*: (S)-malate: NAD^+ oxidoreductase (oxaloacetate decarboxylating); *other names*: malic enzyme; pyruvic-malic carboxylase; malate oxidoreductase; it catalyses the reversible reaction:



Example, mitochondrial enzyme (a tetramer that can use $NADP^+$) from human (precursor): database code MAOM_HUMAN, 584 amino acids (65.37 kDa). (2) Malate dehydrogenase (decarboxylating), EC 1.1.1.39; *systematic name*: (S)-malate: NAD^+ oxidoreductase (decarboxylating); *other names*: malic enzyme; pyruvic-malic carboxylase; it catalyses a reaction between (S)-malate and NAD^+ to form pyruvate, CO_2 , and NADH. (3) Malate dehydrogenase (oxaloacetate decarboxylating) ($NADP^+$), EC 1.1.1.40; *systematic name*: (S)-malate: $NADP^+$ oxidoreductase (oxaloacetate decarboxylating); *other names*: malic enzyme; pyruvic-malic carboxylase; it catalyses a reaction between (S)-malate and $NADP^+$ to form pyruvate, CO_2 , and NADPH. It is an important enzyme in generating NADPH in the cytosol of mammals. Thus types 1 and 2 utilize NAD^+ , while only types 1 and 3 decarboxylate added oxaloacetate, which leads to three activities being distinguished. (4) D-Malate dehydrogenase (decarboxylating), EC 1.1.1.83; *systematic name*: (R)-malate: NAD^+ oxidoreductase (decarboxylating); it catalyses a reaction between (R)-malate and NAD^+ to form pyruvate, CO_2 , and NADH.

malate synthase EC 4.1.3.2; *systematic name*: L-malate glyoxylate-lyase (CoA-acylating); *other names*: malate synthetase; malate condensing enzyme; glyoxylate transacylase. An enzyme that catalyses the formation of L-malate and CoA from acetyl-CoA, H_2O , and glyoxylate. This is the second step in the glyoxylate bypass, an alternative to the complete tricarboxylic-acid cycle (in bacteria and plants). Example from *Escherichia coli*: database code MASY_ECOLI, 533 amino acids (60.21 kDa).

maleate 1 the trivial name for cis-1,2-ethylenedicarboxylate; (Z)-butenedioate; the dianion of maleic acid, the diastereoisomer of fumarate. **2** any mixture of maleic acid and its mono- and dianions. **3** any salt or ester of maleic acid.

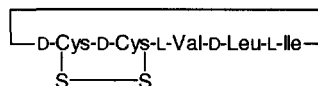


maleoyl *symbol*: Mal; the bivalent cis-acyl group, $-CO-CH=CH-CO-$, derived from maleic acid.

maleyl *symbol*: Mal; the univalent cis-acyl group, $HOOC-CH=CH-CO-$, derived from maleic acid.

maleylation the introduction of one or more maleyl groups into a substance by acylation, e.g. with maleic anhydride. Maleylation of a protein with maleic anhydride is used to acylate its free lysine (and other) amino groups in order to change their charge at neutral pH from positive to negative and to render the adjacent peptide bonds resistant to hydrolysis by trypsin; it can also lead to disaggregation of a multimeric protein. The reaction occurs at pH 8.5 (usually at 2 °C) and the maleyl groups can be removed at pH 3.5 (usually at 60 °C). Maleylation of thiol groups may also occur. Compare **citraconylation**, **succinylation**. —maleylate *vb*.

malformin A a cyclic pentapeptide from *Aspergillus niger* that causes curvature and malformations of various plants. The structure contains D-enantiomers of cysteine and leucine.



malic enzyme or **pyruvic-malic carboxylase** a trivial name commonly applied to any of three enzymes, found in most organisms, that catalyse the oxidative decarboxylation of (S)-malate with concomitant production of pyruvate. See **malate dehydrogenase (decarboxylating)**.



malignant

malignant (in pathology) **1** (of a tumour) invasive, spreading, resistant to therapy. **2** (of a disorder) becoming progressively more serious if not treated.

malnutrition any imbalance, usually a deficiency, between the nutrients taken in by an organism and the amounts necessary to maintain that organism's normal development and health.

malonate **1** the trivial name for propanedioate; methanedicarboxylate; $\text{OOC-CH}_2\text{-COO}^-$, the dianion of malonic acid. It is a potent competitive inhibitor of **succinate dehydrogenase**. **2** any mixture of free malonic acid and its mono- and dianions. **3** any salt or ester of malonic acid.

malonate semialdehyde the recommended trivial name for malonate semialdehyde; formylethanoate; $\text{OHC-CH}_2\text{-COO}^-$. It is an intermediate in the oxidation of propionyl-CoA to acetyl-CoA in the terminal stages of the beta oxidation of fatty acids with an odd number of carbon atoms in plants.

malonyl **1** (in biochemistry) the univalent acyl group, $\text{HOOC-CH}_2\text{-CO-}$, derived from malonic acid by loss of one hydroxyl group. **2** (in chemistry) the bivalent acyl group, $\text{-CO-CH}_2\text{-CO-}$, derived from malonic acid by loss of both hydroxyl groups.

malonyl-CoA a key intermediate in fatty-acid biosynthesis, formed by reaction of carbon dioxide with acetyl-CoA, with hydrolysis of ATP to ADP and inorganic phosphate. This is catalysed by **acetyl-CoA carboxylase**; biotin is an essential cofactor. The plant enzyme also carboxylates propanoyl-CoA and butanoyl-CoA.

malonyl-CoA-acyl carrier protein transacylase EC 2.3.1.39; recommended name: [acyl-carrier-protein] S-malonyltransferase; systematic name: malonyl-CoA:[acyl-carrier-protein] S-malonyltransferase. An enzyme that catalyses a reaction between malonyl-CoA and ACP to form CoA and malonyl-ACP. It is a component of fatty-acid synthase in bacteria (see **fatty-acid synthase complex**). Example from *Escherichia coli*: database code FABD_ECOLI, 308 amino acids (32.25 kDa).

maloyl the bivalent acyl group, $\text{-OC-CH(OH)-CH}_2\text{-CO-}$, derived from malic acid by loss of the hydroxyl groups from both of its carboxyl groups.

Malpighian body or **Malpighian corpuscle** or **renal corpuscle** the part of a nephron comprising a cluster of blood capillaries (i.e. the renal glomerulus) together with the enclosing **Bowman's capsule**. [After Marcello Malpighi (1628–1694), Italian anatomist.]

malt **1** barley or other cereal grain that has been soaked in water, allowed to germinate, and then dried. In this process the starch of the cereal is largely degraded to maltose and other oligosaccharides. **2** to make into or become malt; to make or treat (something) with malt or an extract of malt.

maltase a trivial name for α -glucosidase; see **glucosidase** (def. 2).

malto+ comb. form (in chemical nomenclature) designating an oligosaccharide or polysaccharide (e.g. maltose, maltotriose, maltoheptaose) consisting of D-glucopyranose residues in (α 1 $\rightarrow$ 4)-linkage (as in maltose).

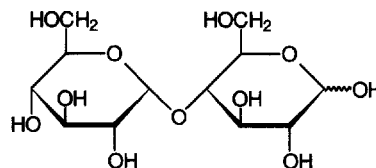
maltoporin a bacterial outer membrane transmembrane protein, with possibly 16 helices, having two roles in *Escherichia coli*: it transports maltodextrins (including maltose) into the cell, and acts as a receptor for lambda and some other phages. Example from *E. coli*: database code LAMB_ECOLI, 446 amino acids (48.86 kDa).

maltose or (formerly) **malt sugar** the trivial name for the disaccharide 4-O- α -D-glucopyranosyl-D-glucopyranose. It is an intermediate in the enzymic breakdown of glycogen and starch. (Illustrated top right.)

maltotriose the trisaccharide $\text{Glc}(\alpha$ 1-4) $\text{Glc}(\alpha$ 1-4) Glc ; it is formed by the action of α -amylase on starch.

malt sugar a former name for maltose.

malyl either of the isomeric univalent acyl groups, $\text{HOOC-CH(OH)-CH}_2\text{-CO-}$ or $\text{HOOC-CH}_2\text{-CH(OH)-CO-}$, derived from malic acid by loss of a hydroxyl group from one or the other of the carboxyl groups.



maltose

mamma (pl. **mammæ**) a mammary gland. —**mammary** *adj.*

mammal any animal of the Mammalia, a large class of warm-blooded tetrapod vertebrates characterized by the possession of sweat glands in the skin and, generally, an insulating layer of hair. Female mammals characteristically suckle their young at modified sweat glands – mammary glands – that secrete milk. Like birds, mammals possess a four-chambered heart, but their thoracic diaphragm is a uniquely mammalian feature. —**mammalian** *adj.*

mammalian artificial chromosome *abbr.*: MAC; a still hypothetical mammalian counterpart of a **yeast artificial chromosome** (YAC).

mammary gland or **mamma** the milk-secreting organ of a mature female mammal. The mammary glands occur in one or more pairs on the ventral surface.

mammotropic or **mammotrophic** an alternative term for **lactotropic**.

mammotropic hormone or **mammotrophic hormone** an older name for **prolactin**.

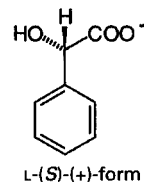
mammotropin or **mammotrophin** an older name for **prolactin**.

Man symbol for a residue (or a molecule) of the aldohexose mannose. In the condensed system of symbolism of sugar residues it signifies specifically a mannose residue with the common configuration and ring size, i.e. D-mannopyranose.

manaic acid see **cyclic fatty acid**.

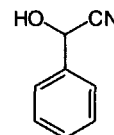
Mancini technique see **single radial immunodiffusion**.

mandelate α -hydroxybenzeneacetate; the anion, $\text{C}_6\text{H}_5\text{-CH(OH)-COO}^-$, derived from mandelic acid, the α -carbon atom of which is chiral. **2** any salt or carboxylic ester of mandelic acid.



L-(S)-(+)-form

mandelonitrile α -hydroxybenzeneacetonitrile; benzaldehyde cyanohydrin; a compound that occurs free and as β -glucosides of (+)-, (–)-, and (±)-mandelonitrile.



manganese *symbol*: Mn; a ferromagnetic, metallic transition element of group 7 of the IUPAC **periodic table**; atomic number 25; relative atomic mass 54.938. There is one stable nuclide of mass number 55. Manganese is widely distributed and abundant.

manifold

dant in the Earth's crust and is an essential component of all living material. It forms manganese(II) (manganous) and less stable manganese(III) (manganic) compounds, including ionic compounds, and complexes; it can also exist in higher oxidation states. Mn^{2+} ions act as activators of various enzymes. A number of artificial radioactive isotopes are known.

manifold a vessel or tube with a number of inlets or outlets used to collect or distribute gases or liquids in an apparatus.

manna a sweet-tasting, white to yellow dried exudate of the manna ash tree (*Fraxinus ornus*) and other plants. *See also* manninotriose.

mannan the main **hemicellulose** of soft (coniferous) wood, made up of D-mannose, D-glucose, and D-galactose in the ratio of 3:1:1. Glucomannans occur in ferns and others occur in certain higher plants and algae where they may function as the main structural component of the cell wall. Highly branched mannans can be extracted from the cell walls of some yeasts.

mannanase or **mannase** *see* **mannosidase** (def. 2).

Mannich reaction any chemical reaction in which an iminium ion of the type $CR'_2=N^+R_2$ reacts with a nucleophilic carbon atom, such as that of an enolate anion derived from a ketone or that of an activated double bond as in a phenol, aryl ether, pyrrole, or indole. Mannich reactions are believed to occur widely in biosynthetic pathways of polycyclic compounds, especially alkaloids and porphyrins. [After Carl Mannich (1877–1947).]

manninotriose the trisaccharide Gal(α 1-6)Gal(α 1-6)Glc; a component of **stachyose**, it also occurs free in **manna** from ash trees.

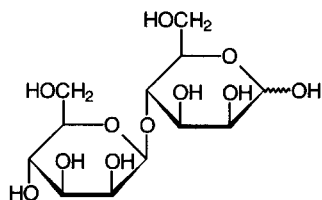
mannite a former name for **mannitol** (use deprecated).

D-mannitol the alditol derived from D-mannose by reduction of the -CHO group. It is widespread in higher plants, plant exudates, and algae, and is obtained from the manna ash (*Fraxinus ornus*) and seaweeds. Mannitol is metabolized by a number of bacteria including many strains of *Escherichia coli* and *Salmonella* spp. Clinically it is administered intravenously as an osmotic diuretic.

manno+ *comb. form* of mannose.

manno- *prefix (in chemical nomenclature)* indicating a particular configuration of a set of (usually) four contiguous >CHOH groups, as in the acyclic forms of D- or L-mannose. *See* **monosaccharide**.

mannobiose the disaccharide Man(β 1-4)Man; a constituent of D-glucosyl-D-mannans isolated from hardwoods, and of polysaccharides from iris, orchid, and lily families.



mannoheptulose *manno*-2-heptulose; the D enantiomer (formerly known as D-manno-ketoheptose) occurs in the avocado.

mannokinase EC 2.7.1.7; an enzyme that catalyses the phosphorylation by ATP of D-mannose to D-mannose 6-phosphate with release of ADP. It initiates the further metabolism of mannose, e.g. after uptake by bacteria. Hexokinase has a wide specificity for sugars and catalyses this reaction, e.g., in mammalian liver.

mannosamine *symbol*: ManN; a *trivial name* for 2-amino-2-deoxymannose.

mannose *symbol*: Man; a *trivial name* for the aldohexose *manno*-hexose, the C-2 epimer of **glucose**. The D-(+)-form is

widely distributed in mannans and hemicelluloses and is of major importance in the core oligosaccharide of N-linked oligosaccharides of glycoproteins, especially at branch points. A commonly occurring structure is Man(α 1-6)[Man(α 1-3)]-Man(β 1-4)GlcNAc(β 1-4)GlcNAc-Asn.

mannose 6-phosphate an intermediate in **mannose** metabolism, formed from mannose by **mannokinase** or from fructose 6-phosphate by **mannose-6-phosphate isomerase**. Apart from acting as a precursor in GDPmannose formation, it is a component of lysosomal enzymes, playing an important role in sorting the enzymes into lysosomes. *See also* **I-cell disease**, **mannose 6-phosphate receptor**.

mannose-1-phosphate guanylyltransferase EC 2.7.7.13; *other name*: GTP-mannose-1-phosphate guanylyltransferase; an enzyme that catalyses the reaction between GTP and α -D-mannose 1-phosphate to form GDPmannose with release of pyrophosphate. Example (fragment) from pig: database code A47415, 23 amino acids (2.49 kDa).

mannose-1-phosphate guanylyltransferase (GDP) EC 2.7.7.22; *systematic name*: GDP:D-mannose-1-phosphate guanylyltransferase. An enzyme involved in M-antigen biosynthesis in salmonellae; it catalyses a reaction between GDP and D-mannose 1-phosphate to form orthophosphate and GDPmannose. Example from *Salmonella choleraesuis*: database code CPSB_SALCH, 471 amino acids (52.57 kDa). This protein belongs to the **mannose-6-phosphate isomerase** family 2.

mannose-6-phosphate isomerase EC 5.3.1.8; *systematic name*: D-mannose-6-phosphate ketol-isomerase; *other names*: phosphomannose isomerase; phosphohexoisomerase; phosphohexomutase. An enzyme that catalyses the reaction:

D-mannose 6-phosphate = D-fructose 6-phosphate;

zinc is a cofactor. The sequences belong to two families. Example (family 1) from *Escherichia coli*: database code MANA_ECOLI, 391 amino acids (42.80 kDa). Most enzymes (including human) belong to this family. Family 2 is represented by an enzyme involved in alginate metabolism; example from *Pseudomonas aeruginosa*: database code ALGC_PSEAE, 462 amino acids (50.23 kDa). *See also* **mannose-1-phosphate guanylyltransferase**.

mannose 6-phosphate receptor a protein on the surface of membranes in the Golgi apparatus and late endosomes that recognizes mannose 6-phosphate residues on lysosomal enzymes and is responsible for sorting them into late endosomes, which then become lysosomes. Lysosomal enzymes bearing phosphomannosyl residues bind specifically to mannose-6-phosphate receptors in the Golgi apparatus, and the resulting receptor-ligand complex is transported to an acidic prelysosomal compartment where the low pH mediates dissociation. There are two types of receptor protein (examples are bovine): (1) the cation-dependent type – these are type I membrane glycoprotein dimers involved in transport of phosphorylated lysosomal enzymes from the Golgi apparatus and the cell surface to lysosomes; example (precursor): database code MPRI_BOVIN, 279 amino acids (31.17 kDa); (2) the cation-independent type – these are similar in action and with some sequence homology to the first type; example (precursor): database code MPRI_BOVIN, 2499 amino acids (274.22 kDa). *See also* **I-cell disease**.

mannosidase 1 α -mannosidase; EC 3.2.1.24; *systematic name*: α -D-mannoside mannohydrolase; an enzyme that catalyses hydrolytic removal of terminal, nonreducing residues in α -D-mannosides. Example from *Saccharomyces cerevisiae*: database code MAN1_YEAST, 1083 amino acids (124.5 kDa). **2** β -mannosidase EC 3.2.1.25; *systematic name*: β -D-mannoside mannohydrolase; *other names*: mannanase; mannase; an enzyme that catalyses hydrolytic removal of terminal, nonreducing residues in β -D-mannosides. Example from *Bos taurus*: database code A5881, 879 amino acids (101.06 kDa). *See also* **ER mannosidase**, **Golgi mannosidase**.

mannoside any **glycoside** in which the sugar moiety is a **mannose** residue.

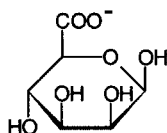


mannosyl

mannosyl any of the glycosyl ligands formed by detaching the anomeric (α - or β -) hydroxyl group, on C-1, from a cyclic form (pyranose or furanose) of mannose.

α -1,3-mannosyl-glycoprotein β -1,2-*N*-acetylglucosaminyltransferase EC 2.4.1.101; *other name*: *N*-glycosyl-oligosaccharide-glycoprotein *N*-acetylglucosaminyltransferase I (*abbr.*: GlcNAc-TI or GNT-I); an enzyme involved in the synthesis of oligosaccharides, particularly those that are N-linked in glycoproteins. It catalyses the reaction between UDP-*N*-acetyl-D-glucosamine and α -D-mannosyl-1,3-(R)- β -D-mannosyl-R' to form *N*-acetyl- β -D-glucosaminyl-1,2- α -D-mannosyl-1,3-(R)- β -D-mannosyl-R' with release of UDP. Example from human: database code GNT1_HUMAN, 445 amino acids (50.80 kDa).

mannuronate 1 the anion of mannuronic acid, the compound formed by oxidation of the C-6 hydroxymethylene group of mannose to a carboxyl group. 2 any salt or ester of mannuronic acid. D-Mannuronic acid is a component of **alginate**.



β -D-pyranose form

manoalide a naturally occurring marine sesterterpenoid. It is an irreversible inhibitor of phospholipases A₂ and C, and a potent inhibitor of Ca²⁺ mobilization.

manometer an instrument for measuring pressure differences. The simplest type consists of a U-tube filled with a suitable liquid (often mercury), and compares a pressure delivered to one side with atmospheric pressure on the other, open, side of the tube. The difference in the level of liquid in the two arms of the tube gives a measure of the unknown pressure. In biochemistry the term is often used to mean a **manometric respirometer**.

manometer constant a predetermined factor, calculated for each individual constant-volume manometric **respirometer**, that when multiplied by the manometer reading immediately gives the volume of gas evolved in cubic millimetres at **standard** temperature and pressure. The factor is itself dependent upon temperature, liquid volume, density of the manometer fluid, and nature of the gas phase.

manometer fluid any fluid used to fill a **manometer**. In biochemistry this usually refers to the fluid used to fill the manometer of a respirometer. Such fluids are generally easily visible, wet the glass walls of the manometer, have a density that will give an easily calculated manometer constant, and have a low vapour pressure. Examples include **Brodie's fluid**, and **Krebs' manometer fluid**.

manometric 1 of, pertaining to, or measured with a **manometer**. 2 (*in biochemistry*) of, or pertaining to, a manometric respirometer. —**manometrically** *adv.*

manometry 1 the measurement of gas pressure by means of a manometer. 2 (*in biochemistry*) the use of manometric respirometers in studies of chemical and biochemical reactions.

Manp *symbol* for a mannopyranose residue.

mantissa the fractional part of any **logarithm**.

MAO *abbr.* for monoamine oxidase.

MAO inhibitor *abbr.* for monoamine oxidase inhibitor.

map 1 any representation of the surface features of an object, especially the spatial relationships of physical or chemical landmarks, as in **electron-density map**, **genetic map**, and **restriction map**. By extension the term is applied to certain two-dimensional maps.

mensional patterns obtained in analysis, e.g. **peptide map**. 2 to make a **map** (def. 1). 3 to occur, or be placed, at a specific position on a **map** (def. 1).

MAP *abbr.* for microtubule-associated protein.

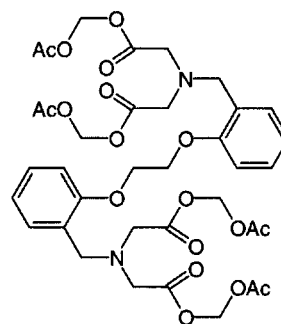
map distance (*in genetics*) the distance, in terms of percentages of crossing-over (i.e. 100 $\times$ no. of recombinants/total no. of progeny), between particular linked genes. *See also* **map unit**.

MAP kinase *abbr.* for mitogen-activated protein kinase; a family of protein kinases that perform a crucial step in relaying signals from the plasma membrane to the nucleus. They are turned on by a wide range of proliferation- or differentiation-inducing signals. Example, MAP kinase FUS3 of *Saccharomyces cerevisiae*, required for the transition from mitosis into conjugation: database code FUS3_YEAST, 353 amino acids (40.77 kDa). The name does not indicate that this is a kinase for MAPs – rather that it is a protein kinase that is mitogen-activated. Nevertheless, some MAPs can be substrates for this enzyme. *See also* **ERK**.

MAP kinase kinase *abbr.*: MAPKK; EC 2.7.1.–; a dual-specificity mitogen-activated protein kinase kinase; *other names*: ERK activator kinase; extracellular signal-regulated kinase; an enzyme that catalyses the concomitant phosphorylation of threonine (T) and tyrosine (Y) residues in a TEY sequence in **MAP kinases**. It requires activation by the serine/threonine kinase, MAP kinase kinase kinase (*see* **RAF**). Examples from human: MAPKK 1, database code MPK1_HUMAN, 392 amino acids (43.28 kDa); MAPKK 2, database code MPK2_HUMAN, 400 amino acids (44.37 kDa).

maple-syrup urine disease or **branched-chain ketonuria** an autosomal recessive disease of humans characterized by neurological signs and the onset of feeding problems in the first week of life, and associated with a maple-syrup odour of the urine. The concentrations of branched-chain amino acids and branched-chain keto acids are elevated in the blood of patients, due to deficiency of the enzyme responsible for the oxidative decarboxylation of branched-chain keto acids.

MAPTAM 1,2-bis-(2-amino-5-methylphenoxy)ethane-*N,N,N',N'*-tetraacetic acid tetraacetoxymethyl ester; a substance that readily penetrates cell membranes and undergoes intracellular hydrolysis with the formation intracellularly of the 5-methyl form of **BAPTA**. It is useful as an indicator of intracellular calcium concentrations.



map unit or **Morgan's unit** a unit of genetic length along a chromosome and defined in terms of recombination frequency, expressed as a percentage; a recombination frequency of 1 per cent (that is one recombinant individual in a hundred progeny) equals one map unit. 1 map unit = 1 centimorgan. *Compare* **morgan**.

marasmus a wasting away of the body, especially of infants, caused by chronic deficiency of intake of energy-producing foodstuffs. *Compare* **kwashiorkor**.

marcks *abbr.* for myristoylated alanine-rich C kinase substrate; a filamentous (F) actin cross-linking protein and the most prominent substrate for protein kinase C. It also binds



margarine

calmodulin, actin, and synapsin and is displaced from the membrane by phosphorylation by protein kinase C. Example from human: database code MACS_HUMAN, 331 amino acids (31.38 kDa).

margarine a processed fat used as a substitute for butter. Modern margarines are prepared from hydrogenated natural fats, the trend in recent years having been to preserve a high content of nutritionally desirable **polyunsaturated fatty acids**. Apart from the oil or fat content, margarine contains a water phase that is 16–20% of content and includes water, cream, whole milk, skim milk (powder), whey, and buttermilk in various ratios; flavouring, which may include diacetyl, fatty acids, and ketones; colouring agents, mainly based on carotenes; additional vitamins A and D; and **antioxidants** such as di-*tert*-butyl *p*-cresol. The process was invented in 1869 by H. Mège-Mouriès on the order of Napoleon III to produce a cheap stable butter substitute. Mège-Mouriès churned a beef tallow fraction, known as **oleomargarine**, with cream.

marijuana *see* **cannabis**.

Mariotte bottle or **Mariotte flask** a device for providing a flow of liquid maintained at a constant pressure head; it is used, e.g., in column chromatography. It consists of a sealed flask, closed to the atmosphere, with an outlet at or near the bottom and an air inlet tube open to the atmosphere. The height of the bottom of the inlet tube is adjustable. When the flask is filled with liquid to a level above the bottom of the inlet tube the pressure of the liquid flowing from the flask is determined by the height of the bottom of the inlet tube above the outlet point. [After Abbé Edmé Mariotte (1620–84), French physicist.]

Mariotte's law *an alternative name (esp. Fr.) for Boyle's law (see gas laws).*

marker **1** an appropriate reference substance or material included in a sample being subjected to a separation procedure such as chromatography, electrophoresis, or density-gradient centrifugation. **2** *see* **biochemical marker**. **3** *see* **genetic marker**. **4** *an alternative term for label* (def. 3).

marker enzyme any enzyme of known intracellular location that can be assayed as an aid in following the isolation and purification of subcellular fractions.

marker gene any gene of known function and known location on a chromosome. *Compare* **genetic marker**.

marker rescue a process whereby one or more genetic markers from a UV-irradiated virus are integrated into an unirradiated helper virus in infected bacteria; such genes are thereby 'rescued' by recombination.

Markham still a simple, all-glass, steam-jacketed apparatus for the rapid and efficient steam distillation of a succession of small samples of volatile substances. It was designed especially for the distillation of ammonia as part of the micro-Kjeldahl procedure for the determination of nitrogen, but has been found to be prone to carry-over from one sample to the next. [After Roy Markham (1916–79), British biochemist.]

Markovnikov's rule a generalization stating that when a molecule of water, or a hydrogen halide, or some other polar substance, is added heterolytically across an asymmetrically substituted carbon-carbon multiple bond, the hydrogen atom or other more electropositive part of the molecule becomes attached usually to that carbon atom carrying the larger number of hydrogen atoms while the hydroxyl, halo, or other more electronegative part of the molecule becomes attached to that carbon atom carrying the smaller number of hydrogen atoms. [After Vladimir Vasilyevich Markovnikov (1838–1904), Russian chemist, who formulated the rule on empirical grounds (1870).]

Maroteaux-Lamy syndrome *an alternative name for mucopolysaccharidosis VI.*

marsh gas *another name for methane.*

Martin, Archer John Porter (1910–), British biochemist, physical chemist, and inventor; Nobel Laureate in Chemistry (1952) jointly with R. L. M. Synge for their invention of partition chromatography.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

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MAS a human transforming gene that encodes a neuronal-type angiotensin III receptor. This is a transmembrane cell surface protein, mediating a G-protein-linked phosphoinositide response. Example: database code MAS_HUMAN, 325 amino acids (37.42 kDa).

maser a device for producing intense beams of microwave radiation of defined frequency. *Compare* **laser**. [From microwave amplification by stimulated emission of radiation.]

masked residue any residue in a biopolymer that is not accessible to, and cannot react with, a reagent specific for that type of residue, either because of the position of the residue or because of the nature of its immediate environment.

maspin a **serpin**-like tumour suppressor of epithelial cells. Example from human: database code HSU04313, 375 amino acids (42.09 kDa).

mass **1** *symbol: m or M*; one of the seven SI base physical quantities; the SI base unit of mass is the **kilogram**. **2** *a general term for the quantity of matter in a body, a characteristic that is both a measure of the resistance of the body to changes in its velocity, i.e. inertial mass, and a measure of the force experienced by the body in a gravitational field, i.e. gravitational mass*. Mass is commonly confused with weight. According to the theory of relativity, inertial and gravitational masses are equal, mass increases with velocity, and mass and energy are interconvertible.

mass action, law of a law stating that, for a homogeneous system, the rate of reaction of a chemical reaction is proportional to the active masses of the reacting substances; i.e. the rate is proportional to the product of the activities of the reactants, each raised to a power equal to the number of reactant molecules of that type that participate in the reaction.

mass-average molar mass *see* **average molar mass**.

mass concentration *see* **concentration** (def. 1a).

mass density *see* **concentration** (def. 1a).

mass fraction *symbol: w*; the mass of a specified component of a system divided by the mass of the system; it is dimensionless.

mass fragmentography *a common but less rigorous name for selected-ion monitoring.*

mass number *an older name for nucleon number.*

mass spec. *abbr. (colloquial) for mass spectrometer, mass-spectrometric, or mass spectrometry.*

mass spectrograph a **mass spectrometer** in which the ions fall on a photographic plate.

mass spectrometer *abbr.: MS or (colloquial) mass spec.*; any instrument for **mass spectrometry**.

mass spectrometry *abbr.: MS or (colloquial) mass spec.*; an analytical method for determining the relative masses and relative abundances of components of a beam of ionized molecules or molecular fragments produced from a sample in a high vacuum. It is applicable to any sample that is gaseous or volatile at low pressure, or that can be so rendered by derivatization. In the commonest version, the ions, generated usually by electron bombardment of the sample, are formed into a beam and then accelerated to a uniform velocity by means of an arrangement of electrically charged slits. The beam is then dispersed into a spectrum of particles of different mass-to-charge ratios, *m/e*, by application of a suitably disposed magnetic field. The ions of any particular *m/e* value may be focused on a collecting electrode by appropriate adjustment of the accelerating potential or magnetic field and the resulting ion currents measured electrometrically. With a suitably designed mass spectrometer the method can be used to determine the relative abundances of isotopes of an element or the relative molecular masses of two or more compounds, or used to aid the identification of a complex compound by the mass spectrum of the fragments generated from it on electron bombardment. *See also* **gas chromatography/mass spectrometry**, **selected-ion monitoring**. —*mass-spectrometric adj.*

mass spectrum a tracing, photographic record, or diagram of the relative numbers of different ions (representing molecules or molecular fragments) obtained (from a specimen) by

mass-to-charge ratio

mass spectrometry, ranked in order of their mass-to-charge ratios.

mass-to-charge ratio (*in mass spectrometry*) the ratio of the mass, m , of an ion to the charge, e , on the ion. *See also* **mass spectrometry**.

mass transfer the spontaneous and irreversible process of transfer of mass across nonhomogeneous fields, e.g. between liquid phases, or from one side of a membrane to another.

mass unit *see* **atomic mass unit**, **dalton**.

mast cell a type of leukocyte, of the subclass granulocyte; it is characterized by having numerous basophilic granules in the cytoplasm; these granules contain relatively large amounts of histamine together with 5-hydroxytryptamine and heparin. Degranulation of mast cells, with concomitant release of the vasoactive amines, occurs by reaction, through a specialized region of the Fc piece, of antigen with homocytotropic antibody (reagin) bound to the cell. This leads to an immediate hypersensitivity reaction.

mast cell degranulating peptide *abbr.*: MCDP; one of the major neurotoxic peptides of bee venom; the 22-residue peptide IKCNCKRHVHKPHICRKICGKN-NH₂. It stimulates degranulation of mast cells, and also selectively blocks the neuronal Ca²⁺-dependent K⁺ channel.

mast cell growth factor *see* **stem cell factor**.

mastoparan a tetradecapeptide toxin from wasp venom: Ile-Asn-Leu-Lys-Ala-Leu-Ala-Ala-Leu-Ala-Lys-Lys-Ile-Leu-NH₂. It is neurotoxic, and functions in a similar way to **melittin** and **mast cell degranulating peptide**, in generating ion channels in membranes.

MAT α 2 yeast mating-type protein α -2 repressor; a regulatory DNA-binding protein whose DNA recognition sequence (CATGTAATT) resembles a homeobox, the recognition sequence of homeobox proteins in higher organisms. MAT α 2 acts as a master switch in yeast differentiation by controlling gene expression in a cell-type-specific fashion. In α haploids and a/a diploids (*see* **mating type**) it represses the transcription of α -cell-specific genes. In a/a diploids it also forms a complex with MAT α 1 protein that represses haploid-specific gene expression. Database code MAT2_YEAST, 210 amino acids (24.25 kDa).

matchmaker proteins *see* **molecular matchmakers**.

maternal-effect gene a gene that is transcribed from the maternal genome during oogenesis, and expresses products at or immediately before fertilization, so that the phenotype of the embryo is determined by the maternal alleles alone. Genes that determine egg polarity, e.g. **oskar** are of this type.

mathematics **1** *functioning as sing.* the group of related sciences concerned with number, space, shape, and quantity, and their interrelationships. **2** *functioning as sing. or pl.* the use of mathematical processes or operations in calculations or the solution of problems.

mating type one of the two types of *Saccharomyces cerevisiae*, α and a , each of which is necessary for the occurrence of mating, a process in which two haploid cells fuse. The resulting diploid cell, a/a , cannot mate, but can produce spores, giving rise to haploid cells by meiosis. The mating type of the haploid cell is determined by a single locus, the mating-type (MAT) locus. In a cells, this encodes a single gene regulatory protein $a1$, and in an α cell it encodes two such proteins, $\alpha1$ and $\alpha2$. In the a/a cell, the regulatory proteins $a1$ and $\alpha2$ combine to generate a completely new pattern of gene expression. *See also* **MAT α 2**.

matrilysin EC 3.4.24.23; *other names*: **matrin**; uterine metallo-endopeptidase; matrix metalloproteinase 7 (*abbr.*: MMP7); putative (or punctuated) metalloproteinase-1 (*abbr.*: PUMP-1); an enzyme that catalyses the cleavage of Ala¹⁴-Leu¹⁵ and Tyr¹⁶-Leu¹⁷ in the B chain of insulin. It has no action on collagen types I, II, IV, or V. It cleaves casein and fibronectin and shows preference for gelatin chain α -2(I) over α -1(I). It is secreted as the precursor, promatrilysin, and is activated by autocatalytic cleavage at Glu⁷⁷-Tyr⁷⁸. Zinc and calcium are

cofactors. Example (precursor) from human uterus: database code COG7_HUMAN, 267 amino acids (29.64 kDa).

matrin or **DNA-binding protein F/G** a zinc-finger DNA-binding protein of the fibrogranular areas of the nucleus. Matrins comprise a family of eight major internal matrix polypeptides of nuclear matrices of rat liver and tissue culture cells, highly conserved between humans and mouse. Example from rat: database code MAFG_RAT, 544 residues (59.67 kDa). The name has sometimes been used as a synonym for **matrilysin** and **matrin F/G** is therefore used to avoid confusion.

matrix (*pl. matrices or matrixes*) **1** a medium or place wherein something is formed, develops, or is embedded or enclosed. **2** the medium that forms, or supports, the stationary phase in certain separative procedures, as in chromatography or zone electrophoresis; the material that forms, or is used to form, the substratum of an immobilized reagent in affinity chromatography. **3** (*in animal histology*) *see* **extracellular matrix**. **4** (*in plant histology*) the meshwork of solubilizable substances, largely polysaccharide in nature, occurring in cell walls and in which insoluble microfibrils of cellulose are embedded. **5** *see* **mitochondrial matrix**. **6** cytoplasmic matrix; *see* **groundplasm**. **7** the substratum, living or nonliving, in or on which a fungus or lichen grows. **8** (*in mathematics*) any rectangular array of numbers or symbols manipulable algebraically as a single entity and used according to specific rules to facilitate solving of certain kinds of problems, e.g. a set of simultaneous linear equations.

matrix Gla protein a protein that mediates association of the organic matrix of bone and cartilage. It contains γ -carboxyglutamate (Gla) residues, formation of which is vitamin-K-dependent. Example (precursor) from human: 103 amino acids (12.31 kDa).

matrix metalloproteinase *abbr.*: MMP; *a name given to* proteolytic enzymes associated with the **extracellular matrix** as the basis of a new nomenclature for these enzymes. The classification includes: MMP1, interstitial collagenase; MMP2, gelatinase A; MMP3, stromelysin-1; MMP8, neutrophil collagenase; MMP10, stromelysin-2; MMP12, metalloelastase. Such metalloproteinases have been implicated in clearing a path through the extracellular matrix for cell migration, and thus have been thought to be involved in the ability of malignant cells to invade tissues and metastasize. They have also been shown to be involved in conversion of tumour necrosis factor (TNF) precursor into active TNF and may be implicated in other diseases such as rheumatoid arthritis or osteoarthritis. For these reasons, there is current interest in matrix metalloproteinase inhibitors (*abbr.*: MMPis) as potentially useful drugs. *See also* **TIMP**.

matrix metalloproteinase inhibitor *see* **MMPI**.

matrix processing peptidase *abbr.*: MPP; *an alternative term for* **mitochondrial processing peptidase**.

matrix protein **1** the major protein component of the outer membrane of *Escherichia coli*. It is a transmembrane protein that forms hydrophilic pores permitting the passage of microsolutes through the outer membrane; in association with **lipopolysaccharide** (def. 4) it forms a receptor for phages. It is a member of the **porin** family, and is a product of the gene **DMPC**: database code DMPC_ECOL1, 367 amino acids (40.39 kDa). **2** any of a class of proteins that coat the inside of the lipid bilayer in the envelope of some groups of enveloped viruses.

matrix space or **(mitochondrial) lumen** the innermost compartment of a mitochondrion, containing the **mitochondrial matrix**.

matrix vesicle any of the minute, membrane-bound, extracellular particles (≈ 100 nm in diameter) occurring in the interstitial matrix of calcifying cartilage. The vesicles contain a number of enzymes involved with local calcification mechanisms.

maturation **1** the process of coming to full development, ripening, or becoming mature. **2** or **maturation division** the processes of formation of gametes (sperm or ova) from primary



maturation-promoting factor

gametocytes involving meiosis of the nucleus with reduction of chromosome number from diploid to haploid. **3** (*in virology*) the overall process leading to the incorporation of a viral genome into a capsid and the development of a complete virion.

maturation-promoting factor an alternative name for **M-phase-promoting factor**.

maturation protein or **A protein** one of the three proteins coded for by certain single-stranded RNA bacteriophages, one molecule of which is necessary for proper encapsulation of the RNA and for binding of the phage to a pilus on the host cell. Example from bacteriophage ϕ r: database code VASS_BPFR, 393 amino acids (43.96 kDa).

maturity-onset diabetes a former name for type 2 diabetes mellitus.

max *abbr.* for maximal or maximum.

MAX the symbol for a gene family coding for DNA-binding proteins that function in cell growth control. The product, Max, forms heterodimers with Myc (*see* **MYC**) that bind specifically to CACGTG; it also forms homodimers of similar specificity. Max contains **helix-turn-helix** and hydrophobic heptad repeats. Database code MAX_HUMAN, 160 amino acids (18.27 kDa).

Maxam-Gilbert method (for DNA sequencing) *see* **chemical cleavage method**.

maxicell or **maxi cell** a bacterial cell that has been heavily irradiated with UV light and in which the chromosomal DNA is extensively degraded. Any plasmid(s) present usually escape damage and the genes they encode can be expressed.

maxicircle DNA any large, closed circular DNA molecule, as in the kinetoplast of *Leishmania tarantolae*.

maxima a plural of **maximum** (def. 1).

maximal *abbr.*: max; of, pertaining to, or reaching a maximum; being the greatest possible in size, duration, etc.; **maximum** (def. 2). —**maximally** *adv.*

maximal agonist effect or **intrinsic activity** *symbol*: α ; the maximal effect that an agonist, whether conventional or inverse, can elicit in a given tissue under particular experimental conditions, expressed as a fraction of that produced by a full agonist acting through the same receptors under the same conditions.

maximal rate or **maximum rate** another name for **limiting rate** (now recommended).

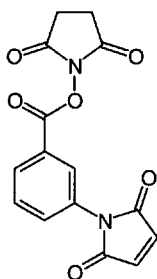
maximal velocity or **maximum velocity** an older name for **limiting rate**.

maximum *abbr.*: max; **1** (*pl. maxima or maximums*) the highest or greatest possible amount, degree, etc.; the highest value of a variable quantity. *Compare* **minimum**. **2** of, being, or exhibiting a maximum or maxima; **maximal**.

Maxwell relation a relation between the dielectric constant, ϵ , and the refractive index, n , of a nonabsorbing, nonmagnetic material such that $\epsilon = n^2$. [After James Clerk Maxwell (1831–79), British physicist.]

Mb **1** *abbr.* and *symbol* for myoglobin; hence MbO₂, oxymyoglobin; MbCO, carbonmonoxymyoglobin; etc. **2** *symbol* for megabase, i.e. = 10⁶ bases. **3** (*in computing*) *abbr.* for megabyte (i.e. 2²⁰ bytes; alternative to MB).

MBS *abbr.* for meta-maleimidobenzoyl-N-hydroxysuccinimide



ester; a heterobifunctional reagent useful, *inter alia*, for coupling an enzyme to an antibody, without appreciable loss of either activity.

MBSA *abbr.* for methylated bovine serum albumin.

mc (formerly) *abbr.* for millicurie (mCi now used).

Mc₃ *symbol* for mucotriose (McOse₃ preferred if space permits).

Mc₄ *symbol* for mucotetraose (McOse₄ preferred if space permits).

MCAD *abbr.* for medium chain acyl-CoA dehydrogenase; *see* **beta-oxidation system**.

McArdle's disease an alternative name for type V **glycogen disease**.

McClintock, Barbara (1902–92), US geneticist noted for her studies on chromosome breakage and reunion and her proposal of the existence of transposable elements, Nobel Laureate in Physiology or Medicine (1983) 'for her discovery of mobile genetic elements'.

MCD *abbr.* for **1** magnetic circular dichroism. **2** mean corpuscular (erythrocyte) diameter.

MCDP *abbr.* for mast cell degranulating peptide.

McFarlane's method a method for the radioiodination of proteins with high efficiency in which the protein to be labelled is reacted with iodine monochloride that has been previously equilibrated with radioiodide. The extent of substitution can be precisely controlled by use of an appropriate quantity of iodine monochloride. [After Arthur Sproul McFarlane (1905–78), British biophysicist.]

mCi *symbol* for millicurie.

McIlwain chopper or **McIlwain-Buddle chopper** a hand- or motor-driven machine for rapidly cutting a piece of animal tissue into slices of uniform, preselected thickness (≥ 0.07 mm) and then, if desired, into prisms by making a second series of cuts. Such prisms can yield easily dispensed suspensions that retain those metabolic characteristics that tend to be lost on homogenization of tissue. [After Henry McIlwain (1912–92), British biochemist, and H. L. Buddle.]

McIntosh and Fildes jar *see* **anaerobic jar**.

McLeod gauge a mercury-in-glass vacuum pressure gauge that is capable of measuring down to 10⁻⁵ mmHg (1.33 × 10⁻³ Pa). A known large volume of gas is compressed into a small volume, in which the now much larger pressure is measured.

McOse₃ *symbol* for mucotriose (preferred alternative to Mc₃).

McOse₄ *symbol* for mucotetraose (preferred alternative to Mc₄).

MCP *abbr.* for methyl-accepting chemotaxis protein.

md *symbol* for mendelevium.

MD (*in clinical chemistry*) *abbr.* for **malate dehydrogenase** (def. 1).

MDa *abbr.* for megadalton.

MDCK cell *abbr.* for Madin–Darby canine kidney cell; a cell of a heteroploid cell line.

MDH *abbr.* for **malate dehydrogenase** (def. 1).

Mdm2 protein a nuclear phosphoprotein that binds to **p53** and inhibits its action. Example MDM2_MOUSE, 489 amino acids (54.54 kDa).

MDMA *abbr.* for 3,4-methylenedioxymethamphetamine.

MDP *abbr.* for muramyl-dipeptide.

MDR *abbr.* for multiple drug resistance or multidrug resistance; the acquired simultaneous resistance to a wide spectrum of drugs arising from the administration, typically over long periods, of one or a limited number of drugs. Such resistance may extend to drugs with little structural or even functional similarity to the original drug(s), and results in reduced efficacy of all the drugs concerned. The phenomenon is seen, e.g., in long-term chemotherapy of cancer patients, and is due to the induced expression of plasma membrane glycoproteins known as P-glycoproteins, or **MDR proteins**.

MDR protein any of several integral transmembrane glycoproteins of the **ABC** type that are involved in (multiple) drug resistance (*see* **MDR**). **MDR 1** (or P-glycoprotein 1) is an energy-

Me

dependent efflux pump responsible for decreased drug accumulation in multidrug-resistant cells. Example from human: database code MDR1_HUMAN, 1280 amino acids (141.34 kDa). MDR 3 (or P-glycoprotein 3) is an energy-dependent efflux pump that causes decreased drug accumulation but is not capable of conferring drug resistance by itself. Example from human: database code MDR3_HUMAN, 1279 amino acids (140.52 kDa). MDR-associated protein participates in the active transport of drugs into subcellular organelles. Example from human: database code MRP_HUMAN, 1531 amino acids (171.47 kDa). MDR protein (or P-glycoprotein) of *Leishmania tarentolae* is a protein responsible for acquired methotrexate resistance, and is encoded by a gene located in the 'H-circle', a 68 kb duplex DNA circle containing a 30 kb inverted repeat. Database code MDR_LEITA, 1548 amino acids (172.04 kDa).

Me symbol for the methyl group, CH₃–.

Mead acid a name sometimes given to (all-Z)-eicosa-5,8,11-trienoic acid. It can be synthesized *de novo* by mammals via palmitate and oleate, and is thought to substitute partially for the long-chain polyunsaturated fatty acids lacking in essential fatty acid deficiency. [After J. F. Mead, who first identified it as an acid found in elevated concentrations in mammals during essential fatty acid deficiency].

mean or arithmetic mean or average the quotient of the arithmetic sum of any set of two or more numbers, measurements, values, etc. and the number of components of the set. *See also* **geometric mean**. *Compare* **median** (def. 1).

mean deviation the arithmetic average of the absolute values of the differences between the individual values of a set of numbers, values, etc. from their mean.

mean free path the average distance travelled by a molecular entity between collisions. In a gas, the mean free path between molecular entities is inversely proportional to the pressure of the gas.

mean ionic activity coefficient symbol: $\gamma_{\pm}$; the average rational activity coefficient of the ions of an electrolyte that dissociates in solution into cations of charge z^+ and anions of charge z^- , given, according to the limiting law of the Debye–Hückel theory, by:

$$-\log \gamma_{\pm} = z^+ z^- A I^{0.5},$$

where A is a constant whose magnitude depends on temperature, dielectric constant, etc., having a value near 0.5, and I is the ionic strength of the solution.

mean life or average life abbr.: τ ; the mean of the lengths of time required for the disintegration of all atoms of a given radionuclide. It is related to the **half-life**, $t_{1/2}$, and to the **disintegration constant**, λ , of the radionuclide by the expressions:

$$\tau = t_{1/2} / \ln 2 = 1.443 t_{1/2},$$

and

$$\tau = 1/\lambda.$$

mean relative residue mass or (formerly) mean residue weight symbol: M_0 ; the quotient of the relative molecular mass of a polymer and the number of residues it contains.

mean residue rotation the average molar optical rotation for a residue in a polymer. It is obtained by substituting the mean relative residue mass, M_0 , for the relative molecular mass, M_r , of the polymer in the equation defining **molar optical rotation**.

mean residue weight a former name for **mean relative residue mass**.

Mec abbr. for the 4-methylcoumarin-7-yl group.

mechanical concerned with, executed or operated by, or as by, a machine; relating to mechanics. —**mechanically** adv.

mechanically gated channel *see* **ion channel**.

mechanics 1 the science concerned with the equilibrium or motion of bodies in a given frame of reference. 2 the science of machines and/or machinery.

mechanism 1 the chain of events in a particular process; the action or process by which an effect is produced or an action

performed. 2 (*in chemistry*) the detailed description of the pathway leading from reactants to products of a reaction, including where possible a characterization of the composition, structure, and other properties of any reaction intermediates and transition states. 3 any system or structure of moving parts that performs some function; a machine.

mechanistic of, pertaining to, or explained by mechanics; of, pertaining to, or resembling a mechanism. *Compare* **stochastic**.

mechanochemical coupling the coupling of a chemical process with a mechanical process in the same structure. *See also* **motor protein**.

meconium antigen *see* **carcinoembryonic antigen**.

Medawar, (Sir) Peter Brian (1915–87), British zoologist and immunologist; Nobel Laureate in Physiology or Medicine (1960) jointly with F. M. Burnet 'for discovery of acquired immunological tolerance'.

media 1 a plural of **medium**. 2 (*in morphology*) or **tunica media** the middle layer of the wall of a blood or lymph vessel.

medial 1 of, situated in, or towards the middle. 2 of, or pertaining to, a **media** (def. 2).

median 1 (*in statistics*) the centre value of any series of numbers, measurements, values, etc. when these are arranged in order from the lowest value to the highest value (with an even-numbered series, the mean of the two central values is taken); i.e. half the members of the series have values greater than that of the median and half have values smaller than that of the median. *Compare* **mean**. 2 of, pertaining to, situated in, or pointing towards the middle. 3 (*in biology*) of or pertaining to the plane that divides an organism, organ, etc. into two symmetrical parts.

median effective dose abbr.: ED₅₀ or ED50; the dose of a drug, agonist, or other agent that is effective in 50% of the test objects.

median infectious dose abbr.: ID₅₀ or ID50; the dose of bacteria, viruses, or other infective agents that produces infection in 50% of the test objects.

median lethal dose abbr.: LD₅₀ or LD50; the dose of bacteria, viruses, toxic substances, or other agents that causes death of 50% of the test objects.

mediated transport the transport of a substance across a (biological) membrane that requires the participation of one or more transporting agents (or **carriers** (def. 3)). *See also* **facilitated diffusion**.

medical of, or pertaining to, the science or practice of medicine.

medicinal pertaining to or possessing therapeutic properties.

medicine 1 the science or practice of preventing, diagnosing, alleviating, and curing disease. 2 a common term for a pharmaceutical substance or mixture.

medium (pl. **media** or **mediums**) 1 the surrounding substance in which organisms or cells are studied or preserved. 2 the continuum in which particular chemical entities and their reactions are studied, commonly the solvent together with any nonreacting solutes. 3 any of the materials in which a chromatographic or an electrophoretic separation is effected. 4 the continuum through which electromagnetic radiation is considered to be transmitted. 5 *see* **culture medium**.

MEDLINE a US computerized science bibliography related to *Index Medicus*, that contains medical and many biochemical and biological references.

medulla (pl. **medullas** or **medullae**) the innermost part of an organ, tissue, or structure; marrow, pith. —**medullary** adj.

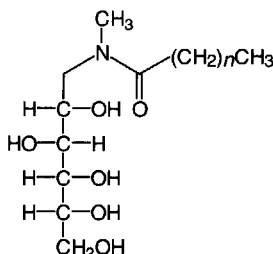
mega- 1 symbol: M; an SI prefix denoting 10⁶ times. 2 a prefix used in computing to indicate a multiple of 2²⁰ (i.e. 1 048 576), as in megabyte. 3 a general prefix denoting large size.

Mega-8 or Omega the octanoyl amide of *N*-methylglucamine (i.e. 1-methylamino-1-deoxy-D-glucitol); a water-soluble, non-denaturing detergent (CMC = 58 mM) useful for membrane research. It is one of a series, which also comprises Mega-9, the nonanoyl derivative (CMC = 19–25 mM), and Mega-10, the decanoyl derivative (CMC = 6–7 mM).

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Mega-8 $n = 6$ Mega-9 $n = 7$ Mega-10 $n = 8$

megabase *symbol:* Mb; a unit of length of a polynucleotide equal to 10^6 base residues in a single-stranded polynucleotide or 10^6 base pairs in a double-stranded polynucleotide.

megadalton *symbol:* MDa; a unit of mass equal to 10^6 daltons.

megakaryocyte a large (diameter 40–150 μm), polyploid cell that gives rise to the blood platelets by budding of its cytoplasm. It contains, on average, four times the diploid number of chromosomes in a large, usually lobulated nucleus. Megakaryocytes are found in bone marrow, lung, and to a lesser extent in other tissues.

megaloblastic anemia any of a group of disorders characterized by an abnormality of red cell development in the bone marrow. There is an accumulation of primitive cells, with failure of nuclear development, while cytoplasmic development, including hemoglobinization, continues; oval macrocytes appear in the blood. Vitamin B₁₂ or folate deficiency, or abnormality of vitamin B₁₂ or folate metabolism, is a frequent cause, but drug treatment (with antifolates, e.g. methotrexate, or drugs that interfere with DNA synthesis, e.g. cytarabine (β -cytosine arabinoside), hydroxyurea, AZT) or certain leukemias may also be causes.

megatomic acid *see* **tetradecadienoic acids**.

meiosis or **reduction division** a process by which the nucleus of a diploid cell divides twice with the formation of four daughter haploid cells, each containing half the number of chromosomes of the parent nucleus. Meiosis occurs during the formation of gametes from diploid cells and at the beginning of haplophase in those organisms that exhibit alternation of generations. Meiosis involves two rounds of cell division. Prophase in the first division proceeds as follows. Chromosomes first become visible as elongated threads during **leptotene**; by this time each chromosome has already replicated to form two identical sister chromatids. During **zygotene** each chromosome pairs with its homologue; the two become aligned and the **synaptonemal complex** develops between them. When synapsis is complete **pachytene** is said to have begun; it is during this stage that **crossing over** occurs between a chromatid in one partner and another chromatid in the homologous chromosome. The points of crossing over, or **chiasmata**, first become visible as the two chromosomes begin to separate in **diplotene**, when the synaptonemal complex dissolves. The final phase of prophase I, **diakinesis**, is the transition into metaphase I, which is analogous to mitotic metaphase and is followed by anaphase I and telophase I, producing two daughter cells each with one copy of each chromosome (each still made up of two sister chromatids). A second division quickly follows, comprising an interphase (without chromosome replication), and mitosis-like prophase II, metaphase II, anaphase II, and telophase II to separate the two chromatids in each chromosome. *Compare* **mitosis**. —*meiotic adj.*

MEK *abbr.* for methyl ethyl ketone.

melanin any of the black or brown pigments of high molecular

mass formed by the action of oxidase enzymes on phenols. They are widely distributed in plants and animals and are usually bound to protein. The most important precursor of melanins is the phenolic amino acid **tyrosine**. Plant melanins are called **catechol melanins** because they yield catechol on alkali fusion; animal melanins, termed **eumelanins**, arise from the formal polymerization of indole-5,6-quinone and thus, in contrast to plant melanins, contain nitrogen. —*melanic adj.*

melanocortin a generic term for either of the peptide hormones **melanotropin** or **corticotropin**, deriving from the composite **proopiomelanocortin**, the polypeptide from which they are formed. Although the two hormones have distinct functions (melanotropin in control of melanocyte growth and pigment formation, corticotropin in stimulating the adrenal cortex to produce glucocorticoids and aldosterone), they also share functions in, e.g., immunomodulation and antipyretic activity; while corticotropin can regulate melanocyte function, it is thus possible also to regard it as a melanotropin.

melanocortin receptor similarities between the receptor structures for **corticotropin** (*abbr.*: ACTH) and **melanotropin** (*abbr.*: MSH) indicate that this is a subfamily of G-protein-coupled, adenylate cyclase-activating, seven-transmembrane-domain receptors. Several types of receptor have been identified: **melanocortin-1 receptor**, *other names*: melanocyte-stimulating hormone receptor; melanotropin receptor; located on melanocytes and adrenocortical tissue; example from human: database code MSHR_HUMAN, 317 amino acids (34.66 kDa); **melanocortin-2 receptor**, *other name*: adrenocorticotrophic hormone receptor; located on melanocytes and adrenocortical tissue; example from *Bos taurus*: database code ACTR_BOVIN, 297 amino acids (33.29 kDa); **melanocortin-3 receptor**, located on brain, placental, and gut tissue; example from human: database code MC3R_HUMAN, 360 amino acids (40.11 kDa); **melanocortin-4 receptor**, located on brain, placental, and gut tissue; example from human: database code MC4R_HUMAN, 332 amino acids (36.95 kDa); **melanocortin-5 receptor**, expressed in brain but not melanocytes; example from human: database code MCSR_HUMAN, 325 amino acids (36.6 kDa). Human ACTH receptor is 39% identical with the human MSH receptor. Receptors 1–3 and 5 respond to α -, β -, and γ -MSH and ACTH, melanocortin-4 receptor being specific to a common heptapeptide core sequence MGHFRWG. However, despite similarities in structure and action on melanocortin receptors, physiologically MSH has no ACTH-like activity. These are among the smallest G-protein-coupled receptors identified to date; they are more closely related to the Δ^9 -tetrahydrocannabinol receptor (*see* **cannabinoid receptor**) than to other peptide and amine G-protein-coupled receptors.

melanocyte a type of pigmented cell that can synthesize and store **melanin** in melanosomes. Melanocytes occur in the skin and in the pigmented layer of the eye. *See also* **melanophore**.

melanocyte regulatory factor an alternative name for **melanotropin regulatory factor**.

melanocyte-stimulating hormone *abbr.*: MSH; any of three peptide hormones, α -MSH, β -MSH, and γ -MSH, produced primarily by the intermediate lobe of the pituitary gland, that cause dispersal of melanosomes in **melanophores** of poikilothermic vertebrates, but whose physiological function in mammals is not understood. Their sequences are all contained within that of **proopiomelanocortin** (the common precursor of corticotropin and β lipotropin). α -MSH is a tridecapeptide amide (usually acetylated on the N-terminal serine residue) that corresponds to residues 1–13 of corticotropin and has the same sequence in almost all animals studied. β -MSH (*formerly called intermedin*), not believed to be present in humans, corresponds to residues 41–58 of β lipotropin and shows greater structural variability. γ -MSH, a putative hormone, is a peptide of variable length that corresponds to part of the so-called cryptic region of proopiomelanocortin. *See also* **melanotropin**.

melanocyte-stimulating hormone receptor *see* **melanotropin receptor**.

melanocyte-stimulating hormone receptor

melanocyte-stimulating-hormone regulatory factor or **melanocyte-stimulating-hormone regulatory hormone** an alternative name for **melanotropin-regulatory factor**.

melanocyte-stimulating-hormone release-inhibiting factor (abbr.: MSHIF or MSHRIF) or **melanocyte-stimulating-hormone release-inhibiting hormone** (abbr.: MSHIH or MSHRIH) an older name for **melanostatin**.

melanocyte-stimulating-hormone releasing factor (abbr.: MSHRF) or **melanocyte-stimulating-hormone releasing hormone** (abbr.: MSHRH). See **MSHRH**.

melanoliberin the recommended name for melanocyte-stimulating-hormone releasing hormone (see **MSHRH**).

melanophore a type of **melanocyte** that occurs especially in the skin of poikilothermic vertebrates. The cell's contractile properties allow rapid internal translocation of melanosomes, thereby bringing about rapid adaptive changes in the skin coloration of such creatures.

melanosome a type of cellular organelle that occurs in **melanocytes**; melanosomes contain tyrosinase and can synthesize and store melanin.

melanostatin a substance identified by its ability to inhibit release of melanotropin; it is now known to be identical with **neuropeptide Y**. Other names: melanotropin release-inhibiting factor (abbr.: MIF or MRIF); melanotropin release-inhibiting hormone (abbr.: MIH or MRIH); melanocyte-stimulating-hormone release-inhibiting factor (abbr.: MSHIF or MSHRIF); melanocyte-stimulating-hormone release-inhibiting hormone (abbr.: MSHIH or MSHRIH).

melanotransferrin see **transferrin**.

melanotropic or **melanotrophic** describing any agent that causes dispersal of melanosomes within **melanophores**.

melanotropin or (formerly) **melanotrophin** a generic term for any melanotropic hormone of the anterior pituitary that is derived from **proopiomelanocortin**. The term embraces α , β , and γ **melanocyte-stimulating hormones**. See also **melanocortin**.

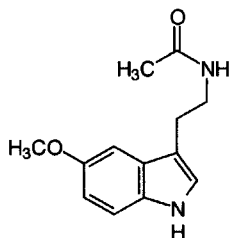
melanotropin receptor any of a family of receptors that bind **melanotropins**. See also **melanocortin receptor**.

melanotropin-regulatory factor or **melanotropin-regulatory hormone** an older generic name for either **MSHRH** or **melanostatin**. Other names: melanocyte-regulatory factor; melanocyte-stimulating-hormone regulatory factor; melanocyte-stimulating-hormone regulatory hormone.

melanotropin release-inhibiting factor (abbr.: MIF or MRIF) or **melanotropin release-inhibiting hormone** (abbr.: MIH or MRIH) an older name for **melanostatin**.

melanotropin-releasing factor (abbr.: MRF) or **melanotropin-releasing hormone** (abbr.: MRH) see **MSHRH**.

melatonin *N*-acetyl-5-methoxytryptamine; a neuroendocrine principle synthesized in the pineal gland from 5-hydroxytryptamine; its synthesis exhibits a circadian photoperiodism. In amphibia, it has an action opposite to that of **melanocyte-stimulating hormone**. It stimulates the aggregation of melanosomes in melanophores, thus lightening the skin. When used as a drug it is reputed to control jet lag.



melezitose the trisaccharide, *O*- α -D-glucopyranosyl-(1 $\rightarrow$ 3)-*O*- β -D-fructofuranosyl-(2 $\rightarrow$ 1)- α -D-glucopyranoside; a component of the sweet exudations from many plants and of the leaves of some trees after insect attack.

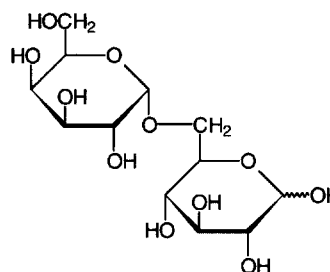
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membrane attack cor

melibiase see **glucosidase**.

melibiose the disaccharide, 6-*O*- α -D-galactopyranosyl-D-glucose; it occurs as a constituent of the trisaccharide, **raffinose**, and in the exudates and nectaries of a number of plants.



melissyl an alternative name for **myricyl**.

melitriose see **raffinose**.

melittin a strongly basic, sulfur-free, hemolytic, 26-residue peptide amide, often formylated on its N-terminal glycine residue. It comprises 40–50% of the dry mass of the venom of the honey bee, *Apis mellifera*. The hemolytic activity of melittin appears to be related to the ordering of its amino-acid residues, which renders the polypeptide markedly surface active. Example from *A. mellifera*, chain A: database code NRL_2MLTA, 26 amino acids (2.84 kDa); chain B: database code NRL_2MLTB, 26 amino acids (2.84 kDa); both 3-D structures are known.

melting 1 the (reversible) act or process of liquefaction of a solid through the action of heat (or change in pressure). 2 the increase (reversible and usually fairly abrupt) of molecular mobility of lipids in a biological membrane or artificial lipid bilayer with increase of temperature above the transition temperature. 3 or **thermal denaturation** the transition (usually fairly abrupt but not necessarily reversible) of a biopolymer from a highly ordered to a less highly ordered state with increase in temperature. Examples are the heat-induced change from a helical to a random coil structure in a protein or a duplex nucleic acid. See also **localized melting**, **melting temperature**.

melting curve an alternative name for **thermal denaturation profile**.

melting point abbr.: m.p.; the temperature at which the solid and liquid phases of a substance are in equilibrium at a given pressure. Melting points are usually given for standard atmospheric pressure, i.e. 101 325 Pa.

melting temperature symbol: T_m or t_m or Θ_m ; the temperature at which a conformational transition brought about by change in temperature, e.g. of a macromolecule in solution, is 50% complete. It follows therefore that $T_m = \Delta H_{ot}/\Delta S_{ot}$, where ΔH_{ot} and ΔS_{ot} are, respectively, the standard enthalpy difference and the standard entropy difference for the transition. In the denaturation of DNA the melting temperature is taken as the midpoint of the **helix-to-coil transition**.

MEM abbr. for 1 minimum essential medium. 2 minimum Eagle's medium.

membrane any thin sheet or layer, often separating two compartments. Membranes may be natural or artificial, and are sometimes soft and pliable. See also **cell membrane**, **semipermeable membrane**, **unit membrane**. —**membranous adj.**

membrane attack complex see **MAC** (def. 1).

membrane attack complex inhibition factor abbr.: MACIF or CD59 glycoprotein precursor; other names: MAC-inhibitory protein, protectin, membrane inhibitor of reactive lysis (abbr.: Mirl); a surface antigen of T cells that potentially inhibits the membrane attack complex (see **MAC**). It is complexed with a GPI anchor. Example from human: database code CD59_HUMAN, 128 amino acids (14.18 kDa).

membrane-bound describing any component that is not readily dissociable from preparations of cellular membranes, in



membrane-bound

vitro and/or that appears to be an integral part of a membrane in an intact cell.

membrane channel a protein complex that specifically facilitates the passive diffusion of small molecules through a membrane; particularly applied to **ion channels**.

membrane electrode any electrode that incorporates a membrane in its structure, e.g. a glass electrode, an ion-exchange electrode, or an oxygen electrode.

membrane filter or **filter membrane** any membrane consisting of a thin polymeric structure with pores of a defined and very uniform size that is used for filtration; a disk or piece of such membrane.

membrane filtration the act or process of separating suspended particles from a liquid, or of separating macromolecules from their solvent, with a **membrane filter**.

membrane-organizing spike protein *see moesin*.

membrane osmometer *see osmometer*.

membrane potential the electric potential existing across any membrane arising from charges in the membrane itself and from the charges present in the media on either side of the membrane. Plant and animal plasma membranes maintain a potential, the cell interior being more negative than the exterior. In plants this arises mainly from the activity of electrogenic pumps, whereas in animals passive ion movements predominate in creating this potential. The resting potential varies from -20 mV to -200 mV according to cell type; for a neuron it is about -70 mV. In plants, **H⁺-transporting ATPases** pump protons out of the cell, thus maintaining a pH gradient across the plasma membrane. This is involved in symport of carbohydrates and amino acids into the cell. In animals, the **Na⁺,K⁺-ATPase** exchanges intracellular Na⁺ for extracellular K⁺, maintaining the high intracellular Na⁺ for extracellular K⁺ concentration and low intracellular Na⁺ concentration. K⁺-leak channels in the plasma membrane allow K⁺ to escape until counteracted by the electrical potential and excess of intracellular negative charge that results. An electrochemical gradient across the mitochondrial inner membrane is created by proton-pumping cytochromes. *See also chemiosmotic coupling hypothesis, Nernst equation.*

membrane protein any protein that is normally closely associated with a cell membrane. Such an association results from a variety of properties of the protein and the method by which it is synthesized. **Integral proteins** contain hydrophobic regions that traverse the lipid bilayer; in some cases, the protein has only one such region, with hydrophilic domains on either side of the membrane; in other cases, where several such regions are present, a single polypeptide chain may traverse the membrane several times (*see seven-transmembrane-domain proteins*). Other proteins are bound to the membrane by a **GPI anchor**, and others by the hydrophobic moiety of a fatty-acyl group (*see myristoylation*) or a prenyl group (*see prenylation*). Other proteins may be more loosely attached, mainly by ionic forces (peripheral proteins).

Membrane-anchored proteins may be classified into a number of types, depending on the number of transmembrane domains, the orientation of the protein, and the presence of other kinds of attachment to the lipid bilayer. **Type I membrane proteins** have a single transmembrane domain, with the C terminus on the cytoplasmic side. **Type II membrane proteins** have a single transmembrane domain with an N terminus that is cytoplasmic. **Type III membrane proteins** have a single polypeptide chain that traverses the membrane several times. **Type IV membrane proteins** are made up of multimers of subunits that when assembled make a transmembrane channel. Proteins attached to the membrane only by covalently bound lipids are classified as **type V**, and those that are anchored by one or more transmembrane domains and also by a GPI anchor are designated **type VI**. The structures of only a few membrane proteins have been solved at high resolution (7 Å or better). Except for these it is not known whether the transmembrane domain is helical.

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membrane receptor any of the many proteins, situated in or on a cell membrane, that bind extracellular signalling molecules (e.g. hormones, growth factors) and transduce the signal to intracellular effects.

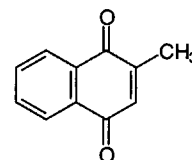
membrane skeleton the protein network of an erythrocyte membrane, defined operationally as the insoluble residue that remains after extraction of either intact red cells or their isolated membranes with the nonionic detergent Triton X-100.

membrane teichoic acid any **teichoic acid** found in the protoplast of a bacterial cell as opposed to any found exclusively in the cell wall or in capsular material. Such substances commonly are glycerol teichoic acids, and they occur, covalently attached to lipid, in the cell membrane and probably also in mesosomes.

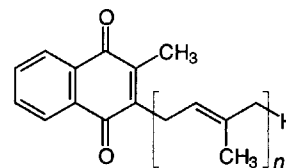
membrane transport any movement of substances across a (biological) membrane.

memory cell any of a population of long-lived **T** or **B lymphocytes** that have previously been stimulated by antigen, and make an accelerated response to that antigen if they encounter it again. Memory B cells carry surface immunoglobulin G as their antigen receptor; this is of higher affinity than on virgin lymphocytes. Memory T cells may be distinguishable by high expression of CD44.

menadiione 2-methyl-1,4-naphthalenedione; 2-methyl-1,4-naphthoquinone; vitamin K₃; a synthetic naphthoquinone derivative possessing the physiological properties of **vitamin K**.



menaquinone *abbr.*: MK; the class name for a series of 2-methyl-3-*all-trans*-polyprenyl-1,4-naphthoquinones, also known as vitamins K₂ (*see vitamin K* for function). They are antihemorrhagic vitamins, present in bacteria, and possess 1–13 prenyl residues, the number of prenyl residues being indicated by a suffixed Arabic number, e.g. menaquinone-6 (MK-6; also known as **farnoquinone**). [From methyl naphthoquinone.]



Mendel, Gregor Johann (1822–84), Austrian monk and botanist. *See also Mendel's laws of heredity.*

Mendeleev or **Mendeleyev** or **Mendeleeff**, Dmitri Ivanovich (1834–1907) Russian chemist who proposed the **periodic law** and **periodic table**.

Mendelian 1 of or relating to **Mendel**, or following his laws of heredity. 2 one who adheres to or supports **Mendel's laws of heredity**.

Mendel's laws of heredity the principles of heredity proposed by Gregor **Mendel** on the basis of his researches on hybridization of peas to explain the behaviour of factors (now called genes) that govern the transmission of inherited characters from generation to generation. On the basis of modern knowledge, Mendel's first law, the **law of segregation**, states that every somatic cell carries a pair of allelic genes for each character and that the two genes in each pair separate during meiosis so that each gamete carries only one gene from each



meniscus

pair. Mendel's second law, the **law of independent assortment**, states that the separation of members of any pair of allelic genes occurs independently of every other pair provided they are not linked.

meniscus (*pl. menisci*) the curved upper surface of a liquid column that is concave when the containing walls are wetted by the liquid (contact angle $<90^\circ$) or convex when not (contact angle $>90^\circ$), sometimes extended to include a flat surface (contact angle = 90°). — **meniscoid** *adj.*

meniscus depletion (sedimentation equilibrium) method or **Yphantis method** a variant of the sedimentation equilibrium method for determining the relative molecular mass of a macromolecule in which centrifugation is effected at a speed high enough to render the concentration of the solute at the meniscus essentially zero; the concentrations of solute along the length of the cell are proportional to the difference in refractive index at any point from that at the meniscus. The method is especially useful for monodisperse systems.

Menke's disease or **kinky-hair syndrome** or **steely-hair syndrome** a genetically determined defect of copper metabolism found in some humans. Most of the clinical and pathological manifestations can be explained on the basis of a profound copper deficiency caused by an inability to absorb copper from the gastrointestinal tract and also, probably, to a reduced ability to absorb copper from the mother during intrauterine life.

menopausal gonadotropin or **human menopausal gonadotropin** (*abbr.*: hMG or HMG); *another name for menotropin*.

menotropin or **menotrophin** *other name*: human menopausal gonadotropin; *abbr.*: hMG or HMG; a gonadotropin with **follicle-stimulating hormone** and **luteinizing hormone** activity, isolated from the urine of postmenopausal women and used to stimulate ovulation in infertility. One proprietary name is Pergonal.

Menten, Maude Leonora (1879–1960), Canadian physician, biochemist, and experimental pathologist; best known for her collaboration with Leonor **Michaelis** in the formulation of Michaelis–Menten kinetics of enzyme action (*see Michaelis kinetics*).

MEP *abbr. for major excreted protease; see cathepsin L*.

meprin A EC 3.4.24.18; *other names*: endopeptidase-2; endopeptidase MEP-1; a metalloendopeptidase of the membranes of kidney and intestinal brush borders. It is an integral protein with the catalytic domain lying towards the exterior. Zinc is a cofactor. It catalyses the hydrolysis of proteins and peptides on the C-terminal side of hydrophobic residues, and is a homotetramer of α or β subunits. Example: precursor of mouse α -subunit (mep-1); database code MEPA_MOUSE, 760 amino acids (85.61 kDa); theoretical 3-D model of this: database code NRL_11AF, 198 amino acids (22.88 kDa).

meq or **mEq** *symbol for milliequivalent* (use deprecated, mmol recommended).

mer or **constitutional unit** the repeating unit of a polymeric molecule.

+mer *suffix forming nouns (in chemical nomenclature)* denoting specific kinds or classes of polymer (e.g. hexamer, oligomer) or isomer (e.g. enantiomer, epimer, tautomer). — **+meric** *adj.*; **+merism** *n.*

mer the symbol for genes of the *mer* operon whose products confer (or regulate) mercury resistance. The following examples are of *mer* products from *Serratia marcescens*. (1) MerA: mercury II reductase (EC 1.16.1.1); *systematic name*: Hg:NADP⁺ oxidoreductase; one of the very rare enzymes with metal substrates, it catalyses the reduction by NADPH of Hg²⁺ to form Hg and NADP⁺. It is a flavoprotein (FAD) with redox-active disulfide bonds; homodimer; database code MERA_SERMA, 460 amino acids (48.53 kDa). This enzyme is responsible for volatilizing mercury as Hg(0). (2) MerB: alkylmercury lyase (EC 4.99.1.2); *systematic name*: alkylmercury mercuric-lyase; it catalyses a reaction between RHg⁺ and H⁺ to form RH and Hg²⁺. This enzyme cleaves the carbon–mercury bond of organomercurials and the Hg²⁺

product is detoxified by MerA. Database code MERB_SERMA, 212 amino acids (23.05 kDa). (3) MerR: this is a **helix-turn-helix** regulatory protein; database code MERR_SERMA, 144 amino acids (16.01 kDa). In the absence of mercury, MerR represses transcription by binding tightly to the *mer* operator region; when mercury is present the dimeric complex binds a single ion and becomes a potent transcriptional activator, while remaining bound to the *mer* site. *mer* operons are encoded by a number of plasmids which have been cloned into *Escherichia coli*, e.g. Tn21 of *Shigella flexneri*, Tn501 from *Pseudomonas aeruginosa* (these two lack the organomercury lyase gene and so confer resistance only against inorganic mercury salts), and pDU1358 from *S. marcescens* (which includes *merB* and thus encodes a broad spectrum *mer* operon).

mer+ *prefix (in coordination compound nomenclature)* denoting a **meridional** (def. 2) arrangement of ligands.

MER *abbr. for medium reiteration frequency repeat; other usage*: medium reiteration frequency interspersed repetitive elements (*abbr.*: MER elements). They consist of repeated base sequences in eukaryotic genomes. More than fifty human repetitive families are known, of which most represent MERs. The latter have copy numbers per genome ranging from hundreds to thousands, and length ranging from tens to a few hundred base pairs. They are of unknown function, but some may play a role in regulation of gene expression and others may affect genome stability and gene rearrangements.

mercapt+ a variant form of **mercapto+** (sometimes before a vowel).

mercaptan an old name for **thiol**.

mercapto+ or (sometimes before a vowel) **mercapt+** *comb. form (in chemical nomenclature)* indicating the presence of an unsubstituted thio group, –SH.

2-mercaptoethanol HO–CH₂–CH₂–SH; 2-hydroxyethylmercaptan; a sulfhydryl reducing agent often used in buffers to maintain protein sulfhydryl groups in the reduced state.

mercaptapurine or **6-mercaptapurine** *abbr.*: 6-MP; *symbol*: Shy; thiohypoxanthine; 6-purinethiol; an immunosuppressive and antineoplastic drug useful in organ transplantation and in the treatment of some leukemias. It is a substrate for guanine-hypoxanthine phosphoribosyltransferase, which converts it to **6-mercaptapurine ribonucleoside** (or 6-thioinosine; *symbol*: Sno or M or SI).

mercapturic acid any *S*-alkyl- or *S*-aryl-*N*-acetylcysteine; such compounds are formed as detoxification products in mammals. The initial reaction involves conjugation with the thiol group of glutathione catalysed by a **glutathione transferase** to form the *S*-radyl-glutathione (*see radyl*). Glutamic acid and glycine are then removed in the kidney, and the resulting mercapturic acid is excreted in the urine.

mercurate to treat or mix with mercury; to introduce mercury into an organic compound. — **mercuration** *n.*

mercurial 1 of, containing, like, or pertaining to mercury. 2 any compound containing mercury, especially one used in medicine and/or agriculture; *see also organomercurial*.

mercuric perchlorate Hg(ClO₄)₂; a substance used in solution to trap ethylene.

mercury II reductase EC 1.16.1.1; product of the *Serratia marcescens merA* gene (*see mer*).

mercury-vapour lamp or **mercury-discharge lamp** or (*informally*) **mercury lamp** a lamp that emits bluish-green light rich in ultraviolet radiation, produced by passing an electric discharge through mercury vapour, often at low pressure. It is useful for provoking or investigating fluorescence, as a light source for a **polarimeter**, and as a source of sterilizing radiation.

meridian 1 any line on the surface of a sphere or spheroid that is orthogonal to its equator, or any corresponding line in a projection of part of the surface of a sphere or spheroid onto a planar or cylindrical surface. 2 in a fibre diagram produced by X-ray diffraction, the vertical axis on the film that corresponds to the *z* axis in reciprocal space.

meridional 1 of, along, pertaining to, or resembling a meri-

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dian. 2 describing the geometrical isomer of a coordination compound of octahedral configuration in which three ligands in the coordination sphere are in such a relationship that one is *cis* to the two others, which are themselves *trans*. Such an isomer is designated by prefixing *mer-* to the chemical name.

meristem a region of tissue in a plant that is composed of one or more undifferentiated cells that are capable of undergoing mitosis and differentiation, and thereby of effecting growth and development of a plant by giving rise either to more meristem or to specialized tissue. In vascular plants, meristems may be apical (located at the tips of growing roots and shoots) or lateral (arranged circumferentially and constituting the cambium). —**meristematic** *adj.*

merlin or **neurofibromatosis type 2 (tumour suppressor)** (*abbr.*: NF2) or **Schwannomin** a membrane-stabilizing protein from fetal brain, and other tissues such as kidney, lung, and breast. It is similar to **ezrin** and **talin**. Example from human: database code MERL_HUMAN, 595 amino acids (69.61 kDa). It is a tumour suppressor of the **TERM** family. In the human, it is encoded by a gene on chromosome 22q12. Mouse complete sequence: database code MUSNF2D, 596 amino acids (69.61 kDa).

mero+ *comb. form* denoting part, partly, or partial.

merocrine 1 describing a type of secretion that does not entail disintegration of the secretory cell. 2 describing a gland made up of such cells. *Compare* **apocrine**, **eccrine**, **holocrine**.

merodiploid a haploid organism that is diploid for a small region of the chromosome, i.e. a partial diploid.

meromyosin either of the two fragments, **heavy meromyosin** and **light meromyosin**, produced by brief treatment of myosin with trypsin or certain other proteinases.

mero-receptor the smallest subunit of a receptor for a (steroid) hormone that contains an intact hormone-binding site. The biological significance of mero-receptors is unclear; they may be artefacts of experimental cellular disruption or products formed by proteolysis associated with the termination of (steroid) hormone action.

merosin a tissue-specific-basement-membrane protein, and the M chain of **laminin**. It is covalently associated with laminin B chains and like laminin is cruciform. It has homology to the C-terminal part of the laminin A chain. Example from human: database code HSLAMMMER, 3110 amino acids (342.38 kDa).

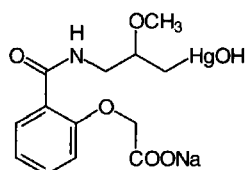
merozygote any bacterial cell that is partly diploid and partly haploid, formed by conjugation, transduction, or transformation. It contains one complete genome, that of the recipient organism, and a fragment of a second genome, derived from the donor.

Merrifield, Robert Bruce (1921–), US peptide chemist and biochemist; Nobel Laureate in Chemistry (1984) 'for his development of methodology for chemical synthesis on a solid matrix'.

Merrifield method an alternative name for **solid-phase peptide synthesis**.

Merrifield resin a beaded reactive synthetic resin designed for use in the Merrifield method (i.e. **solid-phase peptide synthesis**).

mersalyl *pharmaceutical name* of sodium *o*-[3-(hydroxymercurio)-2-methoxypropyl]carbamoyl}phenoxyacetate; a mercury-containing derivative of salicylamide. It is a diuretic drug useful as an inhibitor of the exchange of phosphate and hydroxide ions across the inner mitochondrial membrane.



mes+ a variant form of **meso+** (sometimes before a vowel).

Mes or **MES** *abbr. for and, trivial name of* 2-(*N*-BẠN đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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morpholino)ethanesulfonic acid; a **Good buffer substance**, pK_a (20 °C) = 6.15.

mesaconic acid (*E*)-2-methyl-2-butenedioic acid; methyl-fumaric acid; an intermediate in the fermentation of glutamate to acetyl-CoA by *Clostridium tetanomorphum*.

mescaline 3,4,5-trimethoxyphenylethylamine; a hallucinogenic and sympathomimetic drug isolated from the flowering heads of the mescal cactus (*Lophophora williamsii*), also known as peyote or peyote.

Meselson-Stahl experiment a landmark experiment, reported by Meselson and Stahl in 1958, that provided convincing evidence for semiconservative replication of DNA in bacteria. Cultures of *Escherichia coli* were grown in a medium containing $^{15}\text{NH}_4\text{Cl}$ (96.5% isotopic purity) as sole nitrogen source for many generations so as to label the DNA with heavy nitrogen. The cells were then transferred to a medium containing $^{14}\text{NH}_4\text{Cl}$ and the culture sampled for several subsequent generations. Cell lysates were subjected to density-gradient centrifugation and the positions of the DNA bands in the gradients located by UV absorption. Initially the DNA appeared as a single band corresponding to DNA containing heavy nitrogen; subsequently a second band of DNA, containing equal amounts of heavy and light nitrogen, appeared, and one generation after transfer to natural ammonium chloride all the DNA was this hybrid form. After two generations only hybrid and natural DNA was present. Furthermore, the separated strands of this hybrid DNA proved to contain all heavy nitrogen or all normal nitrogen only. [After Matthews Meselson (b. 1930), US molecular biologist, and Franklin William Stahl (b. 1929), US geneticist.]

mesenchyme immature, unspecialized connective tissue found in the early embryo of animals. It consists of cells embedded in a tenuous extracellular matrix. —**mesenchymal** *adj.*

mesentery 1 the double-layer fold of peritoneum serving to hold the abdominal viscera in position. 2 any of the muscular partitions extending inwards from the body wall in coelenterates. —**mesenteric** *adj.*

Me₃Si *symbol for the trimethylsilyl group.*

meso describing a bridging atom or group between two rings in a polycyclic molecule.

meso+ or (sometimes before a vowel) **mes+** *comb. form* denoting middle, or intermediate.

meso- *abbr.: ms-; prefix (in chemical nomenclature) designating a substance whose individual molecules contain equal numbers of enantiomeric groups, identically linked, but no other chiral group; i.e. a substance whose molecules are optically inactive because of internal compensation; e.g. meso-compound, meso-cystine. Compare racemo-*

meso-carbon atom a term suggested for any prochiral carbon atom in an organic molecule whose valence bonds are linked to two identical groups, *a*, and to two other dissimilar groups, *b* and *d*. The resulting molecule, *Caabd*, has a plane of symmetry and thus the carbon atom is not chiral; however, the two *a* groups are not geometrically equivalent but bear a mirror-image relationship to each other, and they may react at different rates with chiral reagents or in enzymic reactions. The term is now obsolete. See **prochiral**.

mesoderm the middle of the three primary germ layers of a triploblastic animal embryo, lying between the ectoderm and endoderm. It is the source of bone, muscle, connective tissue, the inner layer of the skin, and other structures. —**mesodermal** *adj.*

meso-inositol a former name for **myo-inositol**.

mesomeric effect (in chemistry) an experimentally observable effect (on reaction rates, etc.) of a substituent caused by the overlap of its *p*- or π -orbitals with the *p*- or π -orbitals of the rest of the molecular entity, **resonance** (def. 2) (or mesomerism) being thereby introduced into, or extended within, the molecular entity in question. *Compare inductive effect.*

mesomerism (in chemistry) a term used to imply that the correct representation of the structure of a particular molecular



mesomerism

entity is intermediate between two or more formal valence-bond representations (known as **contributing structures**); **resonance** (def. 2). — **mesomeric adj.**

mesomorphic 1 (*in chemistry*) existing in, designating, or pertaining to an intermediate state between a true liquid and a true solid that exists in a **liquid crystal**. 2 (*in biology*) having a form, size, or structure that is average, normal, or intermediate between extremes.

mesophase any phase of matter intermediate between a true liquid and a true solid that exists in a **liquid crystal**.

mesophil or **mesophile** any organism that thrives at moderate temperatures, between 20 and 40 °C. — **mesophil, mesophile, or mesophilic adj.**

mesophyll the internal parenchyma of a foliage leaf between the two epidermal layers; it is usually photosynthetic. — **mesophyllic adj.**

mesophyte any plant requiring moderate amounts of water for optimum growth. — **mesophytic adj.**

mesopore any pore (in a gel or porous solid) intermediate in size between a micropore and a macropore, i.e. of width in the range roughly 2.0–50 nm. — **mesoporous adj.**

mesosecrin a glycoprotein, M_r 46 000, secreted by human mesoderm-derived and endothelial cells in culture.

mesosome an intracellular, often complex, membranous structure found in many bacteria and probably formed by invagination of the plasma membrane. It is more complex and larger in Gram-positive organisms than in Gram-negative ones. Its function is unclear. — **mesosomal adj.**

mesothelium (*pl. mesothelia*) a layer of flat cells derived from mesoderm and found lining various body cavities. — **mesothelial adj.**

mesotocin [Ile^8]oxytocin; a neurohypophyseal hormone present in some nonmammalian vertebrates, so-named because it is intermediate in structure between **isotocin** and **oxytocin**. It may represent an ancestral molecule, which has been preserved in these groups of vertebrates but has undergone a single residue replacement in mammals.

mesotrophic describing an environment providing a moderate amount of nutrients.

message 1 a specific piece of information, or something containing information, that is transmitted from a donor to a recipient by some means or agent. 2 a jargon term used, sometimes without an article, as the name of a substance, to denote the particular **messenger RNA** (mRNA) of interest in a specified context, or any preparation containing that mRNA. 3 the segment of mRNA (or any particular segment of a polycistronic mRNA) that consists of a sequence of codons carrying the information determining the sequence of amino-acid residues of a specific polypeptide to be synthesized at a ribosome. 4 the part of the information contained within the amino-acid sequence of a hormonal polypeptide, or of its biosynthetic precursors, that elicits a specific physiological response to the hormone. *Compare address* (def. 2).

message-dependent (of a partially purified, cell-free, protein-synthesizing system) capable of a significant response to added mRNA.

message sequence or **message region** the segment (or segments) of the complete sequence of amino-acid residues of a molecule of a hormonal or prohormonal polypeptide in which the **message** (def. 4) resides. *Compare address sequence*.

messenger 1 something, or someone, that bears or carries a **message** (def. 1), or that is able to do so. 2 a molecule or substance that carries genetic information; *see messenger RNA*. 3 a molecule or substance that carries physiological instructions; a **rheosome**. *See also chemical messenger, first messenger, second messenger*.

messenger ribonucleoprotein *abbr.*: mRNP; any complex of messenger RNA (mRNA) and protein. Included are: nuclear ribonucleoprotein that contains mRNA sequences; cytoplasmic ribonucleoprotein that is associated with polyribosomes;

and free cytoplasmic ribonucleoprotein particles, also known as **informosomes**.

messenger RNA or **template RNA** *abbr.*: mRNA; a class of naturally occurring RNA molecules that carry the information embodied in the genes of DNA to the ribosomes, where they direct protein synthesis. Molecules of mRNA are single-stranded, relatively short-lived, and formed by the process of **transcription** from the template strand of genomic DNA. They carry genetic information in the form of a succession of **codons**. At the ribosomes, by the process of **translation**, they orchestrate the synthesis of the corresponding polypeptide(s). The plus strand of certain RNA viruses can act as messenger RNA without transcription, as can a synthetic polyribonucleotide. The primary transcript of DNA is often referred to as **hnRNA** which contains the **introns** which have to be removed by excising and splicing together the **exons** to form the mature mRNA. mRNA may also undergo 'editing' whereby some bases are changed. *See also mRNA binding site, splicing*.

messenger-RNA-interfering complementary RNA *abbr.*: micRNA; an alternative name for **antisense RNA**.

Messing vector any one of certain cloning vectors derived from bacteriophage M13 by insertion of a synthetic multipurpose cloning site and which are especially suitable for the cloning and rapid sequence determination of small (≈ 500 base pair) fragments of DNA. These have developed into the **pUC vectors**. [After Joachim Messing (1946–).]

met+ a variant form of **meta+** (usually before a vowel or letter h).

Met symbol for a residue of the α -amino acid L-methionine (alternative to M).

MET a gene, first identified in an *N*-methyl-*N'*-nitrosoguanidine-treated cell line, that encodes a receptor tyrosine kinase with α and β subunits. The receptor binds **hepatocyte growth factor** and undergoes autophosphorylation of the cytoplasmic domain. Cells co-transfected with **MET** and **HGF** are highly tumorigenic. Example, database code KMET_HUMAN. *See also hepatocyte growth factor receptor*.

meta+ or (*before a vowel or letter h*) **met+** prefix 1 indicating alteration, alternation, or change of condition or position; after, distal to, behind, beyond; more highly specialized form of. 2 (*in inorganic chemistry*) denoting the least hydrated acid of a series of oxoacids formed from any one of certain anhydrides, or the anion or a salt formed from such an acid; *Compare ortho+* (def. 2), **pyro+** (def. 2). It is now approved only for the polymeric acids: metaboric acid, $(\text{HBO}_2)_n$, metaphosphoric acid, $(\text{HPO}_3)_n$, and metasilicic acid, $(\text{HSiO}_3)_n$; and for anions and salts of these acids (*but see metabisulfite*). 3 (*in organic chemistry*) distinguishing a particular polymeric form of the substance named, e.g. metaldehyde, metaformaldehyde.

meta- symbol: *m-*; prefix (*in organic chemical nomenclature*) denoting the isomer of a disubstituted monocyclic aromatic compound in which the substituents are separated by one intervening unsubstituted ring-carbon atom. *Compare ortho-, para-*.

metabisulfite an old name, still frequently used, esp. in commerce, for inorganic salts of the general formula $\text{M}_2\text{S}_2\text{O}_5$, where M is a monovalent metallic element.

metabolic of, pertaining to, occurring in, produced by, or liberated during **metabolism**. — **metabolically adv.**

metabolic acidosis any clinical condition in which hydrogen ions accumulate in the blood plasma through an excessive production of acidic metabolites (as in ketoacidosis or lactic acidosis), the excessive ingestion of acids, or a defect in the excretion of hydrogen ions by the kidneys.

metabolic activation **bioactivation**, usually with particular reference to the conversion of procarcinogens to carcinogens in the body.

metabolic alkalosis any clinical condition in which there is a deficiency of hydrogen ions in the blood plasma caused by the excessive ingestion of alkalis, extrarenal loss of acid (as with vomiting), or therapy with diuretic agents.

metabolic balance *see balance* (def. 1).

metabolic block the complete or partial interruption of a specific reaction in a metabolic pathway caused by the absence or greatly reduced activity of the enzyme catalysing that reaction (for genetic reasons or due to the action of a specific agonist). The consequences of a metabolic block include: (1) the complete or partial absence of the final product of the pathway; (2) the accumulation of a metabolic intermediate of the pathway before the block, for which no significant alternative pathway exists; (3) the accumulation of products of an alternative, normally minor, metabolic pathway. *See also* **inborn error of metabolism**.

metabolic code a model for the evolution of biological regulation and the origin of hormone-mediated intercellular communication. It is characterized by a set of symbols and their corresponding domains. A **symbol** is an intracellular molecule, e.g. a hormone or other messenger, that indicates to the cell the existence of a particular state of the environment, e.g. lack of glucose or of amino acids. The symbol regulates a diverse set of biochemical events and reactions in the cell, the nature and number of which are the **domain** of that symbol.

metabolic compartmentation or **metabolic compartmentalization** the concept that certain metabolic reactions occur in separate compartments of the cell (or organism) and are therefore not necessarily in equilibrium with each other.

metabolic control an alternative term for **metabolic regulation**.

metabolic cycle any closed sequence of metabolic reactions in which an end product is a reactant in the initiation of the cycle, e.g. the **tricarboxylic-acid cycle**.

metabolic disease any disease caused by a recognizable abnormality of metabolism.

metabolic flux the rate of passage of material, or of a specified substance or molecular fragment, through a given metabolic pathway.

metabolic pathway any series of connected (enzymic or other) reactions occurring in a cell or organism, whether anabolic, catabolic, or amphibolic.

metabolic pool the sum total of all the different amounts of a given substance, group of similar substances, or specified molecular fragment, irrespective of their distribution in a cell, organ, tissue, or organism, that are in complete equilibrium with each other in regard to participation in a specified set of metabolic reactions.

metabolic quotient *symbol:* Q_x ; a measure of the rate at which a given substance (x) is taken up by, or discharged from, a cell, organ, tissue, or organism.

metabolic rate the rate of metabolism of an animal measured under given conditions. It may be determined from the dioxygen consumption and carbon dioxide (and urinary nitrogen) excretion, or from the total energy output of the animal. *Compare* **basal metabolic rate**.

metabolic regulation the coordinated action of the many systems that control the activity of metabolic pathways. The systems embrace gene regulation and control of protein biosynthesis, as well as the control of the rate of enzyme reactions by phosphorylation of the enzyme protein or by the action of allosteric effectors. The action of second messengers is important in the transduction of signals from outside a cell to the interior.

metabolic state (of mitochondria) an alternative term for **respiratory state**.

metabolic turnover the continual exchange of cellular components by degradation and synthesis. Even for cell components for which the content and composition remain relatively constant, a continuous breakdown and resynthesis of individual molecules is occurring.

metabolism 1 the totality of the chemical reactions and physical changes that occur in living organisms, comprising **anabolism** and **catabolism**. 2 the totality of the chemical reactions and physical processes undergone by a particular substance, or class of substances, in a living organism.

metabolite any substance, either endogenous or exogenous.

that is formed or changed by metabolism. *See also* **primary metabolite**, **secondary metabolite**.

metabolize or **metabolise** 1 to affect by metabolism. 2 to effect metabolism. —**metabolizable** or **metabolisable** *adj.*

metabolon a supramolecular complex of sequential metabolic enzymes and cellular structural elements.

metabotropic describing a type of receptor (e.g. certain **glutamate receptors**) that mediates its effects by activation of enzymes, especially **adenylate cyclase** or the **phosphatidylinositol cycle**, in distinction from receptors that regulate ion channels. *Compare* **ionotropic**. *See also* **excitatory amino-acid receptor**.

metachromasia or **metachromasy** or **metachromatism** 1 the property exhibited by certain cationic (i.e. basic) dyes of shifting their absorption spectra towards shorter wavelengths in the presence of substances with appropriately arranged anionic groups. 2 the property exhibited by some biological structures or materials, when treated with certain cationic (basic) dyes, of staining with a different colour from that of the dye used. Biological materials showing metachromasia include heparin, chondroitin sulfate, sulfatides, gangliosides, and polyphosphates. —**metachromatic** *adj.*; **metachromatically** *adv.*

metachromatic granule any cytoplasmic granule that exhibits **metachromasia** (def. 2). **Volutin granules** are an example.

metachromatic leukodystrophy any of a group of hereditary diseases of humans characterized by disintegration of myelin and the accumulation of **metachromatic lipid** – galactosyl sulfatide (cerebroside sulfate) – in the white matter of the central nervous system, in peripheral nerves, and in some visceral organs. The accumulation of galactosyl sulfatide is due to deficiency in the activity of **cerebroside sulfatase** (EC 3.1.6.1). The disorder is characterized clinically by progressive paralysis and dementia, usually in the first few years of life, and is transmitted as an autosomal recessive trait. *See also* **saposin**.

metachromism *see* **metachromasia**.

metadrenaline an alternative name for **metanephrine**.

metagon a stable, virus-like RNA particle postulated to occur in certain mate-killer strains of *Paramecium aurelia*, in which it acts like messenger RNA.

metahalone any of a class of metaphase inhibitors having the general formula 5-halopyridine-2-one. [From meta(phase) halone.]

metakentrin an obsolete name for **lutinizing hormone**.

metal 1 any of a large class of chemical elements that, with certain exceptions, are ductile, malleable solids with a characteristic lustre, are good conductors of electricity and heat, and generally form simple cations. 2 any substance or material consisting entirely or predominantly of one or more **metals** (def. 1). 3 consisting of or made of metal. —**metallic** *adj.*

metal-chelate affinity chromatography a method of purifying proteins or other substances that contain, or are dependent on, chelatable metal ions. It consists of a type of **affinity chromatography** in which a chelate-forming adsorbent is used. The adsorbent is prepared by covalently linking molecules of a suitable chelator, e.g. iminodiacetate, via long, stable, hydrophilic spacer arms to a gel-forming matrix, e.g. a preparation of cross-linked agarose, and then saturating it with ions of an appropriate metal, e.g. copper(II); the affinity of the metal ions for the adsorbent must be greater than that for the substance to be separated. Components of a mixture selectively retained by the adsorbent may subsequently be eluted, e.g. with EDTA, by change of pH, or with a gradient of a weak base.

metal-ion buffer a **buffer** (def. 1, 2) for metal ions.

metallo+ *comb. form* denoting metal, or characterized by the presence of metal.

metalloantigen any metal-labelled antigen for use in a **metallo-immunoassay**.

metallocarboxypeptidase *recommended generic name* for any **carboxypeptidase** of the sub-subclass EC 3.4.17, for which a bivalent cation is an essential component of the catalytic mechanism.

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metallocene

metallocene the generic name for π -bonded bis(η -cyclopentadienyl) coordination compounds of certain metals, e.g. ferrocene, bis(η -cyclopentadienyl)iron, $[\text{Fe}(\text{C}_5\text{H}_5)_2]$.

metallochromic describing a dye, indicator, or stain that exhibits a distinctive colour change when complexed with metal ions; the colour change is often specific for a particular metal.

metalloendopeptidase recommended generic name for any endopeptidase of the sub-subclass EC 3.4.24, for which a bivalent cation is an essential part of the catalytic mechanism.

metalloenzyme any enzyme that is a metalloprotein.

metallohaptin any metal-labelled haptin for use in a metallo-immunoassay.

metalloimmunoassay a form of immunoassay in which ions of an appropriate transition metal, tightly complexed to the antigen or hapten in question, are used as the distinguishable label. The amount of metal present in the free and antibody-bound forms is determined by a suitable analytical method such as emission, absorption, or fluorescence spectrometry.

metallopeptidase see **metallocarboxypeptidase**, **metalloendopeptidase**.

metalloporphyrin any compound formed by chelation of a porphyrin with a metal ion, e.g. hemoglobins (with iron), cytochromes (with iron), and chlorophylls (with magnesium).

metalloprotease a former name for metalloendopeptinase (see also **metallocarboxypeptidase**, **matrix metalloproteinase**).

metalloprotein any protein containing one or more molecular proportions of a metal bound to it, either directly or as part of a prosthetic group.

metalloproteinase see **metallocarboxypeptidase**, **metalloendopeptidase**.

metallo-selenonein a synthetic analogue of the copper metallo-thionein peptide of *Neurospora crassa* that contains seleno-cysteine residues in place of cysteine.

metallothionein any of a group of very small proteins with extraordinarily high contents both of one or more metals (cadmium, copper, zinc) and of cysteine. They occur widely, from microorganisms to humans, and have highly conserved structures. Mammalian metallothioneins are of 6.8 kDa, consist of 61 amino-acid residues, including 20 cysteine residues, and contain about 7 g atoms of metal per molecule. The apoprotein may be termed **thionein**.

metal proteinase a former name for metalloproteinase (see **metallocarboxypeptidase**, **metalloendopeptidase**).

metal response element abbr.: MRE; a sequence in the promoter region regulating groups of genes that respond to the presence of metals. See also **response element**.

metamer (in chemistry) a former term for any one of two or more constitutional isomers that are of the same chemical class, e.g. ethylamine, $\text{CH}_3\text{-CH}_2\text{-NH}_2$, and dimethylamine, $\text{CH}_3\text{-NH-CH}_3$.

metamere (in morphology) any of the linearly arranged similar body segments in animals that exhibit **metamerism** (def. 2); a somite.

metameric 1 (in chemistry) of or pertaining to a **metamer**; pertaining to the phenomenon of, or displaying **metamerism** (def. 1). **2** (in morphology) of, or pertaining to, a **metamere**; consisting of metameres; segmented; of, or pertaining to, **metamerism** (def. 2).

metamerism 1 (in chemistry) the occurrence as **metamers** of two or more chemical substances of the same molecular formula. **2** (in morphology) the condition of an animal body characterized by the presence of a series of similar segments or **metameres**; segmentation.

metameter a quantity derived from an observation, and independent of all parameters, that conveys the magnitude of the phenomenon observed. A metameter is frequently used to simplify the expression or analysis of dose-response relationships.

metamorphosis 1 a complete change in physical form; a structural transformation. **2** (in zoology): the rapid, post-embryonic, structural transformation from a larva into an

adult that occurs in certain animals, e.g. the change from a tadpole to a frog. —**metamorphose** vb.; **metamorphic** adj.

metanephrine or **metadrenaline** 3-*O*-methylepinephrine; an inactive metabolite of epinephrine, produced by the action of **catechol *O*-methyltransferase**. Small quantities are normally found in the urine, and larger amounts occur in the urine of patients with catecholamine-secreting tumours.

metaphase the stage in **mitosis**, following **prometaphase**, during which the chromosomes become arranged so that their centromeres all lie in one plane and their longitudinal axes are at right angles to the spindle axis; the corresponding phases in **meiosis**.

metaphosphatase an alternative name for **1** endopolyphosphatase (EC 3.6.1.10). **2** exopolyphosphatase (EC 3.6.1.11).

metaphosphate the traditional name for any anion or any (partial or full) salt or ester derived from the condensed acid metaphosphoric acid, polytrioxophosphoric(v) acid, $(\text{HPO}_3)_n$, where n is a small integer, commonly 3, 4, or 6. Metaphosphoric acid and its derivatives are thought not to exist in the monomeric state, although they are often represented as such for convenience. See also **polyphosphate**.

metaprotein an old term for the products formed by acid or alkaline hydrolysis of proteins that are soluble in weak acids or alkalis but insoluble in neutral solutions.

metarhodopsin either of two intermediates, metarhodopsin I (absorption maximum 480 nm) and metarhodopsin II (absorption maximum 380 nm), formed during the photolysis of **rhodopsin**.

metastable 1 a describing any body or system existing in a state of apparent equilibrium when undisturbed, or subject to very small perturbation, but capable of passing to a more stable state when subjected to greater perturbation. **2** describing an excited state of an atom, ion, nucleus, or other quantum-mechanical system that is relatively long-lived because the transitions to a lower energy state are forbidden transitions. **3** symbol: *m*; describing any comparatively stable excited state of a radioisotope that decays to a more stable state by isomeric transition, with the emission of gamma radiation, e.g. the metastable nuclide $^{99\text{m}}\text{Tc}$, which is formed by decay of ^{99}Mo and which in turn decays by transition to its nuclear isomer ^{99}Tc with a half-life of 6 h and emission of one quantum of gamma radiation.

metastasis (pl. **metastases**) the transfer of disease, especially tumour cells, from one part of the body to another by way of the natural passages (blood vessels, lymphatics) or by direct continuity. —**metastasize** or **metastasise** vb.; **metastatic** adj.

metazoa the plural of **metazoan** (def. 1) or **metazoan**.

Metazoa a subkingdom of animals consisting of multicellular animals whose cells are organized into tissues and their activities coordinated by a nervous system. The group includes all animals except the Protozoa and Parazoa.

metazoal a variant form of **metazoan** (def. 2).

metazoan 1 or **metazoan** (pl. **metazoa**) any animal of the subkingdom Metazoa. **2** or **metazoal** or **metazoic** of, characteristic of, or belonging to the Metazoa.

metazoic a variant form of **metazoan** (def. 2).

metazon (pl. **metazoa**) an alternative term for **metazoan** (def. 1). [**Met**]**jenkephalin** or [**Met**]**jenkephalin** abbr. for [5-methionine]jenkephalin.

meter 1 an instrument or device for measuring (and recording) the quantity of something (either instantaneously or cumulatively). **2** to measure (something) by means of a **meter** (def. 1). **3** to supply (something) in a measured or regulated amount. **4** the US spelling of **metre**.

+meter comb. form denoting an instrument or device for measuring (something indicated or specified). —**+metric** adj.

metestrus see **estrous cycle**.

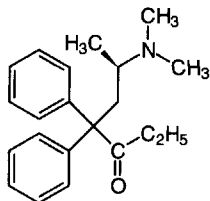
meth+ a variant form of **metho+** (before a vowel).

methadone (hydrochloride) 6-dimethylamino-4,4-diphenyl-3-heptanone hydrochloride; a narcotic analgesic used in cases of opioid dependence.

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R(-)-enantiomer

methaemoglobin a variant spelling (esp. Brit.) of **methemoglobin**.

methaemoglobinaemia a variant spelling (esp. Brit.) of **methemoglobinemia**.

methane or **marsh gas** a colourless, odourless, flammable gas, CH_4 ; it is the simplest alkane and is the principal constituent of natural gas found in association with mineral oil. It is formed by the anaerobic decay of vegetable matter and in large amounts by ruminants and other herbivores.

methano+ *comb. form.* 1 indicating methane. 2 (in chemical nomenclature) denoting the presence of a $-\text{CH}_2-$ bridge in a polycyclic hydrocarbon.

methanoate *see* **formate**.

methanogen any microorganism that produces methane. — **methanogenic** *adj.*

methanogenesis the microbial production of methane by anaerobic breakdown of carbon-containing compounds.

methanogenic the ability to form methane from CO_2 , a property of some anaerobic bacteria. The pathway involves the coenzyme **tetrahydromethanopterin** (*abbr.*: H_4MPT), and involves formation of 5-formyl- H_4MPT , from which successively methenyl- H_4MPT , methylene- H_4MPT , and methyl- H_4MPT are formed in reactions analogous to those for **folate coenzymes**. From the last of these, methyl is transferred to **coenzyme M** to form methyl-coenzyme M, from which methane is formed in a reductive step involving **factor F₄₃₀**. *See also* **coenzyme F₄₂₀**.

methanol methyl alcohol; $\text{CH}_3\text{-OH}$; the lowest member of the alkanols. It is a colourless, flammable, mobile, poisonous liquid, widely used as a solvent. Numerous esters and ethers of methanol occur naturally, especially in plants. *Other names*: wood alcohol; wood spirit; carbinol. — **methanolic** *adj.*

methanol oxidase *see* **alcohol oxidase**.

methanolysis alcoholysis using methanol.

MetHb *abbr.* for **methemoglobin** (i.e. **ferrihemoglobin**).

methemoglobin or (esp. Brit.) **methaemoglobin** *abbr.*: **MetHb**; an older name for **ferrihemoglobin**.

methemoglobinemia or (esp. Brit.) **methaemoglobinaemia** any clinical condition characterized by an excess of **ferrihemoglobin** (i.e. **methemoglobin**) in the erythrocytes. It accompanies a number of **hemoglobinopathies**, and also results from administration of large amounts of sulfonamides. A hereditary methemoglobinemia is associated with a lack of enzymic capacity to reduce ferrihemoglobin within the red cell. Ferrihemoglobin imparts a brown colour to urine.

methene an old name, still commonly used in biochemistry, for the trivalent diatomic group, $=\text{CH}-$, attached to two other groups in an organic molecule. The name now recommended is **methyne**.

methene bridge the **methene** group that links consecutive pyrrole rings in the molecular structure of tetrapyrroles and related compounds.

methenyl+ *prefix (to a trivial biochemical name)* denoting substitution of a **methene** group into the indicated parent compound (at the positions specified by appropriate locants); e.g. 5,10-methenyltetrahydrofolate. *Compare* **methylene+**.

methenyltetrahydrofolate cyclohydrolase EC 3.5.4.9; systematic name: 5,10-methenyltetrahydrofolate 5-hydrolase (de-

cyclizing); one of the enzymes involved in transformations between **folate coenzymes**. It catalyses the hydrolysis of 5,10-methenyltetrahydrofolate to form 10-formyltetrahydrofolate. One of three enzymes (*see also* **methylenetetrahydrofolate dehydrogenase (NADP⁺)** and **formate-tetrahydrofolate ligase**) necessary for the biosynthesis of purines, thymidylate, methionine, histidine, pantothenate, and formyl tRNA-Met. *See* **methylene-tetrahydrofolate dehydrogenase (NADP⁺)** for an example.

5,10-methenyltetrahydrofolate synthetase *see* **5-formyltetrahydrofolate cyclo-ligase**.

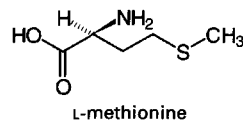
MeTHFA *abbr.* for 5-methyltetrahydrofolate.

methicillin *see* **metcillin (sodium)**.

methine a former name for **methyne**.

methioninate 1 methionine anion; the anion, $\text{CH}_3\text{-S-}[\text{CH}_2]_2\text{-CH(NH}_2\text{)-COO}^-$, derived from **methionine**. 2 any salt containing methionine anion. 3 any ester of methionine.

methionine the trivial name for α -amino- γ -methylmercaptobutyric acid; 2-amino-4-(methylthio)butanoic acid; $\text{CH}_3\text{-S-}[\text{CH}_2]_2\text{-CH(NH}_2\text{)-COOH}$; a chiral α -amino acid. L-Methionine (*symbol*: **M** or **Met**), (S)-2-amino-4-(methylthio)butanoic acid, is a coded amino acid, found in peptide linkage in proteins; codon: AUG (or AUA also in various species of mitochondria). D-Methionine (*symbol*: **D-Met** or **DMet**) is not known to occur naturally. In mammals methionine is glucogenic and is an essential dietary amino acid, although either enantiomer will serve (the D-form being oxidatively deaminated to the α -oxo acid, which is then reaminated with inversion).



L-methionine

methionine adenosyltransferase EC 2.5.1.6; *other names*: S-adenosylmethionine synthetase; AdoMet synthetase; an enzyme that catalyses the formation of S-adenosyl-L-methionine from L-methionine, ATP, and H_2O with the release of pyrophosphate and orthophosphate. Example from *Escherichia coli*: database code METK_ECOLI, 384 amino acids (42.03 kDa). Example from human: database code METK_HUMAN, 395 amino acids (43.61 kDa).

methionine dioxide *symbol*: MetO_2 ; methionine S,S-dioxide; $\text{CH}_3\text{-SO}_2\text{-}[\text{CH}_2]_2\text{-CH(NH}_2\text{)-COOH}$; an oxidation product of methionine, formerly known (incorrectly) as **methionine sulfone**.

[5-methionine]enkephalin *abbr.*: **[Met]enkephalin** or **[Met⁵]enkephalin**; the recommended designation for the trivial name of the pentapeptide Tyr-Gly-Gly-Phe-Met. *See* **enkephalin**.

methionine oxide *symbol*: **MetO**; methionine S-oxide; $\text{CH}_3\text{-SO-}[\text{CH}_2]_2\text{-CH(NH}_2\text{)-COOH}$; an oxidation product of methionine and an intermediate in the formation of **methionine dioxide**. It was formerly known (incorrectly) as **methionine sulf-oxide**.

methionine salvage pathway a metabolic cycle in which methionine is synthesized from carbons originating in the ribose of ATP, the sulfur atom being conserved throughout the cycle. In outline, the cycle consists of reactions in which, starting with L-methionine, S-adenosylmethionine (I) is formed in a reaction with ATP. I is then converted by 1-aminocyclopropane-1-carboxylate synthase, EC 4.4.1.14, to 5'-methylthioadenosine (II) and 1-aminocyclopropane-1-carboxylate, a precursor of ethylene (*see* **ethylene-forming enzyme**). Adenine is then cleaved from methylthioadenosine to form 5-methylthioribose from which 5-methylthioribose-1-phosphate (III) is formed by reaction with ATP. III is then converted through a series of intermediates to 2-keto-4-methylthiobutyrate from which methionine is formed by a transamination reaction.

methionine sulfone an old (incorrect) name for **methionine dioxide**.



methionine sulfone

methionine sulfoxide an old (incorrect) name for **methionine oxide**.

methionine sulfoximine [3-amino-3-carboxypropyl]methyl sulfoximine; an inhibitor of glutamine synthetase (**glutamate-ammonia ligase**) in plants, and identified as the toxic agent in flour treated with the bleaching agent NCl_3 (agene, hence the term **agenized flour**). It interferes with the action of γ -amino-*n*-butyric acid in the brain. This is associated with the ability of methionine sulfoximine to cause fits in a wide variety of animals – ferrets are particularly susceptible. Although methionine sulfoximine has not been clearly proven to be toxic to humans, agene is no longer used for bleaching flour.

methionine synthase see **homocysteine methyltransferase**.

methioninium methionine cation; the cation, $\text{CH}_3\text{-S-}[\text{CH}_2]_2\text{-CH(NH}_3^+)\text{-COOH}$, derived from **methionine**.

methionino the alkylamino group, $\text{CH}_3\text{-S-}[\text{CH}_2]_2\text{-CH(COOH)-NH-}$, derived from **methionine**.

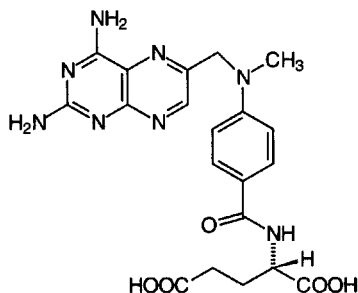
methionyl the acyl group, $\text{CH}_3\text{-S-}[\text{CH}_2]_2\text{-CH(NH}_2\text{)-CO-}$, derived from **methionine**.

methionyllsylbradykinin or **Met-Lys-bradykinin** see **bradykinin**.

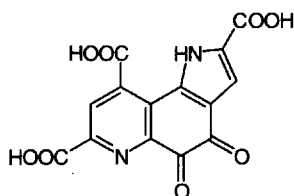
methionyl-tRNA formyltransferase EC 2.1.2.9; *systematic name*: 10-formyltetrahydrofolate:L-methionyl-tRNA *N*-formyltransferase. An enzyme that catalyses a reaction between 10-formyltetrahydrofolate, L-methionyl-tRNA, and H_2O to form tetrahydrofolate and *N*-formylmethionyl-tRNA. *N*-formylmethionyl-tRNA is involved in the initiation of prokaryote translation. Example from *Escherichia coli*: database code FMT_ECOLI, 314 amino acids (34.00 kDa).

metho+ or (before a vowel) **meth+** (comb. form) a variant of **methyl+** in the names of certain chemical compounds.

methotrexate 4-amino-10-methylfolic acid; *other name*: amethopterin; a folic acid analogue that is a potent competitive inhibitor ($K_i < 10^{-9}$ M) of **dihydrofolate reductase**. It is useful as an antineoplastic agent in the treatment of acute leukemia and choriocarcinoma. It acts by blocking nucleotide synthesis and thus also DNA synthesis and has similar activity to **aminopterin**. The structure of a complex of methotrexate and dihydrofolate reductase from *Lactobacillus casei* is known: database code NRL_3DFR.



methoxatin or **pyrroloquinoline quinone** 4,5-dihydro-4,5-dioxo-1*H*-pyrrolo-[2,3-*f*]quinoline-2,7,9-tricarboxylic acid; a heat-stable fluorescent cofactor implicated in the action of

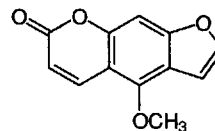


and possibly more generally associated with the oxidation of single-carbon compounds.

methoxy *symbol*: OMe or MeO; the alkoxy group, $\text{CH}_3\text{-O-}$, derived from methanol by loss of a hydrogen atom.

3-methoxy-4-hydroxymandelate acid See **hydroxymethoxymandelate**.

5-methoxypsoralen an intercalating reagent for DNA; it forms covalent cross-links on irradiation. See also **psoralen**.



methoxyvinylglycine *abbr.*: MVG; L-2-amino-4-methoxy-*trans*-3-butenic acid; an inhibitor of ethylene biosynthesis in plants.

methyl *symbol*: Me; the alkyl group, $\text{CH}_3\text{-}$, derived from methane.

methyl- *prefix (in chemical nomenclature)* used as a locant to indicate in or at a methyl group.

methyl-accepting chemotaxis protein *abbr.*: MCP; any of several similar integral plasma membrane proteins, one or more of which may be present in the cell membranes of particular strains of certain motile bacteria, e.g. *Escherichia coli* and *Salmonella typhimurium*, and each of which appears to mediate the responses of these organisms to corresponding classes of chemotactic agent by undergoing enzymic methylation/demethylation of the γ -carboxylate group(s) of one or more glutamic residues (for the enzymes responsible see **che**). The type of ligand determines the nature of the methylation, which in turn determines the action of the receptor on the bacterial flagellar motor, so that the effect can be either attractant or repellent. The following examples are from *E. coli*. MCP1 is a receptor for attractants such as L-serine and other amino acids and is also responsible for chemotaxis away from a wide range of repellents, including leucine, indole, and weak acids; database code MCP1_ECOLI, 551 amino acids (59.38 kDa). MCP2 is a receptor for L-aspartate, dicarboxylic acids, and maltose (via interaction with a periplasmic maltose-binding protein), and is also responsible for chemotaxis away from cobalt and nickel; database code MCP2_ECOLI, 553 amino acids (59.96 kDa). MCP3 interacts with periplasmic ribose- or galactose-binding proteins and mediates taxis towards these sugars; database code MCP3_ECOLI, 535 amino acids (57.99 kDa). MCP4 interacts with periplasmic dipeptide-binding proteins and mediates taxis towards such peptides; database code MCP4_ECOLI, 535 amino acids (57.65 kDa).

m^5 -methyladenine a methylated base of DNA; this and **5-methylcytosine** are the only modified bases in the DNA of cellular organisms.

methyl alcohol a common name for **methanol**.

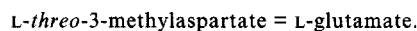
methylase a common name for **methyltransferase**.

methyiaspartase see under **methyiaspartate mutase**.

***N*-methyl-D-aspartate** *abbr.*: NMDA; an excitatory amino acid and a neurotransmitter. It acts as agonist for the *N*-methyl-D-aspartate receptor, a subclass of glutamate ionotropic receptors (see **glutamate receptor**).

methyiaspartate ammonia-lyase recommended name for **methyiaspartase**; see under **methyiaspartate mutase**.

methyiaspartate mutase EC 5.4.99.1; *systematic name*: L-threo-3-methyiaspartate carboxy-aminomethylmutase; *other name*: glutamate mutase. An enzyme that catalyses the reaction:



This is the first step of the glutamate-fermentation pathway of the anaerobic bacteria *Clostridium cochlearium*, *C. tetanomorphum*.

primary alcohol dehydrogenases in methyiotrophic bacteria

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methylate

phum, and others. The product, (2*S*, 3*S*)-3-methylaspartate, is converted to mesaconate by the enzyme methylaspartate ammonia-lyase (EC 4.3.1.2). Methylaspartate mutase requires a cobamide coenzyme and has been studied as an example of such a reaction. The enzyme has two subunits (large and small, sometimes called E and S). Examples from *C. tetanomorphum*: large chain, database code S32433, 485 amino acids (53.68 kDa); small chain, database code S29502, 137 amino acids (14.73 kDa).

methylate 1 to introduce (one or more) methyl groups into (a substance). **2** (*in commerce*) to mix or adulterate with methanol. — **methylation** *n*. See also **methylation of DNA**.

methylated-DNA-protein-cysteine S-methyltransferase EC 2.1.1.63; *other names*: 6-*O*-methylguanine-DNA methyltransferase; *O*⁶-methylguanine-DNA-alkyltransferase. It catalyses the reaction between DNA (containing 6-*O*-methylguanine) and protein L-cysteine to form DNA (without 6-*O*-methylguanine) and protein S-methyl-L-cysteine. As the acceptor is in the enzyme itself this is a 'suicidal' reaction. Bacterial and mammalian enzymes have common motifs. Example from *Escherichia coli*: database code OGT_ECOLI, 171 amino acids (19.06 kDa).

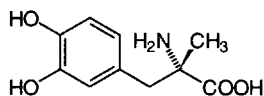
methylation of DNA the occurrence in vertebrate DNA of varying amounts of 5-methylcytosine which arises from methylation of cytosine bases where they occur in the sequence CpG. The methylation status of DNA correlates with its functional activity: inactive genes are more heavily methylated. See also **CpG islands**.

3-methylcholanthrene a carcinogenic hydrocarbon, found in coal tar. It induces synthesis of cytochrome P450 mRNA.

methylcrotonyl-CoA carboxylase EC 6.4.1.4; an enzyme of the pathway for the degradation of leucine. It catalyses the formation of 3-methylglutaconyl-CoA from ATP, 3-methylcrotonyl-CoA, and HCO₃⁻ with release of ADP and orthophosphate. Biotin is a cofactor. The 3-methylglutaconyl-CoA is then converted to 3-hydroxy-3-methylglutaryl-CoA.

5-methylcytosine a methylated base of DNA; this and **N⁶-methyladenine** are the only modified bases in the DNA of cellular organisms. In T-even phages all the C residues of DNA are 5-hydroxymethylcytosine.

methyl dopa L-2-amino-2-methyl-3-(3,4-dihydroxyphenyl)propionic acid; a compound used as an antihypertensive. It is a competitive inhibitor of **aromatic L-amino acid decarboxylase** (dopa decarboxylase), but its effects on blood pressure may result from its metabolism to α -methylnorepinephrine which stimulates presynaptic α_2 receptors, thus decreasing norepinephrine release.



methylene 1 the recommended name for the bivalent triatomic group CH₂ in a molecule, whether attached to two other atoms by single bonds or to one other atom by a double bond. **2** a former name for **carbene** (def. 1).

methylene+ prefix (to a trivial chemical name) denoting substitution of a **methylene** group into the indicated parent compound (at the positions specified by appropriate locants); e.g. 5,10-methylenetetrahydrofolate (which may be shortened to 5,10-CH₂-H₄folate). Compare **methenyl+**.

N,N'-methylenebisacrylamide (CH₂:CHCONH₂)₂CH₂; a cross-linking agent used in a ratio (w/w) of between 1:49 and 1:19 with acrylamide in the preparation of polyacrylamide gels.

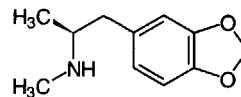
Methylene Blue or **Swiss Blue** or **Basic Blue 9** CI 52015; tetramethylthionine chloride; 3,7-bis(dimethylamino)phenol. Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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methylest

thiazinium chloride; a water- and ethanol-soluble blue dye; $A_{\max}$ (in water) at 663–667 nm. It is reducible to a leuco (i.e. colourless) compound, which is reoxidizable by dioxygen, and hence is useful as a redox indicator; E° (pH 7.0, 30 °C) = +0.011 V. It is useful also as a stain in bacteriology and histology, e.g. being preferentially taken up by melanin-containing cells and can be used to target therapeutic agents to such cells.

3,4-methylenedioxymethamphetamine *abbr.*: MDMA; a phenethylamine derivative and analogue of **mescaline** and **amphetamine** that is a potent CNS stimulant and hallucinogen; popularly known as **ecstasy**. A controlled substance in the US Code of Federal Regulations.



S-(+)-enantiomer

2-methyleneglutarate mutase EC 5.4.99.4; *systematic name*: 2-methyleneglutarate carboxy-methylenemethylmutase. An enzyme that catalyses the reaction:

2-methyleneglutarate = 2-methylene-3-methylsuccinate.

It is a key enzyme in the fermentation of nicotinate to ammonia, propionate, acetate, and CO₂ by the strict anaerobe *Clostridium barkeri*, but has been much investigated as an example of an enzyme mechanism based on the adenosylcobalamin coenzyme. 2-Methylene-3-methylsuccinate is converted to 2,3-dimethylmaleate by methylitaconate Δ -isomerase (EC 5.3.3.6). The enzyme is a homotetramer of 70 kDa subunits: database code CBMGM1, 614 amino acids (66.67 kDa).

methylene-interrupted double bonds double bonds separated by one methylene group. The term is used especially to describe the double bonds in naturally occurring long-chain fatty acids; in many cases these are all of the *Z* configuration, with each pair separated by one methylene group.

methylenetetrahydrofolate dehydrogenase (NADP⁺) EC 1.5.1.5; *systematic name*: 5,10-methylenetetrahydrofolate: NADP⁺ oxidoreductase. *other name*: C1-tetrahydrofolate synthase, cytoplasmic; an enzyme involved in the interconversion of folate coenzymes that catalyses the reaction:

5,10-methylenetetrahydrofolate + NADP⁺ =
5,10-methenyltetrahydrofolate + NADPH.

One of three enzymes (see also **methenyltetrahydrofolate cyclohydrolase**, and **formate-tetrahydrofolate ligase**) necessary for the biosynthesis of purines, thymidylate, methionine, histidine, pantothenate, and formyl tRNA-Met. Example from human, a cytoplasmic trifunctional homodimer: database code C1TC_HUMAN, 934 amino acids (101.31 kDa) – this has an N-terminal domain (amino acids 1–304) with methylenetetrahydrofolate dehydrogenase and cyclohydrolase activity; the C-terminal section (305–934) forms the formyltetrahydrofolate synthetase domain; *other names*: mitochondrial (precursor) NAD-dependent methylenetetrahydrofolate dehydrogenase (EC 1.5.1.5)/methenyltetrahydrofolate cyclohydrolase (EC 3.5.4.9): database code MTDC_HUMAN, 344 amino acids (37.28 kDa).

5,10-methylenetetrahydrofolate reductase (FADH₂) EC 1.7.99.5; an enzyme that catalyses the reaction:

5-methyltetrahydrofolate + acceptor =
5,10-methylenetetrahydrofolate + reduced acceptor.

FAD is a coenzyme. Example from *Escherichia coli*: database code METF_ECOLI, 296 amino acids (33.06 kDa).

methylesterase any of various enzymes that catalyse the hydrolysis of methyl esters. For example, **juvenile-hormone esterase** (EC 3.1.1.59) of insects, *other name*: JH-esterase.



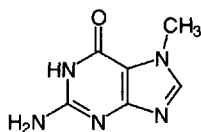
methylglucosamine

catalyses the hydrolysis of methyl (2*E*,6*E*)-(10*R*,11*S*)-10,11-epoxy-3,7,11-trimethyltrideca-2,6-dienoate to form (2*E*,6*E*)-(10*R*,11*S*)-10,11-epoxy-3,7,11-trimethyltrideca-2,6-dienoate and methanol; i.e. it demethylates the insect juvenile hormones JH₁ and JH₂. Example (precursor) from the moth *Heliothis virescens*: database code ESTJ_HELVI, 564 amino acids (62.53 kDa); it is similar to certain other carboxylesterases and lipases. Another example, the **protein-glutamate methyl-esterase** of *Escherichia coli*, is involved in chemotaxis in this organism (see **che**).

methylglucosamine *N*-methyl-L-glucosamine; 2-deoxy-2-(methylamino)-L-glucose; a component of **streptomycin**.

O⁶-methylguanine a modified and, in principle, highly mutagenic base in DNA (it base pairs with T). It has a special detoxication enzyme, **methylated-DNA-protein-cysteine S-methyl-transferase**.

7-methylguanine a modified base found in transfer RNA. See **guanine** (for numbering), **7-methylguanosine**.



7-methylguanosine a modified nucleoside that forms a **cap** at the 5'-terminus of eukaryotic mRNA. See also **7-methylguanine**. **methylidyne** the recommended name for the trivalent diatomic group CH≡ attached to a single other group in a molecule of an organic compound. Compare **methine**.

methylmalonate CH₃CH(COO⁻)₂; isosuccinate 2-methylpropanedioate; the dianion of methylmalonic acid; as **methylmalonyl-CoA** it is an intermediate in the conversion of propionyl-CoA to succinyl-CoA. Note: methylmalonate should be distinguished from methyl malonate (two words), which is propanedioic acid dimethyl ester.

methylmalonic acidemia or **methylmalonic aciduria** any of a group of diseases, inherited as autosomal recessive traits, in which abnormally large amounts of methylmalonate occur in the blood and urine. The condition may be caused by abnormalities in either **methylmalonyl-CoA epimerase** or **methylmalonyl-CoA mutase** apoenzymes, or by three different abnormalities in adenosylcobalamin synthesis.

methylmalonyl-CoA 2-carboxypropanoyl-CoA; it can exist as the enantiomers (*R*)- and (*S*)-methylmalonyl-CoA; (*S*)-methylmalonyl-CoA is an intermediate in the **beta oxidation** of odd-numbered fatty acids in animals. Propionyl-CoA is the major end product; for further metabolism, this is converted to succinyl-CoA by, first, carboxylation to (*S*)-methylmalonyl-CoA, which is converted by **methylmalonyl-CoA epimerase** to (*R*)-methylmalonyl-CoA, which is then converted to succinyl-CoA by **methylmalonyl-CoA mutase**.

methylmalonyl-CoA epimerase EC 5.1.99.1; an enzyme that interconverts (*R*)-methylmalonyl-CoA to (*S*)-methylmalonyl-CoA. It is sometimes incorrectly referred to as methylmalonyl-CoA racemase (chiral centres in the CoA moiety are not affected).

methylmalonyl-CoA mutase EC 5.4.99.2; **systematic name**: (*R*)-2-methyl-3-oxopropanoyl-CoA CoA-carbonylmutase; an enzyme that converts (*R*)-methylmalonyl-CoA to succinyl-CoA; adenosylcobalamin is a cofactor. Example from human, homodimer (precursor): database code MUTA_HUMAN, 750 amino acids (83.01 kDa). This enzyme is required for degradation of several amino acids, odd-chain fatty acids, and cholesterol via propionyl-CoA to the tricarboxylic-acid cycle; deficiency results in **methylmalonic acidemia**. It is a mitochondrial enzyme (the precursor includes a transit peptide in amino acids 1-32). In some bacteria the enzyme is a heterodimer.

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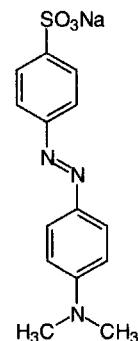
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methylsulfonyl nitrophenyl carb

both subunits have homology with the human protein. The metabolic role is in the synthesis of propionate from tricarboxylic-acid-cycle intermediates. Examples from *Propionobacter freudenreichii shermanii*, α subunit: database code MUTA_PROFR, 638 amino acids (69.39 kDa); β subunit: database code MUTB_PROFR, 728 amino acids (80.09 kDa).

2-methyl-1,4-naphthalenedione 2-methyl-1,4-naphthoquinone; see **menadione**.

Methyl Orange a dye used as an indicator, changing from red to orange-yellow over the pH range 3.1-4.4.



methylotroph any organism that can use certain single-carbon compounds, in a lower oxidation state than CO₂, as sole carbon source. —**methylotrophic** *adj.*; **methylotrophy** *n*.

methylotrophic yeast any yeast belonging to either of the genera *Hansenula* or *Pichia*. They are potentially valuable recombinant-DNA expression hosts as certain oxidase genes (*MOX* and *AOX*) are vastly overexpressed and their promoters are suitable for the expression of foreign proteins. See also **alcohol oxidase**, **methylotroph**.

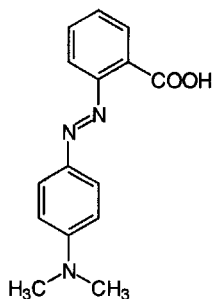
3-methyl-2-oxobutanoate dehydrogenase (lipoamide) EC 1.2.4.4; **other names**: 2-oxoisovalerate dehydrogenase; branched-chain α -keto-acid dehydrogenase; branched-chain α -keto-acid decarboxylase; an enzyme activity present in a complex that catalyses the oxidative decarboxylation of oxo acids derived from branched-chain amino acids. It is similar in mechanism to the **pyruvate dehydrogenase complex**. This enzyme is the E₁ component of the complex, and catalyses the release of CO₂ from the oxo acid and transfer of the resulting acyl group to lipoamide. Thus, for the oxo acid from valine, it catalyses the reaction of 3-methyl-2-oxobutanoate with lipoamide to form *S*-(2-methylpropanoyl) dihydrolipoamide with release of CO₂. Thiamine diphosphate is a coenzyme. Example from human, precursors of α and β subunits respectively: database codes ODBA_HUMAN, 445 amino acids (50.41 kDa); ODBB_HUMAN, 392 amino acids (43.07 kDa). See also **[3-methyl-2-oxobutanoate dehydrogenase (lipoamide)] kinase**.

[3-methyl-2-oxobutanoate dehydrogenase (lipoamide)] kinase EC 2.7.1.115; **other name**: branched-chain α -keto-acid dehydrogenase kinase; an enzyme that catalyses the phosphorylation by ATP of [3-methyl-2-oxobutanoate dehydrogenase (lipoamide)] with release of ADP, thereby inactivating the enzyme complex. Example (precursor) from rat: database code BCKD_RAT, 412 amino acids (46.34 kDa).

[3-methyl-2-oxobutanoate dehydrogenase (lipoamide)]-phosphatase EC 3.1.3.52; **other name**: branched-chain oxo-acid dehydrogenase phosphatase; an enzyme that catalyses the hydrolysis of phosphate from [3-methyl-2-oxobutanoate dehydrogenase (lipoamide)] phosphate.

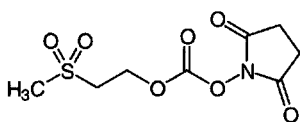
Methyl Red a dye used as an indicator, changing from red to yellow over the pH range 4.2-6.3. (Illustrated on p. 414.)

methylsulfonyl nitrophenyl carbonate 2-(methylsulfonyl)-ethyl 4-nitrophenyl carbonate; a reagent used for the introduction of the **MSOC** amino-protecting group.

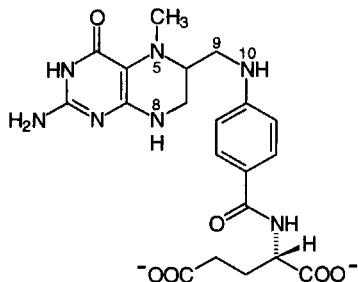


Methyl Red

methylsulfonyl succinimidyl carbonate 2-(methylsulfonyl)-ethyl *N*-succinimidyl carbonate; a reagent used for the introduction of the **MSOC** amino-protecting group.



5-methyltetrahydrofolate *abbr.*: MeTHFA or 5-CH₃-H₄folate; a member of the tetrahydrofolate group of coenzymes. One of its functions is to act as intermediate in the methylation of vitamin B₁₂, which can then react with homocysteine to form methionine. The methyl group of MeTHFA is derived from C-3 of serine via 5,10-methylene H₄folate.



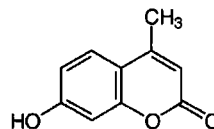
methyltetrahydrofolate trap hypothesis a hypothesis to explain the accumulation of 5-methyltetrahydrofolate (5-CH₃-H₄folate) in the sera of patients with vitamin B₁₂ deficiency. It suggests that, because the only route by which 5-CH₃-H₄folate can be converted to H₄folate (required for single-carbon transfer reactions) is by the vitamin B₁₂-dependent 5-methyltetrahydrofolate-homocysteine *S*-methyltransferase (EC 2.1.1.13) reaction, a large proportion of the body's available folate becomes trapped as 5-CH₃-H₄folate, leading to folate deprivation.

methylthioribose 5-methylthio-D-ribose; an intermediate in the **methionine salvage pathway**.

methyltransferase any of the enzymes of sub-subclass EC 2.1.1, which catalyse the transfer of a methyl group. *See also che, O⁶-methylguanine*.

4-methylumbelliferol *abbr.*: MU; a group that is attached in glycosidic linkage to various sugars to form substrates useful in the assay of glycosidases. The highly fluorescent 4-methylumbelliferone is released on enzyme action, and can be mea-

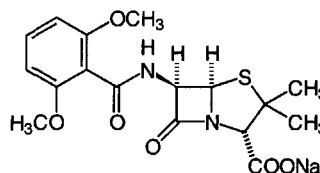
sured in alkaline solution by its fluorescence (max. 448 nm, excitation at 364 nm), which is quenched in the glycosidic substrate. *See also* **MUG**.



4-methylumbelliferone

methyne or (formerly) **methine** the name now recommended for the trivalent diatomic group, =CH-, attached to two other groups in an organic molecule. Such a group is commonly still known in biochemistry as **methene**. *Compare* **methylidene**.

meticillin (sodium) or **methicillin** a semisynthetic antibiotic of the **penicillin** type that is lactamase resistant.



metJ the gene for the repressor (co-repressor is *S*-adenosyl-methionine) of the methionine biosynthetic operon, and also *metJ* itself, in *Escherichia coli*. The repressor protein undergoes dimeric binding to DNA by a motif other than a helix-turn-helix. Database code METJ_ECOLI, 104 amino acids (12.00 kDa).

Met-Lys-bradykinin a short name for methionyllysylbradykinin; *see* **bradykinin**.

MetMb *abbr.* for metmyoglobin (i.e. **ferrimyoglobin**).

metmyoglobin *abbr.*: MetMb; an old name for **ferrimyoglobin**.

metre or (US) **meter** *symbol*: m; the SI base unit of length, equal to the length of the path travelled by light in a vacuum during a time interval of 1/299 792 458 of a second.

metre-kilogram-second system (of electrical units) *abbr.*: mks system or MKS system; a system of electrical (physical) units based on the metre, kilogram, and second as fundamental units of measurement. It has now been superseded by the **SI** system of units.

metric 1 of, or relating to, the metre or the metric system. 2 quantitative.

+metric *see* **+meter, +metry**.

metric system a system of weights and measures, devised in France (1793-95), based on the principle that each of the physical quantities length, volume, and mass should be defined in terms of one unit whose multiples and submultiples are related by powers of ten and that all three units should be simply interrelated. The units were, respectively, the metre, the litre (one cubic decimetre), and the gram (the mass of one cubic centimetre of water at its maximum density). It has now been supplanted by the International System of Units (*see* **SI**).

metridin 1 a toxin from *Metridium senile* (brown sea anemone); database code TXAM_METSE, 36 amino acids (3.97 kDa). 2 or **metridium proteinase A** EC 3.4.21.3; *other name*: sea anemone protease A; an enzyme that catalyses the preferential cleavage: Tyr-I-Xaa > Phe-I-Xaa > Leu-I-Xaa; it has little action on Trp-I-Xaa.

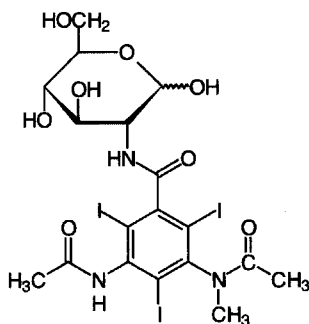
metridium proteinase A *see* **metridin** (def. 2).

metrizamide the trivial name for 2-[3-acetamido-5-(*N*-methylacetamido)-2,4,6-triiodobenzamido]-2-deoxy-D-glucose; a dense,

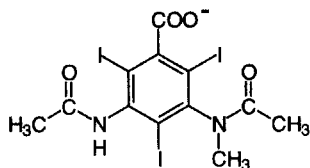


metrizoate

synthetic, nonionic substance, highly soluble in water, that is capable of forming (nonautoclavable) aqueous solutions of densities up to 1.3 g cm^{-3} , useful as media for **isopycnic centrifugation** of biological particles.

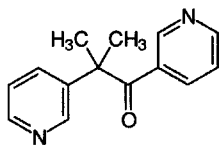


metrizoate the trivial name for the 3-acetamido-5-(N-methylacetamido)-2,4,6-triiodobenzoate anion; a dense, synthetic substance used primarily as its sodium salt, which is highly soluble in water and capable of forming autoclavable aqueous solutions of densities up to 1.4 g cm^{-3} , or more, such solutions being useful as media for **isopycnic centrifugation** of biological particles.



+metry comb. form denoting the process or science of measuring (something indicated or specified). — **+metric adj.**

metryapone 2-methyl-1,2-di(3-pyridyl)propan-1-one; an inhibitor of the enzyme 11β -hydroxylase (steroid 11β -monooxygenase; EC 1.14.15.4); this enzyme is involved in the synthesis of cortisone, cortisol and aldosterone, its inhibition causing a rise in **corticotropin** secretion as a result of release from negative feedback on the pituitary by cortisol. Metyrapone is used in **challenge** tests of hypothalamic-pituitary function and sometimes in the treatment of **Cushing's syndrome**.



MeV symbol for megaelectronvolt (i.e. 10^6 electronvolts).

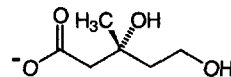
Mevacor a proprietary name for **lovastatin**.

mevaldate 1 the anion $\text{OHC-CH}_2\text{-C(CH}_3\text{)(OH)-CH}_2\text{-COO}^-$ derived from mevaldic acid; its (*R*)-enantiomer occurs as an enzyme-bound intermediate during the conversion of hydroxymethylglutaryl-coenzyme A to mevalonate in the biosynthesis of polyprenyl compounds. 2 any salt or ester of mevaldic acid.

mevalonate 1 the anion, $\text{HO-[CH}_2\text{]}_2\text{-C(CH}_3\text{)(OH)-CH}_2\text{-COO}^-$, derived from mevalonic acid; its (*R*)-enantiomer is a strategic intermediate derived from hydroxymethylglutaryl-CoA in the biosynthesis of polyprenyl compounds. 2 any salt or ester of mevalonic acid.

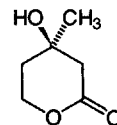
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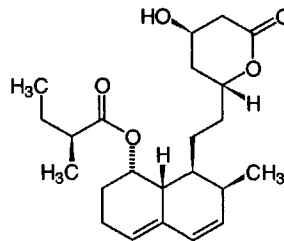
(*R*)-mevalonate

mevalonolactone mevalonic acid lactone; it is hydrolysed in cells to mevalonate, and is often the form in which mevalonate is administered to animals or cells, either radioactively or non-radioactively as a metabolic precursor.



mevalonoyl the acyl group, $\text{HO-[CH}_2\text{]}_2\text{-C(CH}_3\text{)(OH)-CH}_2\text{-CO-}$, derived from mevalonic acid.

mevastatin or **compactin** 6-demethylmevinolin; a substance isolated from *Penicillium citrinum*; it is a potent inhibitor of **HMG-CoA reductase** with actions similar to **lovastatin**.



Mevinacor a proprietary name for **lovastatin**.

mevinolin an alternative name for **lovastatin**.

Meyerhof, Otto Fritz (1884–1951), German physiologist and biochemist; Nobel Laureate in Physiology or Medicine (1922) 'for his discovery of the fixed relationship between the consumption of oxygen and the metabolism of lactic acid in the muscle' [prize shared with A. V. Hill].

Meyerhof pathway see **Embden–Meyerhof pathway**.

Meyerhof quotient an index of the effect of the presence or absence of dioxygen on glycolysis or fermentation. It is: [(rate of anaerobic fermentation) – (rate of aerobic fermentation)]/(rate of dioxygen uptake). Assuming 6 mol ATP are formed per mol dioxygen used, a Meyerhof quotient of 6 would mean that aerobic and anaerobic energy supply are the same; a lower value indicates an increase in ATP-supplying and ATP-requiring processes on transition from anaerobic to aerobic conditions.

mfd abbr. for microfarad.

mg symbol for milligram (i.e. 10^{-3} gram).

Mg symbol for magnesium.

MHC abbr. for major histocompatibility complex.

mho a practical unit of electric conductance equal to the reciprocal of the impedance measured in ohms. For circuits containing only noninductive resistance the **siemens** is the preferred unit.

MHS abbr. for major histocompatibility system (i.e. **major histocompatibility complex**).

MIC abbr. for minimum inhibitory concentration.

mica substrate technique a technique used in preparing



specimens of macromolecules for electron microscopic examination. Freshly cleaved mica is used to support the specimen during shadowing with a metal; a backing film of carbon is then applied by evaporation normal to the shadowed surface, and both films are transferred to the grid.

micelle 1 an aggregate, of colloidal dimensions, of oriented molecules of amphipathic substances existing in equilibrium in solution with the chemical species from which it is formed; micelles are generally electrically charged. In aqueous solution the individual molecules of the micellar aggregate are oriented with their polar groups pointing towards the aqueous medium and their hydrophobic moiety directed into the centre of the micelle. 2 a hypothetical ordered region in a natural fibre such as cellulose. —**micellar adj.**

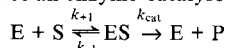
micellization or **micellisation** transformation into micelles.

micF a regulatory gene in the osmoregulatory system in *Escherichia coli*; *micF* RNA and *ompF* mRNA (*omp* = outer membrane protein) are induced by interaction of *ompR* with a common divergent, central promoter region. *micF* RNA is partly complementary to the Shine–Dalgarno region of *ompC* and thus represses its expression by translational control.

Michaelian of, relating to, or described by **Michaelis kinetics**.

Michaelis, Leonor (1875–1949), German physician and biochemist; distinguished for his contributions to the theory of hydrogen ions and to the development of the theory of enzyme–substrate combination. He also worked in Japan and latterly in the USA.

Michaelis kinetics or **Michaelis–Menten kinetics** a model to explain the kinetics of an enzyme-catalysed reaction,



where E is enzyme, S is substrate, ES is an intermediate complex (the **Michaelis complex**), and P is product. The initial velocity of the reaction, v , when $[P] = 0$, is given by the **Michaelis equation** (or **Michaelis–Menten equation**):

$$v = V[S]/(K_m + [S]) = d[P]/dt$$

in which V is the limiting or maximal reaction rate (i.e. that occurring when the enzyme is saturated with substrate), and the **Michaelis constant**, K_m , is given by:

$$K_m = (k_{-1} + k_{cat})/k_{+1}.$$

If $k_{cat} \ll k_{+1}$, then K_m approximates to K_s , the enzyme–substrate dissociation constant (usually known as the **substrate dissociation constant**). K_m and V are known as the **Michaelis parameters**. Compare **Dalziel coefficient**.

Michaelis pH function any one of a set of functions relating hydron concentration, $[H^+]$, to the acid dissociation constants for different stages of dissociation of a given polybasic acid.

Consider first the simplest example, that of a symmetrical dibasic acid, AH_2 (e.g. succinic acid). This can exist in three stages of dissociation, represented by the equilibria



where K_1 and K_2 are the acid dissociation constants for the respective stages of dissociation. For such a dissociation there are three Michaelis pH functions, f_0 , f_1 , and f_2 , which are defined thus:

$$f_0 = 1 + K_1/[H^+] + K_1K_2/[H^+]^2$$

$$f_1 = 1 + [H^+]/K_1 + K_2/[H^+]$$

$$f_2 = 1 + [H^+]/K_2 + [H^+]^2/K_1K_2$$

The total concentration of acid of all forms, $[A]_t$, is given by

$$[A]_t = [AH_2] + [AH^-] + [A^{2-}];$$

by substitution for K_1 and K_2 it can be shown that

$$[A]_t = f_0[AH_2] = f_1[AH^-] = f_2[A^{2-}]$$

Hence it will be apparent that the reciprocals of the respective pH functions give the fractional concentrations of the various ionic forms of the acid at a given pH value. The theory of pH

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functions is independent of the chemical nature of the ionizing groups: the functions apply equally to molecules with identical ionizing groups (e.g. dicarboxylic acids, as above), to ampholytes (e.g. amino acids), and to molecules or parts of molecules in which the nature of the ionizing groups may not be known (e.g. the active centres of enzymes). For an unsymmetrical dibasic acid there are potentially four ionic forms, four dissociation constants, and thus four pH functions, but these are important in practice only if the pK values of the two dissociating groups are fairly close, as could be the situation at an active centre of an enzyme. If, on the other hand, the pK values are widely separated, as is the case with a typical amino acid, then there is effectively only one intermediate form and a set of three pH functions as above may be considered to apply.

Michel, Hartmut (1948–), German biophysicist; Nobel Laureate in Chemistry (1988) jointly with J. Deisenhofer and R. Huber 'for the determination of the three-dimensional structure of a photosynthetic reaction centre'.

micr+ see **micro+** (def. 2).

micra a plural of **micron**.

micRNA *abbr.* for messenger-RNA-interfering complementary RNA (i.e. **antisense RNA**).

micro 1 (concerning something) very small (e.g. micro method). 2 short for a microcomputer.

micro+ 1 *symbol:* μ ; the SI prefix denoting 10^{-6} times. 2 or (sometimes before a vowel) **micr+** denoting very small or concerned with very small things (e.g. microbiology, microsome); abnormally small, underdeveloped (e.g. microcyte); on a small scale or concerned with small quantities (e.g. microanalysis, microbalance); of a small area or volume, or small part of a larger one (e.g. microenvironment). Compare **macro+**. 3 indicating amplification or magnification of something very small (e.g. microphone, microscope). 4 involving the use of a microscope (e.g. micrograph).

microaerophilic 1 describing an aerobic environment with a lower partial pressure of dioxygen than that under normal atmospheric conditions. 2 describing an organism whose maximal rate of growth occurs in such an environment.

microalbuminuria the excretion of plasma albumin in the urine at a rate above the normal range of values ($5\text{--}20 \mu\text{g min}^{-1}$) but below the threshold level ($300 \mu\text{g min}^{-1}$) for detection by routine dipstick methods and measurable only by radioimmunoassay (or other techniques of comparable sensitivity). The condition is associated with a general increase in vascular permeability and forms a sensitive indicator of diabetic nephropathy.

microanalysis any technique whereby very small amounts of substance may be analysed qualitatively or quantitatively. The term was originally used to indicate analysis of substances in the range 1–10 mg. —**microanalytic** or **microanalytical adj.**

microassay any technique whereby very small amounts of substance may be assayed. Compare **microanalysis**.

microbalance any balance designed to weigh very small quantities, typically between 1 μg and 1 mg.

microbe a common term for a microorganism, especially a pathogenic bacterium. —**microbial adj.**

microbial collagenase EC 3.4.24.3; other names: *Clostridium histolyticum* collagenase; clostridiopeptidase A; collagenase A; collagenase I; an enzyme that catalyses the digestion of native collagen in the triple-helical region at Xaa-I-Gly bonds. Some preference is shown for Gly at P3 and P1', Pro and Ala at P2 and P2', and hydroxyproline, Ala, or Arg at P3' (see **peptidase P-sites**). Example (precursor) from *C. perfringens*: database code A36866, 1104 amino acids (125.97 kDa).

microbiological of, pertaining to, or concerned with microbiology; effected by microorganisms.

microbiological assay any assay for a substance, especially an amino acid or vitamin, that is based on the extent of growth of a microorganism for which the analyte in question is an essential growth factor.

microbiology the science concerned with microorganisms, and



microbody

with their interactions with other organisms and the environment. —**microbiologist** *n.*

microbody a cytoplasmic organelle, spherical or oval in shape, and 0.1–1.5 nm across, that is bounded by a single membrane and contains oxidative enzymes, especially those utilizing H_2O_2 . Microbodies include **glyoxysomes** and **peroxisomes**.

microbore high-pressure liquid chromatography or **microbore HPLC** a variant of **high-pressure liquid chromatography** in which a column of only about 1 mm diameter is used. The advantages of the technique include greater sensitivity and a much reduced solvent consumption.

microcalorimeter a **calorimeter** designed to measure continuous power outputs down to 1 μW , or heat pulses down to 200 μJ . —**microcalorimetry** *n.*; **microcalorimetric** *adj.*

microcapsule any very small thin-walled capsule of plastic, wax, etc., containing a liquid, powder, etc., from which the contents may be released (by heat, impact, radiation, dissolution, or other means) in order to fulfil some specific function. *See also* **microencapsulation**.

microcarrier any of a number of preparations of finely particulate, nontoxic, nonrigid, usually transparent material of density close to unity, e.g. dextran-based beads, that enable the growth of animal cells in suspension culture, especially on a large scale. Microcarriers permit the growth of anchorage-dependent cells as a monolayer on the surface of the particles. Generally, the cells may be easily harvested, e.g. by allowing the particles to settle and then treating with a proteinase such as trypsin.

microcell 1 a small, experimentally produced eukaryotic cell-like structure in which a limited amount of genetic material is contained in a micronucleus surrounded by a rim of cytoplasm and an intact cell membrane. It is useful as a vector for the transfer of a small number of chromosomes into a normal cell by virus-induced cell fusion. 2 any small **cell** (def. 2), commonly with a capacity less than 1 mL.

microchemistry chemistry on a very small scale, commonly handling less than 1 mg or 1 mL of material and often requiring special small apparatus. —**microchemical** *adj.*

microcin any of a number of microbial peptide toxins that have antibiotic action. They are of diverse structure and action. Examples include microcin E492, a 6 kDa polypeptide antibiotic. It forms cation-selective channels in planar phospholipid bilayers, leading to a loss of membrane potential. Another example is microcin B17 (precursor): database code MCBA_ECOLI, 69 amino acids (6.01 kDa); residues 1–26 are a propeptide, 27–69 are microcin B17. This is a glycine-rich peptide that inhibits DNA gyrase (*see* (DNA) **topoisomerase** type II).

micrococcal of, pertaining to, or produced by micrococci; being a member of the genus **Micrococcus**.

Micrococcus a genus of Gram-positive, strictly aerobic, usually nonmotile coccoid bacteria of the family Micrococcaceae. The cells are commonly 1–2 μm in diameter.

microconcentrator a commercial device for the concentration and desalting of 'small' volumes (i.e. ≤ 2 mL) of biological samples.

microcrystal any very small crystal, especially one that is visible only under a microscope. —**microcrystalline** *adj.*

microcurie *symbol:* μCi or (formerly) μc ; a unit of (radio)activity or of radioactive material equal to 10^{-6} **curie**.

microdeletion a DNA/chromosomal deletion that is not detectable by conventional techniques such as **Southern blotting** or microscopy.

microdensitometry the measurement of **absorbance** (i.e. optical density) over a very small finite area, such as that viewed by an optical microscope. —**microdensitometric** *adj.*; **microdensitometer** *n.*

microdialyser or (sometimes, esp. US) **microdialyzer** a device for **microdialysis**, especially one designed to minimize loss and dilution of samples.

microdialysis dialysis of a sample of small volume (commonly of less than 1 mL).

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microdiffusion analysis *see* **Conway microdiffusion method**.

microdissection the dissection of small organisms or parts of organisms by means of mechanically controlled instruments viewed through an optical microscope.

microelectrode any electrode in which the sensitive element is very small and also often needle-like. It is useful for investigation of electrical phenomena in individual cells or in very small volumes.

microencapsulation the act or process of enclosing substances, solutions, etc. in **microcapsules**.

microenvironment a very small domain of the environment, such as that surrounding a single molecule or a functional group of a molecule, a single cell or organism, or a small group of cells or organisms. The term is used especially to distinguish zones of local difference in the overall environment.

microfarad *abbr.:* mfd; *symbol:* μF ; a unit of capacitance equal to 10^{-6} **farad**.

microfibre or (US) **microfiber** any fibre having a diameter in the range 0.1–1 μm , such as used in making glass microfibre filters.

microfibril any small fibril occurring in biological material that is distinguishable only by electron microscopy, especially one of a set that make up a fibril, e.g. in cellulose. —**microfibrillar** *adj.*

microfibrillar polysaccharide one of three groups of cell-wall polysaccharides characterized by solubility differences: the main component in the plant kingdom is **cellulose** but **chitin** is found in some fungi, β -1,4-mannans in some green algae (e.g. Codiaceae) and β -1,3-xylans in other green algae (e.g. Bryopsidaceae).

microfilament a type of **actin filament** that contributes to the cytoskeleton of eukaryotic cells. Microfilaments are elongated structures consisting of two strands of globular **actin** monomers twisted into a helix with 13.5 molecules per turn. Microfilaments form the major component of the cell's contractile machinery and are implicated in cell division, cell movement, muscle contraction, nerve outgrowth, tubular gland formation, gastrulation, and neurulation. *Compare* **intermediate filament**, **microtubule**.

microfilter 1 any small filter, especially as used in microchemistry. 2 *see* **microporous filter**.

microfiltration filtration with a **microporous filter**.

microflora 1 a collective name for very small plants. 2 the flora of a **microhabitat**. 3 a collective name for all the microorganisms of a tissue or organ of an animal.

microfluorimeter an instrument for **microfluorimetry**.

microfluorimetry the measurement of intensity of fluorescence over a very small finite area, such as a single cell viewed under an optical microscope. —**microfluorimetric** *adj.*

microforge a device by means of which micropipettes, very small glass tools, etc. can be made while being viewed under a low-power microscope.

microfungus (*pl.* **microfungi**) any fungus whose vegetative thallus or fruiting body is too small to be seen easily with the unaided eye. *Compare* **mould**.

microglobulin 1 any plasma **globulin** molecule or globulin fragment of less than ≈ 40 kDa, e.g. **Bence-Jones protein**, β_2 -**microglobulin**. 2 a term sometimes used to refer to 7S immunoglobulins such as IgG. *Compare* **macroglobulin**.

β_2 -**microglobulin** or **betamicroglobulin** a human plasma protein, of 11.8 kDa and containing 99 amino-acid residues, that in size, primary structure, and tertiary structure strongly resembles a single domain of the constant region of immunoglobulin molecules: it is a constituent of class I **major histocompatibility complexes**. Plasma levels may be elevated in hepatobiliary disease, chronic active hepatitis, and alcohol-induced liver cirrhosis.

microgram 1 *symbol:* μg or (esp. in pharmacy) mcg; a unit of mass equal to 10^{-9} kilogram or 10^{-6} gram; formerly called **gamma** (*symbol:* γ). 2 an alternative name (sometimes) for **micrograph**.



micrograph

micrograph or (sometimes) **microgram** a graphic reproduction of the image of an object as seen through a microscope.

microhabitat any habitat that is small or limited in extent and differs in character from a larger, surrounding habitat.

microheterogeneous describing a preparation of a biopolymer that superficially appears to be homogeneous but that, on refined analysis, shows slight differences in size, sequence, charge, state of aggregation, or other properties of its molecules. Such differences are believed, at least in some instances, to be attributable either to artefactual changes or to genetic differences. —**microheterogeneity** *n.*

microincineration a technique for examining the distribution of minerals in tissue sections or microorganisms. The specimen is placed on a slide, the organic material burned away in a furnace, and the nature and position of the mineral ash determined by microscopy.

microinject to inject liquid into a single cell or nerve fibre using a micropipette. —**microinjection** *n.*

microiontophoresis iontophoresis (def. 2) applied to very small areas of tissue.

micro-Kjeldahl method an adaptation of the **Kjeldahl method** for determining small amounts of nitrogen, in the range 0.1–1.0 mg. A correspondingly small flask, commonly known as a **micro-Kjeldahl flask**, is used for the digestion, and a **Markham still** may be used for the steam distillation.

microlitre or (esp. US) **microliter** *symbol:* μL or μl ; a unit of volume equal to 10^{-6} litre; formerly called **lambda** (*symbol:* λ).

micromanipulation the techniques and practice of performing very delicate operations, such as microdissection, microinjection, or the isolation of single cells, under an optical microscope, usually with the aid of a **micromanipulator**.

micromanipulator an instrument used in conjunction with an optical microscope for holding and manipulating very small instruments and specimens in the field of view.

micrometer 1 any instrument or device for measuring very small distances or angles. 2 or **micrometer gauge** a device for measuring small distances, thicknesses, diameters, etc. with great accuracy. 3 the US spelling of **micrometre**.

micrometer burette a **micrometer syringe** adapted for use as a burette.

micrometer pipette a **micrometer syringe** adapted for use as a pipette.

micrometer screw a screw with a fine thread of accurate and consistent pitch.

micrometer syringe a syringe for the accurate delivery of small volumes of liquid in which the plunger of the syringe is driven by a calibrated micrometer screw.

micrometre or (US) **micrometer** *symbol:* μm ; a unit of length equal to 10^{-6} metre; formerly called **micron** (*symbol:* μ).

micromicro+ *symbol:* $\mu\mu$; a former name for **pico+**.

micromodification alteration of the covalent structure of a macromolecule, especially a histone, by the addition of various groups, e.g. acetyl, methyl, phosphate, after the initial biosynthetic process is terminated.

micromolar *symbol:* μM or μM ; describing a solution containing one micromole of solute per litre of solution, or a specified multiple or fraction thereof; used also of an amount-of-substance concentration similarly expressed.

micromole *symbol:* μmol ; a unit of amount of substance equal to 10^{-6} mole.

micromolecule any (type of) molecule composed of a relatively small number of atoms. In practice the term is usually taken to mean any (type of) molecule that is capable of passing through the pores of dialysis tubing (as generally used) and that thus has a mass less than about 10 kDa. —**micromolecular** *adj.*

micron (pl. **microns** or **micra**) *symbol:* μ ; a former term for **micrometre**.

micronucleus a eukaryotic cell nucleus with less than the normal complement of chromosomes, produced experimentally by incubation of a cell with colchicine. —**micronuclear** *adj.*

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micronutrient any chemical element that is required by living organisms in tiny quantities only, also known as **trace element**. The term is often extended to include organic compounds such as vitamins.

microorganism any noncellular or unicellular (including colonial) organism, most of which are too small to be seen with the unaided eye. Microorganisms comprise bacteria (including cyanobacteria), lichens, microfungi, protozoa, rickettsiae, virinos, viroids, and viruses, and also some algae; all prokaryotes are included. See also **microbe**.

microperoxisome (sometimes) a type of small **peroxisome**, 150–250 nm in diameter, found in all mammalian cells.

micropinocytosis (sometimes) a form of **pinocytosis** occurring in higher animals, especially in the transfer of dissolved macromolecules from the blood to the cytoplasm of cells, in which the pinosomes formed are very small (**micropinosomes**) and can only be observed in the electron microscope.

micropipette or (US) **micropipet** any pipette calibrated to contain or deliver a volume in the range 5–500 μL .

micropore any pore (in a gel or porous solid) whose width does not exceed about 2.0 nm. Compare **macropore**, **mesopore**. —**microporous** *adj.*

microprocessor a single integrated electronic circuit designed to carry out a specific set of operations, e.g. as the central processor of a computer or a control system.

minisatellite a repeating DNA segment that is similar to a **minisatellite** except that the overall range of size is smaller and the base core repeat unit involves a two or four nucleotide pair repeat motif. They are so named because of their resemblance to **satellite DNA** in having tandem repeats. The number of copies of the repeats varies not only between individuals but also between the two chromosomes of a pair.

microscope an instrument consisting of a lens or a combination of lenses that uses light or other electromagnetic radiation to make enlarged images of objects that are invisible to, or not easily seen by, the unaided eye. See also **electron microscope**, **phase-contrast microscopy**.

microscopic 1 of a size that is not visible (in detail) to the unaided eye. 2 concerned with small units. 3 relating to the behaviour of individual molecular entities. 4 or **microscopical** of, pertaining to, or using a microscope. Compare **macroscopic**. —**microscopically** *adv.*

microscopic equilibrium constant any constant describing the equilibrium of the interconversion of two particular molecular entities. Compare **intrinsic constant**, **macroscopic equilibrium constant**.

microscopic reversibility a thermodynamic principle requiring that at equilibrium the reaction pathway for the reverse of any particular chemical reaction is the exact opposite of the pathway for the forward reaction, i.e. both forward and backward reactions must pass through the same transition states or intermediates and occur with equal frequency.

microscopy the art and practice of using a microscope; investigation with a microscope.

microsecond *symbol:* μs ; a unit of time equal to 10^{-6} second.

microsequencer an apparatus for microsequencing (see **microsequence**).

microsequence to sequence (def. 3) a biopolymer using a very small sample, e.g. 5–100 pmol of a protein.

microsolute an operational term used to describe any diffusible solute in a solution that also contains nondiffusible macromolecular solute(s).

microsomal of, pertaining to, or containing **microsomes**; being in the **microsomal fraction** of a homogenate of a cell or tissue.

microsomal fraction an operational term for the subcellular fraction of a homogenate of a eukaryotic cell or tissue that, on differential centrifugation, is sedimented at 10^5 g; the fraction from some tissues, such as liver, although consisting largely of microsomes, commonly also contains free ribosomes and fragments of the plasma membrane, of the Golgi apparatus, of mitochondria, and of other subcellular structures. The morpho-

microsome

logical constituents differ according to the tissue from which the fraction was derived.

microsome any of the small, heterogeneous, artefactual, vesicular particles, 50–150 nm in diameter, that are formed when some eukaryotic cells are homogenized and that sediment on centrifugation at 10^5 g. Microsomes are formed mostly from disrupted endoplasmic reticulum membranes but some arise from the plasma membrane. Those formed from rough endoplasmic reticulum are studded with ribosomes on the outside (**rough microsomes**); those formed from smooth endoplasmic reticulum and from plasma membrane have no adhering ribosomes (**smooth microsomes**).

microspectrophotometer an instrument, consisting of a microscope and a spectrophotometer, used for measuring light of selected frequencies passing through, or reflected from, a very small specimen, e.g. a cell or a subcellular structure.

microspectrophotometry **spectrophotometry** applied to a very small specimen, using a microspectrophotometer. —**microspectrophotometric** *adj.*

microsphere one of a preparation of spherical particles of known and uniform size, usually in the range 10–40 μ m in diameter, and prepared from, e.g., serum albumin or an ion-exchange resin, that may be labelled with an appropriate radionuclide and used in studies of the circulation of the blood in health or disease.

microspore the smaller of the two types of spores produced by the heterosporous plants. In seed plants the microspore is the cell from which the pollen grain develops. —**microsporous** *adj.*

microstructure *an alternative term for fine structure.*

microsyringe a syringe for delivering volumes of gases or liquids in the range 0.1–500 μ L.

microtome an instrument for cutting thin sections of objects, e.g. tissues or cells, preparatory to microscopic observation.

microtrabecular lattice an irregular, three-dimensional lattice of very fine strands, up to 15 nm in diameter, sometimes termed **microtrabeculae**, that can be visualized in the cytoplasmic ground substance of eukaryotic cells by high-voltage electron microscopy. It forms part of the cytoskeleton.

microtubule any of the long, generally straight, hollow tubes of internal diameter 12–15 nm and external diameter 24 nm found in a wide variety of eukaryotic cells. Each consists (usually) of 13 protofilaments of polymeric **tubulin**, staggered in such a manner that the tubulin monomers are arranged in a characteristic helical pattern on the microtubular surface, and with the α/β axes of the tubulin subunits parallel to the long axis of the tubule. Microtubules exist in equilibrium with a pool of tubulin monomers in the cytoplasm and can be rapidly assembled and disassembled in response to various physiological stimuli; there are also a number of **microtubule-associated proteins**. The majority of microtubules in a cell appear to be initiated at **microtubule-organizing centres** and some may also become attached to the **kinetochores**. Microtubules are concerned in force generation in cilia and flagella, where they exist in a characteristic 9 + 2 array (a ring of nine doublet microtubules with two single microtubules in the centre), and the integrity of the microtubule network appears to be necessary for certain aspects of mitotic spindle function and for the saltatory movement of cell organelles. *Compare intermediate filament, microfilament.*

microtubule-associated protein *abbr.*: MAP; any protein that binds to **microtubules** and modifies their properties, including proteins that induce polymerization of purified tubulin and become associated with the newly formed microtubules. Many different kinds have been found, including structural proteins and motor proteins. MAP 1 is a structural protein involved in the filamentous cross-bridging between microtubules and other skeletal elements. Example, MAP 1A from rat: database code MAPA_RAT, 2774 amino acids (299.20 kDa). The role of MAP2 is unclear; in neuronal cells it is confined to the cell body and dendrites. It changes structurally during development by alternative splicing mechanism.

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anisms. Example from rat: database code MAP2_RAT, 1830 amino acids (198.83 kDa). **Dynein** is an example of a motor protein MAP. *Compare tau protein. See also MAP kinase.*

microtubule-organizing centre a region near the centre of a eukaryotic cell consisting typically of two **centrioles** at right angles to each other and surrounded by a complex of associated proteins. The centrioles and their satellite proteins serve as initiating sites for the assembly of microtubules.

microvilli a protein of the microvilli of rat mammary cells of M_r 200 000.

microvillus (*pl. microvilli*) any of the very small, finger-like projections that occur on the exposed surfaces of epithelial cells, especially where cellular function requires a maximal surface area for absorption or secretion, as in the proximal renal tubule and the small intestine, where they constitute the brush border. The projections are about 1–2 μ m long and 100 nm in diameter. The core of a microvillus contains a bundle of about 40 actin filaments of uniform polarity with their 'barbed' (+) ends anchored in a cap of ill-defined amorphous material in the microvillar tip. At the base the actin filaments extend into the perpendicular network, also made up largely of actin, known as the terminal web. In the core the actin filaments are bound together by the actin-binding proteins fimbrin and villin. The renal and intestinal microvilli contain a wide range of hydrolases that are integral membrane proteins; their hydrophilic domains, containing the catalytic sites, face outwards from the cell. —**microvillar** *adj.*

microvolume a very small volume, commonly in the range 1–100 μ L.

microwave radiation electromagnetic radiation of wavelengths in the range 1–300 mm (frequencies 1–300 GHz), lying between infrared radiation and most radiofrequency radiation.

middle molecule *a term sometimes used for* any compound in the range 350–2000 Da that accumulates in the body fluids of persons with uremia.

middle T antigen *see T antigen* (def. 1).

midkine *other names:* neurite outgrowth-promoting protein, midgestation and kidney protein, amphiregulin-associated protein (*abbr.*: ARAP); a heparin-binding growth/differentiation factor with neurite-promoting activity. It is a cell-surface or basement-membrane protein, induced by retinoic acid and similar to **pleiotrophin**. The midkine gene product is primarily detected in the embryonic period, whereas pleiotrophin is expressed most strongly in the early neonatal period. Example from human (precursor): database code MKHUMAN, 143 amino acids (15.58 kDa); residues 1–20 are the signal, 21–143 midkine.

midpoint **1** the point on a line, or a graph, that is equidistant from either end of the line or from two inflections in the graph. **2** a point in time that is half way between the beginning and the end of an event, or a series of events.

midpoint potential the **electrode potential**, at a given pH and temperature and 1 atm H_2 , at the midpoint of a redox titration when the activities of the reductant and the oxidant are equal.

MIF *abbr. for* **1** (macrophage) migration inhibition factor. **2** melanotropin release-inhibiting factor (i.e. **melanostatin**).

mifepristone *see RU-486.*

mutatase *an old name for* **mutase**; e.g. lysolecithin migratase is an old name for lysolecithin acylmutase (EC 5.4.1.1).

migrate (*in chemistry*) (of atoms, ions, or molecules) to move from one region of a solution to another, or from a solution in one compartment to a solution in another compartment, under the influence of an electric or a centrifugal field or by diffusion. —**migration** *n.*

migration inhibition factor *abbr.*: MIF; *see* (macrophage) migration inhibition factor.

MIH *abbr. for* melanotropin release-inhibiting hormone (i.e. **melanostatin**).

MIL the avian homologue of **RAF**; *v-mil* is the oncogene of the avian retrovirus Mill-Hill-2, which also carries *v-myc*. It codes for a serine/threonine kinase of the **RAF/MOS** subfamily of



milieu

unknown function but exhibiting homology with Src and related proteins (see **src**). Example, database code KMIL_CHICK, 647 amino acids (73.12 kDa).

milieu (pl. **milieux** or **milieux**) environment or setting.

milieu intérieur (French) the internal environment of a multicellular organism that surrounds the tissues and cells.

milk 1 the white liquid produced and secreted by the mammary glands of mature female mammals for the nourishment of their young. The milk of certain mammals, especially cows, is used as a food by humans, either as liquid milk or in the production of butter, cheese, etc. 2 any of various natural or manufactured liquids that have the appearance or consistency of milk, e.g. coconut milk, milk of magnesia. 3 to squeeze or draw milk from; to yield milk. —**milky** adj.

milk factor an alternative name for **Bittner factor**.

milk sugar an old name for D-lactose.

Miller index (in crystallography) any of a set of three integers that determine the position of a face, or internal plane, of a crystal in relation to three crystallographic axes. They are determined on the basis of the reciprocals of the intercepts of the face or plane on the crystallographic axes. [After W. H. Miller (1801–80).]

milli- 1 symbol: m; an SI prefix denoting 10^{-3} times. 2 prefix denoting one-thousandth.

millicurie symbol: mCi or (formerly) mc; a unit of (radio)activity or of radioactive material equal to 10^{-3} curie.

milliequivalent symbol: mequiv or meq or mEq; a unit of amount of substance equal to 10^{-3} equivalent (def. 3); (mmol recommended).

milligram symbol: mg; a unit of mass equal to 10^{-6} kilogram or 10^{-3} gram.

millilitre or (US) **milliliter** symbol: ml or mL; a unit of volume equal to 10^{-3} litre. It is now identical to the cubic centimetre; i.e. $1 \text{ mL} = 1 \text{ cm}^3 = 10^{-6} \text{ m}^3$.

millimetre or (US) **millimeter** symbol: mm; a unit of length equal to 10^{-3} metre.

millimetre of mercury symbol: mmHg; a non-SI unit of pressure defined as the pressure exerted by a column exactly 1 mm high of a fluid of density that is exactly $13.5951 \text{ g cm}^{-3}$ in a place where the acceleration of free fall is exactly $980.665 \text{ cm s}^{-2}$. Its use is deprecated. $1 \text{ mmHg} = 1 \text{ torr} = 133.32 \text{ pascal}$.

millimicro- a former name for **nano-**.

millimicron (pl. **millimicrons** or **millimicra**) symbol: mμ; a former name for **nanometre**.

millimolar symbol: mM or MM; describing a solution containing one millimole per litre of solution, or a specified multiple or fraction thereof; the term is also used of an amount-of-substance concentration similarly expressed.

millimole symbol: mmol; a unit of amount of substance equal to 10^{-3} mole.

Millipore 1 the proprietary name for a range of microporous and other types of membrane filters manufactured by the Millipore Corporation. 2 a water purification system, manufactured by the same company and based on reverse osmosis.

millisecond symbol: ms; a unit of time equal to 10^{-3} second.

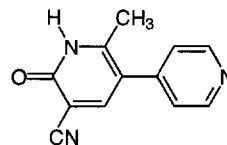
Millon's reaction the formation of red complexes when compounds containing the hydroxybenzene group are heated with **Millon's reagent**. The reaction is given by proteins that contain one or more tyrosine residues. [After Auguste Millon (1812–67), French chemist.]

Millon's reagent a 15% solution of mercuric sulfate in 15% sulfuric acid. This is added to the test sample solution, which is then heated in a boiling water bath for 10 min. After cooling, a 1% aqueous solution of sodium nitrite is added to the cooled reaction mixture. Originally it was a solution of mercuric nitrate in 50% nitric acid containing some nitrous acid. It gives **Millon's reaction** with compounds containing the hydroxybenzene group.

Mills perspective representation or **Mills representation** a diagrammatic way of unambiguously representing the structural formulae of carbohydrates and their polycyclic deriva-

tives to show on a plane surface the relative configurational arrangements of substituents and bridges. It is particularly useful in instances where the **Haworth representation** is inconvenient, e.g. for depicting saccharides with substituents that bridge two positions on the ring. The carbohydrate in question, or its derivative, is drawn with single rings and fused rings in the plane of the surface, and with bonds to substituents or bridges above or below the plane of the rings drawn with dark lines (often wedge-shaped) or broken lines, respectively (as is customary in representations of steroids and terpenes). Hydrogen atoms in the angular positions at ring junctions are always shown but those at points of substitution may be omitted. Like **Fischer projections**, but unlike Haworth representations, Mills representations may be rotated in the plane of the paper without alteration of their significance. It was proposed primarily for carbohydrate structures, but resembles systems used for other types of molecule, e.g. steroids.

milrinone 1,2-dihydro-6-methyl-2-oxo-5-(4-pyridinyl)nicotinonitrile; a phosphodiesterase inhibitor with vasodilating and cardiostimulant activity.



Milstein, César (1927–), Argentinian-born British biochemist and immunologist; Nobel Laureate in Physiology or Medicine (1984) jointly with N. K. Jerne and G. J. F. Köhler 'for theories concerning the specificity in development and control of the immune system and the discovery of the principle for production of monoclonal antibodies'.

mimetic imitating closely.

+mimetic suffix (esp. in pharmacology) denoting having a similar effect to the substance or action indicated.

min symbol for minute.

mince 1 to disrupt tissue with a **tissue grinder** to the level at which there is extensive, but incomplete, rupture of cells. 2 the preparation resulting from the action of a coarse tissue grinder.

mineral 1 any naturally occurring inorganic substance of a type often obtained by mining, e.g. coal, stone, petroleum. 2 any of a class of naturally occurring inorganic substances, of definite chemical composition and physical properties and often crystalline in character. 3 of, pertaining to, or being a **mineral** (def. 1, 2).

mineralize or **mineralise** (of an organic material) to convert into, or impregnate with, a mineral or minerals. —**mineralization** or **mineralisation** n.

mineralocorticoid or **mineralocorticosteroid** any naturally occurring or synthetic hormonal **corticosteroid** that acts primarily on water and electrolyte balance by promoting the renal retention of sodium ions and excretion of potassium ions. Mineralocorticoids are produced in the outer layer (zona glomerulosa) of the adrenal cortex. **Aldosterone** is the most potent of the naturally occurring mineralocorticoids. Compare **glucocorticoid**.

mini- comb. form denoting miniature; of smaller than usual dimensions.

minicell 1 a small cell-like body arising from a polar region of a rod-shaped bacterium by aberrant cell division. It is spherical or near-spherical in shape and devoid of chromosomal DNA; hence it does not grow or divide and is normally unable to synthesize nucleic acid or protein (but may do so if a plasmid is introduced). 2 an experimentally produced eukaryotic cell in which an intact cell nucleus is surrounded by scant cytoplasm and a cell membrane. It is prepared by enucleation of a normal mononucleate cell.

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miniexon

miniexon a small exon, as found in the troponin T gene.

minigastrin a form of the peptide hormone **gastrin** that contains 14 amino-acid residues. It has been isolated from tumour tissue and corresponds to residues 5–17 of human gastrin 1, having the sequence LEEEEAYGWLDHFH-amide.

minima a plural of **minimum**.

minimal of, being, or constituting the minimum; the least possible.

minimal medium a (synthetic) culture medium containing only those compounds essential for the growth of a wild-type organism and that is not able to support growth of an auxotroph.

minimum (*pl.* **minimums** or **minima**) **1** the least quantity, degree, or value possible or permissible. **2** (*in mathematics*) a value of a function that is less than that of any neighbouring value. **3** of, or pertaining to, a minimum or minimums.

minimum inhibitory concentration *abbr.*: MIC; the lowest concentration of a drug, or other substance, that just inhibits the growth of a given test organism (bacterium, etc.) in defined conditions.

minimum lethal dose *abbr.*: MLD; the minimum dose of a toxic substance, or organism, that will kill, within a specified time, all the animals in a test group to which it is administered.

minimum (relative) molecular mass or (*formerly*) **minimum molecular weight** the (relative) molecular mass of a compound as determined by the assay of one of its structural elements, e.g. a metal atom, a ligand, a terminal residue, etc., assuming that there must be at least one such structural element in each molecule of the compound.

minisatellite a repeat DNA segment comprising short head-to-tail tandem repeats giving **variable number tandem repeat**-type polymorphisms, with approximate size between 1 and 30 kb. The name derives from the fact that repeat DNA segments were originally found in **satellite DNA**, but the overall size of the minisatellite segments and the length of individual repeats are smaller, and satellite peaks are not seen on centrifugation.

minivial a very small vial, especially a low-capacity container that can be inserted into a normal-sized sample vial used in liquid scintillation counting, to enable the (radio)activity of small-volume samples to be determined more efficiently.

minor base *see* **minor nucleoside**.

minor groove the smaller and shallower of the two grooves on the surface of a double-helical DNA molecule. *Compare* **major groove**.

minor nucleoside a nucleoside containing any of a group of purine or pyrimidine bases that occur in small amounts in most nucleic acids but in relatively large amounts in transfer RNA molecules; they are sometimes known as 'odd bases'. These are a characteristic feature of **transfer RNA** from all organisms. The modifications range from the simple methylation of bases or ribose sugar residues to very complex substitutions. All modifications take place after transcription of the transfer RNA genes. The importance of the modifications is thought to be related to the stability of the tertiary structure. A typical modified nucleoside is **pseudouridine**. *See also* **wyosine**.

minuend (*in mathematics*) the number (or quantity) from which another number (or quantity), the subtrahend, is to be subtracted.

minus end the end of a microtubule or actin filament at which the addition of monomers occurs least readily; the 'slow-growing' end of a microtubule or actin filament. The minus end of an actin filament is also known as the pointed end. *Compare* **plus end**.

minus 10 region and **minus 35 region** regions in the promoter defined by the number of nucleotides 5' from the first nucleotide transcribed, i.e. on the **coding strand**. The first nucleotide (the start site) of a transcribed DNA sequence is denoted as +1, the second +2; the nucleotide preceding the start site is denoted as -1. The most conserved sequence upstream (towards the 5' terminus) of DNA in *Escherichia coli* is a hexa-

mer centred at the -10 position (*see* **Pribnow box**). It has a consensus sequence of TATAAT in which the leading TA and final T are highly conserved. -35 is the centre point of a consensus sequence involved in RNA polymerase recognition, TTGACA.

minus strand **1** the complementary strand to the parental polynucleotide strand – the **plus strand** (def. 1) – in bacteriophages whose genomes consist of duplex RNA. **2** any polynucleotide strand (RNA or DNA) in a viral genome that is complementary to the virus-specified mRNA – the **plus strand** (def. 2). **3** (*sometimes*) an alternative term for **coding strand** (of genetic duplex DNA).

minute **1** *symbol*: min; a non-SI unit of time equal to 60 seconds. **2** *symbol*: ' ; a unit of plane angle equal to ($\pi/10$ 800) radians, or 0.291 milliradians. There are 60 seconds (of arc) in one minute (of arc), and 60 minutes in one degree (of arc). **3** (*in genetics*) an empirical unit of distance between genes in a bacterial chromosome, being a measure of the time taken, in minutes, for the transfer of a particular gene during bacterial conjugation relative to an arbitrarily standardized selected origin.

7-minute phosphate or **10-minute phosphate** *see* **labile phosphate**.

MIP *abbr.* for macrophage inflammatory protein.

misacylate or **mischarge** to acylate (a transfer RNA molecule) with an aminoacyl group other than the specific one. —**misacylation** *n*.

mischarge an alternative term for **misacylate**.

miscible (of two or more particular liquids) capable of being mixed together to give a homogeneous liquid. Particular liquids may be completely miscible or miscible only in certain proportions. *See also* **proofreading**. —**miscibility** *n*.

mismatch repair a system for the correction of errors introduced during DNA replication when an incorrect base, which cannot form hydrogen bonds with the corresponding base in the parent strand, is incorporated into the daughter strand. **Excinucleases** recognize a pair of non-hydrogen-bonded bases and cause a segment of polynucleotide chain to be excised from the (less methylated) daughter strand, thereby removing the mismatched base. The resulting gap is then filled by the actions of DNA polymerase I and DNA ligase.

mispairing the occurrence of a base in one polynucleotide strand of duplex DNA that is not complementary to the corresponding base in the second polynucleotide strand.

missense codon a particular codon that has been altered so that it codes for an amino-acid residue different from the one for which it normally codes.

missense mutation any mutation causing a base substitution in a gene that results, following transcription and translation, in the incorporation of an incorrect amino-acid residue at a specific point in the polypeptide gene product. *Compare* **non-sense mutation**, **samesense mutation**.

mistranslation the insertion at a specific point in a polypeptide chain of an amino-acid residue that is not indicated by the corresponding codon in the mRNA molecule that is being translated.

MIT *abbr.* for (3-)monoiodotyrosine.

Mitchell, Peter Dennis (1920–92), British biochemist; Nobel Laureate in Chemistry (1978) 'for his contribution to the understanding of biological energy transfer through the formulation of the chemiosmotic theory'.

Mitchell hypothesis the hypothesis that mitochondrial systems can be driven by proton and electrical gradients that build up across membranes, a mechanism now referred to as **chemiosmotic coupling**.

mitochondria the plural of **mitochondrion**.

mitochondriac a biochemist with a chronic and unusually intense interest in mitochondria.

mitochondrial carrier protein any of various proteins involved in the translocation of small molecules across the mitochondrial inner membrane. Many are translocases using an



antiport mechanism, and include carriers for pyruvate (antiport molecule is OH^-), phosphate (malate), citrate (malate), phosphate (OH^-), ADP (ATP), aspartate (glutamate), and malate (2-oxoglutarate); an example is ADP/ATP translocase 1 of yeast: database code ADT1_YEAST, 309 amino acids (34.08 kDa); this is an integral inner membrane protein, comprising a homodimer with three homologous domains in each subunit. See also **OGCP**.

mitochondrial DNA *abbr.*: mtDNA; DNA that is contained, replicated, and expressed in mitochondria. Mitochondrial DNA is double-stranded in almost all organisms studied; that from mammals has a contour length of $\approx 5 \mu\text{m}$, about 15 kilobases, while that from protozoa and fungi is about five times larger. In yeast, mitochondrial DNA codes for two types of rRNA molecules, about 26 types of tRNA molecules, and about 10 proteins (which represent only about 5% of the total mitochondrial protein). The code in mitochondrial DNA differs in some respects from that in nuclear DNA (see **genetic code**).

mitochondrial fraction *an operational term* for the more rapidly sedimenting components (for example 10 000–15 000 g for 30 min) of the **cytoplasmic fraction** of a homogenate of eukaryotic cells or tissue. It generally consists predominantly of mitochondria but may also contain varying numbers of secretion granules, lysosomes, microbodies, or other intracellular organelles.

mitochondrial plasmid DNA a distinct class of genetic elements, found in certain wild-type strains of *Neurospora*, that show virtually no sequence homology with standard mitochondrial DNA (mtDNA) in DNA–DNA hybridization experiments and achieve high copy number without suppressive behaviour towards wild-type mtDNA.

mitochondrial matrix the gel-like material, with considerable fine structure, that lies in the matrix space, or lumen, of a **mitochondrion**. It contains the enzymes concerned with fatty-acid oxidation and enzymes of the tricarboxylic-acid cycle.

mitochondrial processing peptidase EC 3.4.99.41; an enzyme that catalyses the cleavage of the leader peptide from precursor proteins imported into the mitochondrion, typically ones with Arg in position P2 (see **peptidase P-sites**). These enzymes resemble in sequence **preurein-leader peptidase**. Example from *Neurospora crassa*; catalytic subunit (precursor): database code MPP1_NEUCR, 577 amino acids (62.96 kDa); enhancer subunit (precursor): database code MPP2_NEUCR, 476 amino acids (52.50 kDa). Both are transmembrane proteins; the enzyme requires the presence of divalent cations for activity.

mitochondrial RNA *abbr.*: mtRNA; any RNA molecules that are complementary to mitochondrial DNA.

mitochondrial state see **respiratory state (of mitochondria)**.

mitochondrion (*pl.* **mitochondria**) a semiautonomous, self-replicating organelle that occurs in varying numbers, shapes, and sizes in the cytoplasm of virtually all eukaryotic cells. It is notably the site of tissue respiration (see **respiratory chain**, **oxidative phosphorylation**). Conventionally, mitochondria are represented as elongated cylinders with a diameter of 0.5–1.0 μm ; however, in living cells they show great mobility and plasticity, in some cells forming long, moving chains while in other types of cells being fixed in position near sites of high ATP consumption; e.g. in cardiac muscle they are packed between the myofibrils while in a sperm they are tightly wrapped around the flagellum. Mammalian hepatocytes each contain 1000–2000 mitochondria, which occupy roughly 20% of the cell volume. A mitochondrion has two functionally distinct lipid-bilayer membrane systems, the outer one completely surrounding the whole organelle and the inner one being infolded into **cristae**. These membrane systems define two mitochondrial compartments: the mitochondrial matrix and the intermembrane space. The matrix contains enzymes for the oxidation of pyruvate and fatty acids and for the tricarboxylic-acid cycle; it also contains the mitochondrial DNA.

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and the enzymes and structures necessary for expression of the mitochondrial genes. The inner mitochondrial membrane contains the enzymes of the respiratory chain, ATP synthetase, and specific transport proteins. The intermembrane space contains a number of kinases. The outer membrane, which is permeable to molecules of up to 10 kDa, contains monoamine oxidase, cytochrome *b₅*, a number of transferases, and a fatty-acid elongation system. The mitochondrion was formerly known as a **chondriosome** and by numerous other names. See also **cercidosome**. —**mitochondrial** *adj.*

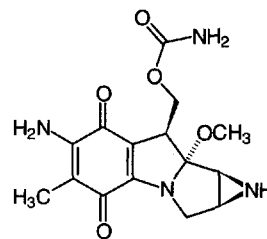
mitochrome a chromoprotein, or mixture of chromoproteins, released from mitochondria *in vitro* after ageing or other degradative processes. They are hemoproteins of $\approx 105 \text{ kDa}$ with 1 atom of Fe per molecule, and are thought to inhibit *in vitro* oxidative phosphorylation.

mitogen any substance or agent that induces or stimulates **mitosis**. —**mitogenic** or **mitogenetic** *adj.*

mitogenic protein *abbr.*: MP; an alternative name for **interleukin 1**.

mitogillin EC 3.1.27.-; *other names*: allergen I/A; IgE-binding ribotoxin; a purine-specific ribonuclease that attacks 28S rRNA. The protein is a powerful allergen, an inhibitor of eukaryotic protein synthesis, and has anti-tumour activity. Its action is similar to **restrictocin** and ribonuclease U2. Example (precursor) from *Aspergillus restrictus* and *A. fumigatus*: database code RNMG_AS PRE, 176 amino acids (19.57 kDa).

mitomycin C a 334 Da antibiotic and the principal member of a group of closely related aziridine antibiotics (mitomycins) produced by *Streptomyces caespitosus*. Mitomycin C is toxic to bacteria and mammalian cells and possesses antineoplastic activity. When administered, mitomycin C undergoes intramolecular rearrangement and reduction within cells to form a bifunctional alkylating agent that irreversibly cross-links the two strands of duplex DNA molecules, thereby inhibiting DNA synthesis.



mitoplast a structure consisting of the inner membrane and the matrix of a **mitochondrion**. It is formed by removing the outer membrane and the intermembrane material by treatment of a mitochondrion with nonionic detergents under defined conditions.

mitoribosome any of the bacterial-like **ribosomes** found in mitochondria.

mitosis or **karyokinesis** the division of a eukaryotic cell nucleus to produce two daughter nuclei that contain identical numbers of chromosomes and that are identical genetically to the parent nucleus except where crossing over or mutation has occurred. It is divided into five main stages: **prophase**, **prometaphase**, **metaphase**, **anaphase**, and **telophase**. Mitosis is normally accompanied by **cytokinesis** leading to division of the cell. Compare **meiosis**. —**mitotic** *adj.*

mitotic coefficient or **mitotic index** the proportion of cells in a population that are in **mitosis** at a given time.

mitotic recombination or **somatic crossing over** crossing over between homologous chromosomes during mitotic division of a diploid somatic cell. The result of such a recombination will be imperceptible if the cell is homozygous at the loci exchanged but may produce a marked effect if the dividing cell is



mitotic spindle

heterozygous at these loci. *See also* **Holliday junction recombination**.

mitotic spindle *see* **spindle**.

mixed acid fermentation a particular mode of bacterial fermentation of glucose that is characteristic of many members of the Enterobacteriaceae, e.g. *Escherichia*, *Salmonella*, and *Shigella*. It yields, as principal products, acetic, formic, lactic, and succinic acids, in varying proportions according to the organism and growth conditions.

mixed anhydride any **acid anhydride** formed between two different acids.

mixed-bed deionizer an apparatus for removing electrolytes from impure water (or an aqueous solution) and containing an intimate mixture of anion- and cation-exchange resins. It is frequently fitted with a device for measuring the electrolytic conductivity of the effluent.

mixed-bed resin a common but imprecise term for an intimate mixture of anion- and cation-exchange resins as used in a mixed-bed deionizer.

mixed-function oxidase or **mixed-function oxygenase** an older term for **monooxygenase**.

mixed indicator **1** a mixture, in definite proportions, of two acid-base indicators, or of an indicator and a dye, chosen so as to produce a sharp colour change over a narrow range of pH values. **2** a mixture, in definite proportions, of two or more acid-base indicators chosen so as to produce a series of colour changes over an extended range of pH values.

mixed inhibition or **mixed-type inhibition** inhibition of the activity of an enzyme that yields complex kinetic data that cannot be clearly categorized as either competitive or non-competitive. This can result e.g. when the enzyme-substrate-inhibitor complex undergoes the catalytic process slowly, or where binding of the inhibitor changes both k_{cat} and K_m .

mixed inhibitor or **mixed-type inhibitor** any inhibitor that produces **mixed inhibition** of the activity of an enzyme.

mixed labelled describing a specifically isotopically labelled compound that has more than one type of modified atom, e.g. $[^{13}\text{C}]\text{H}_3\text{-CH}_2[^{18}\text{O}]\text{H}$.

mixed-lymphocyte reaction or **mixed-leukocyte reaction** the reaction that occurs when leukocytes from two genetically differing individuals are cultured together. Small lymphocytes are transformed into immature blast cells, with synthesis of DNA, to an extent that is related to the degree of disparity between the histocompatibility antigens of the cell donors. This reaction forms the basis of a test for the acceptability of an allograft.

mixed triacylglycerol or (formerly) **mixed triglyceride** any **triacylglycerol** in which individual molecules contain acyl residues of more than one kind.

mixed-type inhibition an alternative term for **mixed inhibition**.

mixed-type inhibitor an alternative term for **mixed inhibitor**.

mixotropic series a series of solvents arranged on the basis of their ability to form hydrogen bonds. It is based on the extent to which two given solvents in this series will mix; near neighbours in the series are **isopartitive**.

mixture (in chemistry) homogeneous or heterogeneous material in any physical state that consists of two or more chemical components that have no firm chemical bonding between them and that are physically separable.

MK *abbr.* for menaquinone.

mks system or **MKS system** *abbr.* for metre-kilogram-second system (of electrical units).

mL or **ml** *symbol* for millilitre.

MLD *abbr.* for minimum lethal dose.

M line *see* **sarcomere**.

mM or **mm** *symbol* for millimolar.

M **macroglobulin** a paraprotein of IgM type that is produced by individuals with Waldenström's macroglobulinemia.

mmHg *symbol* for the conventional millimetre of mercury.

mmol *symbol* for millimole.

MMP *abbr.* for matrix metalloproteinase.

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MMPI *abbr.* for matrix metalloproteinase inhibitor.

MMTV *abbr.* for mouse mammary tumour virus.

Mn *symbol* for manganese.

mnemonic enzyme mechanism or **mnemonical enzyme**

mechanism an enzyme mechanism proposed to explain cooperativity towards substrate shown by monomeric enzymes with only one substrate-binding site. In this mechanism free enzyme is considered to exist in two conformations, I and II, that are only relatively slowly interconvertible and that differ in their substrate-binding affinities; the binding of substrate induces a change of the enzyme to a third conformation, III, which reverts to conformation I when product is released. The behaviour of the system depends on the rate of interconversion of I and II and on the relative rates at which these forms can bind substrate. If the binding by I is more rapid than that by II, the system will show positive cooperativity, i.e. the enzyme has 'remembered' the catalytically more effective conformation; if the converse, the system will show negative cooperativity.

MNSs system one of the blood-group systems of humans, in which the antigens are the **glycophorins** of the erythrocytes. The MN system was the second human-blood-group system to be discovered, the first being the ABH system (*see* **ABH antigens**). The M and N antigenic sites are poorly immunogenic and antibodies to M and N antigens do not normally occur in the blood. The determinants are located at the N terminus of glycophorins A and B (GPA and GPB); polymorphism between M and N antigens on GPA is attributed to amino-acid substitutions at residues 1 (Ser/Leu) and 5 (Gly/Glu), but the N-terminal Ser¹ and Leu¹ are the primary determinants for the M and N antigens respectively. GPB carries 'N' (N-like activity), S/s, and U blood-group antigens; 'N' activity resides at the N terminus of GPB, the sequences of GPA and GPB between residues 1 and 26 being identical; the S/s antigenic determinants arise from Met/Thr polymorphism at residue 29 of GPB; and the U antigen, present in all Caucasians and most Blacks, resides between residues 33 and 39.

Mo *symbol* for molybdenum.

mob the genetic notation for genes involved in the mobilization of a **plasmid** from cell to cell.

mobile element an alternative term for **transposon**.

mobile phase the phase that moves in a chromatographic system; it can include the portion of the starting sample present in this phase at any given time.

mobility **1** the property of being mobile. **2** *see* **electrophoretic mobility**.

mobilization or **mobilisation** **1** the act or process of mobilizing. **2** the process by which a conjugative plasmid brings about the transfer from one cell to another of DNA to which it is not stably and covalently linked.

mobilize or **mobilise** to bring (resources or reserves) into use for a particular purpose, e.g. to mobilize fat by lipolysis or glycogen by glycogenolysis.

mobilizing plasmid or **mobilising plasmid** an alternative name for **conjugative plasmid**.

Möbius strip or **Moebius strip** any one-sided continuous surface with a single bounding curve made by twisting a ribbon through 180° and joining the ends together. When divided lengthwise it forms another Möbius strip of double the length, and is a model for the replicative behaviour of a circular chromosome that contains a single twist. [After its inventor, August Ferdinand Möbius (1790–1868), German mathematician.]

Mo cell a relatively small cell, containing round, osmophilic granules, that is believed to produce **motilin**. Mo cells occur in the upper small intestine of humans, mainly in the crypts; they are sparse in the lower small intestine.

modal class (in a statistical distribution) the class that contains the **mode** (def. 2).

mode **1** any particular way or manner of existing or of doing something. **2** (in a statistical distribution) the value of the variable occurring with greatest frequency. **3** any of the sev-



eral wave frequencies that an oscillator can generate, or to which a resonator can be tuned to respond. —**modal adj.**; **modality n.**

modeccin a 57 kDa lectin present in the turnip-like root of the southern African plant, *Modecca digitata*. It is highly toxic to animals, its prime action being on the liver, and to animal cells in culture. It consists of two polypeptide subunits in disulfide linkage, the A (smaller) subunit being biologically indistinguishable from the A subunits of ricin and abrin. *See also* **ribosome-inactivating protein**.

model **1** a three-dimensional representation of an object, structure, etc., sized to be conveniently seen, handled, and studied. **2** a conceptual representation of a particular phenomenon, system, or set of experimental observations as an aid to understanding and as an object for test or for further experimentation. **3** an animal that, often as a result of mutation, mimics a pathological condition and can be used to study the pathogenesis of the condition. *See also* **transgenic**. **4** a mathematical representation of a particular phenomenon, system, or set of experimental observations as an aid to calculation and prediction. **5** to construct or create a **model** (def. 1, 2, 3). —**modelled or (sometimes US) modeled adj.**; —**modelling or (sometimes US) modeling n.**

model system any biological or biochemical system that is used for study because it is considered to be representative of one or more other (often more complex) systems in which similar phenomena occur or are believed to occur.

modem an electronic device, consisting of a combined modulator and demodulator, that is used to convert an output signal from one type of equipment (e.g. a computer) into a form suitable for input into another type, and vice versa. [From modulator–demodulator.]

moderator any substance that changes the rate of an enzymic or other reaction, either positively or negatively.

modification **1** (in biochemistry) *see* **covalent modification**. **2** (in genetics) any nonheritable change in the phenotype of an organism in response to variation of its environment; it includes **host-controlled modification**. *Compare* **mutation**.

modification enzyme any enzyme that effects **covalent modification**. *See also* **restriction–modification system**.

modification gene *see* **restriction–modification system**.

modification methylase a common name for any of the enzymes that methylate specific sites on a DNA molecule.

modification–restriction system an alternative name for **restriction–modification system**.

modifier an alternative term for **effector**.

modifier gene or **modifying gene** any gene that affects the phenotypic expression of a gene or genes at other loci.

modify **1** to alter the structure, character, activity, etc. of (something). **2** to convert a chemical compound into an analogue, especially by substitution. *See also* **modification**.

modular primer an unligated tandem arrangement of **primers** (usually two or three) used for the **polymerase chain reaction** and related experiments.

modulating codon any codon that controls the frequency with which a **cistron** or cistrons are transcribed.

modulation **1** adjustment or regulation of the degree or activity of something. **2** regulation of the frequency with which a **cistron** or cistrons is transcribed. **3** control of the activity of a regulatory enzyme by an effector. **4** (in physics) the process of changing the amplitude, frequency, or phase of a wave by combination with another wave.

modulator **1** something that modulates or brings about modulation. **2** any molecule that acts as an **effector**. **3** (in pharmacology) a ligand that increases or decreases the action of an agonist by combining with a distinct (allosteric) site on the receptor macromolecule.

modulator protein one provisional name for **calmodulin**.

module a standardized or self-contained part or unit of construction, especially one that can be used in alternative combinations with others; e.g. one of a set of such assemblies of

electronic components that comprise a larger piece of equipment. *See also* **Appendix E**. —**modular adj.**

moesin *abbr. for* membrane-organizing extension spike protein; a protein, isolated from bovine uteri, with one domain similar to one in **ezrin**, band 4.1, and related proteins. Moesin occurs in many cell types and tissues and is involved in connections of major cytoskeletal structures to the plasma membrane. Example from human: database code MOES_HUMAN, 576 amino acids (67.61 kDa).

Moffitt–Yang equation or **Moffitt equation** a phenomenological equation, related to the **Drude equation**, that describes the anomalous variation of optical rotatory dispersion with wavelength shown by polypeptides containing helical sections. It is used to provide an estimate of the alpha-helical content of polypeptides and proteins. If λ is the wavelength of observation and $[m']_\lambda$ is the reduced **mean residue rotation** at that wavelength,

$$[m']_\lambda = \{a_0\lambda_0^2/(\lambda^2 - \lambda_0^2)\} + \{b_0\lambda_0^4/(\lambda^2 - \lambda_0^2)^2\},$$

where a_0 , b_0 , and λ_0 are empirical constants; a_0 is a function of contributions to the optical rotation from both the helix and the residue side-chains, as well as from the solvent, and b_0 and λ_0 are related principally to the amount of helical structure in the molecule. A value of λ_0 is chosen to give the best straight line when $[m'](\lambda^2 - \lambda_0^2)$ is plotted against $1/(\lambda^2 - \lambda_0^2)$, a good fit usually being obtained if $\lambda_0 = 212$ nm when $\lambda = 350$ –600 nm. The slope of this line, $b_0\lambda_0^4$, gives a value of b_0 , whence by reference to the values found for nonhelical and fully helical synthetic polypeptides an estimate may be made of the alpha-helical content of a polypeptide or protein under investigation. [After William E. Moffitt (1925–58), British physical chemist and Jen Tsi Yang (1922–95), who formulated it in 1956.]

Mohr pipette a cylindrical pipette calibrated for delivery from the zero mark to a particular chosen graduation mark. [After Carl Friedrich Mohr (1806–1879), German chemist.]

Mohr's salt ammonium ferrous sulfate; di-ammonium iron(II) sulfate hexahydrate; $(\text{NH}_4)_2\text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$.

moiety either of two, usually distinctive, component parts of a complex molecule; e.g. the steroid and saccharide moieties of a cardiac glycoside.

mol *symbol for* mole.

mol. *abbr. for* 1 molecular. **2** molecule.

μmol *symbol for* micromole.

molal **1** denoting the concentration of a solute (or a solution as a whole) expressed as the amount of substance of the solute divided by the mass of solvent. It is measured in moles per kilogram. **2** denoting an amount-of-substance **concentration** measured in such terms. *Compare* **molar**.

molality *symbol: m or b*; a measure of concentration expressed as the number of moles (of a specified solute) per kilogram of solvent. Molality is independent of temperature. *See also* **molal**.

molar **1** denoting an extensive physical quantity that is measured per mole (of a specified substance). It is usually denoted by the subscript m in the symbol; e.g. **molar volume** is the volume of one mole of a substance, and the symbol is V_m . *Compare* **specific**. **2** denoting an intensive physical quantity that is measured per mole (of a specified substance); i.e. divided by concentration. Use of the term molar in this sense is no longer recommended, except in the terms **molar absorption coefficient** and **molar conductivity**. **3** denoting the concentration of a solute (or a solution as a whole), expressed as the amount of substance of the solute divided by the mass of the solution. The recommended units are moles per litre but the convenient symbols M and m are frequently used. *Compare* **molal**.

molar absorptancy index a former name for **molar absorption coefficient**.

molar absorption coefficient either the **molar decadic absorption coefficient** (*symbol: ε*) or the **molar Napierian absorption coefficient** (*symbol: κ*); these are, respectively, the decadic **absorbance** or the Napierian absorbance of a substance, or of a



molar absorptivity

solute in solution, (at a specified wavelength (or frequency) and temperature) divided by the optical path length (in cm) of the medium and by the amount-of-substance concentration of the substance; i.e. the decadic or Napierian absorption coefficients of a medium whose concentration is expressed in moles per litre. *Former names:* molar absorptancy index; molar absorptivity; molar extinction coefficient; molecular extinction coefficient. *See also* **absorption coefficient**.

molar absorptivity a former name for **molar absorption coefficient**.

molar activity an **activity** (def. 1) of a substance or material divided by its amount of substance (in moles). *Compare* **specific activity**.

molar catalytic activity the **catalytic activity** of an enzyme, under specified conditions, divided by the amount of substance (in moles), either of enzyme catalytic centres or of multi-centre molecules; in the latter case it is important to specify whether the measurements refer to catalytic centres or to molecules. It may be expressed in katal per mole of enzyme. If the conditions specified are such that the enzyme is saturated with substrate, the expression (which then replaces the obsolete term turnover number) will be numerically equal to the **catalytic constant**, in s^{-1} , for decomposition of the enzyme-substrate complex into enzyme and products, although its interpretation may be complex. *Compare* **molar enzymic activity**, **specific catalytic activity**.

molar concentration an old but now incorrect term for **amount-of-substance concentration** measured in moles per litre. *See* **molar** (def. 2).

molar conductivity *symbol:* Λ ; the **electrolytic conductivity** of a (specified) electrolyte divided by its amount-of-substance concentration. The SI derived unit is the $S\ m^{-1}$.

molar decadic absorption coefficient *symbol:* ϵ ; *see* **molar absorption coefficient**.

molar ellipticity or (formerly) **molecular ellipticity** *symbol:* $[\theta]$; an expression of circular dichroism, $\Delta\epsilon$, at a specified wavelength, λ . It is given by the relation: $[\theta]_{\lambda} = 3299\ \Delta\epsilon_{\lambda}$.

molar enzymic activity (*obsolete*) the phenomenological coefficient relating the **enzymic activity** (under specified conditions) to the molar amount of enzyme substance; it is expressed in katal per mole of enzyme. It has now been superseded by **molar catalytic activity**.

molar extinction coefficient a former name for **molar absorption coefficient**.

molarity an older (still widely used) term for **amount-of-substance concentration** expressed in mol dm^{-3} or mol L^{-1} . *See* **concentration** (def. 1b), **molar** (def. 3).

molar mass the mass of a substance divided by its **amount of substance**; it is commonly expressed in grams (sometimes kilograms) per mole. *Compare* **molecular mass**, **relative molecular mass**.

molar Napierian absorption coefficient *symbol:* κ ; *see* **molar absorption coefficient**.

molar optical rotation *symbol:* $[m]$ or $[M]$; the observed optical rotation, α_{obs} , of a substance, or of a solute in solution, (at a specified wavelength (or frequency) and temperature) divided by the optical path length (in dm), l , of the medium and by its amount-of-substance concentration (in mol dm^{-3}), c . It is given by:

$$[m] = \alpha_{\text{obs}}/cl = [\alpha]_{\lambda}^0 M_r/100$$

in $\text{deg dm}^2\ \text{g}^{-1}$, where $[\alpha]_{\lambda}^0$ is the **specific optical rotation**.

molar optical rotatory power *symbol:* α_m ; it is given by the relation $\alpha_m = \alpha/cl$ in $\text{rad m}^2\ \text{mol}^{-1}$, where α is the observed rotation in radians, c is the amount-of-substance concentration in mol dm^{-3} , and l is the optical path length of the sample in m. *See also* **specific optical rotatory power**.

molar volume *symbol:* V_m ; the volume occupied by a substance, under specified conditions of temperature and pressure, divided by its **amount of substance**; it is commonly expressed in litres per mole.

mold the US spelling of **mould**.

mole *symbol:* mol; the SI base unit of **amount of substance**, defined as the amount of substance of a system that contains as many elementary entities as there are atoms in 0.012 kilogram of carbon-12; the elementary entities must be specified and may be atoms, molecules, ions, electrons, other particles, or specified groups of such particles. One mole of a compound has a mass equal to its **relative molecular mass** in grams.

molecular 1 of or pertaining to a molecule or molecules. 2 consisting of or existing as molecules; (especially of certain gaseous elements) present or reacting as intact molecules rather than as atoms, ions, etc., e.g. molecular oxygen (i.e. dioxygen). 3 of dimensions comparable to those of molecules. 4 (in physical chemistry) an obsolete term meaning divided by molecular weight or expressed per gram-molecule; **molar** now used. 5 considered in terms of or at the level of molecules, or in terms of chemical substances or structures and their inter-conversions; e.g. molecular biology, molecular medicine.

+molecular comb. form indicating the **molecularity** of a chemical reaction (when qualified by a multiplicative prefix), e.g. unimolecular, bimolecular, termolecular.

molecular biology the science that strives to understand the chemical and physical basis of biological specificity and variation, especially with regard to the structure, replication, and expression of genes, and to the structure, interaction, and physiological function of gene products. —**molecular biologist** n.

molecular chaperone *see* **chaperone**.

molecular cloning or **DNA cloning** the isolation in a bacterial cell of a fragment of any heterologous DNA in covalent linkage with a replicon (plasmid, phage, etc.) in order to obtain a homogeneous population (i.e. clone) of DNA molecules from the progeny of such a cell. The DNA to be cloned may be a fragment of genomic DNA obtained by restriction endonuclease digestion, or a complementary DNA (cDNA). The foreign DNA is inserted into the **cloning vector** which has been cleaved with a restriction enzyme; the gaps between vector and insert DNA are sealed with DNA ligase. In cloning using plasmid vectors, plasmids are then reintroduced into host cells by **transformation**; markers on the cloning vector allow selection of plasmid-containing cells and identification of those cells that contain recombinant plasmids (rather than 'empty' vectors). To identify clones containing particular sequences, DNA from individual colonies may be screened with a labelled DNA probe by **colony hybridization**. In cloning using bacteriophage vectors, the ligated insert and vector are packaged into phage capsids *in vitro*, and then these are used to infect host bacteria.

molecular disease any disease caused by an abnormality in a single protein, usually an enzyme. This component may have an abnormal structure that makes it functionally less efficient or harmful to the organism, or it may have a normal structure but be present in reduced amount. An example is **sickle-cell anemia**. Overproduction of proteins, e.g. of oncogene products, may also lead to a pathological situation.

molecular distillation distillation at very low pressures (at or below $1.3\ \text{N m}^{-2}$) when the mean free path of the molecules is of the same order as the distance between the surface of the liquid being distilled and the distillate; the molecules can therefore travel rapidly from the liquid to the distillate with relatively few collisions, and with a reduced risk of thermal denaturation or decomposition. The technique is useful in the separation and purification of fat-soluble vitamins and other natural products.

molecular dynamics the study of the relative intramolecular differences in the motions of the components of (macro)molecules.

molecular ellipticity a former term for **molar ellipticity**.

molecular endocrinology endocrinology at the molecular level.

molecular entity any chemically or isotopically distinct atom, molecule, ion, ion pair, radical, radical ion, complex, con-

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molecular enzymology

former, etc., capable of existence as a separately distinguishable entity.

molecular enzymology (*sometimes*) the aspects of enzymology concerned with kinetics and mechanisms.

molecular exclusion chromatography a less common name for gel-permeation chromatography.

molecular extinction coefficient a former name for molar absorption coefficient.

molecular formula a chemical formula giving the kinds and number of atoms in one molecule of a specified substance.

molecular fragmentography an alternative (less satisfactory) name for selected-ion monitoring.

molecular genetics the branch of genetics that attempts to describe genetic events in molecular terms.

molecular graphics *functioning as sing.* the use of computer programs for visualizing (macro)molecules, refining structures and molecular design, protein engineering, etc. Suites of programs that are commonly used include Insight, Ribbons, Midas, and O.

molecular hybridization an alternative name for hybridization analysis.

molecular ion any ion produced, especially in a mass spectrometer, when a molecule loses or gains an electron, forming a molecular cation or molecular anion, respectively.

molecularity the number of molecular entities involved in a simple, one-step, chemical reaction (it is not used to describe an overall reaction with more than one step). In the two-step reaction: $A + B \rightarrow C \rightarrow D$, the molecularity of the first step is two, i.e. it is bimolecular, and the molecularity of the second step is one, i.e. it is unimolecular.

molecular mass the mass of one molecule of a specified substance expressed in (unified) atomic mass units or daltons. Compare **relative molecular mass**, **molecular weight**.

molecular matchmaker any of a class of proteins, also called matchmaker proteins, that cause a conformational change in one or both components of a DNA-protein binding pair to promote formation of a metastable DNA-protein complex. ATP hydrolysis occurs in the process. Such matchmaking is used extensively in repair, replication, and transcription of DNA.

molecular medicine (*sometimes*) the branch of medicine concerned with the biochemical mechanisms of disease processes and therapy.

molecular orbital a wave equation for one electron describing its movement in the effective field provided by the nuclei and all other electrons of a molecular entity of more than one atom.

molecular pathology the use of techniques of molecular biology to enhance the understanding, diagnosis, and therapy of disease (mainly with respect to humans).

molecular pharmacology the use of techniques of molecular biology to enhance the understanding of the mechanism of action of existing drugs, and with the help of molecular graphics to predict the structure of novel drugs, especially compounds that might bind to proteins of known structure.

molecular pK any one of the **pK** values of an acid with two or more dissociations. Compare **group pK**, **titration pK**.

molecular recognition the specific recognition of one molecule by another; it often includes **chiral recognition**.

molecular ruler 1 any rigid oligoproline crosslinking reagent of defined length. Such reagents are useful in probing the structures of biological macromolecules in solution. 2 any molecule that regulates length, e.g. **nebulin**.

molecular sieve any of a class of substances or materials, whether naturally occurring or artificial, that contain pores of molecular dimensions and of a closely defined range of diameters. They are useful for the separation of substances within a particular range of molecular dimensions on the basis of selective exclusion.

molecular-sieve chromatography any form of chromatography in which a separation is based mainly upon differences in molecular size and/or shape of the substances being separated.

It includes **gel-permeation chromatography** and separations using **zeolites**.

molecular-sieve electrophoresis (*sometimes*) an alternative term for **polyacrylamide gel electrophoresis**.

molecular weight *abbr.*: mol. wt. or M.W.; a former name for **relative molecular mass**.

molecular weight cut-off or **cut-off** a jargon term for the nominal maximal value of the relative molecular mass of a macromolecule that can pass through an ultrafiltration membrane of a particular grade.

molecular weight marker or **molecular weight standard** a jargon term for any soluble polymer of accurately known relative molecular mass that serves as a reference in estimations of M_r values of polymers by techniques such as gel permeation chromatography or gel electrophoresis. Generally a set of such substances, covering a range of M_r values, is used.

molecule a structural unit of matter consisting of one or more atoms; the smallest discrete part of a specified element or compound that retains its chemical identity and exhibits all its chemical properties.

mole fraction an alternative name for amount-of-substance fraction. See **concentration**.

mole percent the quantity mole fraction multiplied by 10^{-2} ; usage no longer recommended.

Molisch test a qualitative test for free or combined aldohexoses. It depends on the formation of a purple colour when the furfural derivatives, formed when aldohexoses are heated with strong acids (e.g. H_2SO_4), react with sulfonated 1-naphthol. [After Hans Molisch (1856–1937), Austrian chemist.]

mollicute see **mycoplasma**.

molt the US spelling of **moult**.

molting hormone the US spelling of **moulting hormone** (i.e. **ecdysterone**).

mol. wt. *abbr.* for **molecular weight** (alternative to M.W.).

molybdate 1 the bivalent anion, MoO_4^{2-} , derived from molybdic acid. 2 any salt of molybdic acid.

molybdenum *symbol*: Mo; a metallic transition element of group 6 of the (IUPAC) **periodic table**; atomic number 42; relative atomic mass 95.94. It consists of a mixture of eight stable nuclides of which the commonest is that of mass number 98. It forms molybdates (molybdenum(VI)) and can also exist in various lower oxidation states. Molybdenum is of low abundance in the Earth's crust and is an essential trace component of living material, where it participates in numerous redox reactions. It is a constituent of component I (or molybdoferredoxin) of the **nitrogenase** (EC 1.18.6.1) complex from legume nodule bacteria (bacteroids), which converts nitrogen into ammonia. There are many artificial radionuclides of molybdenum, of which the most widely used is molybdenum-99.

molybdo+ *comb. form* denoting containing molybdenum, e.g. molybdoprotein.

molybdoferredoxin an alternative name for component I of **nitrogenase**.

molybdophosphoric acid or **phosphomolybdic acid** a heteropolyacid of approximate formula $24MoO_3 \cdot P_2O_5 \cdot xH_2O$, where x is approximately 50. It is useful in reagents for alkaloids, xanthine, uric acid, creatinine, etc.

mon+ a variant form of **mono+** (sometimes before a vowel).

Mon *symbol* for the montanoyl group.

monactin see **nonactin**.

monatomic or **monoatomic** (of a molecular entity) composed of only one atom.

monellin an ≈ 11 kDa protein consisting of two polypeptide chains of the same size but different sequence, linked noncovalently, isolated from the fresh fruits of the African shrub *Dioscoreophyllum cumminsii*, known as serendipity berry. It is intensely sweet-tasting, being 3000 times sweeter than sucrose on a mass basis, and has sequence similarities with **thaumatin**. A chain: database code MONA_DIOCU, 50 amino acids (5.83 kDa); B chain: database code MONB_DIOCU, 50 amino acids (5.83 kDa).

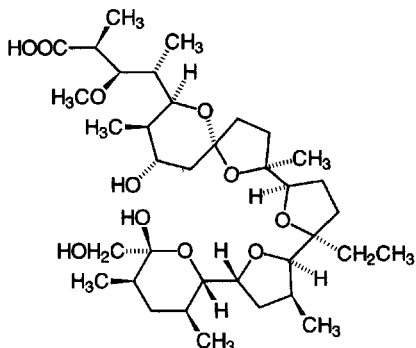
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monensin a 671 Da, open-chain macrotetrolide-like antibiotic, with many chiral centres, isolated from *Streptomyces cinnamonensis*. It forms stable complexes with monovalent cations, especially with Na^+ , and exhibits high Na^+/K^+ selectivity. These complexes are soluble in organic solvents and thus monensin can act as an **ionophore** for monovalent cations. It is used as a digestion-modifying feed additive in ruminants.



Monera the kingdom containing all prokaryotic organisms (i.e. bacteria and cyanobacteria) in the five-kingdom system of classification. —**moneran** *adj.*, *n*.

mongolism an older name for **Down's syndrome**.

monitor **1** any instrument or device for checking, controlling, measuring, or keeping a record of some varying biological activity or chemical or physical property, e.g. absorbance, (radio)activity, concentration, pH, respiration. **2** to serve as a monitor of (some activity or property); to use a monitor.

mono- or (sometimes before a vowel) **mon+** *comb. form* **1** denoting one, single, onefold, once, alone; **uni+**. **2** (in adjectives describing physical attributes) homogeneous; e.g. monochromatic, monodisperse, monoenergetic. **3** (in generic names of chemical compounds) denoting that member of a series (a) having only one specified substituent or other chemical modification, e.g. monoacylglycerol, monoamine; or (b) comprised of only one constitutional unit, e.g. monomer, monosaccharide. **4** informal term (in specific names of chemical compounds) indicating or emphasizing the presence in a molecule of a single specified atom, group, or residue; e.g. moniodotyrosine, adenosine monophosphate, monosodium dihydrogen phosphate. **5** (in adjectives describing compounds or substances) having a single function of the kind specified; e.g. monoacidic, monobasic, monodentate, monohydric. **6** (in names of processes or enzymes) concerned with or acting on, one or only one chemical group of the kind specified; e.g. moniodination, monoesterase.

monoacidic (of a **base** (def. 1)) able to accept only one proton per molecule in solution.

monoacylglycerol or (formerly) **monoglyceride** any ester of glycerol in which any one of its hydroxyl groups has been acylated with a fatty acid, the other being non-esterified.

monoacylglycerol pathway a pathway in which diacylglycerols and triacylglycerols are formed by sequential acylation of monoacylglycerol by acyl-CoA. It was initially studied especially in cells of mammalian intestine, and is important during fat absorption since monoacylglycerol is then extensively absorbed from the gut; however, it is also present in a large number of other mammalian tissues. There is little evidence for this pathway in plants.

monoamine oxidase *abbr.*: MAO; a common name for amine oxidase (flavin-containing), EC 1.4.3.4; an enzyme, widely distributed in animal tissues, that catalyses the oxidative deamination of primary amines by a reaction between dioxygen and $\text{R-CH}_2\text{-NH}_2$ to form R-CHO , NH_3 , and H_2O_2 . It also acts on secondary and tertiary amines. It is important in the catabol-

ism of neuroactive amines, e.g. epinephrine, norepinephrine, serotonin (5-hydroxytryptamine), and tyramine. Example from *Rattus norvegicus*: database code AOFA_RAT, 526 amino acids (59.51 kDa).

monoamine oxidase inhibitors compounds that inhibit **monoamine oxidase** (*abbr.*: MAO) and are useful as thymoleptic (antidepressant) drugs. They act by increasing the concentrations of **norepinephrine** and **serotonin**, which, because of the block on their metabolism, accumulate in the synaptic gap and exert an increased postsynaptic effect. They were discovered as a result of the observation of the euphoric side effect of the antitubercular drug **isoniazid**. The main groups of these drugs are (1) the **cyclopropylamines**, including tranlycypromine (*trans*-phenyl-cyclopropylamine); and (2) the **propargylamines**, including pargyline (*N*-benzyl-*N*-methyl-2-propynylamine). **Harmine** and related carboline alkaloids are reversible MAO inhibitors but are not used therapeutically.

monoatomic a variant spelling of **monatomic**.

monobactam any of a family of *N*-sulfonated monocyclic β -lactam antibiotics produced by a range of bacterial species.

monobasic **1** (of an acid) able to yield only one hydrogen ion per molecule in solution. **2** (of a salt or ester of a polybasic acid) derived by the replacement of only one of its dissociable hydrogen atoms per molecule.

monochromatic **1** or **homochromatic** involving, possessing, or relating to a single colour; consisting of, relating to, or emitting light (or other electromagnetic radiation) of a single frequency or wavelength. **2** an alternative term for **monoenergetic**. Compare **heterochromatic** (def. 1), **polychromatic**.

monochromator a device used for producing light (or other electromagnetic radiation) of essentially one frequency or wavelength, which may be fixed or selectable.

monocistronic containing as much genetic information as is contained in a single **cistron**.

monocistronic messenger RNA any messenger RNA molecule that carries one initiation site only and encodes the synthesis of one polypeptide chain only. It is characteristic of eukaryotes, which do not possess **polycistronic messenger RNA**.

monoclonal **1** of, pertaining to, being, formed by, or derived from a single **clone** (def. 1). **2** (sometimes) an informal name for **monoclonal antibody**.

monoclonal antibody or (sometimes, informal) **monoclonal** an immunoglobulin secreted by a single clone of antibody-producing cells, either *in vivo* or in culture. Such antibodies are **monospecific**; they can thus be useful as immunochemical reagents for the detection or assay of particular antigens. See also **hybridoma**.

monocomponent insulin a commercial preparation of natural insulin that has been highly purified by chromatography and is thus made essentially free of proinsulin, intermediates in the conversion of proinsulin to insulin, and breakdown products of insulin, as well as any unrelated contaminants.

monocotyledon or **monocot** any flowering plant having a single cotyledon (seed leaf) in the embryo. Such plants comprise the class **Monocotyledonae**.

monocyclic having a single ring of atoms in the molecule.

monocyte a large leukocyte of the **mononuclear phagocyte system**, 16–22 μm in diameter (in humans), with pale-staining cytoplasm and a reniform or horseshoe-shaped nucleus. Found in bone marrow and the bloodstream, monocytes are derived from pluripotent stem cells and become macrophages when they enter the tissues.

Monod, Jacques Lucien (1910–76), French microbiologist and molecular biologist; Nobel Laureate in Physiology or Medicine (1965) jointly with F. Jacob and A. M. Lwoff 'for their discoveries concerning genetic control of enzyme and virus synthesis'.

monodisperse describing a colloidal system in which the dispersed phase consists of particles all of (nearly) the same size. —**monodispersity** *n*.

Monod-Wyman-Changeux model (for **allosteric interactions**)

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abbr.: MWC model; a model describing the nature of **allosteric** interactions in proteins. It requires an allosteric protein to be an oligomer, the protomers of which are associated in such a way that they all occupy equivalent positions. Each protomer has only one stereospecific binding site for each ligand. The protein can exist in either of two conformational states, the T form (**tense form**), the predominant form when unligated, and the R form (**relaxed form**); these states are in equilibrium. The affinity of the R form for ligand is higher than that of the T form. All binding sites in each state are deemed to be equivalent and to have identical dissociation constants, K_R and K_T for the R and T forms, respectively. The sigmoidal binding curve for any allosteric protein and a given ligand can be calculated from the allosteric constant, L , equal to the ratio $[T \text{ form}]/[R \text{ form}]$ for the unligated states and the two dissociation constants. [After J. **Monod**, Jeffries Wyman (1901–95), US biochemist, and Jean-Pierre Changeux (1936–), French biochemist.] See also **mnemonic enzyme mechanism**.

monoenergetic or **monochromatic** (of moving particles and sometimes also photons) all having the same kinetic energy.

monoenoic denoting any alkenyl carboxylic acid containing one carbon–carbon double bond per molecule.

monoester any simple ester; i.e. any **ester** formed by condensation of one molecular proportion of an alcohol or phenol with one of an oxoacid.

monoesterase see **phosphomonoesterase**.

monofunctional having only one function or one reactive chemical group.

monoglyceride a former name for **monoacylglycerol**; its use is discouraged as it does not convey the intended meaning.

monognatobiotic an alternative term for **axenic**.

monohydric or **monohydroxy** describing any chemical compound containing one hydroxyl group per molecule. It is used especially of alcohols.

moniodinated reacted with or containing only one atom of iodine per molecule.

monokine an alternative name for a **cytokine** produced by monocytes.

monolayer 1 an alternative name for **monomolecular layer**. 2 a single layer of cells grown or growing in culture.

monomer 1 any substance that can provide one or more (in number or species) of the monomeric units of an **oligomer** (def. 1) or of a **polymer** (def. 1); a molecule of such a substance. 2 a loose term for any of the component molecules (identical or nonidentical) formed by the complete dissociation of a macromolecule with quaternary structure. 3 (in molecular biology) a any protein that is made up of nonidentical structural units. b any of the structural units formed by dissociation of an oligomeric protein and corresponding to a **protomer** in the undissociated protein. —**monomeric** *adj.*

monomeric unit a group of atoms, derived from a molecule of a given **monomer** (def. 1), that comprises any one species of constitutional unit of a polymer.

monomolecular 1 relating to, consisting of, or involving a singular **molecular entity**. 2 (of a reaction) having a **molecularity** of one; unimolecular.

monomolecular layer or **monolayer** a layer of a substance or substances that is one molecule thick.

mononuclear 1 (of a cell) having one nucleus. 2 (of a metal–ligand complex) containing a single central metal atom.

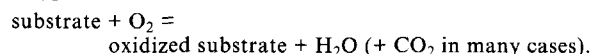
mononuclear phagocyte system a cell system in higher animals that comprehends all the highly phagocytic mononuclear cells and their precursors. As well as the free macrophages, it includes precursor cells and promonocytes of the bone marrow, monocytes of the bone marrow and the blood, and the tissue macrophages. The term has been proposed as a replacement for **reticuloendothelial system**, now held to lack precision.

mononucleosis see **Epstein–Barr virus**.

mononucleotide any **nucleotide** (def. 1); the term is used especially as the generic name of the constitutional repeating unit of all oligonucleotides, polynucleotides, and nucleic acids. A

mononucleotide consists of the (3′- or 5′)-phosphate of a single ribonucleoside or deoxyribonucleoside. The term is extended generally to include: (1) nucleoside oligophosphates, e.g. adenosine 5′-triphosphate (ATP); (2) nucleoside diphosphate sugars, e.g. uridine 5′-diphosphate glucose (UDPG); (3) nucleoside 2′,3′- and 3′,5′-(cyclic)phosphates, e.g. adenosine 3′,5′-phosphate (cyclic AMP); (4) nucleoside phosphates derived from artificial heterocyclic bases, e.g. 5-iodo-2′-deoxycytidine 5′-triphosphate, or from ones that do not occur naturally in nucleic acids, e.g. inosine 5′-phosphate (i.e. inosinic acid; hypoxanthine riboside 5′-phosphate; IMP); and (5) certain analogous compounds such as those containing, in place of a residue of ribose or deoxyribose, a residue of ribitol, e.g. flavin mononucleotide (i.e. riboflavin 5′-phosphate; FMN), of dideoxyribose, e.g. 2′,3′-dideoxyadenosine 5′-triphosphate (i.e. ddATP), or of another pentose, e.g. cytosine arabinoside 5′-triphosphate (i.e. ara-CTP). Compare **oligonucleotide**, **polynucleotide**.

monooxygenase any oxidoreductase enzyme that brings about the incorporation of only one atom of oxygen from dioxygen into the donor; such enzymes catalyse reactions of the type:



The following are examples. (1) Lysine 2-monooxygenase; EC 1.13.12.2; it catalyses the oxidation by dioxygen of L-lysine to 5-aminopentanamide, CO_2 , and H_2O ; FAD is a coenzyme. (2) Lysine 6-monooxygenase; EC 1.13.12.10; it catalyses the oxidation by dioxygen of L-lysine to N^6 -hydroxy-L-lysine and H_2O . (3) Tryptophan 2-monooxygenase; EC 1.13.12.3; it catalyses the oxidation by dioxygen of L-tryptophan to indole-3-acetamide, CO_2 , and H_2O . (4) *myo*-Inositol oxygenase; EC 1.13.99.1; it catalyses the oxidation by dioxygen of *myo*-inositol to D-glucuronate and H_2O ; iron is a cofactor. (5) Phenol 2-monooxygenase; EC 1.14.13.7; *other name*: phenol hydroxylase; it catalyses a reaction between phenol, NADPH, and O_2 to form catechol, NADP^+ , and H_2O ; FAD is a coenzyme. It is an enzyme of bacterial aromatic substrate utilization; example from *Pseudomonas pickettii*: database code: TBUD_PSEPI, 670 amino acids (72.79 kDa). (6) *trans*-Cinnamate 4-monooxygenase; EC 1.14.13.11; *other names*: cinnamic acid 4-hydroxylase; cinnamate 4-hydroxylase; CA4H; it catalyses a reaction between *trans*-cinnamate, NADPH, and O_2 to form 4-hydroxycinnamate, NADP^+ , and H_2O . It is a cytochrome P450–thiolate enzyme; example from Jerusalem artichoke: database code: TCMO_HELTU, 505 amino acids (57.85 kDa). See also **monophenol monooxygenase**.

monophenol monooxygenase EC 1.14.18.1; *other names*: tyrosinase; phenolase; monophenol oxidase; cresolase. An enzyme involved in the formation of melanins and other polyphenolic pigments, etc. It catalyses a reaction between L-tyrosine, L-dopa, and O_2 to form L-dopa, dopaquinone, and H_2O ; copper is a cofactor. It has four motifs, three for the Cu site. Example from human (precursor): TYRO_HUMAN, 529 amino acids (60.33 kDa). Deficiency results in **albinism**.

monophosphatidyl glycerol any [1,2-diacyl-*sn*-glycero-3-(phospho-1-*sn*-glycerol)] lipid present in chloroplasts of higher plants.

monoploid (of cells or individuals) having a single set of chromosomes; i.e. in a polyploid series, having the fundamental **haploid** chromosome number; true haploid.

monosaccharide the generic name of the simplest carbohydrates. Monosaccharides cannot be hydrolysed to give smaller carbohydrates. They are polyhydric alcohols containing either (in **aldoses**) an aldehyde group or (in **ketoses**) a keto group and with from three to ten or more carbon atoms. Monosaccharides form the constitutional repeating units of oligo- and polysaccharides. The names and structures of the common aldoses (from triose to hexose) form the basis for prefixes used to describe other compounds containing a set of



monose

>CHOH groups. For structures, see illustration (note that if $X = \text{CHO}$ and $Y = \text{CH}_2\text{OH}$ these structures also depict the common D aldoses.) An example of prefix use is the ketoheptonic acid phosphate, DAHP, **3-deoxy-D-arabino-(2)-heptulosonate 7-phosphate**. By extension the stereochemical relationships are applied to substituents other than OH, e.g. in **threonine**. See also **glycose** (def. 1), **heptose**, **hexose**, **pentose**, **tetrose**, **triose**.

monose an obsolete term for **monosaccharide**.

monosodium glutamate *abbr.*: MSG; the monosodium salt of L-glutamic acid, widely used as a flavouring agent for its meaty taste, and the most active component of soy sauce. Consumption by adults of more than about 3.0 g can cause **Chinese restaurant syndrome**, characterized by a feeling of tightness, facial pressure, burning sensation, and headache. The D-enantiomer does not have the characteristic taste and up to 7.0 g can be consumed without ill effect.

monosome **1** any chromosome in a functionally diploid chromosome set that lacks a homologue. **2** a structure consisting of a single ribosome bound to a molecule of messenger RNA.

monosomic (of cells, tissues, or individuals) having one or more **monosomes** (def. 1).

monospecific (of an antibody or an antiserum) only able to react with a single specified antigen or antigenic determinant.

monoterpene any **terpene** containing or derived from two isoprene units and thus containing ten carbon atoms. Monoterpenes may be acyclic or mono- or dicyclic.

monotonic (of a mathematical function) varying in such a manner that it continuously either increases or decreases as the independent variable increases. —**monotonically** *adv.*

monotopic describing a protein molecule that does not cross a cellular membrane.

monovalent an alternative term for **univalent**.

monoxenic *see* **synxenic**.

monozygotic **1** (of twins, triplets, etc.) derived from a single fertilized ovum. **2** any one of a set of monozygotic individuals.

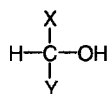
Compare **dizygotic**.

montanate **1** the trivial name for octacosanoate, $\text{CH}_3\text{-(CH}_2\text{)}_{26}\text{-COO-}$, the anion derived from montanic acid (i.e. octacosanoic acid), a saturated straight-chain higher fatty acid. **2** any mixture of free montanic acid and its anion. **3** any salt or ester of montanic acid. Montanate esters of long-chain alcohols occur in montan wax, beeswax, and some other natural waxes.

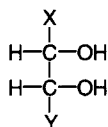
montanoyl *symbol*: Mon; the trivial name for octacosanoyl, $\text{CH}_3\text{-(CH}_2\text{)}_{26}\text{-CO-}$, the acyl group derived from montanic acid (i.e. octacosanoic acid).

Monte Carlo method a way of solving a mathematical problem by sampling methods. The procedure involves the construction of a stochastic model of the mathematical process and the performance of sampling experiments upon it. Such methods have application in molecular graphics aimed at extracting the fundamental principles of macromolecular conformation. [After the resort in Monaco famous for its casino.]

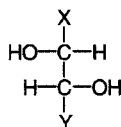
Moore, Stanford (1913–82), US protein chemist; Nobel Laureate in Chemistry (1972) jointly with W. H. Stein 'for their contribution to the understanding of the connection between chemical structure and catalytic activity of the active centre of the ribonuclease molecule' [prize shared with C. B. Anfinsen]; also noted for other work (*see* **Stein**).



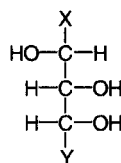
D-glycero



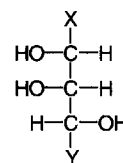
D-erythro



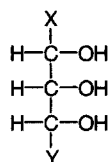
D-threo



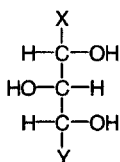
D-arabino



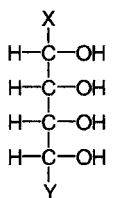
D-lyxo



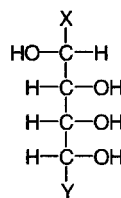
D-ribo



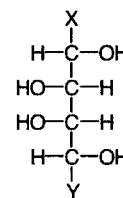
D-xylo



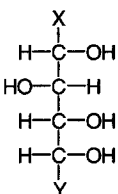
D-allo



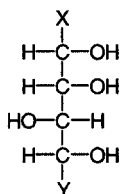
D-altro



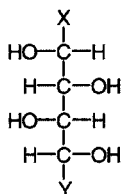
D-galacto



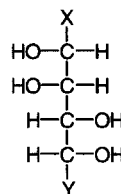
D-gluco



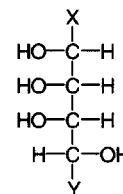
D-gulo



D-ido



D-manno



D-talo

monosaccharide structures

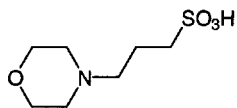
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Mops

Mops or **MOPS** *abbr. for* 3-(*N*-morpholino)propanesulfonic acid; a **Good buffer substance**, pK_a (20 °C) = 7.2.



morgan a unit of genetic length along a chromosome, defined as the distance between two loci that show, on average, one crossover per meiotic event; it is equal to 100 **map units** (or Morgan's units).

Morgan, Thomas Hunt (1866–1945), US geneticist and zoologist noted for developing *Drosophila* mutants and for gene mapping; Nobel Laureate in Physiology or Medicine (1933) 'for his discoveries concerning the role played by the chromosome in heredity'.

Morgan's unit an alternative name for map unit. Compare **morgan**.

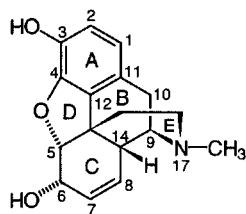
Mörner's test a qualitative test for tyrosine in which a green coloration is produced when a tyrosine-containing sample is boiled in a solution of formaldehyde in strong sulfuric acid. [After K.A.H. Mörner (1855–1917), Swedish chemist.]

morph+ a variant form of **morpho+** (before a vowel).

+morph *comb. form* indicating of a specified form, shape, or structure. — **+morphic** or **+morphous** *adj.*; **+morph** *n.*

morphiceptin a tetrapeptide amide fragment of the milk protein β -casein, structure Tyr-Pro-Phe-Pro-NH₂. It relates to the N-terminal tetrapeptide of casomorphin. It is a potent and specific agonist for μ -opiate receptors, its effects on the receptor being blocked by naloxone.

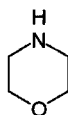
morphine 7,8-didehydro-4,5-epoxy-17-methylmorphinan-3,6-diol; an opiate and the most important opium alkaloid; it is a powerful, habit-forming narcotic analgesic. The diacetate ester, diacetylmorphine, is known as **diamorphine** or **heroin**. This is strongly habit forming and its manufacture, importation, and use are controlled in a number of countries.



morpho+ or (before a vowel) **morph+** *comb. form* indicating form or structure.

morphogenesis the ensemble of the processes of development of a part, organ, or organism. — **morphogenetic** or **morphogenic** *adj.*

morpholine tetrahydro-1,4-oxazine; a heterocyclic compound with a six-membered oxygen- and nitrogen-containing ring. A number of *N*-substituted morpholines are useful as biological buffers, e.g. **Mes**, **Mops**.



morphology the branch of biology concerned with the outer form and inner structure of living organisms.

Morquio syndrome another name for mucopolysaccharidosis IV. See **mucopolysaccharidosis**.

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mortalin other name: 75 kDa glucose regulated protein (*abbr.*: GRP 75); a protein that resembles **heat-shock proteins** in sequence, and which may be a **chaperone**. It is a member of the heat-shock protein 70 family, which functions as a mortality marker in fibroblasts. Mortal fibroblasts contain a protein, MOT-1, uniformly distributed in the cytosol, while in immortal fibroblasts another form of the protein, MOT-2, is concentrated in perinuclear or juxtannuclear regions. In MOT-2 there are differences at residues 618 (Val to Met) and 624 (Arg to Gly). The cytosolic form induces cellular senescence, the perinuclear form does not. Example (precursor), MOT-1 from mouse: database code GR75_MOUSE, 679 amino acids (73.45 kDa).

MOS a gene family encoding serine/threonine kinases, e.g. the human and mouse protein pp39^{mos}; *v-mos* is the oncogene of the acutely transforming murine Moloney sarcoma virus (hence 'mos' from mo + s). *MOS* mRNA is detectable only in germ cells in testes and ovaries, and *MOS* expression is implicated in meiosis; directly or indirectly it stabilizes **M-phase promoting factor** and activates **MAP kinase**. The protein has homology with Src (*see* **src**) but is not a tyrosine kinase. Example from human: database code KMOS_HUMAN, 346 amino acids (37.82 kDa).

mosaic 1 a design or pattern on a surface, made up numerous separate pieces of more or less the same area but differing in some characteristic, e.g. colour, shape, texture. 2 an alternative name for **chimera** (used especially of animals).

mosaic model *see* **fluid-mosaic model**.

MOT-1, MOT-2 *see* **mortalin**.

mother liquor the liquid remaining after a substance has been crystallized or precipitated from a solution.

motif (in relation to protein structure) 1 a locally ordered region within the hydrophobic core of a globular protein molecule, formed by three-dimensional interaction between two or three segments of the secondary structure (α helix and/or β strand) that are near one another along the polypeptide chain. The most important types are: ($\alpha\alpha$), ($\alpha\beta$), ($\beta\beta\beta$), and ($\beta\alpha\beta$). *See also* **domain**. 2 or **rule** any set of amino acids forming part of the primary sequence of a protein, either contiguous or capable of being aligned to certain positions that are invariant or conserved, that is associated with a particular function; e.g. the motif GXGXXG is associated with nucleotide-binding sites. *See also* **Appendix E**.

motile (of organisms or cells) capable of independent locomotion; (of cilia, flagella, etc.) capable of spontaneous movement. — **motility** *n.*

motilin a 22-residue peptide produced by **Mo cells** of the mucosa of the upper small intestine and having putative hormonal activity. Its plasma level rises when food or alkali is placed in the duodenum and it stimulates contractions of the stomach and the secretion of pepsin, but its effects show marked species variation. Example from dog: database code MOTI_CANFA, sequence

FVPIFTHSELQKIREKERNKGQ

(2.685 kDa).

motor (*in biology*) pertaining to movement of an organ or part; pertaining to nerves or neurons involved in muscular activity.

motor protein a protein, e.g. **dynein**, **kinesin**, **myosin**, that propels itself along a filament or polymeric molecule. Nucleoside triphosphate hydrolysis is usually necessarily involved. A similar process is thought to be involved in the action of elongation factor G (*see* **EF-G**). These processes are referred to as **chemical-mechanical coupling** (or mechanochemical coupling).

Motrin a proprietary name for **ibuprofen**.

mould or (*US*) **mold** 1 any **microfungus** having a distinctive mycelium; the common name for any superficial growth of fungal mycelium. 2 a matrix for shaping a cast.

moult or (*US*) **molt** 1 to cast off periodically an outer covering of cuticle, feathers, hair, horns, skin, etc. 2 the act, process, or condition of moulting; in invertebrates it is known as ecdysis.



moulting hormone

moulting hormone or (*US*) **molting hormone** a common name for **ecdysterone**.

mouse-convulsion method a sensitive method for the detection and biological assay of insulin, used primarily to control the strength of pharmaceutical preparations of insulin. It is based on the finding that small doses of insulin injected into mice (0.1–1.0 µg per animal) cause convulsions. When the method is used for assay, large equal numbers of animals are injected with either of two dilutions of the test solution or of a standard and the results are analysed statistically.

mouse mammary tumour virus *abbr.*: MMTV; a DNA virus that causes breast cancer in mice. The virus contains a powerful enhancer DNA sequence in the terminal repeats of the genome. This activates an oncogene called *wnt-1* (formerly called *int-1*) which is closely homologous to the *Drosophila* gene *wingless*. MMTV is often used as an **expression vector** especially in transgenic mice.

moving-boundary electrophoresis an analytical method of electrophoresis carried out using a **Tiselius apparatus**, formerly used for measuring electrophoretic mobility directly.

MOX *abbr.* for methanol oxidase (see **alcohol oxidase**).

m.p. *abbr.* for melting point.

MP *abbr.* for mitogenic protein (i.e. **interleukin 1**).

6-MP *abbr.* for 6-mercaptopurine.

MPF *abbr.* for M-phase-promoting factor.

M phase the period of mitosis and cytokinesis in the eukaryotic **cell-division cycle**. During this phase the nucleus and cytoplasm divide.

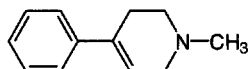
M-phase-promoting factor or **maturation-promoting factor** or **p34^{cdc}** *abbr.*: MPF; a protein complex, containing **cyclin** and a protein kinase, that triggers a cell to enter M phase of the **cell-division cycle**. Example, protein kinase of meiotic maturation of *Xenopus* oocytes: database code A44349, 302 amino acids (34.47 kDa).

MPP *abbr.* for mitochondrial processing peptidase.

M6P receptor *abbr.* for **mannose 6-phosphate receptor**.

M protein 1 a structural protein that occurs in the M lines of myofibrils of striated muscle. 2 an alternative name for galactoside permease of *Escherichia coli*. 3 a cell-surface protein of Gram-positive cocci, associated with virulence; it renders the organism resistant to phagocytosis. It is a cell-wall component, consisting of a coiled coil with membrane anchor. Example from *Streptococcus pyogenes* serotype 6 (precursor): database code M6_STRPY, 483 amino acids (53.41 kDa).

MPTP *abbr.* for 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; a toxic by-product of the synthesis of heroin; it produces severe and permanent parkinsonism in primates, resulting from its oxidation by monoamine oxidase to the quaternary 1-methyl-4-phenyl-pyridinium (MPP⁺) ion. The latter is a potent inhibitor of complex I of the mitochondrial electron-transport chain and a selective neurotoxin for dopaminergic neurons, by which it is taken up in the substantia nigra via the normal neuronal dopamine re-uptake system.



MRC *abbr.* for Medical Research Council (of the UK).

MRE *abbr.* for metal response element; see also **response element**.

MRF *abbr.* for melanotropin-releasing factor (see **MSHRH**).

MRF4 *abbr.* for muscle specific regulatory factor 4; other name: myogenic factor 6; a DNA-binding, probably **helix-turn-helix** protein of the **MYC** family. It induces fibroblasts to differentiate into myoblasts. Example from chicken: database code MYF6_CHICK, 242 amino acids (26.32 kDa).

MRH *abbr.* for melanotropin-releasing hormone (see **MSHRH**).

MRI *abbr.* for magnetic resonance imaging.

MRIF *abbr.* for melanotropin release-inhibiting factor (i.e. **melanostatin**).

MRIH *abbr.* for melanotropin-release inhibiting hormone (i.e. **melanostatin**).

mRNA *abbr.* for messenger RNA.

mRNA binding site (ribosome) a ribosomal binding region for mRNA. In bacteria this involves **ribosomal proteins** S1, S3, S4, S5, S12, S18, S21, and the 3' end of 16S rRNA.

mRNA splicing the process in which excision of **introns** from the primary transcript of **messenger RNA** (mRNA) is followed by ligation of the two **exon** termini exposed by removal of each intron, so that mRNA consisting only of the joined exons is produced. See **splicing** (def. 2).

mRNP *abbr.* for messenger ribonucleoprotein

MRSA *abbr.* for **methicillin-resistant Staphylococcus aureus**, an organism resistant to antibiotics owing to the inability of penicillins to enter the cell (and not directly to **β-lactamase**). Compare **VRE**.

MS *abbr.* for 1 mass spectrometer or mass spectrometry. 2 magic spot (compound). 3 multiple sclerosis.

MS- *abbr.* for meso-.

MSA *abbr.* for multiplication-stimulating activity.

msDNA *abbr.* for multicopy single-strand DNA.

MSEL-neurophysin see **neurophysin II**.

MSG *abbr.* for monosodium glutamate.

MSH *abbr.* for melanocyte-stimulating hormone.

MSHIF *abbr.* for melanocyte-stimulating-hormone release-inhibiting factor (i.e. **melanostatin**).

MSHII *abbr.* for melanocyte-stimulating-hormone release-inhibiting hormone (i.e. **melanostatin**).

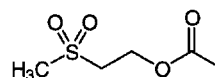
MSHRF *abbr.* for melanocyte-stimulating-hormone releasing factor (i.e. **MSHRH**).

MSHRH *abbr.* for melanocyte-stimulating-hormone releasing hormone; a pentapeptide, possibly identical to the N-terminal moiety of **oxytocin**, that is produced in the hypothalamus. In animals that show an unambiguous response of their melanophores to **melanotropin** this regulating factor is believed to promote melanotropin release. *Recommended name*: melanoliberin; *other names*: melanotropin-releasing factor (*abbr.*: MRF); melanotropin-releasing hormone (*abbr.*: MRH); melanocyte-stimulating-hormone releasing factor (*abbr.*: MSHRF).

MSHRIF *abbr.* for melanocyte-stimulating-hormone release-inhibiting factor (i.e. **melanostatin**).

MSHRII *abbr.* for melanocyte-stimulating-hormone release-inhibiting hormone (i.e. **melanostatin**).

MSOC *abbr.* for the *N*-(2-methanesulfonyl)ethoxycarbonyl group; it is used as an amino-protecting group, its stability to acid, lability to alkali, and resistance to hydrogenation contributing to its usefulness. It is introduced using such reagents as **methylsulfonyl nitrophenyl carbonate** and **methylsulfonyl succinimidyl carbonate**.



MSX-1 or **Hox-7** a **homeobox** protein. Example from human: database code HMX1_HUMAN, 297 amino acids (30.96 kDa).

mtDNA *abbr.* for mitochondrial DNA.

MTOC *abbr.* for microtubule organizing centre.

mtRNA *abbr.* for mitochondrial RNA.

mu symbol: μ (lower case) or M (upper case); the twelfth letter of the Greek alphabet. For uses see **Appendix A**.

MU *abbr.* for 4-methylumbelliferyl.

mu chain or **μ chain** the heavy chain characterizing **immunoglobulins** belonging to the class IgM.

mucic acid an old name for galactaric acid, the *meso*-aldaric acid derived from both D- and L-galactose.

mucilage any of numerous proteoheteroglycans, frequently containing uronic acids, that occur widely in plants, especially

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mucin

in seed coats. Mucilages are hard when dry, and in water they swell, but do not dissolve, to form slimy masses with adhesive properties. *Compare gum. —mucilaginous adj.*

mucin any of a group of widely distributed glycoproteins of high M_r that are secreted by various animal mucous cells and glands; they occur, e.g., in saliva, gastric juice, and intestinal juice. Mucins form viscous solutions that act as lubricants and protectants of the linings of body cavities and of the skin. Example from human intestine: database code HUMMUC2X, 5179 amino acids (539.70 kDa).

mucina an alternative name for **hyaluronidase**.

mucos *comb. form* of mucus or mucin; mucoid.

mucoid 1 any mucin-like substance or mucoprotein. 2 of the nature of or resembling **mucus**. 3 describing a bacterial colony that has a gummy consistency, resulting from the abundant synthesis of capsular material.

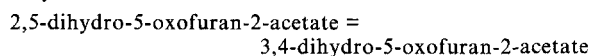
mucolipid an alternative name for **ganglioside**.

mucopolipidosis (*pl. mucopolipidoses*) any of a group of inherited autosomal recessive diseases of humans characterized by deficiency of a variety of lysosomal hydrolases in connective tissue cells of patients, and in fibroblasts cultured from them. Mucopolipidosis I, commonly known as **sialidosis**, is due to a deficiency of neuraminidase that leads to accumulation of sialyl oligosaccharides, and excretion of these in the urine. In mucopolipidosis II, the cytoplasm of cultured cells from affected patients contains numerous highly refractile inclusions, visible by phase-contrast microscopy. These are very large lysosomes containing heterogeneous material, mucopolysaccharides, glycolipids, and whorls of membranes, giving this the name **I-cell** (inclusion cell) **disease**. Mucopolipidosis III is due to a deficiency of N-acetylglucosamine 1-phosphotransferase.

mucolytic having the ability to disrupt mucus.

muconate a trivial name for any one of the three stereoisomers (*cis,cis*, *cis,trans*, or *trans,trans*) of 2,4-hexadienedioate; 1,3-butadiene-1,4-dicarboxylate; $\text{OOC}-\text{CH}=\text{CH}=\text{CH}-\text{COO}^-$, the dianion derived from muconic acid. The *cis,cis* isomer is an intermediate in the degradation of catechol by microorganisms.

muconolactone Δ -isomerase EC 5.3.3.4; an enzyme that catalyses the reaction:



in the pathway for the degradation of catechol to acetyl-CoA. Example from *Pseudomonas putida*: database code NRL_1ML1, 96 amino acids (11.15 kDa); 3-D structure known.

mucopeptide (*sometimes*) an alternative name for **peptidoglycan**.

mucopolysaccharide any of a group of acid heteroglycans built up of characteristic repeating disaccharide units each consisting of an N-acylated hexosamine residue and a uronic acid residue; many mucopolysaccharides also contain sulfate. The group includes hyaluronic acid (*see hyaluronate*), **chondroitin sulfate**, **dermatan sulfate**, **keratan sulfate**, **heparin**, and some **blood-group substances**.

mucopolysaccharide storage disease any of a group of clinically progressive, hereditary diseases of humans characterized by the accumulation of mucopolysaccharides in various tissues. They include the **mucopolysaccharidoses** and the **mucopolipidoses**.

mucopolysaccharidosis (*pl. mucopolysaccharidoses*) any of a group of clinically progressive, hereditary diseases of humans each of which is characterized by the deficiency of a single lysosomal hydrolase concerned in the degradation of mucopolysaccharides. They have been classified into a number of types. *See individual entries below.*

mucopolysaccharidosis I H or **Hurler syndrome** a **mucopolysaccharidosis** caused by a deficiency in lysosomal L-iduronidase (EC 3.2.1.76) that affects the degradation of both dermatan sulfate and heparan sulfate. It is inherited as an autosomal recessive disorder. Subjects excrete excessive amounts

of both dermatan sulfate and heparan sulfate in the urine in the ratio of approximately 7:3. The biochemical lesion is the same as in mucopolysaccharidoses I H/S and I S, from which differentiation can only be made by clinical criteria.

mucopolysaccharidosis I H/S or **Hurler-Scheie compound syndrome** a **mucopolysaccharidosis** caused by the same deficiency in lysosomal L-iduronidase (EC 3.2.1.76) as is found in mucopolysaccharidoses I H and I S. It is inherited as an autosomal recessive disorder and it is believed to be a phenotype intermediate between these two syndromes, one mutant allele being inherited from the father and a different mutant allele from the mother. The condition can only be differentiated from mucopolysaccharidoses I H and I S by clinical criteria.

mucopolysaccharidosis I S or **Scheie syndrome** or (*formerly*) **mucopolysaccharidosis V** a **mucopolysaccharidosis** caused by the same deficiency in lysosomal L-iduronidase (EC 3.2.1.76) as is found in mucopolysaccharidoses I H and I H/S. It is inherited as an autosomal recessive disorder and it has a milder clinical course than mucopolysaccharidoses I H and I H/S, from which it can be differentiated only by clinical criteria.

mucopolysaccharidosis II or **Hunter syndrome** a **mucopolysaccharidosis** caused by a deficiency in iduronate sulfatase that affects the degradation of both dermatan sulfate and heparan sulfate. It is inherited as an X-linked recessive disorder, and several different mutations of the same X-chromosome locus probably account for the different severities of the disease seen in various families. Subjects excrete excessive and roughly equal amounts of both dermatan sulfate and heparan sulfate in the urine.

mucopolysaccharidosis III A or **Sanfilippo syndrome A** a **mucopolysaccharidosis** caused by a deficiency in heparan N-sulfatase that affects the degradation of heparan sulfate. It is inherited as an autosomal recessive disorder and is indistinguishable clinically from mucopolysaccharidosis III B. Subjects excrete excessive amounts of heparan sulfate in the urine.

mucopolysaccharidosis III B or **Sanfilippo syndrome B** a **mucopolysaccharidosis** caused by deficiency in α -N-acetyl-D-glucosaminidase (EC 3.2.1.50) that affects the degradation of heparan sulfate. It is inherited as an autosomal recessive disorder and is indistinguishable clinically from mucopolysaccharidosis III A. Subjects excrete excessive amounts of heparan sulfate in the urine.

mucopolysaccharidosis IV A (*or IVa*) or **Morquio A disease** a **mucopolysaccharidosis** caused by a deficiency of lysosomal N-acetylglucosamine-6-sulfatase, leading to an accumulation of keratan sulfate and chondroitin 6-sulfate. It is inherited as an autosomal recessive disorder.

mucopolysaccharidosis IV B (*or IVb*) or **Morquio B disease** a **mucopolysaccharidosis** caused by a deficiency of β -galactosidase leading to accumulation of keratan sulfate. It is inherited as an autosomal recessive disorder.

mucopolysaccharidosis V a former name for **mucopolysaccharidosis I S**.

mucopolysaccharidosis VI or **Maroteaux-Lamy syndrome** a **mucopolysaccharidosis** caused by a deficiency of lysosomal N-acetylglucosamine 4-sulfatase that affects the degradation of dermatan sulfate. Three clinically distinguishable forms are known: severe, intermediate, and mild; all are inherited as autosomal recessive disorders, with probably two or more mutant alleles at the mucopolysaccharidosis VI locus, one of which produces the severe form when homozygous and the other the mild form. Subjects excrete excessive amounts of dermatan sulfate in the urine.

mucopolysaccharidosis VII a **mucopolysaccharidosis** caused by a deficiency of lysosomal β -D-glucuronidase (EC 3.2.1.31) that affects the degradation of both dermatan sulfate and heparan sulfate. It is inherited as an autosomal recessive disorder.

mucoprotein any glycoprotein containing relatively large amounts of acidic polysaccharide.

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mucosa

mucosa (*pl.* **mucosae**) or **mucous membrane** any of the moist membranes that line the alimentary canal, the glandular ducts, and the respiratory, urinary, and genital passages; it may be smooth, corrugated, or covered with villi. Mucous membranes consist of a surface layer of epithelium containing mucus-secreting glands, and an underlying layer of connective and muscular tissue. *Compare* **serosa**. — **mucosal** *adj.*

mucotetraose *symbol:* McOse₄ or Mc₄ (if space is limited); the tetrasaccharide Gal(β1-3)Gal(β1-4)Gal(β1-4)Glc.

mucotriose *symbol:* McOse₃ or Mc₃ (if space is limited); the trisaccharide Gal(β1-4)Gal(β1-4)Glc.

mucous relating to, resembling, or secreting **mucus**.

mucous cell any cell that secretes **mucus**.

mucous gland any exocrine gland consisting predominantly of mucus-secreting cells.

mucous membrane *an alternative name for* **mucosa**. *Compare* **serous membrane**. — **mucomembranous** *adj.*

mucus the slimy, sticky, viscous secretion, consisting predominantly of **mucins**, that is secreted by mucous cells of the mucosae and of the external body surface of some animals.

MUG *abbr. for* **1** either 4-methylumbelliferyl-glucuronide or 4-methylumbelliferyl-β-D-galactopyranoside. Both compounds are fluorogenic enzyme substrates that yield 4-methylumbelliferone. *See also* 4-methylumbelliferyl. **2** *abbr. for* murinoglobulin.

MUGal *abbr. for* 4-methylumbelliferyl-β-D-galactopyranoside (alternative to **MUG**).

mull a suspension of very finely ground solid in an inert liquid, especially paraffin or **Nujol**, that is used in infrared spectroscopy of the solid.

Muller, Hermann Joseph (1890–1967), US geneticist; Nobel Laureate in Physiology or Medicine (1946) 'for the discovery of the production of mutations by means of X-ray irradiation'.

Mullis, Kary B. (1944–), US biochemist; Nobel Laureate in Chemistry (1993) 'for his invention of the polymerase chain reaction (PCR) method' [prize (awarded 'for contributions to the developments of methods within DNA-based chemistry') shared with M. Smith].

multi+ *comb. form* denoting more than one, several. *See also* **poly+**.

multicatalytic endopeptidase complex *see* **proteasome**.

multicellular having, or consisting of, several or many cells.

multichain (of a protein) consisting of two or more polypeptide chains connected by nonpeptide links.

multichannel analyser any instrument or device for sorting a heterogeneous input into a number of ranges, e.g. of frequency or energy, and determining the relative proportions in each range.

multicopy single-strand DNA *abbr.:* msDNA; DNA generated by bacterial reverse transcriptase and found in certain Gram-negative bacteria. The molecules are unique in being the only known naturally occurring examples of nucleic acids containing an RNA–DNA covalent link.

multidentate (of a ligand) having two or more groups through which it can be attached to a central atom. Such a ligand is able to form a bridge or a chelate.

multidimensional chromatography a technique in which a single component separated from a mixture by chromatography on one type of column may be automatically transferred, as a sample, for chromatography on another type of column in order to effect further resolution.

multienzyme *the recommended class name for* any protein that possesses two or more autonomous catalytic functions due to separate catalytic domains (*see* **domain**, def. 3), whether contributed by distinct subunits (*see* **multienzyme complex**), by distinct parts or an individual polypeptide chain (*see* **multienzyme polypeptide**), or by both. Excluded is any enzyme that can catalyse different reactions at the same catalytic centre. *Compare* **enzyme complex**, **multienzyme system**.

multienzyme cluster *see* **enzyme cluster**.

multienzyme complex *the recommended subclass name for* any multienzyme in which catalytic domains occur on more than one polypeptide chain. *Compare* **multienzyme polypeptide**.

than one type of polypeptide chain. A multienzyme complex may itself contain a **multienzyme polypeptide**.

multienzyme polypeptide *the recommended subclass name for* any multienzyme in which at least two demonstrably distinct catalytic domains of different types occur on a single polypeptide chain and are encoded by a single gene. A multienzyme polypeptide may itself be a component of a multienzyme complex. *Compare* **multifunctional protein**.

multienzyme system *an operational term for* any system of two or more enzymes functioning sequentially to catalyse the reactions of a metabolic pathway, so that the product of the first enzyme becomes the substrate of the second and so on. Three levels of organization of multienzyme systems may be discerned. First, the individual enzymes are independently in solution in the cytoplasm and are not directly associated with each other when acting. Second, the individual enzymes are physically associated and function together as an **enzyme complex**, or they occur as a **multienzyme**. Third and most highly organized, the individual enzymes are associated with large, supramolecular structures such as membranes or ribosomes. *See also* **enzyme cluster**, **metabolon**.

multidrug resistance *see* **MDR**.

multifactorial **1** involving or dependent on more than one factor or causes. **2** (*in genetics*) inheritance depending on more than one gene.

multifunctional protein any protein that combines two or more autonomous functions on one polypeptide chain, the functions being both independently measurable and assignable to separate domains of the protein. Such functions may or may not be catalytic (*compare* **multifunctional protein ade2**). Excluded are allosteric enzymes, proenzymes, and enzymes that can catalyse different reactions at a single reaction centre.

multifunctional protein ade2 a protein that contains **phosphoribosylaminoimidazolesuccinocarboxamide synthase** and **phosphoribosylaminoimidazole carboxylase** and is involved in the sixth and seventh steps in *de novo* purine biosynthesis. Example from human: database code PUR6_HUMAN, 425 amino acids (47.03 kDa); the synthase domain resides in residues 1–260, the carboxylase domain in residues 261–425.

multigenic describing character differences that are controlled by the alleles of several genes.

multiiodinated describing any compound that has been reacted so as to contain two or more atoms of iodine per molecule.

multilamellar occurring as, or made up of, several discrete layers.

multilayered film analyser a device for the rapid determination of any of a range of components present in small volumes of biological fluids. It depends on a single-use reagent slide on which the various chemicals required are present in dry form as a series of thin films. These consist successively of an indicator layer, a semipermeable layer, a reagent layer, and a spreading layer, all mounted between two transparent supports. Typically, a 10 μL sample is applied to the spreading layer, a chemical or enzymatic reaction occurs in the reagent layer, and the semipermeable layer withholds macromolecular substances from the indicator layer, where usually a dye combines with a product of the reaction to form a coloured compound. The intensity of the colour, which is proportional to the quantity of the analyte in the specimen, is measured by reflectance spectrophotometry. The concept has been extended to the use of potentiometry for the measurement of sodium and potassium.

multilocus enzyme any of a set of enzymes, with similar catalytic properties and existing in the same organism, whose amino-acid sequences are encoded in nonallelic structural genes at two or more separate loci. There may be marked variations in the expression of the individual loci in different tissues. A class of **isoenzyme**.

multimer any specific aggregate of (identical or nonidentical) molecular entities held together by noncovalent bonds.



multinucleate

—**multimeric** *adj.*; **multimerize** or **multimerise** *vb.*; **multimerization** or **multimerisation** *n.*

multinucleate or **multinuclear** (of a cell) containing several or many nuclei.

multiphase having several phases; polyphase.

multiple codon recognition the recognition by a single tRNA molecule of more than one type of codon in mRNA.

multiple development technique a chromatographic technique in which the chromatogram is developed more than once with either the same solvent system or a different one with the aim of improving the separation.

multiple drug resistance *see* MDR.

multiple forms (of an enzyme) *the recommended collective term to describe* all proteins that possess the same (specified) enzymic activity and occur naturally in a single species. It includes, but is not limited to, **isoenzymes**; the term isoenzyme should be applied only to multiple forms that arise from genetically determined differences in primary structure, and some limit this to those arising from a single gene locus.

multiple myeloma or **multiple myelomatosis** *see* myeloma.

multiple sclerosis *abbr.*: MS; a chronic and usually progressive disease of the nervous system, also known as disseminated sclerosis, in which the myelin sheaths surrounding the axons are lost; the lesions are confined to the central nervous system. The cause is unknown, but plaques are found in which activated lymphocytes and monocytes are present; these plaques have a discrete edge and range in size from microscopic to macroscopic. The disease is characterized by a course of relapses and remissions over a number of years.

multiplet any group of two or more associated spectral lines. In electronic spectra the lines of a multiplet are very closely spaced and appear at low resolution as a single line, whereas in nuclear magnetic resonance spectra the separation may be considerable.

multiplicand (*in mathematics*) a number to be multiplied by another number – the multiplier – in an arithmetic multiplication.

multiplication-stimulating activity *abbr.*: MSA; an activity found in mammalian serum and produced by some animal cells that is required for the multiplication of animal cells in culture.

multiplicity (*in physics*) the number of components in a multiplet.

multiplicity of infection the ratio of the number of virus particles to the number of susceptible cells in a given system.

multiplicity reactivation the reconstruction of viable genomes in a cell that is multiply infected with otherwise non-viable viruses.

multiplier 1 (*in mathematics*) any number by which another number – the multiplicand – is multiplied. 2 any instrument or device for multiplying an effect, e.g. photomultiplier.

multiply labelled describing an isotopically substituted compound that has more than one modified atom of the same element at the same position or at different positions.

multipotent (of a cell or tissue) capable of giving rise to several kinds of cells, tissues, or structures. *Compare* pluripotent (def. 2).

multisubstrate enzyme any enzyme that requires two or more substrates in order to catalyse a particular reaction.

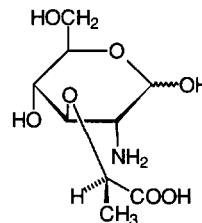
multivalent *an alternative term for* polyvalent.

multivalent phosphorylation the phosphorylation of any hormonally regulated enzyme at a plurality of sites by two or more distinct kinase enzymes, with a resultant increase in the regulatory potential of the enzyme.

multivesicular body an intracellular structure that is lined by a single membrane and contains a number of inner vesicles that are approximately 50 nm in diameter. It is a form of secondary lysosome.

Mur symbol for a residue (or sometimes a molecule) of **muramic acid**.

muramic acid *symbol*: Mur; 3-*O*- α -carboxyethyl-D-glucosamine; the 3-*O*-D-lactyl ether derivative of D-glucosamine. It occurs naturally in peptidoglycan of bacterial cell walls.



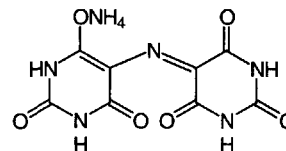
muramidase *an alternative name for* lysozyme.

muramyl the acyl group derived from **muramic acid**.

muramyl-dipeptide *abbr.*: MDP; *N*-acetyl-muramyl-D-alanyl-D-isoglutamine; a synthetic, water-soluble, **peptidoglycan** derivative that is capable of replacing whole mycobacteria in complete **Freund's adjuvant**. It is highly active in stimulating antibody production against an antigen injected simultaneously.

murein *an alternative name for* peptidoglycan.

murexide ammonium purpurate; 5,5'-nitridodibarbituric acid monoammonium salt; a compound that forms a purple-red solution in water. It is useful as an indicator in complexometric titration of calcium, nickel, cobalt, copper, etc. with EDTA.



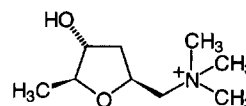
murexide test a colour test for uric acid and some other purines. The (solid) sample is first treated with conc. nitric acid, which is slowly evaporated away; subsequent addition of ammonia solution gives a purple colour if uric acid was present, due to formation of murexide, or a yellow colour that turns to red on heating if xanthine was present.

murine of, belonging to, characteristic of, affecting, transmitted by, or being a member of the Muridae, a family of small rodents that includes mice and rats; of or relating to the mouse genus, *Mus*.

murinoglobulin *abbr.*: MUG; a monomeric proteinase inhibitor found in blood plasma of vertebrates and invertebrates and in egg white of birds and reptiles. It has a characteristic **bait region**. Example, murinoglobulin 2 (MUG2) precursor from mouse: database code A2MH_MOUSE, 1451 amino acids (162.18 kDa). *Compare* α_2 macroglobulin.

muropeptide *an alternative name for* peptidoglycan.

muscarine a toxic alkaloid (quaternary ammonium base) found in the red variety of *Amanita muscaria* (fly agaric) and some other gill fungi. It is an agonist for one type of **cholinoceptor**.

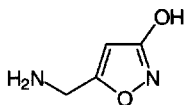


(+)-enantiomer

muscarinic receptor *see* **cholinoceptor**.

muscimol

muscimol 5-aminomethyl-3-hydroxyisoxazole; an isoxazole isolated from the fungus *Amanita muscaria*. A hallucinogen and potent CNS depressant, it is a selective agonist of GABA_A receptors.



muscle 1 a tissue made up of various elongated cells that are specialized to contract and thus to produce movement and mechanical work. 2 an organ consisting principally of muscle tissue. *See* **cardiac muscle**, **smooth muscle**, **striated muscle**. —**muscular** *adj.*

muscle contraction *see* **sliding filament model**.

muscle hemoglobin a former name for **myoglobin**.

muscle sugar a former name for **inositol** (def. 1).

muscular dystrophy any of a number of genetically determined diseases of humans in which there is progressive muscular weakness and evidence of muscle degeneration. All are characterized by creatinuria, and serum creatine phosphokinase activity is elevated in several of them. *See also* **dystrophin**.

mushroom sugar a former name for **trehalose**.

mustard gas sulfur mustard; bis(2-chloroethyl)sulfide; a colourless oily liquid with a very low vapour pressure. It is a powerful vesicant and also a mutagen and carcinogen. *See also* **nitrogen mustard**.

mustard oil any alkyl isothiocyanate, of general formula R-N=C=S. Such compounds have a pungent mustard-like smell and are present in the tissues of a number of plants, either free or as thioglycosides.

mut any of a family of genes whose products are involved in **mismatch repair**. All the following examples are from *Escherichia coli*. The product of *mutH* is a repair endonuclease that is specific for unmethylated GATC sequences; database code MUTH_ECOLI, 228 amino acids (25.36 kDa). The protein encoded by *mutL* is also involved in repair of mismatches in DNA and is required for activity of the *dam* product; database code MUTL_ECOLI, 615 amino acids (67.85 kDa). The protein encoded by *mutS*, again involved in mismatch repair (possibly the recognition step), has weak ATPase activity; database code MUTS_ECOLI, 853 amino acids (95.14 kDa).

mutable able or tending to undergo change or **mutation**. —**mutability** *n.*

mutable gene any gene that spontaneously mutates at a high rate.

mutable site any site on a chromosome at which mutations can occur.

mutagen any physical or chemical agent that is capable of increasing the frequency of **mutation** above the spontaneous, background level. —**mutagenic** *adj.*; **mutagenicity** *n.*

mutagenesis the production of mutations.

mutagenize or **mutagenise** to treat (cells or organisms) with a mutagen. —**mutagenized** or **mutagenised** *adj.*

mutant 1 a any organism that has arisen by or has undergone **mutation**, or one that carries a mutant gene that is expressed in the phenotype of that organism. **b** a mutant gene. 2 produced by or following a mutation; having the attributes of a **mutant** (def. 1).

mutarotation the spontaneous change in the optical rotation of a freshly prepared solution of a pure stereoisomer, especially a carbohydrate, that is caused by epimerization, or some other structural change.

mutase any enzyme that catalyses (or apparently catalyses) an intramolecular transfer of a chemical group. Such enzymes belong to one of the following EC sub-subclasses: EC 5.4.1, transferring acyl groups; EC 5.4.2, transferring phosphoryl groups; EC 5.4.3, transferring amino groups; EC 5.4.99, transferring other groups.

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mutate to undergo or cause to undergo mutation.

mutation 1 the process by which genetic material undergoes a detectable and heritable structural change, or the result of such a change. Three categories are recognized: **genome mutations**, in which addition or subtraction of one or more whole chromosomes occurs; **chromosome mutations**, in which the structure of one or more chromosomes is affected; and **gene mutations**, in which the structure of a gene is altered at the molecular level. 2 any modified gene arising from a **mutation** (def. 1). 3 a **mutant** (def. 1). *See also* **deletion**, **inversion** (def. 4), **point mutation**, **translocation** (def. 2).

mutational heteroduplex *see* **heteroduplex**.

mutation frequency the frequency at which a mutation occurs in a population. It may or may not reflect the **mutation rate**, with which it is sometimes used interchangeably.

mutation frequency decline the significant and irreversible reduction in the frequency of some UV-induced mutations in bacteria during the temporary post-irradiation inhibition of protein synthesis.

mutation pressure the continued production of an allele by mutation that tends to increase the allele's frequency in the gene pool.

mutation rate 1 the probability that a particular mutation will occur per virus, cell, or organism during a reproductive cycle. 2 the frequency at which any mutation or mutational class occurs in a given population; **mutation frequency**.

mutator gene any gene that produces an increase in spontaneous mutation rates. Their action may be limited to certain alleles or to certain mutational pathways.

mutetin a mutant protein produced by site-specific mutagenesis or other recombinant DNA technique.

muton the smallest unit of genetic material whose alteration may give rise to a mutation; commonly supposed to be a single nucleotide (pair).

MVA *abbr.* for mevalonic acid (use discouraged).

MVG *abbr.* for methoxyvinylglycine.

M.W. *abbr.* for molecular weight (alternative to mol. wt.).

MWC model *abbr.* for Monod-Wyman-Changeux model.

my+ a variant form of **myo+** (before a vowel).

MYB a gene family that encodes DNA-binding proteins; *v-myb* is the oncogene of the acutely transforming avian myeloblastosis virus. It is highly conserved between mammalian species. The product, Myb, contains DNA-binding, transcriptional activation, and negative regulatory domains; it recognizes YAAC(G/T)G. It is notably associated with cells of the hemopoietic lineage and is essential for normal hemopoiesis. Example from human: database code MYB_HUMAN, 640 amino acids (72.38 kDa). [From 'myeloblastosis'.]

MYC a family of genes encoding short-lived nuclear regulatory proteins that act as transcription factors, e.g. pp64/pp67 (based on expressed mass). The oncogene, *v-myc*, was originally identified in avian myelocytomatosis virus, MC29, but is also carried by the avian retroviruses CM11 and OK-10. The protooncogene, *c-myc*, is a member of a family of at least seven related genes: *myc*, *B-myc*, *N-myc*, *L-myc*, *P-myc*, *R-myc*, *S-myc*; *B-Myc* lacks the basic helix-loop-helix leucine zipper (bHLHLZ) domains of other Myc proteins. The product of *c-myc* binds nonselectively to DNA and (with higher affinity) to CAC[GA]TG. In normal cells *myc* expression is strictly dependent on mitogenic stimuli and is required for cell proliferation as well as to prevent cell differentiation. Induction of *c-myc* in the absence of growth factors is sufficient to drive rodent fibroblasts into the cell cycle but concomitantly induces apoptosis unless specific survival cytokines are present. Structurally, the product Myc contains an N-terminal transactivation domain that activates transcription of the gene with which it interacts, and a C-terminal bHLHLZ motif, which is known to mediate Myc dimerization and sequence-specific DNA binding. Although Myc-Myc dimers can form in solution, Myc homodimers are not detected *in vivo*, dimers forming with other partners, e.g. Max, which has a bHLHLZ

myc+

motif but no transactivation domain (see **MAX**). *In vitro*, Myc-Max (and Myc-Myc) dimers have been found to bind DNA with a core sequence of CACGTG. Unlike Myc, Max is a stable protein expressed constitutively in growing, resting, and differentiating cells, and is capable of binding further proteins, e.g. Mad and Mxi, which compete with Myc for binding. The *myc* gene may be activated to become a transforming gene by proviral insertion, chromosomal translocation, or gene amplification, but generally protein structure is not affected; **Philadelphia chromosome** is one example. The *myc* gene has been implicated in various neoplasms in birds, mice, and humans, oncogenic activation occurring mainly through constitutive and elevated expression of the Myc protein. Example from human (six motifs): database code MYC_HUMAN, 439 amino acids (48.75 kDa). [From 'myelocytomatosis'.]

myc+ a variant form of **myco+** (before a vowel).

mycarose 2,6-dideoxy-3-C-methyl-L-ribo-hexose; a component of several **macrolide** antibiotics.

mycelia the plural of **mycelium**.

mycelium (pl. **mycelia**) the network or mass of discrete hyphae that forms the body (thallus) of a fungus. —**mycelial** adj.

myceto+ or (before a vowel) **mycet+** comb. form denoting fungus.

myco+ or (before a vowel) **myc+** comb. form denoting fungus.

mycobacterium (pl. **mycobacteria**) any bacterium belonging to the genus **Mycobacterium**.

Mycobacterium a genus of Gram-positive, aerobic, non-motile, non-spore-forming, acid-fast bacteria of the family Mycobacteriaceae. The cells are rodlike and can form filaments or branched structures; their cell walls characteristically contain **mycolic acid**. Some species are pathogenic to humans and/or other animals, e.g. *M. tuberculosis* (the tubercle bacillus) and *M. leprae* (the leprosy bacillus).

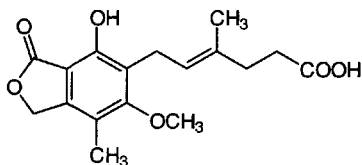
mycobactin any of a family of **siderochromes** of the hydroxamic acid-derivative category, produced by various species of *Mycobacterium*. At least nine have been characterized.

mycolic acid any of a family of α -substituted β -hydroxylated very-long-chain fatty acids of general formula $R_2\text{-CH(OH)-CHR}_1\text{-COOH}$, where R_1 is a $C_{20}\text{-}C_{24}$ linear alkyl group and R_2 is a complex $C_{30}\text{-}C_{60}$ group of very variable structure. They occur naturally as esters of arabinose in the cell walls of *Mycobacterium* spp., *Nocardia* spp., and *Corynebacterium* spp. and as esters of trehalose in **cord factor**. See also **cyclic fatty acids**.

mycology the scientific study of fungi. —**mycological** or **mycologic** adj.; **mycologist** n.

mycolyl the generic name for acyl groups derived from **mycolic acids**.

mycophenolic acid 6-(1,3-dihydro-4-hydroxy-6-methoxy-7-methyl-3-oxo-5-isobenzofuranyl)-4-methyl-4-hexenoic acid; an antibiotic and antitumour material from *Penicillium brevicompactum*. It was first isolated in 1896, and said by **Florey** to be 'the first antibiotic produced by a mould to be crystallized'. As the morpholinoethyl ester (mycophenolate mofetil) it is an approved immunosuppressant in renal transplantation.



mycoplasma or (formerly) **pleuropneumonia-like organism** (abbr.: **PPLO**) any member of the Mycoplasmatales, an order of minute, Gram-negative, prokaryotic microorganisms that are the smallest known free-living organisms. They have no cell walls (except for some saprophytic species), are variable in form, and can pass through bacteria-retaining filters. Some are saprophytic while others are pathogenic to humans, other

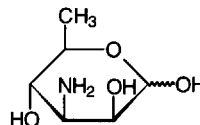
myelin-proteolipid O-palmitoyltransferase

animals, and plants. Mycoplasmas frequently infect tissue cultures. They were once thought to be the same as **L forms**. A newer classification replaces mycoplasma as a trivial name for members of the entire class by the term **mollicute**, and the class is called **Mollicutes**. The trivial name 'mycoplasma' is then restricted to members of the genus *Mycoplasma*.

Mycoplasma the main genus of **mycoplasmas**.

mycoprotein a commercially produced protein-containing foodstuff derived from moulds grown on carbohydrate substrates.

mycosamine 3-amino-3-deoxy-D-rhamnose; 3,6-dideoxy-3-amino-D-mannose; an amino sugar that contributes the nitrogen-containing moiety of the polyene antibiotics **amphotericin B** and **nystatin**.



mycose a common name for **trehalose**.

mycostatic describing any agent that inhibits the growth of fungi; fungistatic.

mycotoxin any poisonous substance produced by a fungus.

myelo+ a variant form of **myelo+** (before a vowel).

myelin the material that constitutes the **myelin sheaths** surrounding vertebrate nerve axons. It is rich in lipid, particularly in glycolipid. See also **myelin protein**.

myelination the process by which a nerve axon acquires a **myelin sheath**.

myelin protein any of the proteins that form part of the **myelin sheath**. In the following descriptions, all examples are of human precursor sequences. **Myelin-associated glycoprotein**, a cell adhesion molecule for postnatal neural development, is a member of the immunoglobulin superfamily. It is an integral membrane protein. Database code MAG_HUMAN, 626 amino acids (68.99 kDa). **Myelin basic protein**, a protein of the cytoplasmic face of the myelin membrane, is a major protein component of normal myelin of the central nervous system. Very basic in nature (pI > 10.5), it is a single-chain protein of known sequence (human and bovine) and having little tertiary structure. Administration of myelin basic protein in complete **Freund's adjuvant** in various experimental animals elicits experimental allergic encephalitis due to induction of delayed hypersensitivity to the protein, which is suspected of having an important role in the causation of demyelinating diseases in humans. Three forms are produced by alternative splicing. Database code MBP_HUMAN, 196 amino acids (21 arginines, 14 lysines) (21.34 kDa). **P2** is probably a lipid transport protein; it is found in Schwann cells, associated with myelin basic protein. Database code MYP2_HUMAN, 131 amino acids (14.76 kDa). **P0** is an immunoglobulin-like membrane protein of Schwann cells of the peripheral nervous system. It is associated with a hereditary polyneuropathy (Charcot-Marie-Tooth disease type 1b). Database code MYP2_HUMAN, 131 amino acids (14.76 kDa). **Myelin proteolipid protein (PLP or lipophilin)** is the major myelin protein from the central nervous system, playing an important role in the formation or maintenance of the multilamellar structure of myelin. It is an integral membrane protein. Alternative splicing leads to a shorter variant. Database code MYPR_HUMAN, 276 amino acids (29.91 kDa); identical in mouse, rat, and human. Mutations are associated with Pelizaeus-Merzbacher disease, and other mutations result in dysmyelination, e.g. *jimpy* and *rumpshaker* (mouse) and MD (rat). See also **Folch-Lees protein**, **Wolfgang proteolipid protein**.

myelin-proteolipid O-palmitoyltransferase EC 2.3.1.100; other names: myelin PLP acyltransferase; acyl-protein synthase; an enzyme that catalyses a reaction between palmitoyl-

myelin sheath

CoA and [myelin proteolipid] to form [myelin proteolipid] O-palmitoylprotein and CoA.

myelin sheath the sheath that surrounds the axons of vertebrate nerves. It is formed by Schwann cells in peripheral nerves and by oligodendrocytes in the central nervous system; these cells wrap up to 100 concentric layers of their plasma membrane in a tight spiral around the axons. Myelin sheaths prevent almost all electric current leakage across the covered portion of the axon membrane, thereby contributing to the rapid transmission of nerve impulses along the axons. *See also myelin, myelin protein.*

myelo+ or (*before a vowel*) **myel+** *comb. form* 1 denoting bone marrow. 2 denoting the spinal cord. 3 denoting **myelin**.

myeloblast a cell of the bone marrow that is the precursor of the **myelocyte**.

myelocyte a nucleated cell of the bone marrow that is a precursor of the **granulocytes** of the blood.

myeloid 1 of or pertaining to bone marrow. 2 of or pertaining to the spinal cord. 3 of or resembling myelocytes.

myeloma or **myelomatosis** a tumour caused by the uncontrolled proliferation of a clone of immunoglobulin-producing plasma cells. It leads to multiple deposits of tumour cells in the bone marrow and an excessive production of a single type of (i.e. monoclonal) immunoglobulin molecule, which may belong to any of the major classes of immunoglobulins. In some cases either the light or the heavy chains may be produced in excess or exclusively. *See also Bence-Jones protein, Waldenström's macroglobulinemia.*

myeloma protein an immunoglobulin that is overproduced by a **myeloma**. They are single immunoglobulin types and form the basis of monoclonal antibody technology. Human examples include **Bence-Jones protein**. *See also hybridoma.*

myeloperoxidase a special name for **peroxidase** (def. 2) from polymorphonuclear leukocytes, so named following its isolation from the blood of patients with myeloid leukemia. It is concerned in the bactericidal action of these cells. It was originally named **verdoperoxidase** on account of its green colour.

Mylar the proprietary name for a type of polyethylene terephthalate polyester that forms very strong films of good transparency and high electrical resistance.

myo+ or (*before a vowel*) **my+** *comb. form* denoting muscle.

myo- prefix used to distinguish common **inositol**, 1,2,3,5/4,6-cyclohexanexanol (*formerly meso-inositol*) from the other eight possible inositol stereoisomers.

myoblast any of the mononucleate precursor cells that fuse to form a multinucleate skeletal muscle cell.

myocardium the middle, muscular, layer of the heart wall. —**myocardial** *adj.*

myocyte 1 a muscle cell of any type. 2 any contractile cell.

myoD the gene for myoblast determination protein, it is related to **MYC**; the human equivalent is **MYF3**. Its product has similar activity to **MRF4**. Example from mouse: database code **MYOD_MOUSE**, 318 amino acids (34.18 kDa).

myofibre or (*esp. US*) **myofiber** a skeletal muscle fibre; it consists of a single, long, multinucleate cell, composed predominantly of a large number of myofibrils (typically 1000 or more).

myofibril any of the long, cylindrical, contractile elements, 1–2 µm in diameter, that comprise the major component of the cytoplasm of the **myofibre** and extend for its entire length. Each myofibril is composed in turn of numerous **myofilaments**. —**microfibrillar** *adj.*

myofilament either of two kinds of thin, contractile, thread-like, interdigitating structures within the myofibrils of **striated muscle**. The so-called thick filaments are about 15 nm in diameter and 1.6 µm in length, and the so-called thin filaments are about 8 nm in diameter and 1.0 µm in length.

myogen a name sometimes used for a number of different protein mixtures extractable with water from skeletal muscle of various species of animal. **Myogen A** is a crystallizable complex of **aldolase** and **glycerol-3-phosphate dehydrogenase** that can be prepared from rabbit muscle.

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myosin-heavy-chain k

myogenic factor any of various proteins involved in muscle differentiation. They are normally produced only in muscle cells, but have been shown to activate muscle differentiation mechanisms when introduced into other cells. They include **MyoD** (*see myoD*), **Myf5**, and **myogenin**. Example from human: database code **MYF4_HUMAN**, 224 amino acids (25.01 kDa).

myogenin a protein involved in muscle differentiation (i.e. a **myogenic factor**) that induces fibroblasts to differentiate into myoblasts. It is probably a sequence-specific DNA-binding protein, similar to other members of the Myc family of **helix-turn-helix** transcription factors (*see MYC*). Example from rat: database code **MYOG_RAT**, 287 amino acids (32.47 kDa).

myoglobin or (*formerly*) **muscle hemoglobin** or **myohemoglobin** *abbr.:* Mb; a 17.5 kDa oxygen-carrying hemoprotein that occurs in muscle, particularly in red fibres, and consists of a single polypeptide chain of 153 amino-acid residues to which a single ferroheme prosthetic group is noncovalently bound. Myoglobin binds dioxygen noncooperatively with an affinity between that of hemoglobin and that of cytochrome oxidase. It thus can accept dioxygen from hemoglobin and store it in the muscle cell for release to cytochrome oxidase when the supply becomes limiting. It also assists the direct transfer of dioxygen from the surface of the myocyte to the mitochondria. **Sperm-whale myoglobin** was the first protein whose three-dimensional structure was determined by X-ray crystallography. Example from *Sus scrofa domestica* (domestic pig): database code **NRL_1PMBA**, 153 amino acids (16.93 kDa); 3-D structure known.

myoglobinuria the excretion of myoglobin in the urine.

myohaematin a historical term, used by MacMunn for the 'histohaematin' (i.e. the cytochromes) of striated muscle.

myo-inositol an alternative name for **inositol** (def. 1) (*formerly known as muscle sugar*). *See also myo-*.

1D-myo-inositol-trisphosphate 3-kinase EC 2.7.1.127; an enzyme that catalyses a reaction between ATP and 1D-myo-inositol 1,4,5-trisphosphate to form 1D-myo-inositol 1,3,4,5-tetrakisphosphate with release of ADP. The enzyme requires calcium. Examples of two human isoforms: database code **IP3K_HUMAN**, 461 amino acids (50.95 kDa); database code **IP3L_HUMAN**, 472 amino acids (53.41 kDa). *See also inositol phosphates.*

myokinase an alternative name for **adenylate kinase**.

myosin any of a group of related proteins with ATPase activity that occur in the contractile apparatus of both muscle and nonmuscle cells; the myosin of striated muscle is the major component of the thick filaments. Individual myosin molecules consist of two identical **heavy chains** of ≈200 kDa and two pairs of light chains (*see light chain* (def. 2)) with distinct masses that vary within the range 15–27 kDa according to the source. The N-terminal region of each heavy chain consists of a globular head with which the light chains are associated, and the remaining three-quarters of each heavy chain consists predominantly of an alpha helix, which is intertwined with its partner to form a long tail to the molecule. The head region contains binding sites for actin and for ATP, whereas aggregation with other myosin molecules to form filaments occurs at the tail region. Example from human muscle protein is a hexamer of two heavy chains and four light chains. Different isoforms exist for different muscle types. Light chain 1, skeletal muscle isoform: database code **MLE1_HUMAN**, 193 amino acids (20.99 kDa); light chain has an **EF hand** motif. Heavy chain, cardiac muscle β isoform: database code **MYSB_HUMAN**, 1934 amino acids (222.73 kDa); five motifs. Some early preparations of myosin were denoted by letters. Thus myosin A and myosin B were actin-poor and actin-rich preparations respectively, while myosin T and L-myosin were among the first actin-free preparations. The name myosin was originally given by F. W. Kühne in 1859 to the substance in muscle press juice which on standing at room temperature set to a gel.

myosin-heavy-chain kinase EC 2.7.1.129; an enzyme that



myosin-light-chain kinase

catalyses the phosphorylation by ATP of specific threonines of myosin heavy chain with release of ADP, during chemotaxis of *Dictyostelium discoideum* (slime mould). It inhibits myosin thick filament formation. Example from *D. discoideum*: database code KMHC_DICDI, 783 amino acids (89.05 kDa).

myosin-light-chain kinase EC 2.7.1.117; *other names*: myosin kinase; smooth-muscle-myosin-light-chain kinase; an enzyme that catalyses the phosphorylation by ATP of myosin light chain with release of ADP. It was one of the first Ca^{2+} /calmodulin-dependent protein kinases to be discovered. Its specificity is restricted to myosin light chain. In skeletal muscle, light-chain phosphorylation modulates tension during contraction. In smooth muscle, it initiates contraction. Examples from chicken: nonmuscle, database code KML2_CHICK, 1258 amino acids (139.56 kDa); muscle, database code KMLC_CHICK, 648 amino acids (73.09 kDa).

myosin-light-chain-phosphatase EC 3.1.3.53; *other names*: myosin light chain kinase phosphatase; myosin phosphatase; an enzyme that catalyses the hydrolysis of phosphate from myosin light-chain phosphate. Example from chicken (large chain, long form): database code A55142, 1004, amino acids (111.48 kDa).

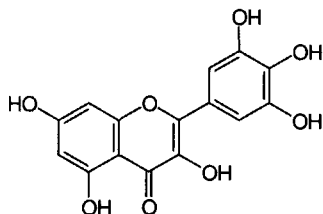
myotonia any genetically distinct hereditary disease characterized by specific muscle malformations.

myotrophin a 12 kDa protein of hypertrophied myocardium of spontaneously hypertensive rats that stimulates protein synthesis in cultured adult rat myocytes *in vitro*. It contains putative consensus phosphorylation sites for protein kinase C and casein kinase II, and binds to **NF- κ B**. Example (fragments): database code A37902, 38 amino acids (3.96 kDa).

myotube the multinucleate structure formed by the fusion of proliferating myoblasts in culture under certain conditions. It is characterized by the presence of certain muscle-specific proteins that are not found in the myoblast and may represent a stage in the development of skeletal muscle cells.

Myr symbol for the **myristoyl** group.

myricetin 3,3',4',5',7-hexahydroxyflavone; a **flavone** that is widely distributed in plants. The name originates from *Myrica*, a genus of shrubs. Myricetin inhibits a number of enzymes, including α -glucosidase, xanthine oxidase, and glyoxalase.



myricyl or **melissyl** the common name for triacontyl, $\text{CH}_3\text{--}[\text{CH}_2]_{28}\text{--CH}_2\text{--}$, the alkyl group derived from the C_{30} straight-chain alkane, triacontane. Myricyl palmitate is the major component of beeswax.

myristate 1 the trivial name for tetradecanoate, $\text{CH}_3\text{--}[\text{CH}_2]_{12}\text{--COO}^-$, the anion derived from myristic acid (i.e. tetradecanoic acid), a saturated straight-chain higher fatty

acid and a major component of many plant and animal fats. 2 any mixture of free myristic acid and its anion. 3 any salt or ester of myristic acid.

myristin an old name for any of the glyceryl esters of myristic acid, especially the triester, **trimyristin**, or trimyristoylglycerol.

myristoleic acid see **tetradecenoic acid**.

myristoyl or (formerly) **myristyl** symbol: Myr; the trivial name for tetradecanoyl, $\text{CH}_3\text{--}[\text{CH}_2]_{12}\text{--CO--}$, the acyl group derived from myristic acid (i.e. tetradecanoic acid). It occurs in most animal and plant fats and oils as acylglycerols and in phospholipids. The oils of the seeds of nutmeg (*Myristica fragrans*) and other members of the Myristicaceae are particularly rich sources. Compare **myristyl**.

myristoylation the introduction of one or more **myristoyl** groups into an organic compound. Protein *N*-myristoylation refers to the covalent attachment of a **myristoyl** group via an amide bond to the N-terminal Gly residue of a nascent polypeptide. This **post-translational modification** has only been observed in eukaryotes and is irreversible. The *N*-myristoyl proteins are numerous with diverse functions. In some cases the modification is essential for the biological role of the protein, while in other cases it is not essential. The process is catalysed by glycylpeptide *N*-tetradecanoyltransferase (EC 2.3.1.97).

myristyl 1 the common name for tetradecyl, $\text{CH}_3\text{--}[\text{CH}_2]_{12}\text{--CH}_2\text{--}$, the alkyl group derived from tetradecane; use deprecated. 2 a former trivial name for tetradecanoyl (**myristoyl** now used).

myrosinase an alternative name for **thioglucosidase**.

myt5 the gene for bacteriophage T5 5'-exodeoxynuclease (EC 3.1.11.3). It performs exonucleolytic cleavage in the 5' to 3' direction to yield 5'-phosphomononucleotides. Database code EXO5_BPT5, 291 amino acids (33.41 kDa). See also **deoxyribonuclease**.

myxamoeba or (esp. US) **myxameba** a nonflagellated, naked, amoeboid cell that occurs in the life cycle of some slime moulds.

myxoedema or (esp. Brit.) **myxoedema** the syndrome caused by hypothyroidism in adults. It is characterized by mucoid infiltration and coarsening of the skin, weight gain, intolerance to cold, and mental sluggishness.

myxo+ or (usually before a vowel) **myx+** comb. form indicating mucoid, mucus, or slime.

Myxobacterales the myxobacteria; an order of Gram-negative bacteria that bear extracellular slime and have a characteristic swarming habit that leads to the formation of stalked fruiting bodies. A more recent classification has suggested replacement of Myxobacterales by Myxococcales.

myxobacteria bacteria belonging to the order **Myxobacterales**.

myxoedema a variant spelling (esp. Brit.) of **myxedema**.

myxomatosis a highly contagious disease of rabbits, caused by a myxoma virus, part of a subgenus of poxvirus. It is characterized by proliferative changes in the connective tissues.

Myxomycetes a large class of very primitive organisms, also known as the true or acellular slime moulds, found living saprophytically in damp conditions. Their major vegetative stage is a multinucleate migratory mass of protoplasm, from which (usually macroscopic) fruiting bodies arise.

myxovirus any of a group of negative-strand RNA animal viruses, now divided into **orthomyxoviruses** and **paramyxoviruses**.

Nn

n symbol for 1 nano+ (SI prefix denoting 10^{-9} times). 2 neutron; may be written as n^0 to indicate its charge.

n symbol for 1 any (unknown or unspecified) number, especially an integer. 2 amount of substance; the substance in question may be indicated as a suffix in parentheses or by a subscript. 3 refractive index (of a nonabsorbing material); n_D symbol for refractive index measured with light of the sodium D-lines (589.0 and 589.5 nm). 4 order of reaction. 5 order of reflection. 6 haploid number of chromosomes. 7 number density.

n- prefix indicating normal (isomer); no longer used.

n_H symbol for Hill coefficient (alternative to h).

N symbol for 1 nitrogen. 2 newton. 3 a residue of the α -amino acid L-asparagine (alternative to Asn). 4 a residue of an incompletely specified base in a nucleic-acid sequence that may be adenine, cytosine, guanine, or either thymine (in DNA) or uracil (in RNA). 5 an undetermined, unidentified, or unspecified nucleoside residue (alternative to Nuc). 6 (when suffixed to the symbol for a sugar molecule or residue) aminodeoxy+; e.g. GlcN is the symbol for 2-amino-2-deoxy-D-glucose (i.e. D-glucosamine). 7 normal (def. 1).

N symbol for 1 the SI derived physical quantity named number of entities (e.g. molecules, atoms, ions, formula units). 2 neutron number. 3 Avogadro constant (N_A or L now recommended). 4 the total number of receptors per unit area of membrane, per cell, or per unit mass of protein.

N- prefix (in chemical nomenclature) indicating substitution on nitrogen; a particular nitrogen atom may be specified by a right superscript.

N_A symbol for Avogadro constant (alternative to L).

N_L symbol for Loschmidt constant.

N_a symbol for sodium.

NA abbr. for 1 nicotinic acid. 2 noradrenaline (i.e. norepinephrine). 3 nucleic acid. 4 numerical aperture.

NAC symbol for (when suffixed to the symbol for a sugar molecule or residue) an acetylated aminodeoxy sugar; e.g. GlcNAc is the symbol for N-acetylglucosamine or 2-acetamido-2-deoxy-D-glucose.

nactin any of a group of **macrotetrolide** antibiotics produced by various strains of *Actinomyces*. The group includes **nonactin**, **monactin**, **dinactin**, and **trinactin**.

NAD abbr. for nicotinamide-adenine dinucleotide (when its oxidation state is unknown or unspecified). See also **NAD⁺**, **NADH**, **NADP**, **NAD(P)**.

NAD⁺ abbr. for oxidized nicotinamide-adenine dinucleotide.

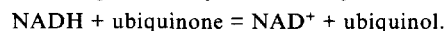
NAD⁺ ADP-ribosyltransferase see **poly(ADP)-ribosyl transferase**.

NADH abbr. for nicotinamide-adenine dinucleotide (reduced). Use of **NADH₂** to represent **NADH** is not recommended.

NADH dehydrogenase EC 1.6.99.3; **systematic name**: NADH:(acceptor) oxidoreductase; **other names**: cytochrome *c* reductase; type I dehydrogenase. An enzyme that catalyses a reaction between **NADH** and an acceptor to form **NAD⁺** and a reduced acceptor; **FAD** is a coenzyme. The protein contains iron-sulfur complexes. This is a form of **NADH dehydrogenase (ubiquinone)**, the catalytic activity of **complex I** of the mitochondrial respiratory chain that results from isolation procedures. The activity is modified in a way that causes the enzyme to react directly with cytochrome *c* as the acceptor. Example from *Escherichia coli*, membrane protein: database code DHNA_ECOLI, 433 amino acids (47.17 kDa). The enzyme in which the acceptor is a quinone is EC 1.6.99.5; **recommended name**: **NADH dehydrogenase (quinone)**; **systematic name**: **NADH:(quinone-acceptor) oxidoreductase**; it has no cofactors. In mitochondria, the enzyme is one activity of **NADH-ubiquinone oxidoreductase**, EC 1.6.5.3; **recommended name**: **NADH dehydrogenase (ubiquinone)**; **systematic name**:

NADH:ubiquinone oxidoreductase; **other names**: ubiquinone reductase, type I dehydrogenase, complex I dehydrogenase. This **FAD**-containing enzyme catalyses the oxidation of **NADH** by ubiquinone to form **NAD⁺** and ubiquinol. The protein also contains iron-sulfur complexes. This is the catalytic activity of mitochondrial **complex I**. Example from *Neurospora crassa*, **NADH-ubiquinone oxidoreductase** 9.6 kDa subunit and **NADH dehydrogenase**: database code ACP_NEUCR, 134 amino acids (14.44 kDa).

NADH dehydrogenase (ubiquinone) EC 1.6.5.3; **other names**: ubiquinone reductase; type I dehydrogenase; complex I dehydrogenase; a flavoprotein enzyme that catalyses the reaction:



The flavin is **FAD**, and the protein also contains iron-sulfur complexes. This is the catalytic activity of mitochondrial **complex I**. Example from *Neurospora crassa*, **NADH-ubiquinone oxidoreductase** 9.6 kDa subunit and **NADH dehydrogenase**: database code ACP_NEUCR, 134 amino acids (14.44 kDa).

NADH kinase EC 2.7.1.86; an enzyme that catalyses the formation of **NADPH** from **ATP** and **NADH** with release of **ADP**.

Nadi reagent a mixture of 1-naphthol and *N,N*-dimethyl-*p*-phenylene diamine that can be oxidized by cytochrome oxidase to form indophenol blue.

NAD⁺ kinase EC 2.7.1.23; **other name (now obsolete)**: **DPN kinase**; an enzyme that catalyses the formation of **NADP⁺** from **ATP** and **NAD⁺** with release of **ADP**.

NADP abbr. for nicotinamide-adenine dinucleotide phosphate (when its oxidation state is unknown or unspecified). See also **NAD**, **NADP⁺**, **NAD(P)**, **NADPH**.

NADP⁺ abbr. for oxidized nicotinamide-adenine dinucleotide.

NAD(P) abbr. for a coenzyme that may be **NAD** or **NADP**.

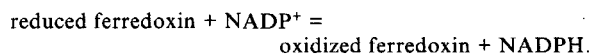
NAD(P)⁺ abbr. for a coenzyme that may be **NAD⁺** or **NADP⁺**.

NAD(P)⁺-arginine ADP-ribosyltransferase EC 2.4.2.31; **other names**: **ADP-ribosyltransferase**; **mono(ADP-ribosyl) transferase**; a sarcoplasmic reticulum enzyme that catalyses the formation of **N²-(ADP-D-ribosyl)-L-arginine** from **NAD⁺** and **L-arginine** with release of **nicotinamide**. It couples **poly(ADP-ribose)** to arginine residues in proteins. Example from rabbit: database code NART_RABIT, 327 amino acids (36.09 kDa).

NADPH abbr. for nicotinamide-adenine dinucleotide phosphate (reduced). Use of **NADPH₂** to represent **NADPH** is not recommended.

NAD(P)H abbr. for a coenzyme that may be either **NADH** or **NADPH**. Use of **NAD(P)H₂** to represent **NAD(P)H** is not recommended.

NADPH:adrenodoxin oxidoreductase EC 1.18.1.2; **other names**: **adrenodoxin reductase**; **recommended name**: **ferredoxin-NADP⁺ reductase**; a flavoprotein enzyme that catalyses the reaction:



The flavin is **FAD**. 3-D structure known; it has much α -helical structure and a **beta barrel**. The location of the cofactor binding site is known. Examples (human): database code ADRO_HUMAN, 497 amino acids (54.39 kDa); 3-D structure with 2'-phospho-5'-AMP from spinach: database code NRL_2FNR, 296 amino acids (33.47 kDa).

NAD(P)⁺ stereospecificity in oxidation-reduction reactions involving the nicotinamide coenzymes, **NAD⁺** and **NADP⁺** reversibly accept hydrogen at the 4 position of the nicotinamide ring. In **NADH** and **NADPH** this position is **prochiral**, and the faces of the nicotinamide ring and the C-4 hydrogens of the dihydronicotinamide ring may be labelled according to the

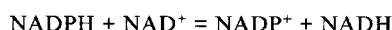


NAD(P)⁺ transhydrogenase

Re/Si and *pro-R/pro-S* conventions, respectively. Then, if the ring is drawn with the $-\text{CONH}_2$ group pointing to the right, the hydrogen projecting out from the plane of the paper is the *pro-R* hydrogen (or H_R) and that projecting back the *pro-S* hydrogen (or H_S). Alcohol dehydrogenases (EC 1.1.1.1) transfer to the *Re* face (i.e. that from which H_R points out) and use H_R . In biochemistry the *pro-R* hydrogen region is also labelled as H_A , and the *pro-S* hydrogen as H_B . Enzymes that use the *pro-R* hydrogen (H_A) are described as A-specific, while those that use the *pro-S* hydrogen (H_B) are called B-specific. See also *pro-R/pro-S* convention and *Re/Si* convention.

NAD(P)⁺ transhydrogenase (AB-specific) EC 1.6.1.2; *other names*: pyridine nucleotide transhydrogenase; transhydrogenase. An enzyme that catalyses the same reaction as **NAD(P)⁺ transhydrogenase (B-specific)**; however, preparations from heart mitochondria are A-specific with respect to NAD⁺ and B-specific with respect to NADP⁺. See also **NAD(P)⁺ stereospecificity**.

NAD(P)⁺ transhydrogenase (B-specific) EC 1.6.1.1; *other names*: pyridine nucleotide transhydrogenase; nicotinamide nucleotide transhydrogenase; a flavoprotein enzyme that catalyses the reaction



(with specificity for H_B of NADPH and NADH). The flavin is FAD. The enzyme is coupled to respiration and acts as a **hydrogen-ion pump**. Example, bovine mitochondrial precursor: NNTM_BOVIN, 1086 amino acids (113.91 kDa). Example, two subunits of enzyme from *Escherichia coli*: α , database code PNTA_ECOLI, 510 amino acids (54.56 kDa); β , database code PNTB_ECOLI, 462 amino acids (48.67 kDa). See also **NAD(P)⁺ stereospecificity**.

NAD⁺ pyrophosphorylase see **nicotinamide-nucleotide adenyltransferase**.

NAD⁺ synthase EC 6.3.1.5; *other name*: NAD⁺ synthetase; an enzyme that catalyses the formation of NAD⁺ from deamido-NAD⁺, ATP, and NH_3 with release of AMP and pyrophosphate. See also **NAD⁺ synthase (glutamine-hydrolysing)**; **nicotinamide-nucleotide adenyltransferase**.

NAD⁺ synthase (glutamine-hydrolysing) EC 6.3.5.1; *other name*: NAD⁺ synthetase (glutamine-hydrolysing); an enzyme that catalyses the formation of NAD⁺ from deamido-NAD⁺, L-glutamine, ATP, and H_2O with release of AMP, pyrophosphate, and L-glutamate. Example from *Escherichia coli* (also called nitrogen-regulatory protein): database code NADE_ECOLI, 274 amino acids (30.56 kDa). See also **nicotinamide-nucleotide adenyltransferase**.

NAF *abbr.* for Neu-activating factor (see **neu**).

NAG *abbr.* for N-acetylglucosamine.

NAGA *abbr.* for N-acetylgalactosamine.

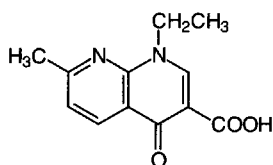
Nagase (proteinase) *another name* for a species variant of **subtilisin**, EC 3.4.21.62, from *Bacillus subtilis*; an endopeptidase with a broad spectrum of proteolytic activity.

Nagase a *pseudonym* for β -N-acetylglucosaminidase (formerly EC 3.2.1.30). This enzyme is now included with EC 3.2.1.52, β -N-acetyl-hexosaminidase.

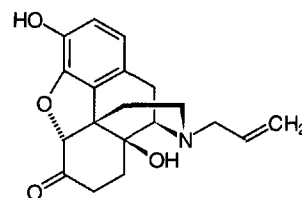
NAIBS *abbr.* for nonadrenergic imidazoline binding site; see **imidazoline receptors**.

Na⁺,K⁺-ATPase see **sodium/potassium ATPase**.

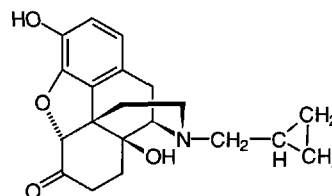
nalidixic acid 1-ethyl-7-methyl-1,8-naphthyridin-4-one-3-carboxylic acid; a synthetic **4-quinolone antibiotic** that is active against many Gram-negative bacteria. It specifically inhibits bacterial DNA gyrase thereby interfering with DNA synthesis. It has no apparent effect on eukaryotic DNA synthesis.



naloxone 1-N-allyl-7,8-dihydro-14-hydroxynormorphinone; one of the most effective opiate antagonists. It binds tightly to opiate receptors in the nervous system, its action depending on the 3-carbon side-chain substitution on the nitrogen. It is used as an antidote to overdosage of narcotics.



naltrexone 1-N-cyclopropylmethyl-7,8-dihydro-14-hydroxynormorphinone; the cyclopropylmethyl analogue of **naloxone**; it has similar properties but is longer acting.



NANA *abbr.* for N-acetylneuraminic acid (i.e. **sialic acid**).

nano+ *symbol*: n; SI prefix denoting 10⁻⁹ times. It was formerly known as millimicro (*symbol*: mμ).

nanogram *symbol*: ng; a unit of mass equal to 10⁻⁹ gram, or 10⁻¹² kilogram.

nanometre or (US) **nanometer** *symbol*: nm; a unit of length equal to 10⁻⁹ metre. It was formerly known as millimicron (*symbol*: mμ).

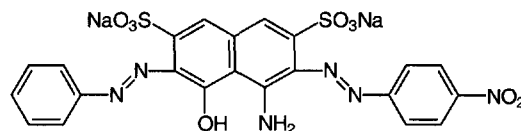
nanomolar *symbol*: nM or nM; describing a solution containing one **nanomole** per litre of solution, or a specified multiple or fraction thereof; describing an amount-of-substance concentration similarly expressed.

nanomole *symbol*: nmol; a unit of amount of substance equal to 10⁻⁹ mole.

nanosecond *symbol*: ns; a unit of time equal to 10⁻⁹ second.

NAP-2 *abbr.* for neutrophil activating peptide 2; one of several peptides derived from **platelet basic protein precursor**. It consists of LA-PF4 without the first 15 N-terminal amino acids. It is a potent chemoattractant and activator for neutrophils.

Naphthalene Black or **Naphthol Blue Black** or **Amido Black** a dye used for staining protein on cellulose acetate or on starch or agarose gels.



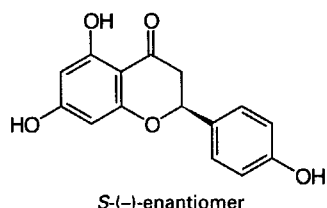
Naphthol Blue Black an alternative name for **Naphthalene Black**.

Napierian logarithm *symbol*: log_e or ln; an alternative name for natural **logarithm**. [After John Napier (1550–1617), Scots mathematician who invented logarithms.]

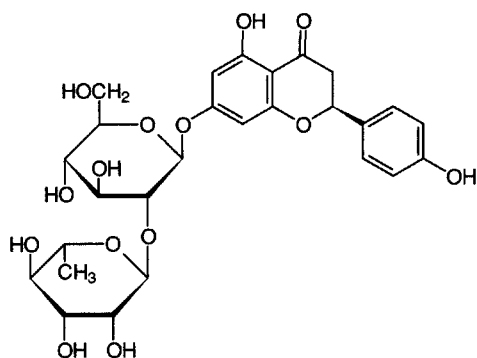
narcosis a state of partial or complete loss of consciousness induced by a drug or drugs that have a depressant action on the central nervous system.



narcotic 1 any of a heterogeneous group of drugs that is capable of inducing narcosis and noted for their analgesic action. **2** (*esp. US*) the trivial name for any illegal drug or abused drug. **3** of, or relating to, narcosis; of, relating to, or denoting a narcotic drug. **naringenin** 4',5,7-trihydroxyflavanone; the aglycon of **naringin**.



naringin naringenin-7-rhamnoglucoside; a compound that occurs in the flowers, fruit, and rind of the grapefruit, and is the main bitter component of grapefruit juice. *See also* **naringenin**.



nascent just formed or in the process of being formed, especially in a chemical or biochemical reaction.

Nathans, Daniel (1928–), US microbiologist and molecular biologist; Nobel Laureate in Physiology or Medicine (1978) jointly with W. Arber and H. O. Smith 'for the discovery of restriction enzymes and their application to problems of molecular genetics'.

native (*of a biopolymer*) undenatured, i.e. having the structure and biological activity possessed in living material.

native state of a protein or nucleic acid, the form in which it occurs in the intact cell, its three-dimensional structure depending on formation of the appropriate hydrogen bonds.

natriuresis the excretion of sodium ions in the urine.

natriuretic 1 describing any agent that promotes the excretion of sodium ions in the urine. **2** such an agent.

natriuretic hormone any of various substances present in plasma or urine that are capable of stimulating sodium excretion by the kidney and of inhibiting membrane Na^+/K^+ -ATPase in cells from kidney and some other tissues. *See* **natriuretic peptide**.

natriuretic peptide any of several peptides, secreted by various tissues, that stimulate natriuresis. The first of the group to be identified was **atrial natriuretic peptide** (or atrial natriuretic factor) (*abbr.*: ANP; *other name*: type A natriuretic peptide; *former names*: atriopeptin; vasoactive peptide). Its discovery arose from the observation that extracts of atrial granules could bring about **natriuresis**. Subsequently, **brain natriuretic peptide** (*abbr.*: BNP; *other name*: type B natriuretic peptide) and **type C natriuretic peptide** (CNP) were identified. Members of this family share many common features; they are produced primarily in the heart and specialized areas of the central nervous system, and induce diuresis, natriuresis, and vasorelaxation. The natriuretic effect results from direct inhibition of

sodium absorption in the collecting duct, increased glomerular filtration, and inhibition of aldosterone production and secretion. The major constituent of atrial granules, ANP, is synthesized from a precursor, prepronatriodilatin (152 amino acids), which is cleaved to proANP (126 amino acids). The latter is strikingly homologous between many species; proANP is cleaved into an N-terminal fragment, amino-acid residues 1–98, and the biologically active hormone, residues 99–126; the amino-acid sequence of ANP (proANP 99–126) is identical in all mammalian species except at residue 12 (110 of proANP), which is methionine in humans, dogs, and cows but isoleucine in rats, mice, and rabbits. There are two genes for the precursor (prepronatriodilatin); cleavage of the latter leads to two peptides, ANP:

SLRRSSCFGGRMDRIGASGLGCNSFRY

and cardiodilatin-related peptide (CDP):

NPMYNAVSNAIDLMDFKNLLDHLEEKMPLED.

The synthesis of BNP follows a very similar course. In humans and pig a 32-amino-acid hormone is the circulating form of BNP, sequence (human)

SPKMQVQSGCGFRKMDRISSSSGLGCKVLRH;

in rodents BNP contains 45 amino acids and is known as iso-ANP. A 17-member ring, formed by a disulfide bridge between the two cysteine residues, is essential for biological activity of ANP and BNP. *See also* **natriuretic-peptide receptor**.

natriuretic-peptide receptor any of several membrane proteins that bind **natriuretic peptides** and mediate their intracellular effects. Three receptors have been identified by molecular cloning: ANP_A (previously known as GC-A, ANP-R_1 , or ANP_B); ANP_B (previously known as GC-B); and ANP_C . Human ANP_A and ANP_B each have one transmembrane domain and intrinsic guanylate cyclase activity in an intracellular domain, which is highly conserved (88% identity). At the ANP_A receptor, atrial natriuretic peptide is about ten times more potent than brain natriuretic peptide, whereas type C natriuretic peptide is 50–500 times more potent than other agonists at the ANP_B receptor. ANP_C is a disulfide-linked homodimer with one transmembrane domain and no guanylate cyclase domain; it was once thought to be involved in clearance of natriuretic peptides, but may act through other second messenger systems. Examples from human; ANP_A receptor: database code ANPA_HUMAN, 1061 amino acids (117.78 kDa); ANP_B receptor: database code ANPB_HUMAN, 1047 amino acids (116.88 kDa); ANP_C receptor: database code ANPC_HUMAN, 540 amino acids (59.70 kDa). All are type I membrane glycoproteins.

natural abundance the number of atoms of a given nuclide in any naturally occurring mixture of the nuclides of a particular element, expressed as a percentage or fraction of the total number.

natural killer cell *see* **NK cell**.

natural logarithm or **Napierian logarithm** *symbol*: $\log_e$ or $\ln$; *see* **logarithm**.

natural product any organic substance of biological origin, as opposed to one obtainable only by chemical synthesis. In practice the term is often restricted to mean **secondary metabolite**.

natural selection the principle that the best competitors in any given population of organisms have the best chance of breeding success and thus of transmitting their characteristics to subsequent generations. The members of any population show individual differences – anatomical, physiological, or metabolic – that affect their functional efficiency in a given environment. The less efficient members tend to die out or produce fewer offspring than the more efficient members, which are better adapted to compete for food or other resources, and so produce relatively more offspring. The principle is fundamental to modern concepts of evolution, and was first articulated, independently, by the British naturalists Alfred Russel Wallace (in 1858) and Charles Darwin (in 1859).

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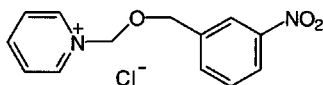
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

natural-selection theory of immunity

natural-selection theory of immunity or **Jerne's theory of immunity** a theory of antibody production that postulates the spontaneous presence, in the blood of an animal, of small numbers of antibody molecules against all antigens to which the animal can respond, and delegates to the antigen the sole role of carrying such specific globulin molecules from the bloodstream into cells in which these molecules can induce proliferation of a particular antibody.

Nb *symbol for niobium.*

NBPC *abbr. for 1-[(m-nitrobenzyloxy)methyl]pyridinium chloride; a reagent useful for the immobilization of single-stranded nucleic acids on cellulose.*



NBS *abbr. for 1 National Bureau of Standards (of the USA) (now National Institute for Standards and Technology). 2 N-bromosuccinimide.*

NCAM *abbr. for neural cell adhesion molecule.*

N cell a type of cell, found mainly in the upper small intestine in humans, that contains round, homogeneous granules in which **neurotensin** is stored. Such cells are sparse in the lower small intestine.

NC-IUB *abbr. for Nomenclature Committee of the International Union of Biochemistry; redesignated (in 1992) NC-IUBMB.*

NC-IUBMB *abbr. for Nomenclature Committee of the International Union of Biochemistry and Molecular Biology. Compare JCBN.*

NCS *the proprietary name for a 0.6 M solution in toluene of a mixture of quaternary ammonium compounds of general formula [(CH₃)₂R'R''N]⁺OY⁻, where R' and R'' are unbranched alkyl groups containing 6 to 20 carbon atoms and Y is a hydrogen atom or a lower alkyl group. It is useful for solubilizing a wide range of specimens of biological origin in organic solvents for liquid scintillation counting.*

Nd *symbol for neodymium.*

nDNA *abbr. for nuclear DNA.*

NDP *abbr. for any (unknown or unspecified) nucleoside diphosphate in which the diphosphate moiety is esterified at the 5' position of the glycoside.*

Ne *symbol for neon.*

NE *abbr. for norepinephrine.*

nearest-neighbour sequence analysis a technique that employs a 5'-³²P-labelled deoxyribonucleotide substrate in the enzymic synthesis of DNA followed by enzymic degradation of the DNA specifically to 3'-deoxyribonucleotides. From knowledge of the labelled substrate used and of the labelled degradation products formed, the **nearest neighbour frequency**, i.e. the frequency with which a particular nucleotide is found next to a specific neighbour nucleotide in the polynucleotide chain, can be calculated.

near ultraviolet radiation **ultraviolet radiation** in the wavelength range 200–400 nm.

nebulin a giant protein (800 kDa) that is specific for the skeletal muscle of vertebrates; it acts possibly as a 'molecular ruler' for length regulation of the thin filament. The protein appears to have small repeats of approximately 35 amino acids. Example, partial sequence (241 amino acids) from human: database code NEBU_HUMAN.

nebulize or **nebulise** to convert a liquid into a mist or a spray of fine droplets. —**nebulization** or **nebulisation** *n.*

nebulizer or **nebuliser** a device for nebulizing a liquid (*see nebulize*); an atomizer.

necrobiosis the death of a cell or a group of cells within a tissue, whether a normal occurrence or due to a pathological process.

necropsy *another name for autopsy.*

necrosis the death of a portion of a tissue as a result of disease or injury.

nectary a gland that secretes the sugary fluid called nectar. Nectaries occur in many animal-pollinated flowers, to attract insects, birds, and other pollinating agents.

Nedd *abbr. for NPC-expressed, developmentally down-regulated genes (NPC is an abbreviation of neural precursor cells). These genes were isolated as DNA corresponding to mRNA, the levels of which are down-regulated during mouse brain development. The products are proteins presumed to be involved in embryonic development and differentiation of the nervous system. Nedd-2 protein is a cysteine endopeptidase that may be involved in controlling cell death. Example, Nedd-2 from mouse: database code NED2_MOUSE, 171 amino acids (19.22 kDa).*

Needham, Joseph (1901–95); British biochemist, embryologist, Sinologist, and science historian, noted for his multivolume *Science and Civilization in China* (now being continued by the Needham Research Institute, Cambridge, UK). During World War II, he proposed the formation of an international science agency, and was credited with putting the S (science) into UNESCO.

NEFA *abbr. for nonesterified fatty acid. Compare FFA.*

negative 1 (of an electric charge) having the same polarity as an electron. 2 (of a chemical entity) having a negative electric charge; having an excess of electrons. 3 (of a point in an electric circuit) having a lower potential than some other point to which a potential of zero is assigned. 4 (*in photography*) an exposed and developed photographic film or plate showing an image in reversed tones, or complementary colours, to those of the object photographed. 5 a quantity less than zero. —**negativeness** or **negativity** *n.*

negative chromatography a term sometimes used to denote methods of purification of substances by **affinity chromatography** in which a specific contaminant in a sample interacts with and is selectively retained by the adsorbent, especially as opposed to **positive chromatography**.

negative-contrast technique a technique used in preparing specimens for electron microscopic examination in which the specimen is mixed with an electron-dense material that penetrates the interstices of the specimen but not the material of the specimen itself. The specimen thus appears transparent against an opaque background. It is commonly called **negative staining**.

negative control the prevention of some biological activity caused by the presence of a specific chemical entity.

negative cooperativity a form of **cooperative ligand binding** in which binding of one ligand to one site on a (macro)molecule decreases the affinity at other sites for the subsequent binding of other ligands.

negative effector *see effector.*

negative feedback *see feedback* (def. 1).

negative-ion mass spectrometry a version of **mass spectrometry** in which negative rather than positive ions, generated from the sample by chemical ionization, are separated and measured in an appropriate instrument.

negative staining *see negative-contrast technique.*

negative strand *another name for minus strand* (def. 2); in some RNA viruses, RNA is a minus strand which does not directly code for protein. Such viruses have a replicase packaged into the capsid to enable them to synthesize the plus strands necessary for replication.

negative-strand virus or **negative-stranded virus** any RNA virus in which the genome consists of single-stranded RNA (i.e. minus strand) of base sequence complementary to that of the virus-specified mRNA (which is the positive, or plus, strand). Such viruses, which comprise class V in the **Baltimore classification**, must make a positive RNA strand before making viral proteins. *See also minus strand* (def. 2), **plus strand** (def. 2).

Neher, Erwin (1944–), German biophysical chemist; Nobel Laureate in Physiology or Medicine (1991) jointly with B. Sakmann 'for their discoveries concerning the function of single ion channels in cells'.

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NEM *abbr. for N-ethylmaleimide.*

nematic (of a substance) being in or having a mesophilic state in which the molecules are arranged as in a series of parallel threads but not in layers. The molecules can rotate about their axes and can move in the plane orthogonal to the line of the thread. *Compare smectic. See also liquid crystal.*

Nembutal a proprietary name for pentobarbitone sodium; pentobarbital sodium.

N-end rule a rule stating that the *in vivo* half-life of a protein is a function of its N-terminal residues.

NENTREZ a version of **ENTREZ** for use on individual personal computers. It is available on a licensing basis free of charge to academic researchers.

neo+ *comb. form 1* denoting new, recent. **2** (in chemistry) denoting an isomer of new (and therefore initially unknown) structure.

neo-Darwinism the reworking of evolutionary theory in the light of discoveries about genetics as a result of the work of Fisher, Haldane, Sewall Wright, and others. The behaviour of genes in natural populations is understood in terms of the four fundamental evolutionary forces: **natural selection**, **genetic drift**, gene flow, and **mutations**.

neoendorphin any of a small group of [Leu⁵]enkephalin-related peptides. α Neendorphin is the decapeptide [Leu⁵]enkephalyl-Arg-Lys-Tyr-Pro-Lys, and β neendorphin is the nonapeptide [Leu⁵]enkephalyl-Arg-Lys-Tyr-Pro; both are hypothalamic opioid peptides. *See also proenkephalin.*

neoflavone any of a group of 4-phenyl-1,2-benzopyrones found in certain plants. Their occurrence is probably restricted to the Guttiferae and Leguminosae. —**neoflavonoid** *n.*

neoglycoprotein a proposed generic name for any originally carbohydrate-free protein that has been chemically derivatized by covalent attachment of carbohydrate.

neokestose *O*- β -D-fructofuranosyl-(2 $\rightarrow$ 6)- α -D-glucopyranosyl- β -D-fructofuranoside; an isomer of **kestose** isolated from the sap of the sugar maple and produced artificially, together with kestose, by the action of yeast or mould invertase preparations on sucrose.

neolignan any of a class of phenylpropanoid dimers in which the phenylpropane units are linked head-to-tail instead of tail-to-tail as in **lignans**; e.g. eusiderin from the heartwood of members of the Magnoliaceae.

neomycin an aminocyclitol antibiotic complex produced by *Streptomyces fradiae*. It is made up of three components: neomycin A (or **neamine**; a degradation product of neomycin B and neomycin C), neomycin B, and neomycin C. Neomycin causes misreading during protein synthesis in bacteria but its target site on the ribosome is apparently different from that of streptomycin. *See also neosamine C.*

neonatal of, relating to, or affecting a newborn animal. For human infants the neonatal period is considered to be the first month after birth.

neonate any newborn animal.

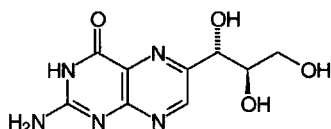
neoplasia **1** the formation of a **neoplasm** or neoplasms. **2** the formation of new tissue.

neoplasm any new and morbid formation of tissue; a tumour.

neoplastic of, relating to, neoplasia or to a neoplasm.

neoprene poly(2-chloro-1,3-butadiene); a synthetic rubber that is resistant to oils and organic solvents and is useful for the manufacture of flexible tubing, washers, etc. for laboratory and many other uses.

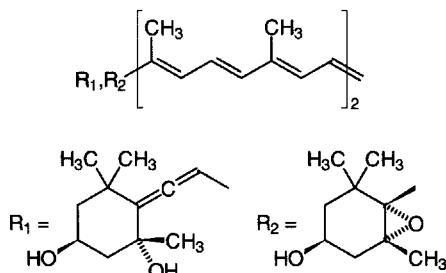
neopterin the *D*-erythro enantiomer of **biopterin**; its biosynthetic precursor.



neosamine C 2,6-diamino-2,6-dideoxy-D-glucose; a constituent of **neomycin C** and some other antibiotics including **kanamycin**.

neosome the suggested secondary particulate biogenic precursor of the ribosome, intermediate between the **eosome** and the completed ribosome. Neosomes in *Escherichia coli* are particles of at least two kinds, with sedimentation coefficients of 43S and 30S.

neoxanthin a carotenoid **xanthophyll** present in the leaves of all higher plants.



nephelometry a technique in which the light dispersed by a suspension is measured orthogonally to the direction of the incident light; the amount of scattered light is dependent on the number and size of the particles in the light path. *Compare turbidimetry. —nephelometric adj.; nephelometrically adv.*

nephro+ a variant form of **nephro-** (before a vowel).

nephrectomy the surgical removal of one or both kidneys. —**nephrectomize** *vb.*

nephro+ or (before a vowel) **nephro+** *comb. form* denoting kidney or kidneys.

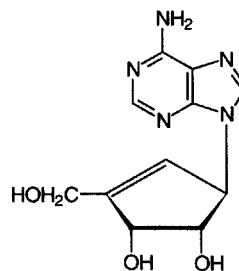
nephrocalcin or **osteocalcin-related protein precursor** a calcium-binding protein found in bone. It is expressed only in kidney and is also found as a urinary acidic glycoprotein of humans and other mammals that strongly inhibits formation of calcium oxalate crystals in renal tubules. It contains γ -carboxyglutamic acid residues; a modified protein lacking these is found in patients with calcium oxalate renal calculi. Example from *Mus musculus*: database code OSTR_MOUSE, 95 amino acids (10.46 kDa).

nephrogenic diabetes insipidus a rare X-linked recessive genetic disorder involving renal insensitivity to vasopressin resulting in polyuria and polydipsia; it can also be acquired as a result of toxic damage to the tubules. The genetic form is associated with mutations in the V_2 receptor (*see vasopressin receptor*), two examples being substitution of Asp for Ala at position 132 and a frameshift mutation at position 246 leading to a premature stop codon and a truncated receptor.

nephron any of the structural and functional urine-secreting units that occur in the kidney. Typically each nephron is made up of a glomerulus, a proximal convoluted tubule, a loop of Henle, and a distal convoluted tubule.

nephropontin *see osteopontin.*

neplanocin A a cyclopentenyl analogue of adenosine; a natural





antibiotic that inhibits *S*-adenosylhomocysteine hydrolase and vaccinia virus multiplication.

neprilysin EC 3.4.24.11; *other names*: neutral endopeptidase; endopeptidase 24.11; enkephalinase; a zinc endopeptidase widely distributed on the outer surfaces of cells.

Ner *symbol* for the nervonoyl (i.e. (Z)-tetracos-15-enoyl) group.

Nernst, Walther Hermann (1864–1941), German physical chemist distinguished for his work on thermochemistry and chemical thermodynamics; Nobel Laureate in Chemistry (1920) 'in recognition of his work in thermochemistry'.

Nernst distribution law when, at a constant temperature, a solute distributes itself between two immiscible phases, then the ratio of its concentrations in the two phases is constant, and is described by the relation $C_1/C_2 = K$, where C_1 and C_2 are, respectively, the amount-of-substance concentrations of the solute in phases 1 and 2 at equilibrium, and K is the distribution constant. The law only applies in dilute solution.

Nernst equation a the potential, E , of an electrode reversible to one ion in a solution containing that ion is given by:

$$E = E^\circ + (RT/zF)\ln a,$$

where E° is the standard potential of the electrode (relative to the standard hydrogen electrode), a is the activity of the ion to which the electrode is reversible, z is the charge number of the ion, R is the (molar) gas constant, F is the Faraday constant, and T is the thermodynamic temperature. This equation can be extended to the e.m.f. of an electrolytic cell. **b** For a redox reaction, the potential of an electrode (usually platinum or gold) immersed in a redox solution, E_h , is given by:

$$E_h = E^\circ + (RT/nF)\ln(a_{ox}/a_{red}),$$

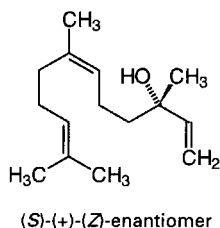
where E° is the standard redox potential of the system at a given pH (usually pH 7.0), n is the number of electrons transferred in the reaction, and a_{ox} and a_{red} are respectively the activities of the oxidized and reduced species in the solution. **c** If two solutions are separated by a membrane permeable to only one of two (or more) ions and the activities of the ions are different on either side of the membrane then a membrane potential, E_A , will be generated where

$$E_A = (RT/zF)\ln(a_L/a_R),$$

where a_L and a_R are the activities respectively of the permeable ion on the left and right sides of the membrane.

Nernst heat theorem *another name* for a particular formulation of the third law of thermodynamics (see **thermodynamics**).

nerolidol peruvicol; 3,7,11-trimethyl-1,6,10-dodecatrien-3-ol; a widespread acyclic sesquiterpene.



nerve any of the cordlike bundles, consisting of **nerve fibres** and **glia** encased in a connective-tissue sheath, that connect the central nervous system to other parts of the body. They may be motor, sensory, or of mixed function.

nerve cell *an alternative name* for **neuron**.

nerve fibre the axonal process of a **neuron** together with its covering sheath. Bundles of nerve fibres running together make up a **nerve**.

nerve gas any of various poisonous gases or volatile liquids that act by inhibiting the passage of impulses through the nervous system and neuromuscular junctions. They are commonly irreversible inhibitors of **acetylcholinesterase**.

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nerve growth factor 1 any of a number of polypeptides that exert a trophic effect on neurons. They play a part in the development and maintenance of sensory neurons in dorsal root ganglia and sympathetic neurons in peripheral sympathetic ganglia. 2 *abbr.*: NGF; the first known member of a family of polypeptides that act as growth factors for neurons. In addition to the general properties of such nerve growth factors, it also stimulates growth and differentiation of B lymphocytes, and stimulates histamine release from mast cells. The biologically active form of NGF contains two identical polypeptide chains of 120 amino-acid residues and known sequence and exerts its action through specific receptors in the neuronal plasma membrane. It is produced by neurons, astrocytes, and Schwann cells, but also by fibroblasts, epithelial cells, activated macrophages, and smooth muscle cells. It exists as an inactive complex of two α subunits, two β subunits, and two γ subunits. The active form consists of the two β subunits, and is known as β NGF. Example: β NGF precursor, a homodimer, from human: database code NGF_HUMAN, 241 amino acids (26.96 kDa); four motifs; residues 122–241 comprise the homodimer subunit. Related proteins subsequently discovered include **brain-derived neurotrophic factor** and **neurotrophins**.

nerve growth factor receptor any of a family of plasma membrane integral proteins that bind nerve growth factors. Low-affinity and high-affinity binding is observed; high affinity binding, and action on effector systems, requires the presence of both the *trk* protooncogene product (p140^{prototrk}) and a 75 kDa low affinity receptor glycoprotein (p75^{NGFR}). The latter can bind **nerve growth factor** (def. 2), **brain-derived neurotrophic factor**, and **neurotrophins** 3 and 4, but it is apparent that considerable specificity exists in transduction of the signal depending on which member of the nerve growth factor family and which cell type is involved. Example: p75^{NGFR}, human precursor, homodimer linked by disulfide bonds, *N*- and *O*-glycosylated; database code NGFR_HUMAN, 427 amino acids (45.13 kDa, increased by glycosylation).

nerve impulse *see impulse* (def. 3).

nerve or **nerve** the **galactocerebroside** in which the acyl group is **nervonoyl**. It was originally isolated from nervous tissue.

nervonate 1 *numerical symbol*: 24:1 (15); *the trivial name* for (Z)-tetracos-15-enoate, $\text{CH}_3\text{--}[\text{CH}_2]_7\text{--CH=CH--}[\text{CH}_2]_{13}\text{--COO}^-$ (*cis* isomer), the anion derived from nervonic acid, (Z)-tetracos-15-enoic acid a monounsaturated straight-chain higher fatty acid. 2 any mixture of free nervonic acid and its anion. 3 any salt or ester of nervonic acid. *See also nervonoyl*.

nervonoyl *symbol*: Ner; *the trivial name* for (Z)-tetracos-15-enoyl, $\text{CH}_3\text{--}[\text{CH}_2]_7\text{--CH=CH--}[\text{CH}_2]_{13}\text{--CO-}$ (*cis* isomer), the acyl group derived from nervonic acid, (Z)-tetracos-15-enoic acid. It occurs as the acyl moiety of the cerebroside **nervon** in brain, as a major constituent of milk phosphatidylcholines and sphingolipids in, e.g., pig and human, and in sphingolipids of other tissues.

nervous of, pertaining to, mediated by, or containing nerves; neural.

nervous system the extensive network of cells specialized to carry information, in the form of nerve impulses, to and from all parts of the body. Nervous systems occur in all orders of multicellular animals other than sponges.

nervous tissue tissue from any part of the nervous system.

nesidioblast a precursor cell of a pancreatic islet cell.

nesidioblastoma a tumour of nesidioblasts.

nesidioblastosis a pathological condition in which individual blastic pancreatic duct cells differentiate abnormally into islet cells scattered among the exocrine tissue. The condition is sometimes associated with severe infantile hyperinsulinism and hypoglycemia.

nesslerize or **Nesslerize** to treat (a sample) with **Nessler's reagent**; to test for or estimate (ammonia) by such means. *Also*: **nesslerise**, **Nesslerise**. —**nesslerization** or **nesslerisation** *n*.

Nessler's reagent an alkaline aqueous solution of potassium tetraiodomercurate(II). It gives a yellow or brown colour or



nestin

precipitate when added to solutions containing ammonia or ammonium ions and is useful in the detection and quantitative estimation of microgram quantities of ammonia in aqueous solution. [After Julius Nessler (1827–1905), German agricultural chemist.]

nestin an intermediate-filament protein whose name derives from neuroepithelial stem cells, in which it is specifically expressed. It has significant homology, including heptad repeats, to other intermediate-filament proteins. Example from rat: database code NEST_RAT, 1805 amino acids (198.53 kDa).

netrin any of a family of proteins first isolated from chick brain as diffusible chemoattractants that guide commissural axons to their targets during development. Netrin 1 is expressed at high levels in the floor plate during the period when commissural axons are growing towards the floor plate, while netrin 2 is similarly expressed at lower levels in the ventral two-thirds of the spinal cord. D netrin is a *Drosophila* homologue, expressed in midline cells of the central nervous system when commissures are forming. The proteins are related to Unc proteins (e.g. see *unc-86*) of the nematode *Caenorhabditis elegans*.

netropsin a *Streptomyces*-derived basic peptide antibiotic that selectively binds to A–T base pairs in the small groove of right-handed B conformations of DNA. It induces A to B conformational changes in DNA.

net sensitivity see **flux control coefficient**.

net synthesis increase in the total mass (of a reaction product), as opposed to any apparent increase estimated by incorporation of labelled precursor (which may arise from exchange).

neu an oncogene first described in association with a neuroblastoma after treatment of rats with ethylnitrosourea. The *neu* protooncogene (also known as *c-erbB2*, *NGL*, and *HER2*) encodes a 185 kDa transmembrane glycoprotein with intrinsic tyrosine kinase activity, related to but distinct from the **epidermal growth factor receptor** (EGFR). Example from rat: database code NEU_RAT, 1260 amino acids (139.03 kDa). Like other members of this receptor family, the *neu* gene product, Neu, possesses a cysteine-rich extracellular domain, a transmembrane domain, and an intracellular domain with tyrosine kinase activity. EGF does not bind to Neu, but a factor known as **Neu-activating factor** (NAF) binds specifically to Neu and will bind to EGFR. Neu was originally identified in brain, and may play a critical role in neurogenesis. The rat *neu* oncogene is activated via a single point mutation in the transmembrane domain, which results in constitutive activation of the tyrosine kinase domain. A similar mutation has not been found in the human *neu* oncogene, the association of which with human cancer through gene amplification, or overexpression of the protein product, has been supported by work with breast and ovarian cancer.

Neu 1 symbol for a residue (or sometimes a molecule) of **neuraminic acid**. **2** the product of the *neu* gene.

NeuAc or (formerly) **AcNeu** symbol for a residue (or sometimes a molecule) of any isomer of acetylneuraminic acid, i.e. **sialic acid**. See also **NeuNAC**, **NeuOAc**, **neuraminic acid**.

Neu-activating factor see **neu**.

Neuberg, Carl Alexander (1877–1956); German biochemist after whom three forms of fermentation are named.

Neuberg ester an old name for D-fructose 6-phosphate. See **fructose 6-phosphate**.

Neuberg's first form of fermentation the normal anaerobic **alcoholic fermentation** of carbohydrate by yeast to yield two moles each of ethanol and carbon dioxide per mole of glucose fermented.

Neuberg's second form of fermentation the anaerobic fermentation of carbohydrate by yeast in the presence of sodium hydrogen sulfite. The acetaldehyde formed normally, which would act as a hydrogen acceptor for the reoxidation of NADH, is trapped as its bisulfite addition product, and its place is taken by dihydroxyacetone phosphate (i.e. glycerone phosphate).

phosphate), which is reduced to *sn*-glycerol 3-phosphate; this is then dephosphorylated to glycerol. Thus one mole each of glycerol, acetaldehyde-bisulfite complex, and carbon dioxide is formed per mole of glucose fermented.

Neuberg's third form of fermentation the anaerobic fermentation of carbohydrate by yeast in alkaline conditions. The acetaldehyde formed normally, which would act as a hydrogen acceptor for the reoxidation of NADH, undergoes a **Cannizzaro reaction** to form acetic acid and ethanol, and its place is taken by dihydroxyacetone phosphate (e.g. glycerone phosphate), which is reduced to *sn*-glycerol 3-phosphate; this is then dephosphorylated to glycerol. Thus one mole each of glycerol and carbon dioxide, and one half mole each of ethanol and acetic acid, are formed per mole of glucose fermented.

NeuGc symbol for a residue (or sometimes a molecule) of *N*-glycolylneuraminic acid. See also **sialic acid**.

NeuNAC or **Neu5Ac** symbol for a residue (or sometimes a molecule) of *N*-acetylneuraminic acid (i.e. **sialic acid**). It is used as an alternative to NeuAc to differentiate the *N*-acetyl from the *O*-acetyl derivative of neuraminic acid.

NeuOAc symbol for a residue (or sometimes a molecule) of *O*-acetylneuraminic acid. It is used as an alternative to NeuAc to differentiate the *O*-acetyl from the *N*-acetyl derivative of neuraminic acid. See also **sialic acid**.

neur+ a variant form of **neuro+** (before a vowel).

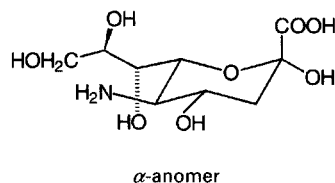
neural of, or pertaining to, a nerve or nerves, or to the nervous system.

neural cell adhesion molecule *abbr.*: NCAM; any of a family of type I membrane glycoproteins of the immunoglobulin superfamily that are involved in neuron–neuron adhesion, neurite fasciculation, outgrowth of neurites, etc. Example (precursor) of NCAM-L1 from human: database code CAML_HUMAN, 1257 amino acids (139.77 kDa); it shows strong homology with a *Drosophila* protein, **neuroglian**.

neural crest a group of embryonic cells, derived from the roof of the neural tube, that migrate to different locations and give rise to various types of adult cells.

neuraminatone 1 the anion derived from **neuraminic acid**. **2** any salt or ester of neuraminic acid.

neuraminic acid *symbol*: Neu; the monosaccharide 5-amino-3,5-dideoxy-D-glycero-D-galacto-nonulosonic acid; it may be regarded as the aldol-condensation product of pyruvic acid and *N*-acetyl-D-mannosamine. It is not found free in nature, but is the parent structure of a family of aminodeoxysugars containing nine or more carbon atoms that occur widely, especially in glycoproteins and gangliosides; they include the **sialic acids**.



neuraminidase a former name for **sialidase**.

neuraminosyl a trivial name for the glycosyl group derived from **neuraminic acid** by removal of the hydroxyl group from the anomeric carbon atom of the cyclic structure.

neuraminoyl a trivial name for the acyl group derived from **neuraminic acid** by removal of the hydroxyl group from the carboxyl group.

neuregulin any member of a family of closely related proteins implicated as regulators of neural and muscle development, and the differentiation and oncogenic transformation of mammalian epithelia. They include the Neu differentiation factor (see *neu*) and the **heregulins**. They exert their effects through



neurexin

binding to and activating receptors of the ErbB2/Neu receptor family. Neuregulins stimulate proliferation of Schwann cells, increase the rate of synthesis of acetylcholine receptors in cultured muscle cells, and are concentrated at nerve-muscle synapses.

neurexin any of a family of cell-surface receptors expressed from two genes in the brain and concentrated in neurons. They are cell-surface proteins implicated in axon guidance and synaptogenesis. Some are probably signalling molecules. They are highly polymorphic and include secreted variants. They contain single transmembrane regions and extracellular domains with repeat sequences similar to those in laminin A (see **laminin**) and **agrin**. Each gene contains two independent promoters so that transcription produces two classes of mRNA, which encode proteins with different N termini, the larger being referred to as α -neurexins and the shorter as β -neurexins. The former contain three epidermal growth factor domains, an O-linked sugar domain, a transmembrane domain, and a short cytoplasmic tail. The α - and β -neurexins differ in their N-terminal sequences but have identical C termini, including the sugar-rich, transmembrane, and cytoplasmic domains. Example, neurexin III- α from female Sprague-Dawley rat brain: database code RATNIIIA5, 1399 amino acids (154.09 kDa).

neurilemma or **neurolemma** the thin outer sheath of cells surrounding the axon (and myelin sheath) of larger peripheral nerves or surrounding the axoplasm of unmyelinated nerves.

neurite a general term for a dendrite or an axon.

neuro+ or (before a vowel) **neur+** comb. form denoting a nerve or the nervous system.

neuroactive (of an agonist or antagonist) having an effect on the nervous system.

neurocalcin an EF-hand Ca^{2+} -binding protein involved in the calcium-dependent regulation of rhodopsin phosphorylation and found in retina and brain neurons. Example (bovine): database code NECX_BOVIN, 192 amino acids (22.09 kDa).

neurochemistry the study of the chemical composition of, and the chemical reactions occurring in, nervous tissue. —**neurochemical** adj.

neurocrine 1 describing or relating to a neuron that secretes an agonist into a synapse with another neuron; describing or relating to such an agonist. 2 such an agonist, e.g. a **neurotransmitter**.

neuroendocrine 1 describing or relating to a neuron that secretes an agonist into the bloodstream; describing or relating to such an agonist. 2 such an agonist; a **neurohormone**.

neuroendocrine convertase EC 3.4.21.93; *abbr.*: NEC; *other name* prohormone convertase. Any of the enzymes that act at paired dibasic amino acids in precursors of hormones and other polypeptides, to cleave the precursor into individual mature proteins. The catalytic activity depends on the Asp-His-Ser charge-relay system. Substrates for NEC1 include **proopiomelanocortin**, **renin**, **enkephalin**, **dynorphin**, **somatostatin**, and **insulin**. NEC2 and NEC3 have also been described. Example from human (NEC1 precursor): database code NEC1_HUMAN, 753 amino acids (84.12 kDa).

neuroendocrinology the study of the interrelations between the nervous system and the endocrine system. —**neuroendocrinological** adj.

neurofascin a chick neurite-associated glycoprotein implicated in axon extension; it is an axonal surface recognition molecule of the immunoglobulin superfamily. At the N terminus it has six immunoglobulin-like motifs, followed by four fibronectin type III motifs, a proline-alanine-threonine-rich region, a transmembrane domain, and a 113-residue cytoplasmic domain. Example from chicken: database code S26180, 1272 amino acids (142.08 kDa).

neurofibril any of the very fine fibres, visible by light microscopy, that occur in the cell bodies, axons, and dendrites of neurons.

neurofibromatosis any of a number of disorders associated with the presence of multiple **neurofibromas**, benign tumours

consisting of a mixture of Schwann cells and fibroblasts. See also **neurofibromin**.

neurofibromin *other name*: neurofibromatosis-related protein (NF-1); a GTPase activating protein (see **GAP** (def. 2)), the product of the *Nf1* tumour-suppressor gene, that stimulates the GAPase activity of the **RAS** oncogene product, thus inactivating it. It appears to be implicated in the pathogenesis of **neurofibromatosis**. Example from *Mus musculus*: database code NF1_MOUSE, 28.41 amino acids (319.59 kDa).

neurofilament a type of **intermediate filament** found in neurons.

neuroglia an alternative name for **glia**.

neuroglian a probable cell-adhesion molecule, with structural similarities to **neural cell-adhesion molecule** L1. It is found in neurons and glia of the developing nervous system. It has a single transmembrane domain, with extracellular immunoglobulin-like and fibronectin-type domains. Example from *Drosophila melanogaster* (presursor): database code NRGDROME, 1239 amino acids (138.28 kDa); residues 1–23 are the signal, 24–1239 neuroglian.

neurohormone any organic compound produced by neurons and released into the circulation to act as a chemical messenger; the term is sometimes extended to all **neurotransmitters** whether released or not. —**neurohormonal** adj.

neurohumour or (US) **neurohumor** a neurohormone or a neurotransmitter. —**neurohumoral** adj.

neurohypophysis (*pl.* **neurohypophyses**) the posterior lobe of the pituitary gland, also called **neural lobe** or **pars nervosa**. It secretes the peptide hormones oxytocin and vasopressin, and also possibly some others. Compare **adenohypophysis**. —**neurohypophyseal** or **neurohypophysial** adj.

neurokinin A or **neuromedin L** or **substance K** the peptide hormone

His-Lys-Thr-Asp-Ser-Phe-Val-Gly-Leu-Met

derived from **protachykinin β precursor**. It is a member of the **tachykinin** family and is a potent bronchoconstrictor. The related peptide [β Ala⁸]neurokinin A 4–10,

H-Asp-Ser-Phe-Val- β Ala-Leu-Met-NH₂,

is a potent and selective **substance K receptor** agonist.

neurokinin B or **neuromedin K** a member of the **tachykinin** family that has neuroendocrine and neuroexcitatory functions, especially in the olfactory, gustatory, and visceral systems. It is a potent bronchioconstrictor. It has the sequence

Asp-Met-His-Asp-Phe-Phe-Val-Gly-Leu-Met-NH₂

database code TKNK_PIG, 10 amino acids (1.211 kDa). Neurokinin B acts through a G-protein-coupled receptor, known as NK-3, linked to the phosphatidylinositol cycle.

neurolemma a variant spelling of **neurilemma**.

neuroleptic any drug used to treat major psychosis, e.g. schizophrenia or mania. Such drugs include phenothiazines, e.g. **chlorpromazine**, and butyrophenones, e.g. **haloperidol**.

neurolysin EC 3.4.24.16; *other name*: neurotensin endopeptidase; an enzyme that catalyses the preferential cleavage in neurotensin of Pro¹⁰-|-Tyr¹¹.

neuromedin B a decapeptide first isolated from porcine spinal cord. Its structure,

Gly-Asn-Leu-Trp-Ala-Thr-Gly-His-Phe-Met,

has close sequence homology to **bombesin**, with which it shares mitogenic and growth-promoting properties. In rat brain its concentration is highest in the olfactory bulb, but is also high in the hypothalamus, hippocampus, and spinal cord. The B in its name indicates its similarity to bombesin. Example of precursor from human (which also includes oxytocin): database code NEUI_HUMAN, 125 amino acids (12.71 kDa).

neuromedin C a neuropeptide similar in its structure,

Gly-Asn-His-Trp-Ala-Val-Gly-His-Leu-Met,

and in its actions to **neuromedin B** and **bombesin**.

neuromedin L an alternative name for **neurokinin A**.

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neuromedin N

neuromedin N a hexapeptide whose C-terminal tetrapeptide is identical to that of **neurotensin**.

neuromodulator any endogenous agent that influences the function of a **neurotransmitter** but has no direct excitatory or inhibitory postsynaptic actions. The term usually implies long-latency and long-lasting effects (i.e. seconds to days).

neuromuscular of, pertaining to, or affecting both nervous and muscular tissue.

neuromuscular junction the place of contact between the motor end-plate of the fibre of a motor neuron and the membrane of a muscle fibre supplied by the neuron. Impulses are transmitted across the gap by diffusion of a neurotransmitter. *Compare synapse.*

neuron or **neurone** or **nerve cell** a cell that is specialized for the transmission of nerve impulses. Neurons are the structural and functional units of the nervous system. Each consists of an enlarged portion, the cell body or perikaryon, containing the nucleus and from which project a variable number of thread-like processes. Of these the short branched dendrites convey impulses towards the cell body, while a longer single nerve fibre, or axon, conveys impulses away from the cell body. The axon is unbranched except at the nerve ending. —**neuronal** *adj.*

neuronatin a protein that is highly expressed in neonate mammalian brain and which may be involved in terminal brain differentiation. Example from rat: database code RNU08290, 81 amino acids (9.21 kDa).

neuropeptide (*sometimes*) any peptide with neuroendocrine activity.

neuropeptide K a polypeptide hormone, DADSSIEKQVALL-KALYGHGQISHKRHKTDSEVGLM-NH₂, derived from **protachykinin β precursor**. It exhibits a wide spectrum of activities overlapping those of other members of the **tachykinin** family, with which it shares the C-terminal sequence FXLM-NH₂.

neuropeptide Y or (*formerly*) **melanostatin**; *abbr.*: NPY; a 36-residue peptide neurotransmitter, structure YPSKPDNPGE DAPAEDMARYYSALRHYNLITRQRY-amide (human, porcine has Leu¹⁷), found in relatively high concentrations in adrenal glands, brain, and heart. It is one of the most abundant neuropeptides. It is a potent stimulator of feeding; leptin inhibits NPY gene expression and release. NPY also regulates secretion of **gonadotropin releasing hormone**. It shows considerable sequence homology with **peptide YY** and **pancreatic polypeptide**. The active peptide is derived from a precursor; example (precursor) from human: database code NEUY_HUMAN, 97 amino acids (10.84 kDa). [Leu³¹, Pro³⁴]NPY is a selective agonist for a Y₁ receptor subclass. *See also neuropeptide-Y receptor.*

neuropeptide-Y receptor *abbr.*: NPY receptor or Y receptor; a membrane protein that binds **neuropeptide Y** (NPY) and mediates its intracellular effects. Effector pathways involve the inhibition of adenylate cyclase and of Ca²⁺ channels. The NPY receptors are seven-transmembrane-helix G-protein-coupled single polypeptides. Four types have been recognized that vary in their affinities for NPY analogues. Example of type 4 from human: database code NY4R_HUMAN, 375 amino acids (42.19 kDa). A fifth type, Y5, is suggested to be the receptor by which NPY exerts its effects on feeding.

neurophysin any of a small number (usually two in any given animal species) of cystine-rich, single-chain polypeptides present in the neurohypophysis. Each binds specifically and stoichiometrically either to oxytocin or to vasopressin (or to certain of their analogues) and is released with the respective hormone. Neurophysin I binds **oxytocin** and shares a common precursor; example from *Rattus norvegicus*; database code NEU1_RAT, 125 amino acids (12.8 kDa); residues 1–19 are the signal peptide, 20–28 oxytocin, 32–125 neurophysin I. Neurophysin II binds **vasopressin** and shares a common precursor; example from *R. norvegicus*; database code NEU2_RAT, 168 amino acids (17.92 kDa); residues 1–23, signal; 24–32, Arg-vasopressin; 36–128, neurophysin II. *See also vasopressin-neurophysin 2-copeptin precursor.*

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neurophysin II another name for **MSEL-neurophysin**.

neuroscience any of the sciences dealing with the nervous system; such sciences collectively.

neurosecretion 1 the process of secretion of a substance by a (specially adapted) nerve cell. 2 a product of such a secretion; a **neurocrine** (def. 2) or **neuroendocrine** (def. 2). —**neurosecretory** *adj.*

neurosome (*sometimes*) a mitochondrion of a nerve cell.

Neurospora a genus of ascomycete microfungi. The species *N. crassa* has been widely used in biochemical and genetic studies. It is commonly found as a mould on bread that has been kept for too long.

neurotensin a tridecapeptide, Glp-Leu-Tyr-Glu-Asn-Lys-Pro-Arg-Arg-Pro-Tyr-Ile-Leu-OH, found in mammalian brain and gut, especially in **N cells**, packaged within secretory vesicles. It is released from the hypothalamus into the circulation. Although initially shown to induce hypotension, it is now known to have a wide variety of pharmacological actions, including muscle contraction; however, its true physiological role remains unclear. Example (precursor) from bovine: database code NEUT_BOVIN, 169 amino acids (19.69 kDa). This precursor gives rise to neurotensin and two other derivatives, **neuromedin** and **neuromedin N**.

neurotensin endopeptidase *see* **neurolysin**.

neurotoxin any toxin that acts specifically or primarily on nervous tissue. —**neurotoxic** *adj.*

neurotransmission the transmission of nerve impulses from a neuron to another excitable cell via a neurotransmitter.

neurotransmitter any chemical substance released at the distal end of the axon of a neuron in response to the arrival of a nerve impulse that, by diffusing across a **synapse** or other junction, is capable of transmitting the impulse to another neuron, to a muscle cell, or to another excitable cell or, in the case of inhibitory neurotransmitters, of inhibiting the transmission. The neurotransmitter is stored in synaptic vesicles in the axon terminal; it is released into the synaptic cleft on arrival of the action potential and diffuses to and stimulates receptors in the membrane of the post-synaptic cell. An essential component of the mechanism is a means to inactivate the neurotransmitter, e.g. by acetylcholinesterase action on acetylcholine. Both excitatory (*see* **cholinoceptor**, **purinoceptor**, and **glutamate receptor**) and inhibitory (*see* **GABA receptor** and **glycine receptor**) neurotransmitters are known.

neurotrophic factor *see* **brain-derived neurotrophic factor**.

neurotrophin *abbr.*: NT; any of a group of growth factors, similar to **nerve growth factor** but with different tissue specificities. The first to be discovered was named neurotrophin 3, as two proteins with neurotrophic properties had already been discovered (nerve growth factor and **brain-derived neurotrophic factor**). Neurotrophins have properties similar to these latter. Neurotrophins 3 to 6 have been described. Examples (precursors, human): NT3, database code NT3_HUMAN, 257 amino acids (29.32 kDa); NT4 (formerly NT5), database code NT4_HUMAN, 210 amino acids (22.40 kDa).

neurotropic having a particular affinity for, or affecting nervous tissue.

neutral 1 being neither acidic nor basic (or alkaline). 2 having no net charge or electric potential.

neutral amino acid any amino acid containing equal numbers of potentially anionic and cationic groups, especially as distinct from acidic amino acids and basic amino acids.

neutral density filter an alternative term for **neutral filter**.

neutral fat *see* **neutral lipid**.

neutral filter or **neutral density filter** (*in photography and photometry*) a filter that attenuates all frequencies of light uniformly so that the relative spectral energy distribution of the transmitted light is the same as that of the incident light.

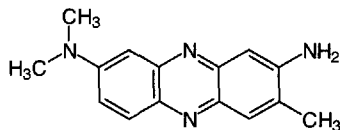
neutrality the condition, quality, or state of being **neutral** (def. 1, 2) or intermediate.

neutralize or **neutralise** to render **neutral** (def. 1, 2) or ineffective; to counteract. —**neutralization** or **neutralisation** *n.*

neutral lipid

neutral lipid an operational term for any lipid that is soluble only in solvents of very low polarity. Neutral lipids are divided into two main groups: (1) acylglycerols (glycerides), i.e. fatty-acid esters of glycerol; and (2) waxes, i.e. fatty-acid esters of long-chain monohydroxy alcohols.

Neutral Red an indicator dye that changes colour from red to orange-brown over the pH range 6.8–8.0.



neutral solution any aqueous solution in which the activities of the hydrogen and hydroxide ions are equal. Such a solution in which water is the sole solvent has a pH of 7.00 at 25° C.

neutron symbol: *n*; an elementary particle of zero charge; nucleon number 1; rest mass 1.6749×10^{-27} kg (1.0086 Da); spin $\frac{1}{2}$. Neutrons are present in the nuclei of all atoms (except the hydrogen atom), where they are stable; when free, a neutron has a mean life of 932 s, decaying to a proton and an electron.

neutron activation analysis an alternative name for **radioactivation analysis**.

neutron diffraction a technique that is similar in principle to **X-ray diffraction** by a crystal, and is applicable particularly to biopolymers and particulate structures. Because hydrogen atoms make a much more substantial contribution to the scattering of neutrons than to X-ray scattering the technique is useful for refining partially determined structures obtained by the use of X-rays. In rare cases it has been used for determination of absolute configuration.

neutron flux (density) the number of free neutrons passing through unit volume multiplied by their mean velocity.

neutron number symbol: *N*; a dimensionless physical quantity, having an integral value for any given nuclide, equal to the number of neutrons contained in the nucleus of an atom of the nuclide. Hence, it is equal to the arithmetical difference between the nucleon number, *A*, of the nuclide and the proton number, *Z*, of the relevant chemical element; i.e. $N = A - Z$.

neutron scattering see **small-angle scattering**.

neutrophil or **neutrophil leukocyte** or **neutrophilic leukocyte** the most numerous **polymorphonuclear leukocyte** found in the blood; a phagocytic cell of the myeloid series that is distinguished by the presence of cytoplasmic azurophil granules and other granules that take up neither acidic nor basic dyes. It plays a major role in the inflammatory response, undergoing chemotaxis towards sites of infection or wounding.

Newman projection a two-dimensional representation of a molecule in which the atoms are arranged as they would be seen if the molecule were viewed from one end along the carbon-carbon bond nearest to the observer. See also **conformation**.

newton symbol: *N*; the SI derived unit of force, defined as the force required to give a mass of one kilogram an acceleration of one metre per second per second; i.e.

$$1 \text{ N} = 1 \text{ kg m s}^{-2} = 1 \text{ J m}^{-1} (= 10^5 \text{ dynes}).$$

Newton, (Sir) Isaac (1642–1727), English astronomer, mathematician, philosopher, and physicist. —**Newtonian** *adj.*

Newtonian flow the type of flow displayed by a **Newtonian fluid**.

Newtonian fluid any fluid whose (coefficient of) viscosity is independent of flow rate, conforming to **Newton's law of viscosity**.

Newton's law of viscosity a law that describes the flow of almost all fluids of low relative molecular mass and also that of some solutions of macromolecules. A fluid moving between two parallel plates, in the *x* direction with a velocity *v*, is thought of as a number of infinitesimal layers, each of which slides along the adjacent one, the frictional resistance between

adjacent layers generating an orthogonal velocity gradient, in the *y* direction. The frictional force, *f*, between the fluid layers is then proportional to the area, *A*, of the layers and to the velocity gradient between them, such that:

$$f = \eta A (dv/dy),$$

where the constant η is known as the **coefficient of viscosity** (or **viscosity** for short) at the temperature of measurement; its reciprocal, $1/\eta$, is the **fluidity**.

new yellow enzyme an obsolete name for the enzyme **D-amino-acid oxidase** (EC 1.4.3.3), a yellow flavoprotein (FAD) with wide specificity for D-amino acids and acting also on glycine. It was so-named to distinguish it from **old yellow enzyme**. See **amino-acid oxidase**.

New Zealand obese mouse *abbr.*: NZO mouse; one of a number of strains of genetically obese mice that are used in metabolic and endocrine research. The obesity is moderate in character and there is an associated hyperinsulinemia and hyperglycemia.

nexin a protein found in the **axoneme** of eukaryotic cilia and flagella. It forms interconnections between the microtubule outer doublets that surround the inner central pair of microtubules. See also **glial-derived nexin**.

nexin 1 other names: protease nexin I (*abbr.*: PN-I); a glycoprotein that promotes neurite extension and is an extracellular serpin-like serine protease inhibitor. Example, residues 20–398 of the human glial derived nexin precursor (GDN): database code GDN_HUMAN, 398 amino acids (43.95 kDa).

nexin 2 the secreted isoform of the Alzheimer's amyloid β -protein precursor (see **β -amyloid peptide**). See also **Alzheimer's disease**.

nexus (in cytology) an alternative name for **gap junction**.

NF-1 see **CTF, neurofibromin**.

NF2 *abbr.* for neurofibromatosis type-2 tumour suppressor; see **merlin**.

NF- κ B a transcription factor for eukaryotic RNA polymerase II promoters. In humans the active protein is a heterodimer of two DNA-binding subunits (p50) which belong to the same family as the **dorsal protein**. It is found in cytoplasm as an inactive complex with an inhibitory subunit, **I κ B**; activation of NF- κ B involves its release from its inhibitory subunit. A 105 kDa protein is the precursor of the p50 subunits. These bind to the **kappa-B** consensus sequence, GGRNNYYCC, e.g. GGGGACTTTC in mouse and human. Example from *Gallus gallus*: database code KBF1_CHICK, 984 amino acids (108.17 kDa).

NF-E1 see **GATA-1**.

ng symbol for nanogram.

NGF *abbr.* for nerve growth factor.

L-NHA *abbr.* for N^G -hydroxy-L-arginine.

Ni symbol for nickel.

niacin an alternative name for **nicotinic acid**.

niacinamide an alternative name for **nicotinamide**.

NICE element *abbr.* for nodule infected cell expression element, a name given to an AT-rich element in *Sesbania rostrata* leghemoglobin glb3.

NICER elements a name suggested for a family of genes that are nerve growth factor-inducible, cyclic AMP-extinguishable, and retrovirus-like, hence the acronym.

nick 1 an interruption in the covalent continuity of one strand of a double-stranded nucleic-acid molecule. 2 to produce a nick in a nucleic-acid molecule. See also **nick translation**.

nickase an endodeoxyribonuclease present in phage-infected *Escherichia coli* cells that gives rise to single-strand breaks (nicks) in duplex DNA.

nickel symbol: *Ni*; it is a ferromagnetic, metallic transition element of group 10 of the IUPAC **periodic table**; atomic number 28; relative atomic mass 58.69. It consists of a mixture of five stable nuclides of which the two most abundant are those of mass numbers 58 and 60; its commonest valence is II. Nickel's abundance in the Earth's crust is 0.018% and it is an essential



nicking-closing enzyme

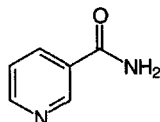
trace element for living organisms. Several artificial radionuclides of nickel are known, including nickel-56 (γ -emitter; half-life 6.1 days) and nickel-63 (emits β^- radiation, 0.066 MeV; half-life 100 years).

nicking-closing enzyme or **nicking-and-closing enzyme** see type I DNA topoisomerase.

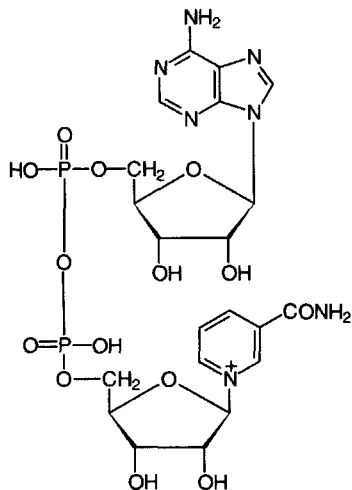
nick translation a procedure for preparing radioactively labelled DNA, especially for use as probes in hybridization experiments. Single-stranded nicks are introduced into unlabelled DNA, usually by limited treatment with deoxyribonuclease I, exposing 3'-hydroxyl termini. *Escherichia coli* DNA polymerase I (at low temperature, $\approx 15^\circ\text{C}$) is then used successively to incorporate radioactive nucleotides (present as dNTPs in the reaction mixture) to form a new, radioactively labelled polynucleotide strand while the 5' $\rightarrow$ 3' exonuclease activity of the polymerase concomitantly hydrolyses the 5' termini ahead of the growing strand, liberating 5'-mononucleotides. The reaction thus progressively replaces nucleotide residues by their labelled counterparts in a DNA duplex, which is otherwise unchanged except for the translation of the nick, i.e. its progressive displacement towards the 3' end of the strand. (The word 'translation' in the name may be misleading as the function involved is replication.)

Nicol prism an optical device constructed of two prisms of calcite cemented together with Canada balsam, used for producing plane polarized light and/or determining its orientation. It has now largely been superseded by Polaroid filters, etc. [After William Nicol (1768–1851), British physicist.]

nicotinamide the recommended trivial name for pyridine-3-carboxamide, the amide of **nicotinic acid**; also called **niacinamide**. It is a member of the B complex of vitamins, and is nutritionally equivalent to nicotinic acid. It occurs widely in living organisms, usually in combined form.

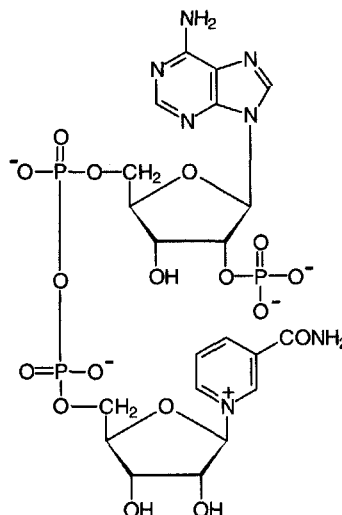


nicotinamide-adenine dinucleotide *abbr.*: NAD⁺ or NAD; if the oxidation state is not to be specified; nicotinamide(1)- β -D-ribose(5')diphospho(5')adenosine; an atypical dinucleotide containing a phosphoric anhydride linkage in place of the usual phosphodiester linkage between the component mononucleotide units, i.e. between nicotinamide mononucleotide and adenylic acid. It is the specific coenzyme in



numerous oxidoreductase enzyme reactions, although in a number it functions interchangeably with **nicotinamide-adenine dinucleotide phosphate**. In these reactions it stereospecifically accepts a hydrogen atom at position 4 of the pyridine ring, plus an electron, to form **nicotinamide-adenine dinucleotide (reduced)**. It acts also as a donor in the ADP-ribosylation of some eukaryotic and prokaryotic proteins. The α anomer is inactive as a coenzyme. *Former names*: codehydrogenase I; coenzyme I (Co I); cozymase; diphosphopyridine nucleotide (DPN). See also **nicotinamide-adenine dinucleotide (reduced)**.

nicotinamide-adenine dinucleotide phosphate *abbr.*: NADP⁺; nicotinamide(1)- β -D-ribose(5')diphospho(5')adenosine 2'-phosphate; a coenzyme with a structure and coenzymic mechanism related to those of **nicotinamide-adenine dinucleotide**, although it functions as the sole coenzyme for fewer oxidoreductases and tends to be involved more in synthetic than in energy-yielding reactions. *Former names*: codehydrogenase II; coenzyme II (Co II); phosphocozymase; triphosphopyridine nucleotide (TPN). See also **nicotinamide-adenine dinucleotide phosphate (reduced)**.



nicotinamide-adenine dinucleotide phosphate (reduced)

abbr.: NADPH; 3-aminocarbonyl-1,4-dihydropyridine(1)- β -D-ribose(5')diphospho(5')adenosine 2'-phosphate; the reduced form of the coenzyme **nicotinamide-adenine dinucleotide phosphate**. As in **nicotinamide-adenine dinucleotide (reduced)** the carbon atom at position 4 of the 1,4-dihydropyridine ring is prochiral. See also **NAD(P)⁺ stereospecificity**. *Former names*: reduced codehydrogenase II; reduced coenzyme II (Co I(H₂)); reduced phosphocozymase; reduced triphosphopyridine nucleotide (TPNH).

nicotinamide-adenine dinucleotide (reduced) *abbr.*:

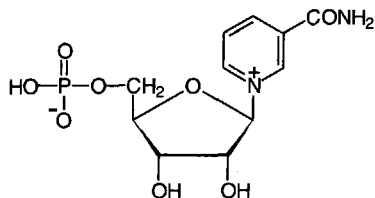
NADH; 3-aminocarbonyl-1,4-dihydropyridine(1)- β -D-ribose(5')diphospho(5')adenosine; the reduced form of the coenzyme **nicotinamide-adenine dinucleotide**. The carbon atom at position 4 of the dihydropyridine ring is prochiral; removal of one of the two hydrogen atoms from position 4 (or introduction of a hydrogen atom at this position into the oxidized form) through the agency of an NAD-linked oxidoreductase is stereospecific, the two hydrogen atoms being designated *pro-R* (or H_A) or *pro-S* (or H_B), and the two corresponding aspects of the (dihydro)pyridine ring frequently being known as the *A* face and the *B* face. The α anomer is inactive as a coenzyme. *Former names*: reduced codehydrogenase I; reduced coenzyme I (Co I(H₂)); reduced cozymase; reduced diphosphopyridine nucleotide (DPNH). See also **NAD(P)⁺ stereospecificity**.

nicotinamide mononucleotide *symbol*: NMN; **nicotin-**



nicotinamide nucleotide

amide(1)- β -D-ribose 5'-phosphate; the mononucleotide produced when **nicotinamide-adenine dinucleotide** acts as a donor of an adenylyl group in the activation of DNA ligase.



nicotinamide mononucleotide

nicotinamide nucleotide any nucleotide that contains combined nicotinamide, e.g. nicotinamide-adenine dinucleotide (reduced), nicotinamide-adenine dinucleotide phosphate (reduced).

nicotinamide-nucleotide adenylyltransferase EC 2.7.7.1; *other name*: NAD⁺ pyrophosphorylase; an enzyme that catalyses the formation of NAD⁺ from ATP and nicotinamide ribonucleotide with release of pyrophosphate and NAD⁺. See also **NAD⁺ synthase**, **NAD⁺ synthase (glutamine-hydrolysing)**.

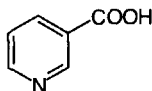
nicotinate pyridine 3-carboxylate; the anion derived from **nicotinic acid**.

nicotinate mononucleotide 3-carboxylatopyridine(1)- β -D-ribose 5'-phosphate; an intermediate in the biosynthesis of nicotinamide-adenine dinucleotide. It reacts with ATP to form desamido-NAD, which is subsequently converted to NAD by a glutamine-dependent amidation of the carboxyl group.

nicotine (S)(-)-3-(1-methyl-2-pyrrolidinyl)pyridine; the principal example of a number of related very poisonous and addictive alkaloids extractable from the dried leaves of various species of tobacco, especially *Nicotiana tabacum*. It is an agonist of the nicotinic class of **cholinoceptors**. —**nicotinic adj.**

nicotinic acetylcholine receptor see **cholinoceptor**.

nicotinic acid *abbr.*: NA; the recommended trivial name for pyridine 3-carboxylic acid, also known as **niacin**. It is a member of the B complex of vitamins, nutritionally equivalent to **nicotinamide**. At physiological pH values it exists virtually exclusively as the nicotinate anion.



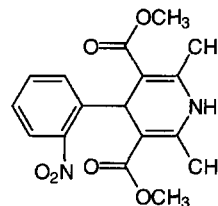
nidogen or **entactin** a sulfated glycoprotein that is widely distributed in basement membranes and tightly associated with **laminin**. Example (precursor) from human: database code NIDO_HUMAN, 1247 amino acids (136.34 kDa).

Niemann-Pick disease an alternative name for **sphingomyelin lipidosis**. [After Albert Niemann (1880–1920), German physician.]

nif the symbol for any gene controlling bacterial nitrogen fixation; individual genes are distinguished by suffixed capital letters, e.g. *nifB*, *nifQ*. In *Klebsiella pneumoniae* the *nif* cluster has at least 18 genes, arranged in seven distinct operons with no other interspersed genes; the total length of the *nif* region is $\approx$ 24 kb. The two polypeptide subunits of component I of **nitrogenase** and the polypeptide subunit of component II are each controlled by a single *nif* gene; the production and insertion of iron-molybdenum cofactor (FeMo-co) into component I appears to require three to six genes; the production of mature component II is controlled by two to four genes; two electron-transporting proteins are coded for by one gene each; and three or more genes are concerned with regulating expression of others.

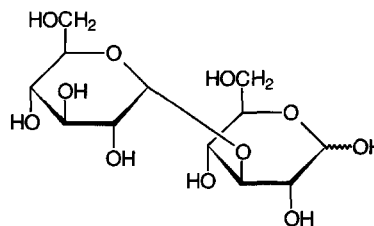
nifedipine 1,4-dihydro-2,6-dimethyl-4-(2-nitrophenyl)-3,5-

pyridinedicarboxylic acid dimethyl ester; one of the dihydropyridine calcium antagonists. It is a calcium channel blocking agent inhibiting the cellular influx of calcium associated with the plateau phase of the action potential. It acts as a vasodilator, reducing blood pressure and increasing coronary blood flow.



nigericin a 725 Da, open-chain macrotetrolide-like antibiotic, with many chiral centres, isolated from *Streptomyces hygroscopicus*. It forms stable complexes with monovalent cations, especially with K⁺; these complexes are soluble in organic solvents, and thus nigericin can act as an ionophore for monovalent cations. It has some structural resemblance to **monensin**.

nigerose the disaccharide 3-O- α -D-glucopyranosyl-D-glucose.



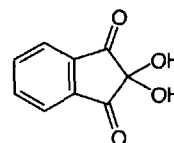
night blindness or **nyctalopia** the inability to see in dim light or at night. It is due to degeneration of the **rod cells** in the retina of the eye, which primarily mediate vision in dim light.

NIH *abbr.* for National Institutes of Health (of the USA).

NIH shift or **hydroxylation-induced migration** an intramolecular migration that often occurs in the hydroxylation reactions of alkylaromatic compounds catalysed by certain mono- or dioxygenases, whereby a hydrogen atom or a substituent atom or group at the position of hydroxylation moves to an adjacent carbon atom in the aromatic ring. For example, in the hydroxylation of phenylalanine to tyrosine, catalysed by phenylalanine 4-monooxygenase (EC 1.14.16.1), the hydrogen atom at C-4 in the ring can be shown by tritium labelling to migrate to C-3. [After NIH, i.e. National Institutes of Health, where the discoverers were based.]

NIMR *abbr.* for National Institute for Medical Research (of the UK).

ninhydrin the trivial name of triketohydrindene hydrate; 2,2-dihydroxy-1,3-indanedione; a pale-yellow crystalline substance, useful as a sensitive chromogenic reagent for the detection and quantification of compounds containing free amino or imino groups. The common amino acids (and the free amino groups of peptides and proteins, and various other amino compounds) give a red, violet, or blue coloration, the shade depending somewhat on conditions, whereas the imino acids proline and





hydroxyproline give a yellow colour. The partially reduced ninhydrin derivative, hydrindantin, is often added to the reagent solution, as it reacts with free ammonia that is formed as a side product, and in combination with ninhydrin forms the coloured product, thus enhancing the sensitivity of the reaction.

Nir symbol for the ribonucleoside ribosylnicotinamide.

Nirenberg, Marshall Warren (1927–), US biochemist and molecular biologist; Nobel Laureate in Physiology or Medicine (1968) jointly with R. W. Holley and H. G. Khorana 'for their interpretation of the genetic code and its function in protein synthesis'.

Nissl substance a collective term for the dense regions of granular endoplasmic reticulum and free polysomes observable by microscopy within the cytoplasm of the cell body of vertebrate neurons. [After Franz Nissl (1860–1919), German neuropathologist, who first observed it.]

nitrate 1 the NO_3^- ion, derived from nitric acid. 2 any salt or ester of nitric acid.

nitrate assimilation an alternative term for assimilatory nitrate reduction; see **nitrate reduction**.

nitrate dissimilation an alternative term for dissimilatory nitrate reduction; see **nitrate reduction**.

nitrate reductase 1 any of the three enzymes designated **nitrate reductase (NADH)** (EC 1.6.6.1), **nitrate reductase (NAD(P)H)** (EC 1.6.6.2), or **nitrate reductase (NADPH)** (EC 1.6.6.3) that reduce nitrate to nitrite coupled to the oxidation of the coenzymes indicated. They are of widespread occurrence in plants and microorganisms, where they participate in assimilatory nitrate reduction. All are flavoproteins, usually containing one atom of molybdenum per molecule. Example from *Arabidopsis thaliana*: database code N1A1_ARATH, 917 amino acids (103.12 kDa). 2 the microbial enzyme, designated **nitrate reductase** (EC 1.7.99.4), that reduces nitrate to nitrite coupled to the oxidation of various electron donors in dissimilatory nitrate reduction. Example from *Bacillus subtilis*: database code NARG_BACSU, 1228 amino acids (139.08 kDa). 3 the microbial enzyme, designated **nitrate reductase (cytochrome)** (EC 1.9.6.1), that reduces nitrate to nitrite coupled to the oxidation of a ferrocycytochrome to the corresponding ferri-cycytochrome. See also **nitrate reduction**.

nitrate reduction any reaction or sequence of reactions in which nitrate ion is converted to other less highly oxidized inorganic nitrogenous substances, e.g. to nitrite ion, nitrous oxide, dinitrogen, ammonium ion, or amino acids. In enzymology the term is restricted to the enzymic reduction of nitrate ion to nitrite ion, catalysed by a **nitrate reductase** enzyme. In biology, nitrate reduction is of two types, distinguishable by the purposes served and by the enzymes involved. **Assimilatory nitrate reduction** is the process occurring in higher plants, algae, and fungi, in which nitrogen from nitrate is converted into organic nitrogen-containing compounds, via the probable reaction sequence: nitrate → nitrite → [hyponitrite] → hydroxylamine → ammonium → amide group of glutamine → amino acids, etc. **Dissimilatory nitrate reduction** or **nitrate respiration** is the process occurring in some microorganisms under anaerobic conditions in which nitrate functions as a terminal electron acceptor in respiratory metabolism. The nitrogen may either be converted to ammonium (which is not assimilated), a process designated **ammonification of nitrate**, or be converted to various gaseous compounds such as nitric oxide, nitrous oxide, and/or dinitrogen, a process designated **denitrification**. Compare **nitrification**.

nitrene the chemical entity NH or any of its substitution derivatives. It contains an electrically neutral mono-coordinate nitrogen atom with four nonbonding electrons; two of these electrons are paired while the other two have spins that may be either antiparallel (singlet state) or parallel (triplet state). Other names: azacarbene; azene; imene; imine radical.

nitric oxide nitrogen monoxide; NO ; a colourless gas, only slightly soluble in water, that reacts readily with O_2 to form

nitrogen dioxide, NO_2 ; it can form the stable cation NO^+ . It has been found to account for the action of endothelium-derived relaxing factor (EDRF). Nitric oxide is a biogenic messenger with potent actions causing *inter alia* vasodilation, inhibition of platelet aggregation, and reduction of endothelium adhesion; it also has anticoagulant and fibrinolytic actions. It is formed biosynthetically from L-arginine by **nitric-oxide synthase** in most mammalian tissues in varying amounts, including vascular endothelial cells, the central and peripheral nervous system, immune cells, lung, or liver. It activates guanylate cyclase, producing cyclic GMP. It is oxidized to nitrite, then to nitrate, which is present in the blood.

nitric-oxide synthase EC 1.14.13.39; *abbr.*: NOS; *systematic name*: L-arginine,NADPH:oxygen oxidoreductase (nitric-oxide-forming). An enzyme that catalyses a reaction between L-arginine, NADPH, and O_2 in which citrulline, NADP^+ , and nitric oxide are formed; heme, FAD, FMN, and tetrahydrobiopterin are required as cofactors; the heme centre has the spectral properties of a cytochrome P450; N^{ω} -hydroxy-L-arginine is an intermediate, and acts as a substrate for the enzyme. Three isoforms are known: **nNOS**, originally expressed in neuronal tissues, and **eNOS**, first identified in vascular endothelial cells, are both expressed constitutively and are Ca^{2+} -dependent; **iNOS**, first identified as an enzyme inducible in macrophages and liver cells, is not Ca^{2+} -dependent, but binds calmodulin with extremely high efficiency. All have consensus sequences for the binding of NADPH, FAD, FMN, and calmodulin, and a phosphorylation site for cyclic-AMP-dependent protein kinase A; eNOS has an N-terminal consensus sequence for myristoylation. Inhibitors include L- N^5 -(1-iminoethyl)ornithine, N^G -monomethyl-L-arginine, N^G -nitro-L-arginine. Example, from human endothelial cells: database code NOS3_HUMAN, 1202 amino acids (133.15 kDa).

nitrification the process by which certain soil bacteria (nitrifying bacteria) oxidize ammonium ions in decomposing organic material into nitrate ions. *Nitrosomonas* spp. first convert ammonium to nitrite, and *Nitrobacter* spp. then oxidize nitrite to nitrate. The process is important because most higher plants can assimilate nitrate much more readily than they can ammonium. Compare **denitrification**.

nitrify 1 (of nitrifying bacteria) to oxidize ammonium to nitrate. 2 to treat (soil) with nitrates. —**nitrified** or **nitrifying** *adj.*

nitrifying bacteria obligately aerobic bacteria that are capable of oxidizing ammonia to nitrite, and nitrite to nitrate. See **nitrification**. Compare **denitrifying bacteria**.

nitrite 1 the NO_2^- ion, derived from nitrous acid. 2 any salt or ester of nitrous acid.

nitrite reductase 1 the enzyme designated **nitrite reductase (NAD(P)H)** (EC 1.6.6.4) that reduces nitrite to ammonium coupled to the oxidation of either NADH or NADPH. It is a flavin-dependent enzyme that also contains **siroheme**; it participates in assimilatory **nitrate reduction** in higher plants, algae, fungi and bacteria. Example from *Bacillus subtilis*: database code NASD_BACSU, 805 amino acids (88.43 kDa). 2 the enzyme designated **nitrite reductase (cytochrome)** (EC 1.7.2.1) that reduces nitrite to nitric oxide coupled to the oxidation of ferrocycytochrome *c* to ferricytochrome *c*. It is a cuproprotein and participates in denitrification processes in some microorganisms. 3 the enzyme designated **ferredoxin-nitrite reductase** (EC 1.7.7.1) that reduces nitrite to ammonium coupled to the oxidation of reduced ferredoxin; the enzyme in spinach leaves contains one molecule of siroheme and one iron-sulfur (Fe_2S_2) cluster per molecule. It participates in assimilatory nitrate reduction, the reduced ferredoxin being derived from the light reactions of photosynthesis. Example from *Zea mays*: database code NIR_MAIZE, 569 amino acids (63.34 kDa). 4 the microbial enzyme, designated **nitrite reductase** (EC 1.7.99.3), that reduces nitrite to nitric oxide coupled to the oxidation of various electron donors in dissimilatory nitrate reduction; the enzymes from *Achromobacter cycloclastes* and *Pseudomonas*

nitro

nitrogen regulatory pr

denitrificans contain copper: database code NIR_ACHCY, 340 amino acids (37.01 kDa).

nitro the chemical group $-\text{NO}_2$ in a molecule of an organic compound.

nitro+ prefix (in chemical nomenclature) 1 indicating the presence of a nitro group in an organic compound. 2 (in chemical technology) indicating that a chemical compound is a nitrate ester.

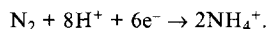
Nitroblue Tetrazolium see **Nitrotetrazolium Blue**.

nitrogen 1 symbol: N; a nonmetallic element of group 15 of the IUPAC periodic table; atomic number 7; relative atomic mass 14.0067. There are two natural stable nuclides of relative masses 14 (99.635 atom percent) and 15 (see **nitrogen-15**); its commonest valences are III and IV. Nitrogen occurs naturally in the form of diatomic molecules of the gas dinitrogen (see below) and in combined form in numberless organic molecules in all forms of life. See also **nitrogen-13**. 2 **dinitrogen** or **molecular nitrogen** symbol: N_2 ; a diatomic nonflammable gas. It constitutes 75.5% by mass, or 75.06% by volume, of the Earth's atmosphere, and undergoes exchange with organically combined nitrogen via the processes of the **nitrogen cycle**. Liquid nitrogen, b.p. 77.36 K, is commonly used as a coolant or freezing agent.

nitrogen-13 the artificial radioactive nitrogen nuclide ^{13}N . It emits a positron (β^+ -particle) (1.19 MeV max.) and no gamma radiation; it has a half-life of 9.96 min.

nitrogen-15 heavy nitrogen; the stable, naturally occurring nitrogen nuclide ^{15}N ; relative abundance in nitrogen 0.365 atom percent. It is extensively used as a tracer in studies of nitrogen metabolism and in **nuclear magnetic resonance**.

nitrogenase an enzyme complex that catalyses the ATP-dependent process of biological fixation of atmospheric nitrogen, summarized by the equation:



The electrons may be donated by either 3 reduced ferredoxin (this enzyme is EC 1.18.6.1) or 6 reduced flavodoxin (this enzyme is EC 1.19.6.1), depending on the bacterial species, the following summary being based on the systems utilizing ferredoxin. Nitrogenase consists of two oxygen-sensitive components, known as **component I** (other names: dinitrogenase; MoFe protein; molybdoferredoxin) and **component II** (other names: dinitrogenase reductase; Fe protein; azoferredoxin), in the molar ratio 1:2. Component I is a tetramer made up of two different polypeptide chains of about 50 and 60 kDa, each present twice; the tetramer also contains two Mo atoms, around 30 Fe atoms and about 30 acid-labile sulfide groups, some of the Mo, Fe, and S being present as iron-molybdenum cofactor (abbr.: FeMo-co). The extremely oxygen-sensitive component II is a dimer of identical, ≈ 30 kDa, polypeptide chains; each dimer contains four Fe atoms and four acid-labile sulfide groups. Electrons from a donor, which may be NADPH, are transferred, via reduced ferredoxin, to component II; then, in the presence of Mg^{2+} and ATP, they are passed to component I where Mo(VI) atoms are reversibly reduced to Mo(IV), the electrons ultimately being used in three two-electron steps to reduce dinitrogen to ammonium with concomitant hydrolysis of 4–5 molecules of ATP per two electrons transferred. Nitrogenase also catalyses ATP-dependent partial reductions of some other small-molecule substances that are isoelectronic with dinitrogen, e.g. acetylene, azide, cyanide, and nitrous oxide. The biosynthesis of nitrogenase is controlled by the **nif** cluster of genes. Example from *Klebsiella pneumoniae*: the subunits are named from the **nif** genes; the enzyme consists of (i) a Mo-Fe protein (component I) with two α (NifD) and two β (NifK) subunits with 30–32 Fe, 2 Mo, and S (inorganic); database codes NIFD_KLEPN, 483 amino acids (54.10 kDa) and NIFK_KLEPN, 520 amino acids (58.34 kDa); (ii) an Fe protein (NifH, component II, four motifs) homodimer (one 4Fe–4S cluster per dimer); database code NIFH_KLEPN, 292 amino acids (31.74 kDa). Associated proteins are NifE (data-

base code NIFE_KLEPN, 457 amino acids (50.10 kDa) and NifN (database code NIFN_KLEPN, 461 amino acids (50.51 kDa)) – these are involved in FeMo-co biosynthesis; NifM (database code NIFM_KLEPN, 266 amino acids (30.58 kDa)) is involved in synthesis/activation of NifH.

nitrogenase reductase another name for dinitrogenase reductase; i.e. component II of **nitrogenase**.

nitrogen balance 1 (in physiology) the **balance** (def. 1) of the nitrogen content of the nutrients and the excretory products of an organism. 2 (in agriculture) the net loss or gain of the nitrogen content of soil that results from removal of nitrogen by cropping, leaching, etc. and addition of nitrogen by nitrogen fixation or addition of nitrogenous fertilizers.

nitrogen base an alternative term for **nitrogenous base**.

nitrogen cavitation an alternative name for **pressure homogenization**.

nitrogen cycle the complex series of chemical processes by which various inorganic and organic nitrogenous compounds are interconverted and maintained in equilibrium in the biosphere. It consists essentially of **nitrogen fixation**, **nitrification**, and **denitrification**, together with assimilatory and dissimilatory **nitrate reduction** and the interconversion of nitrogenous organic matter and ammonium. Individual reactions in the series may be effected by microorganisms, plants, or animals, with contributions from nonbiological processes such as manufacture of nitrogenous fertilizers, the action of lightning, and emission of combustion products.

nitrogen equilibrium the state of **nitrogen balance** (def. 1) when nitrogen intake equals nitrogen excretion.

nitrogen fixation any biological or nonbiological process that converts atmospheric dinitrogen into inorganic nitrogenous compounds; the meaning is often extended to include the biological assimilation of ammonium so produced. Biological nitrogen fixation occurs by the action of the enzyme **nitrogenase**, found in certain species of bacteria – the nitrogen-fixing bacteria.

nitrogen mustard any of a group of bis(2-chloroalkyl)amines, as typified by the volatile base bis(2-chloroethyl)ethylamine, $\text{CH}_3\text{CH}_2\text{N}(\text{CH}_2\text{CH}_2\text{Cl})_2$, a nitrogen-containing analogue of **mustard gas**. They are potent alkylating agents, vesicants, and mutagens. They cross-link DNA chains and have been used as immunosuppressive and antineoplastic agents.

nitrogenous 1 of or relating to **nitrogen**. 2 of or relating to any substance containing nitrogen in combined form.

nitrogenous base or **nitrogen base** a generic term for ammonia or any trivalent compound of nitrogen that can be regarded as a derivative of ammonia; all except quaternary ammonium compounds are weakly basic in character. In biochemistry, the term usually denotes any purine or pyrimidine; now generally shortened to **base**.

nitrogen regulatory protein abbr.: NR protein; any of several proteins of a regulatory system for transcriptional control of the glutamine-ammonia ligase, and for its regulation at the structural level. Transcriptional regulation is mediated by the proteins NR_I , that when phosphorylated acts as a transcriptional activator of the gene *glnA* and NR_{II} , a kinase that converts NR_I to $\text{NR}_I\text{-P}$, the phosphorylated, active form. Another protein P_{II} , prevents this phosphorylation, thereby preventing transcriptional activation. P_{II} can be uridylylated at Tyr⁵¹ to $\text{P}_{II}\text{-UMP}$ by a **uridylyl transferase**, and then does not inhibit NR_I phosphorylation, so activating transcription. This occurs when the level of nitrogen is low. Regulation at the structural level of glutamate-ammonia ligase involves adenylation by adenylyl transferase, which inactivates it. Adenylyl transferase is activated by P_{II} . In the presence of $\text{P}_{II}\text{-UMP}$, adenylyl transferase is inactivated, and glutamate-ammonia ligase reverts to a non-adenylylated, active form. Uridylylation of P_{II} thus results in activation of transcription of glutamate-ammonia ligase, and its activation at the structural level. Example of P_{II} from *Escherichia coli*, a homotrimer: database code GLNB_ECOLI, 112 amino acids (12.41 kDa).

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nitroglycerin

nitroglycerin 1,2,3-propanetriol trinitrate; a compound that is used industrially in the manufacture of dynamite, and pharmacologically as an antianginal and coronary vasodilator. Its action in the latter case results from its conversion to **nitric oxide**.

nitrophenyl a group that is linked to a variety of compounds to form artificial substrates for enzymes. The nitrophenol released by the enzyme action can then be measured spectrophotometrically at 400 nm (4-nitrophenol) or 420 nm (2-nitrophenol) to follow the enzyme reaction, the esterified form having a maximal absorption at a much lower wavelength. Typical substrates are 4-nitrophenylacetate (for carboxylesterase, lipase), 4-nitrophenyl-*N*-acetylgalactosaminide (galactosaminidase), 4-nitrophenyl-*N*-acetylglucosaminide (glucosaminidase), and 2-nitrophenylbutyrate (cholinesterase).

nitroprusside test a test for free thiol groups based on the red coloration given on reaction of thiol-containing compounds with ammoniacal sodium nitroprusside, sodium nitrosopentacyanoferrate(III), $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$. Nitroprusside also reacts with ketones and is used in tests for ketone bodies in urine. The proprietary Acetest tablets contain a mixture of glycine, sodium nitroprusside, disodium phosphate, and lactose. In the presence of glycine, acetoacetate or acetone forms a blue-purple colour. A positive reaction indicates the presence of ≥ 5 mg dL^{-1} and 10 mg dL^{-1} for urine and blood respectively.

nitroso+ a variant form of **nitroso+** (before a vowel).

***N*-nitrosamide** any organic compound in which a **nitroso** group is attached to an amide nitrogen, e.g. *N*-methyl-*N*-nitroso-urea. See also ***N*-nitrosamine**.

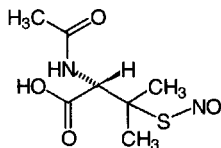
***N*-nitrosamidine** any organic compound in which a **nitroso** group is attached to an amidine nitrogen, e.g. *N*-methyl-*N*-nitroso-*N*-nitrosoguanidine. See also ***N*-nitrosamine**.

***N*-nitrosamine** any organic compound of general formula $\text{R}_1\text{R}_2\text{NNO}$. They are stable compounds formed by the action of nitrous acid on secondary amines with concurrent mono-dealkylation, and also on some tertiary amines. Most nitrosamines are strongly carcinogenic and their metabolites are strong mutagens. ***N*-nitrosamides** and ***N*-nitrosamidines** are unstable nitroso compounds that decompose in neutral or alkaline conditions to produce powerful alkylating agents. They are widely used as experimental mutagens. They are carcinogenic in any tissue with which they come into contact.

nitroso the chemical group $-\text{NO}$ in a chemical compound.

nitroso+ or (before a vowel) **nitros+** prefix (in chemical nomenclature) indicating the presence of a **nitroso** group in a chemical compound; in *C*-nitroso compounds the $-\text{NO}$ is attached to carbon (e.g. nitrosobenzene) and in *N*-nitroso compounds it is attached to nitrogen.

***S*-nitroso-*N*-acetylpenicillamine** *abbr.*: SNAP; an *S*-nitrosothiol reagent that stimulates guanylate cyclase and mimics the action of nitric oxide.



S-nitroso-*N*-acetyl-D-penicillamine

Nitrotetrazolium Blue or **Nitroblue Tetrazolium** a dye very similar in structure to Tetrazolium Blue (see **tetrazolium salt**), differing only in that each of the two tetrazolium groups has a nitrophenyl group in place of a phenyl group.

nitrous acid the very unstable monobasic acid HNO_2 . It reacts with amines, including purine and pyrimidine amines, to form the corresponding hydroxy compounds, and is thus mutagenic.

nitroxide the divalent group $=\text{N}-\text{O}^\cdot$ in a molecule of an *or*-

ganic compound. It has an unpaired electron, and is useful in spin-labelling, its derivatives being stable free radicals.

NK cell *abbr.* for natural killer cell; a large granular lymphocyte that recognizes structures on the surface of virally infected cells, which it kills, probably through the mediation of a **perforin**.

NKSF *abbr.* for NK stimulating factor. See **interleukin 12**.

Nle symbol for the α -amino acid norleucine (now known as 2-aminohexanoic acid, *symbol*: Ahx).

NLS *abbr.* for nuclear localization signal.

nm symbol for nanometre.

nM or **nM** symbol for nanomolar.

NMDA *abbr.* for *N*-methyl-D-aspartate.

NMDA receptor see **glutamate receptor**.

NMN *abbr.* for nicotinamide mononucleotide.

nmol symbol for nanomole.

NMP *abbr.* for any (unknown or unspecified) nucleoside monophosphate in which the phosphate group is esterified at the 5' position of the glycoside moiety.

NMR or **nmr** *abbr.* for nuclear magnetic resonance.

Nn symbol for the nonyl group.

Nno symbol for the nonanoyl group.

nNOS see **nitric-oxide synthase**.

No symbol for nobelium.

nociceptive causing pain or injury; reacting to a painful stimulus.

nociceptor any receptor that reacts to painful (i.e. nociceptive) stimuli; a pain-sense organ.

node 1 a knob, lump, or swelling; a constriction, especially when serially repeated (along the length of something). 2 (*in botany*) the point on a stem to which a leaf or a lateral branch is attached. 3 (*in physics*) a point of zero or minimum displacement in a standing wave or a system of such waves. See also **node of Ranvier**. —**nodal** *adj.*

node of Ranvier any of the constrictions in the myelin sheath that occur at regular intervals along the length of a myelinated nerve fibre. [After Louis Antoine Ranvier (1835–1922), French pathologist.]

NOD mice *abbr.* for nonobese diabetic mice.

nodoc a name sometimes used for **anticodon** (formed by reversing the letters of the word codon).

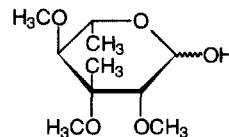
nodule bacterium or **nodule-forming bacterium** a common name for any bacterium of the genus *Rhizobium*, found as endosymbionts in root nodules. The bacterium specifically infects root cells of Leguminosae, and causes the development of **root nodules**, in which it loses its outer membrane and becomes dependent on plant metabolism. It is then morphologically differentiated from the free-living form and known as a bacteroid. In return it fixes nitrogen for the plant. See **nitrogenase**.

NOE *abbr.* for nuclear Overhauser effect.

NOESY *abbr.* for nuclear Overhauser effect spectroscopy.

nogalamycin an antitumour antibiotic material obtained from *Streptomyces nogalater*. It contains the carbohydrate component **nogalose**.

nogalose 6-deoxy-3-*C*-methyl-L-mannose 2,3,4-trimethyl ether; a carbohydrate component of **nogalamycin**.



noggin a dorsalizing factor localized to **Spemann's organizer** in *Xenopus laevis* embryos. Database code A43343, 222 amino acids (25.77 kDa).

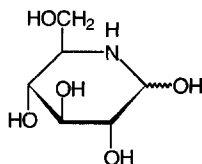
noise any undesired disturbance in the signal from an instru-



ment, or other system, that degrades the amount of useful information extractable. *Compare* **signal-to-noise ratio**.

noise analysis the technique whereby the amplitude and frequency of random fluctuations in the membrane potential of an excitable cell are used to investigate changes in conductance of single ion channels induced, e.g., by a drug or neurotransmitter.

nojirimycin 5-amino-5-deoxy-D-glucose; an antibiotic isolated from several strains of *Streptomyces*. It occurs as a pyranose-like structure with -NH- replacing the usual -O- . It inhibits intestinal α -glucosidase and pancreatic α -amylase.



nominal describing an approximate value assigned to some quantity as a guide to its real value (which may vary according to circumstances).

nomogram or **nomograph** any chart or diagram consisting of scaled lines and used for facilitating calculations, especially one consisting of three lines scaled for different related variables such that a straight line joining chosen values of two variables will intersect the third scale line at the related value of the third variable. —**nomography** *n.*; **nomographic** *adj.*

non+ 1 *prefix* denoting negation; absence of; opposite or reverse of; e.g. nonabsorbable, nonacylated, nonconductor, noncyclic, nonmetal, nonspecific. 2 *a variant form of nona+* (sometimes before a vowel).

nona+ or (sometimes before a vowel) **non+** *comb. form* denoting nine, ninefold, nine times.

nonacos+ *comb. form* denoting twenty-nine, twenty-nine fold, twenty-nine times.

nonacosane a linear C_{29} hydrocarbon, $\text{CH}_3\text{-(CH}_2\text{)}_{27}\text{-CH}_3$.

nonactin a neutral macrotetrolide antibiotic produced by several *Streptomyces* spp. It transports alkali cations through membranes with high selectivity for K^+ , but also for NH_4^+ . It is one of a related series including monactin, dinactin, and trinactin.

nonadeca+ *comb. form* denoting nineteen, nineteen fold, nineteen times.

nonaethyleneglycol mono-*n*-dodecyl ether a nonionic C_{12}E_9 detergent; see **C_xE_y**.

nonanoate or **pelargonoate** 1 the anion $\text{CH}_3\text{-(CH}_2\text{)}_7\text{-COO}^-$, derived from nonanoic acid (i.e. pelargonic acid). 2 any salt or ester of nonanoic acid.

nonanoyl or **pelargonoyl** *symbol*: Nno; the univalent acyl group $\text{CH}_3\text{-(CH}_2\text{)}_7\text{-CO-}$, derived from nonanoic acid (i.e. pelargonic acid).

nonaqueous lacking water; applied especially to solvents whether polar or nonpolar.

nonbonding electrons an alternative name for **lone pair**.

noncoded amino acid or **uncoded amino acid** any α -amino acid residue occurring in a polypeptide or protein for which no codon exists, e.g. γ -carboxyglutamic acid, hydroxylysine, hydroxyproline, oxyproline. Residues of these amino acids are formed from residues of the corresponding coded amino acids by post-translational modification of the polypeptide chain.

noncoding region or **noncoding sequence** any segment of a nucleic-acid molecule for which there is no recognized gene product; it may sometimes play a structural role in the organization of the molecule. *See also* **intron**.

noncoding strand or **template strand** a term that is variously, and confusingly, applied to either of the two strands of genomic duplex DNA. It is now the term recommended for de-

noting the strand that is complementary in its sequence of bases to messenger RNA (mRNA) transcribed from the DNA; it thus functions as the template for synthesis of mRNA, which then contains the sequence present in the **coding strand** of duplex DNA (except for uracil in RNA serving in place of thymine in DNA). In some bacteria and viruses, alternate segments of both strands of duplex DNA may be noncoding. *Other names*: antisense DNA; antisense strand; complementary strand; minus strand; transcribing strand. *Compare* **complementary DNA**.

noncompetitive antagonism (in pharmacology) the phenomenon occurring when agonist and antagonist can be bound simultaneously, in which antagonist binding reduces or prevents the action of the agonist; e.g. channel block of the nicotinic receptor, or inhibition by adrenoceptor antagonists of the response to tyramine.

noncompetitive inhibition inhibition of an enzyme in which inhibition results from binding of the inhibitor at a different site on the enzyme surface from that at which the substrate binds, without interference with substrate binding.

nonconjugative plasmid any plasmid that cannot bring about transfer of DNA by **conjugation**.

nonconservative replication of DNA or **dispersive replication of DNA** *see* **replication of DNA**.

noncoupled solute translocation the translocation of a solute across a membrane that is not coupled to the translocation of another solute, in either the same or the opposite direction. *See* **facilitated diffusion**, **uniport**.

noncovalent describing an interaction between atoms and/or molecules that does not involve the formation of a covalent bond between them.

noncyclic electron flow (in photosynthesis) *see* **photophosphorylation**.

noncyclic photophosphorylation *see* **photophosphorylation**.

nondiffusible material material that is not diffusible in the restricted sense of its being incapable of passing through a diffusion barrier, especially through a semipermeable membrane used in dialysis. *Alternative terms recommended by IUPAC*: retentate, dialysis residue, or residue.

nondissymmetric a term formerly used to describe an achiral molecule that possesses an alternating axis of symmetry but that (unlike a symmetric molecule) lacks a simple axis of symmetry; the converse of **dissymmetric**. An important example is **citric acid**, which has a one-fold alternating axis but no simple axis; thus its pair of carboxymethyl groups on carbon-3 are not geometrically equivalent. *See also* **prochiral**.

nonelectrolyte any substance that is neither an electrolyte nor a conductor of electrons.

nonessential amino acid any amino acid that can be synthesized in the body in an amount corresponding to need, and hence for which there is not an absolute dietary requirement. The extent to which a particular amino acid is nonessential in any one species depends on the stage of development and physiological state of the subject. The following L-amino acids are nonessential for the maintenance of nitrogen equilibrium in an adult human: alanine, arginine, asparagine, aspartic acid, cysteine, glutamic acid, glutamine, glycine, histidine, hydroxyproline, proline, serine, and tyrosine. Arginine and histidine are, however, essential in growing children.

nonesterified fatty acid *abbr.*: NEFA; 1 *an alternative name for free fatty acid*. 2 *a collective term for the total of free fatty acids in a sample, especially of blood*.

nonheme iron or (*esp. Brit.*) **nonhaem iron** any iron in a biological sample or structure that is not chelated in heme groups. In nonheme-iron proteins the iron is often present in iron-sulfur clusters.

nonhistone protein or **nonhistone chromosome protein** any member of a highly heterogeneous group of 10–150 kDa tissue-specific proteins associated with DNA in cell nuclei. They are thought to play a part in the control of gene expression. *See also* **high mobility group proteins**.

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nonhomologous

nonhomologous (of chromosomes or chromosome segments) not **homologous** (def. 4).

nonideal describing any gas or solution that does not behave like an ideal gas or ideal solution, respectively. —**nonideality** *n*.

Nonidet the proprietary name for a range of nonionic detergents that are essentially octyl- or nonyl-phenoxy-polyethoxyethanols. For example, see **NP-40**. Compare **Triton**.

non-insulin-dependent diabetes an alternative name for type 2 diabetes mellitus.

noninvasive 1 not **invasive**. 2 (of a clinical investigative technique or procedure) carried out without penetrating the skin.

nonionic having or yielding no ions; the term is used especially of **nonionic detergents**.

nonionic detergent any detergent whose molecules cannot ionize.

non-lactose-fermenter a jargon term for any genus, species, or strain of microorganism that does not ferment lactose.

nonmediated transport an alternative name for **unmediated transport**.

non-Newtonian fluid any fluid that does not conform to **Newton's law of viscosity**, i.e. one whose (coefficient of) viscosity is not independent of flow rate.

nonose any **aldose** having a chain of nine carbon atoms in the molecule.

nonoverlapping code see **overlapping code**.

nonpermissive restrictive; not **permissive**.

nonpermissive cell any cell that does not support infection by a particular lytic virus; in a small percentage of instances the virus may transform a nonpermissive cell.

nonpolar 1 lacking polarity. 2 or **apolar** (of a chemical group or substance) lacking a significant permanent dipole moment.

nonpolar bond 1 any covalent bond that is not appreciably polarized. 2 (sometimes) an alternative term for **hydrophobic interaction**.

nonpolar solvent a loose term denoting any solvent with low ionizing power and hence low solvating power.

nonproductive binding or **dystopic binding** the binding of a substrate in an unreactive mode at the active site of an enzyme, leading to the formation of a nonproductive, or **abortive complex**.

nonproductive complex an alternative name for **abortive complex**.

nonprotein amino acid any amino acid that does not naturally occur as a constituent residue of proteins. Over 200 such amino acids are known to be produced in higher plants. Although they often contribute to the nitrogen pool of the plants, their specific function is unknown. Nonprotein amino acids also occur in animals but to much less degree, the most prominent examples being **argininosuccinate**, **citrulline**, and **ornithine**.

nonprotein nitrogen *abbr.*: **NPN**; the part of the nitrogen content of a substance (or fluid) that is not contained in the proteins present. The term is used especially of blood or serum, wherein urea accounts for about half the total **NPN**.

nonradioactive having no radioactivity; in practice the term is often used to refer to substances having a normal or natural nuclidic composition, ignoring any contribution from minimal proportions of naturally occurring radionuclides such as the carbon-14 content of natural organic materials.

nonradioactive tracer (sometimes) a labelled substance that can be detected and estimated by virtue of some property other than radioactivity (such as colour, fluorescence, or a difference in stable-isotope ratio).

nonreducing end the end of a linear oligosaccharide or polysaccharide that does not carry a potential hemiacetal or hemiketal (i.e. reducing) group.

nonsecretor see **secretor gene**.

nonselectively labelled describing a labelled compound in which the position(s) and the number of the labelling nucleide(s) are both undefined.

nonsense codon or **nonsense triplet** any one of the nucleotide triplets **UAA** (ochre), **UAG** (amber), and **UGA** (opal) that sig-

nal termination of translation and release of the polypeptide chain from the site of synthesis. A codon that is nonsense in one strain of an organism may nevertheless code for an amino acid in another strain containing an appropriate suppressor gene (see **nonsense suppressor**). **UGA** sometimes codes for selenocysteine and, in mitochondria, for Trp. In mitochondria **AGA** is a nonsense codon rather than **UGA**. Compare **missense codon**.

nonsense mutation any mutation in a DNA molecule that results in an alteration of a sense codon to a **nonsense codon** in the mRNA transcript, the translation of which leads to the formation of a truncated polypeptide gene product. Compare **missense mutation**.

nonsense suppressor any tRNA molecule that is able to insert an amino-acid residue into a growing polypeptide chain at a nonsense codon in an mRNA molecule. It has an altered anticodon, produced as a result of a **suppressor mutation**.

nonshivering thermogenesis *abbr.*: **NST**; a mechanism of rapid heat production in response to cold that does not involve muscular contractions. It is important during the neonatal period, during cold adaptation of a number of small mammals, and during the arousal period of animals that hibernate. Heat is produced in the mitochondria of brown adipose tissue in response to the presence in the inner mitochondrial membrane of a specific 32 kDa protein. This protein uncouples the respiratory chain and is inducible by norepinephrine. It is known as **brown fat uncoupling protein**.

nonsuppressible insulin-like activity *abbr.*: **NSILA**; **insulin-like activity** that is not suppressible by specific antisera to insulin. The activity in serum was once thought to be in two protein fractions: a minor component (*abbr.*: **NSILA_s**), soluble in acidic aqueous ethanol; and a major component (*abbr.*: **NSILA_p**), precipitable by acidic aqueous ethanol and now known to be an artefact. The activity is now attributed to **insulin-like growth factors**.

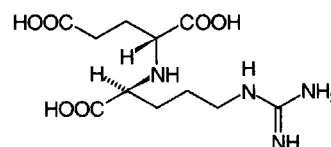
nontranscribing strand (of duplex DNA) see **coding strand**.

nonulose any **ketose** having a chain of nine carbon atoms in the molecule.

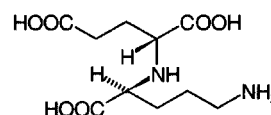
nonyl *symbol*: **Nn**; the alkyl group $\text{CH}_3(\text{CH}_2)_7\text{CH}_2-$, derived from nonane.

nonyl glucoside *n*-nonyl- β -D-glucopyranoside; a mild, non-ionic detergent, CMC 6.5 mM. Compare **heptyl glucoside**, **hexyl glucoside**, **octyl glucoside**.

nopaline *N*-(1-carboxy-4-guanidinobutyl)glutamic acid; *N*²-(1,3-dicarboxypropyl)arginine; a rare amino-acid derivative that is produced by a certain type of crown-gall tissue (see **crown-gall disease**); one of a group of **opines**. The genes responsible for the synthesis of nopaline are part of the **T-DNA** from a **Ti-plasmid**.



nopalinic acid or **ornaline** *N*-(1-carboxy-4-aminobutyl)glutamic acid; *N*²-(1,3-dicarboxypropyl)ornithine; one of a group of **opines** found in plant tumours occurring in **crown-gall disease**.



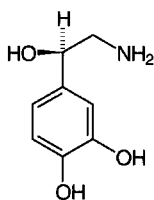
nor+

nor+ *prefix (in chemical nomenclature)* denoting 1 replacement of one side-chain methyl group by a hydrogen atom, e.g. norepinephrine, 19-nortestosterone. Replacement of two or three methyl groups is denoted by the prefix dinor+ or trinor+, respectively (in terpene nomenclature the prefix nor+ formerly denoted removal of all replaceable methyl groups; hence, e.g., 8,9,10-trinorbornane was formerly known as norbornane). 2 elimination of one methylene group from a chain or ring of carbon atoms; it is used especially for steroids, e.g. 23-nor-5 β -cholanoic acid, A-nor-5 β -androstande, norophthalmic acid. Elimination of two or three methylene groups is denoted by the prefix dinor+ or trinor+, respectively. 3 an isomer with an unbranched hydrocarbon chain, e.g. norleucine, norvaline; use discouraged.

noradrenaline *an alternative name for norepinephrine.*

noradrenergic (*sometimes*) 1 an (adrenergic) nerve cell that itself liberates noradrenaline (i.e. norepinephrine). 2 a cell or receptor that is stimulated preferentially by norepinephrine. —**noradrenergically** *adv.*

norepinephrine or **noradrenaline** or **levarterenol** (–)-(3,4-dihydroxyphenyl)-2-aminoethanol; a hormone secreted by the adrenal medulla, and a neurotransmitter in the sympathetic peripheral nervous system and in some tracts in the central nervous system. It is also the demethylated biosynthetic precursor of epinephrine in the adrenal medulla. It has a powerful lipolytic action, mediated by cyclic AMP, and is an agonist for **adrenoceptors**, inducing arteriolar constriction and raised systolic and diastolic arterial blood pressure.



R(–)-enantiomer

Norit *the proprietary name for a commercially produced activated carbon, useful as an adsorbent.*

norleucine *symbol:* Ahx or (formerly) Nle; 2-aminoheptanoic acid; a nonprotein α -amino acid. *See also* nor+ (def. 3).

norm+ *a variant form of normo+ (before a vowel).*

normal 1 not deviating from an established pattern; approximating to an average, or within the usual range (of values). 2 (*in organic chemistry*) *symbol:* N; (of a solution) containing one **equivalent** (def. 4) of a specified substance (of specified equivalence factor) in one litre of solution. Use now not recommended. 3 (*in organic chemistry*) a chemical compound that has an unbranched chain of carbon atoms; the names of such compounds may be prefixed by 'n', as in *n*-butane. However, this is not the currently recommended terminology. 4 (*in biology*) not diseased or not having been exposed to an experimental procedure. 5 (*in mathematics*) (of a straight line or plane) at a right angle (to another straight line or plane).

normal distribution or **Gaussian distribution** any continuous frequency distribution of events, observations, etc. characterized by a symmetrical bell-shaped curve of the frequency of x , $f(x)$, against the value of x , that is represented by the equation:

$$f(x) = [1/\sigma(2\pi)^{0.5}]e^{-(x-\mu)^2/2\sigma^2}$$

where μ is the mean and σ is the standard deviation.

normality (*in chemistry*) the chemical concentration of a solution expressed as the number of **equivalents** (def. 4) of a specified substance (of specified equivalence factor) per litre. Use now not recommended.

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normal range (*in clinical chemistry*) the range of values of a particular characteristic within which 95 percent of observations made on presumed normal healthy individuals fall. It is useful as a reference in the investigation of disease.

normal saline *an alternative and less suitable name for physiological saline.*

normal solution a solution in which the amount-of-substance concentration of the **equivalent** (def. 3) of the reagent is one mole per litre; the equivalence factor of the reagent should be specified. Not now recommended. *See also* normal (def. 2).

normal temperature and pressure *abbr.:* NTP or N.T.P. or ntp; *a former term for standard temperature and pressure.*

normetadrenaline *an alternative name for normetanephrine.*

normetanephrine or **normetadrenaline** 3-*O*-methylnorepinephrine; an inactive metabolite of norepinephrine produced by the action of **catechol *O*-methyltransferase**.

normo+ or (*before a vowel*) **norm+** *comb. form* denoting normal; used especially in biology and medicine to indicate within the normal range. *See also* eu+.

normoblast a normal precursor cell to an erythrocyte, characterized by condensation of the nucleus into a homogeneous densely staining body. Normoblasts are normally found only in hemopoietic tissue. —**normoblastic** *adj.*

normocyte an erythrocyte of normal colour, shape, and size. —**normocytic** *adj.*

normoglycemia or (*esp. Brit.*) **normoglycaemia** *an alternative term for euglycemia.* —**normoglycemic** or (*esp. Brit.*) **normoglycaemic** *adj.*

normotensive having an arterial blood pressure within the normal range for the particular group to which the subject belongs.

Norrish, Ronald George Wreyford (1897–1978), British physical chemist; Nobel Laureate in Chemistry (1967) jointly with G. Porter, the prize being shared with M. Eigen 'for their studies of extremely fast chemical reactions, effected by disturbing the equilibrium by means of very short pulses of energy'.

norsteroid any steroid that, relative to a parent compound, lacks either one axial methyl group or one methylene group in a side chain, or that has one ring contracted by one carbon atom. *See also* nor+.

Northern blotting or **Northern transfer** any **blotting** in which the primary electrophoresis is performed with RNA, which is subsequently hybridized to radioactive DNA. The term was coined to distinguish the process from **Southern blotting**, which is performed with DNA.

Northrop, John Howard (1891–1987), US biochemist and protein chemist; Nobel Laureate in Chemistry (1946) jointly with W. M. Stanley 'for their preparation of enzymes and virus proteins in a pure form' [prize shared with J. B. Sumner].

norvaline *symbol:* Avl (from aminovaleric) or Ape (from aminopentanoic) or (formerly) Nva; 2-aminovaleric acid (now the preferred trivial name); 2-aminopentanoic acid; a nonprotein α -amino acid. *See also* nor+ (def. 3).

NOS *abbr. for nitric-oxide synthase.*

noso+ or (*before a vowel*) **nos+** *comb. form* indicating disease.

nosocomial 1 associated with a hospital. 2 a disorder (e.g. an infection) acquired during a hospital stay that is unrelated to the primary condition of the patient. A nosocomial infection can refer to hospital workers as well.

nosogenesis *an alternative term for pathogenesis.* —**nosogenic** or **nosogenetic** *adj.*

nosography a written systematic classification and description of diseases.

nosology the branch of medicine dealing with the classification of diseases.

notatin *a name sometimes still used for glucose oxidase.* Notatin was so called because it is produced in considerable quantity by some microfungi, e.g. *Penicillium notatum*.

notch a gene, found in *Drosophila*, that encodes a protein essential for the proper differentiation of ectoderm. Expression of the *notch* gene is under the control of neurogenic genes. The

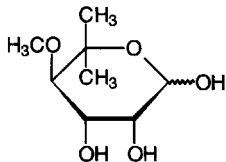


notexin

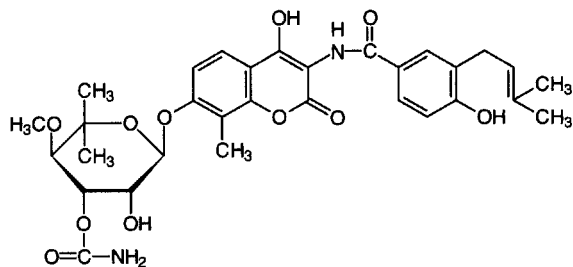
gene product is a transmembrane protein with 36 EGF-like repeats; database code NOTC_DROME, 2703 amino acids (288.81 kDa).

notexin a phospholipase A₂ (EC 3.1.1.4) from snake venom that acts as a presynaptic neurotoxin. The reaction is Ca²⁺-dependent. Example (precursor) from the Australian tiger snake, *Notechis scutatus scutatus*: database code PA2_3_NOTSC, 119 amino acids (13.66 kDa); 3-D structure known.

noviose 6-deoxy-5-C-methyl-L-lyxo-hexose 4-methyl ether; a carbohydrate component of **novobiocin**.



novobiocin an antibiotic produced by *Streptomyces spheroides*, *S. niveus*, and certain other species; it is active against Gram-positive and some Gram-negative bacteria. In susceptible organisms it is a specific inhibitor of (DNA) gyrase, interacting with the B subunit, and thereby interfering with DNA synthesis.



noxa (*pl. noxae*) anything that is harmful to living organisms. —noxious *adj.*

Np symbol for neptunium.

NP-40 polyethyleneglycol-*p*-isooctylphenyl ether; a polyoxyethylene nonionic detergent, CMC 0.05–0.3 mM. *Proprietary name*: Nonidet P40.

NPC *abbr.* for neural precursor cell (see **Nedd**).

NPN *abbr.* for nonprotein nitrogen.

NPY *abbr.* for neuropeptide Y.

nRNA *abbr.* for nuclear RNA.

NR protein *abbr.* for nitrogen regulatory protein.

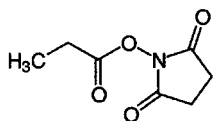
ns symbol for nanosecond (i.e. 10⁻⁹ s).

NSF *abbr.* for National Science Foundation (of the USA).

NSF protein *abbr.* for N-ethylmaleimide-sensitive fusion protein.

NSILA *abbr.* for nonsuppressible insulin-like activity.

NSP *abbr.* for N-succinimidyl propionate, a propionylating agent.



NST *abbr.* for nonshivering thermogenesis.

NT *abbr.* for neurotrophin.

N terminus or sometimes **N terminal** the end of any peptide chain in any (poly)peptide or protein at which the 2-amino function of a constituent α -amino acid (or the 2-imino function of a constituent α -keto acid) is attached. *Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:*

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tion of an α -imino carboxylic acid) is not attached in peptide linkage to another amino-acid residue. This function may bear an acyl group or other function in nonpeptide linkage; when it is a free amino group (and also protonated and positively charged at neutral or acidic pH values), the chain end may then be alternatively termed the **amino terminus** (or NH₂ terminus) or **amino terminal** (or NH₂ terminal). However, 'amino terminus' is often used loosely for any N terminus. *Compare C terminus.* —**N-terminal** *adj.*

NTP 1 *abbr.* for any (unknown or unspecified) nucleoside (5'-)triphosphate. 2 (in clinical biochemistry) *abbr.* for 5'-nucleotidase (EC 3.1.3.5). 3 or N.T.P. or ntp; *abbr.* for normal temperature and pressure; stp (i.e. standard temperature and pressure) now used.

NTPase *abbr.* for nucleoside triphosphatase.

NTP polymerase see **polynucleotide adenyltransferase**.

nu symbol: ν (lower case) or N (upper case); the thirteenth letter of the Greek alphabet. For uses see **Appendix A**.

nu body or **ν body** or **deoxyribosome** a type of spheroid particle seen on electron-microscopic examination of chromatin; now thought to be the same as a **nucleosome**.

Nuc symbol for an undetermined, unidentified, or unspecified nucleoside residue; alternative to **N**.

nucleo+ a variant for **nucleo+** (before a vowel).

nuclear 1 of, pertaining to, or contained within the nucleus of a cell. 2 of, pertaining to, or concerning an atomic nucleus. 3 of, pertaining to, or operated by energy derived from fission or fusion of atomic nuclei.

nuclear area an alternative name for **nucleoid** (def. 2).

nuclear atom or **central atom** (of a coordination entity) the atom to which all other atoms are directly attached.

nuclear body an alternative name for **nucleoid** (def. 2).

nuclear division or **nuclear fission** the division of the nucleus of a cell, especially by **mitosis** or **meiosis**.

nuclear emulsion a fine-grained photographic emulsion designed for recording the tracks of subatomic particles within it.

nuclear envelope or **nuclear membrane** the membrane that surrounds the nucleus of a eukaryotic cell. It is a double membrane composed of two lipid bilayers separated by a gap of width 20–40 nm (the perinuclear space). The outer nuclear membrane is continuous with the endoplasmic reticulum of the cell and is sometimes studded with ribosomes. At specialized regions (**nuclear pores**) the outer nuclear membrane is connected to the inner nuclear membrane.

nuclear filament-related protein *abbr.*: NUF1; an essential component of the nucleoskeleton with a role in crosslinking filaments or anchoring other molecules. Example from yeast: database code NUF1_YEAST, 944 amino acids (111.66 kDa).

nuclear fission 1 (in biology) an alternative term for **nuclear division**. 2 (in physics) see **fission** (def. 1).

nuclear fraction an operational term for the material deposited by low-speed centrifugation of a homogenate of eukaryotic cells or tissue. It consists predominantly of nuclei together with unbroken cells.

nuclear fusion 1 (in biology) the **fusion** (def. 3) of the nuclei of a heterokaryon leading to the formation of a hybrid cell. 2 (in physics) see **fusion** (def. 4).

nuclear isomer (in physics) see **isomer** (def. 2).

nuclear lamina the electron-dense layer lying on the nucleoplasmic side of the inner membrane of a cell nucleus. The polypeptides of the lamina are thought to be concerned in the dissolution of the nuclear envelope and its re-formation during mitosis.

nuclear localization signal *abbr.*: NLS; a sequence of amino acids seven to nine residues long within a protein, normally at sites away from the N or C terminus, that acts as a signal for the localization of the protein within the nucleus. Characteristically, NLSs are rich in basic residues, but include also one or more prolines.

nuclear magnetic resonance *abbr.*: NMR or nmr; the phenomenon that occurs when atomic nuclei that possess a magnetic moment are placed in a magnetic field.